From Cantor to Ito

Combined definition and theorem reference

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Mathematical Logic and Proof

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0.4 Syntax

Definition 0.4.1 (Truth Value)**.** The set of truth values is

$$\mathbb{B} := \{\text{False}, \text{True}\}.$$

Some sources identify this set with $\{0, 1\}$, with 0 read as false and 1 read as true.

Definition 0.4.2 (Propositional Language)**.** A propositional language consists of:

1. a set Prop of propositional variables;
2. logical connectives;
3. punctuation symbols such as parentheses;
4. recursive formation rules specifying the well-formed formulas.

Definition 0.4.3 (Propositional Variable)**.** A propositional variable is an atomic symbol intended to stand for a proposition. A standard supply of variables is

$$\mathsf{Prop} = \{P_1, P_2, P_3, \ldots\}.$$

Informally, variables are often written $P, Q, R, S, \ldots$.

Definition 0.4.4 (Logical Connective)**.** The standard primitive propositional connectives are

$$\neg, \qquad \wedge, \qquad \vee, \qquad \to, \qquad \leftrightarrow .$$

The connective $\neg$ is unary. The connectives $\wedge, \vee, \to, \leftrightarrow$ are binary.

Symbol	Name	Arity	Formation pattern	Reading
$\neg$	negation	unary	$\neg\varphi$	not φ
$\wedge$	conjunction	binary	$(\varphi \wedge \psi)$	φ and ψ
$\vee$	disjunction	binary	$(\varphi \vee \psi)$	φ or ψ
$\to$	conditional	binary	$(\varphi \to \psi)$	if φ, then ψ
$\leftrightarrow$	biconditional	binary	$(\varphi \leftrightarrow \psi)$	φ iff ψ

Definition 0.4.5 (Well-Formed Formula)**.** The set WFF of well-formed formulas is the smallest set of strings satisfying the following formation rules:

1. Every propositional variable $P \in \mathsf{Prop}$ belongs to WFF.
2. If $\varphi \in \mathsf{WFF}$, then $\neg\varphi \in \mathsf{WFF}$.
3. If $\varphi, \psi \in \mathsf{WFF}$ and $\circ \in \{\wedge, \vee, \to, \leftrightarrow\}$, then $(\varphi \circ \psi) \in \mathsf{WFF}$.
4. No string is a well-formed formula unless it is obtained by finitely many applications of the previous clauses.

Lemma 0.4.6 (Closure Under Formation)**.** *The set* WFF *is closed under the propositional formation rules. That is, if* $\varphi \in \mathsf{WFF}$*, then*

$$\neg\varphi \in \mathsf{WFF}.$$

If $\varphi, \psi \in \mathsf{WFF}$ *and*

$$\circ \in \{\wedge, \vee, \to, \leftrightarrow\},$$

then

$$(\varphi \circ \psi) \in \mathsf{WFF}.$$

Go to proof.

Theorem 0.4.7 (Minimality of Well-Formed Formulas)**.** *Let S be a set of strings over the propositional alphabet. Suppose:*

1. $\mathsf{Prop} \subseteq S$*;*
2. *if* $\varphi \in S$*, then* $\neg\varphi \in S$*;*
3. *if* $\varphi, \psi \in S$ *and* $\circ \in \{\wedge, \vee, \to, \leftrightarrow\}$*, then* $(\varphi \circ \psi) \in S$*.*

Then

$$\mathsf{WFF} \subseteq S.$$

Go to proof.

Definition 0.4.8 (Atomic Formula)**.** An atomic formula is a well-formed formula that is a propositional variable.

Definition 0.4.9 (Molecular Formula)**.** A molecular formula is a well-formed formula that is not atomic. Equivalently, it is a well-formed formula formed using at least one logical connective.

Definition 0.4.10 (Main Connective)**.** The main connective of a molecular formula is the connective introduced at the final formation step:

1. for $\neg\varphi$, the main connective is $\neg$;
2. for $(\varphi \circ \psi)$, the main connective is $\circ$.

Atomic formulas have no main connective.

Theorem 0.4.11 (Structural Induction for Propositional Formulas)**.** *Let $A(\varphi)$ be a property of well-formed formulas. Suppose:*

1. $A(P)$ *holds for every* $P \in \mathsf{Prop}$*;*
2. *if* $A(\varphi)$ *holds, then* $A(\neg\varphi)$ *holds;*
3. *if* $A(\varphi)$ *and* $A(\psi)$ *hold, then* $A((\varphi \circ \psi))$ *holds for every* $\circ \in \{\wedge, \vee, \to, \leftrightarrow\}$*.*

Then $A(\theta)$ holds for every $\theta \in \mathsf{WFF}$*.*
Go to proof.

Theorem 0.4.12 (Structural Recursion for Propositional Formulas)**.** *To define a function F on* WFF*, it suffices to specify:*

1. *$F(P)$ for every $P \in \mathsf{Prop}$;*
2. *$F(\neg\varphi)$ in terms of $F(\varphi)$;*
3. *$F((\varphi \circ \psi))$ in terms of $F(\varphi)$, $F(\psi)$, and $\circ$.*

When these clauses respect the formation rules, they determine a unique function on WFF*.*
Go to proof.

Lemma 0.4.13 (Constructor Disjointness for Propositional Formulas)**.** *The three outer formation cases for propositional formulas are disjoint:*

1. *no propositional variable is a negation formula;*
2. *no propositional variable is a binary formula;*
3. *no negation formula is a binary formula.*

Go to proof.

Lemma 0.4.14 (Constructor Injectivity for Propositional Formulas)**.** *The propositional formula constructors are injective:*

1. *if $\neg\varphi = \neg\psi$, then $\varphi = \psi$;*
2. *if $(\varphi \circ \psi) = (\alpha \star \beta)$, then $\varphi = \alpha$, $\psi = \beta$, and $\circ = \star$.*

Go to proof.

Theorem 0.4.15 (Unique Decomposition for Propositional Formulas)**.** *Every $\varphi \in \mathsf{WFF}$ has exactly one outermost form:*

1. *$\varphi = P$ for a unique $P \in \mathsf{Prop}$;*
2. *$\varphi = \neg\psi$ for a unique $\psi \in \mathsf{WFF}$;*
3. *$\varphi = (\psi \circ \chi)$ for unique $\psi, \chi \in \mathsf{WFF}$ and a unique binary connective $\circ \in \{\wedge, \vee, \rightarrow, \leftrightarrow\}$.*

Go to proof.

Definition 0.4.16 (Parse Tree)**.** The parse tree, or syntax tree, of a well-formed formula is the tree representation of its recursive construction:

1. a propositional variable is represented by a single leaf;
2. a formula $\neg\varphi$ is represented by a root labeled $\neg$ with one child, the parse tree of φ;
3. a formula $(\varphi \circ \psi)$ is represented by a root labeled $\circ$ with two children, the parse trees of φ and ψ.

Theorem 0.4.17 (Unique Syntax Tree for Propositional Formulas)**.** *Every propositional well-formed formula has a unique parse tree.*
Go to proof.

Definition 0.4.18 (Variable Set of a Formula)**.** The variable set $\mathrm{Var}(\varphi)$ of a formula φ is defined recursively:

1. if $\varphi = P \in \mathsf{Prop}$, then $\mathrm{Var}(\varphi) = \{P\}$;
2. if $\varphi = \neg\psi$, then $\mathrm{Var}(\varphi) = \mathrm{Var}(\psi)$;

3. if $\varphi = (\psi \circ \chi)$, then
$$\mathrm{Var}(\varphi) = \mathrm{Var}(\psi) \cup \mathrm{Var}(\chi).$$

Definition 0.4.19 (Subformula)**.** The set $\mathrm{Sub}(\varphi)$ of subformulas of a formula φ is defined recursively:

1. if φ is atomic, then $\mathrm{Sub}(\varphi) = \{\varphi\}$;
2. if $\varphi = \neg\psi$, then
$$\mathrm{Sub}(\varphi) = \{\varphi\} \cup \mathrm{Sub}(\psi);$$
3. if $\varphi = (\psi \circ \chi)$, then
$$\mathrm{Sub}(\varphi) = \{\varphi\} \cup \mathrm{Sub}(\psi) \cup \mathrm{Sub}(\chi).$$

Definition 0.4.20 (Proper Subformula)**.** Let $\varphi \in \mathsf{WFF}$. A *proper subformula* of φ is a formula ψ such that
$$\psi \in \mathrm{Sub}(\varphi) \quad \text{and} \quad \psi \neq \varphi.$$

Definition 0.4.21 (Formula Depth)**.** The depth of a formula φ, denoted $\mathrm{depth}(\varphi)$, is defined recursively:

1. if φ is atomic, then $\mathrm{depth}(\varphi) = 0$;
2. if $\varphi = \neg\psi$, then $\mathrm{depth}(\varphi) = \mathrm{depth}(\psi) + 1$;
3. if $\varphi = (\psi \circ \chi)$, then
$$\mathrm{depth}(\varphi) = \max\{\mathrm{depth}(\psi), \mathrm{depth}(\chi)\} + 1.$$

Definition 0.4.22 (Connective Count)**.** The connective count of a formula φ, denoted $\mathrm{conn}(\varphi)$, is defined recursively:

1. if φ is atomic, then $\mathrm{conn}(\varphi) = 0$;
2. if $\varphi = \neg\psi$, then $\mathrm{conn}(\varphi) = \mathrm{conn}(\psi) + 1$;
3. if $\varphi = (\psi \circ \chi)$, then
$$\mathrm{conn}(\varphi) = \mathrm{conn}(\psi) + \mathrm{conn}(\chi) + 1.$$

Lemma 0.4.23 (Proper Subformula Depth)**.** *If ψ is a proper subformula of φ, then*
$$\mathrm{depth}(\psi) < \mathrm{depth}(\varphi).$$

Go to proof.

Lemma 0.4.24 (Finiteness of Variables in a Formula)**.** *For every $\varphi \in \mathsf{WFF}$, the set $\mathrm{Var}(\varphi)$ is finite.*
Go to proof.

Lemma 0.4.25 (Finiteness of Subformulas)**.** *For every $\varphi \in \mathsf{WFF}$, the set $\mathrm{Sub}(\varphi)$ is finite.*
Go to proof.

0.5 Semantics

Truth Assignments

Definition 0.5.1 (Truth Assignment)**.** A *truth assignment* is a function
$$v : \mathsf{Prop} \to \mathbb{B}.$$

For each propositional variable $P \in \mathsf{Prop}$, the value $v(P)$ is the truth value assigned to P.

Definition 0.5.2 (Domain of a Truth Assignment)**.** The *domain* of a truth assignment
$$v : \mathsf{Prop} \to \mathbb{B}$$

is the set Prop of propositional variables.

Extension of a Truth Assignment

Definition 0.5.3 (Extension of a Truth Assignment to Formulas)**.** Let

$$v : \mathsf{Prop} \to \mathbb{B}$$

be a truth assignment. The *extension* of v to well-formed formulas is the function

$$\widehat{v} : \mathsf{WFF} \to \mathbb{B}$$

defined recursively as follows:

$$\widehat{v}(P) = v(P) \quad \text{for every } P \in \mathsf{Prop},$$
$$\widehat{v}(\neg\varphi) = \mathsf{True} \iff \widehat{v}(\varphi) = \mathsf{False},$$
$$\widehat{v}(\varphi \wedge \psi) = \mathsf{True} \iff \widehat{v}(\varphi) = \mathsf{True} \text{ and } \widehat{v}(\psi) = \mathsf{True},$$
$$\widehat{v}(\varphi \vee \psi) = \mathsf{True} \iff \widehat{v}(\varphi) = \mathsf{True} \text{ or } \widehat{v}(\psi) = \mathsf{True},$$
$$\widehat{v}(\varphi \to \psi) = \mathsf{False} \iff \widehat{v}(\varphi) = \mathsf{True} \text{ and } \widehat{v}(\psi) = \mathsf{False},$$

and

$$\widehat{v}(\varphi \leftrightarrow \psi) = \mathsf{True} \iff \widehat{v}(\varphi) = \widehat{v}(\psi).$$

Theorem 0.5.4 (Unique Extension of a Truth Assignment)**.** *Every truth assignment*

$$v : \mathsf{Prop} \to \mathbb{B}$$

has a unique extension

$$\widehat{v} : \mathsf{WFF} \to \mathbb{B}$$

satisfying the recursive truth clauses for propositional variables and the primitive connectives.
Go to proof.

Definition 0.5.5 (Satisfaction of a Formula)**.** Let v be a truth assignment and let $\varphi \in \mathsf{WFF}$. We say that v satisfies φ, written

$$v \models \varphi,$$

if

$$\widehat{v}(\varphi) = \text{True}.$$

Definition 0.5.6 (Satisfaction of a Set of Formulas)**.** Let $\Gamma \subseteq \mathsf{WFF}$. A truth assignment v satisfies Γ, written

$$v \models \Gamma,$$

if $v \models \gamma$ for every $\gamma \in \Gamma$.

Definition 0.5.7 (Truth Table)**.** Let $\varphi \in \mathsf{WFF}$ and suppose the variables occurring in φ are among $P_1, \ldots, P_n$. A truth table for φ is a finite table with one row for each assignment

$$a : \{P_1, \ldots, P_n\} \to \mathbb{B}$$

and a final column recording the value of $\widehat{v}(\varphi)$ for any truth assignment v extending a.

Definition 0.5.8 (Tautology)**.** A formula $\varphi \in \mathsf{WFF}$ is a tautology, written

$$\models \varphi,$$

if every truth assignment satisfies φ.

Definition 0.5.9 (Contradiction)**.** A formula $\varphi \in \mathsf{WFF}$ is a contradiction if no truth assignment satisfies φ.

Definition 0.5.10 (Satisfiable Formula)**.** A formula $\varphi \in \mathsf{WFF}$ is satisfiable if at least one truth assignment satisfies it.

Definition 0.5.11 (Contingency)**.** A formula $\varphi \in \mathsf{WFF}$ is a contingency if it is satisfiable but not a tautology.

Definition 0.5.12 (Logical Consequence)**.** Let $\Gamma \subseteq \mathsf{WFF}$ and $\varphi \in \mathsf{WFF}$. The formula φ is a logical consequence of Γ, written

$$\Gamma \models \varphi,$$

if every truth assignment satisfying Γ also satisfies φ.

Definition 0.5.13 (Logical Equivalence)**.** Formulas $\varphi, \psi \in \mathsf{WFF}$ are logically equivalent, written

$$\varphi \equiv \psi,$$

if they have the same truth value under every truth assignment.

Lemma 0.5.14 (Coincidence Lemma for Propositional Logic)**.** *Let $v, w : \mathsf{Prop} \to \mathbb{B}$ be truth assignments. If v and w agree on every propositional variable occurring in φ, then they assign the same truth value to φ:*

$$v|_{\mathrm{Var}(\varphi)} = w|_{\mathrm{Var}(\varphi)} \Longrightarrow \widehat{v}(\varphi) = \widehat{w}(\varphi).$$

Go to proof.

Theorem 0.5.15 (Finite Truth-Table Theorem)**.** *Every propositional formula has a finite truth table. More precisely, if $\mathrm{Var}(\varphi)$ has n elements, then φ has a truth table with 2^n rows.*
Go to proof.

Definition 0.5.16 (Unary Boolean Function)**.** A unary Boolean function is a function

$$f : \mathbb{B} \to \mathbb{B}.$$

Definition 0.5.17 (Binary Boolean Function)**.** A binary Boolean function is a function

$$f : \mathbb{B}^2 \to \mathbb{B},$$

equivalently a function

$$f : \mathbb{B} \times \mathbb{B} \to \mathbb{B}.$$

Definition 0.5.18 (Boolean Function)**.** An n-ary Boolean function is a function

$$f : \mathbb{B}^n \to \mathbb{B}.$$

Definition 0.5.19 (Truth Function Induced by a Formula)**.** Let $\varphi \in \mathsf{WFF}$, and suppose $\mathrm{Var}(\varphi) \subseteq \{P_1, \ldots, P_n\}$. The truth function induced by φ is the Boolean function

$$f_\varphi : \mathbb{B}^n \to \mathbb{B}$$

defined by

$$f_\varphi(b_1, \ldots, b_n) = \widehat{v}(\varphi),$$

where $v(P_i) = b_i$ for each i.

Theorem 0.5.20 (Counting Boolean Functions)**.** *There are exactly*

$$2^{2^n}$$

Boolean functions $\mathbb{B}^n \to \mathbb{B}$.
Go to proof.

0.6 Algebra of Propositions

Definition 0.6.1 (Propositional Equivalence Law)**.** A propositional equivalence law is a valid schema of the form

$$\varphi \equiv \psi,$$

with $\varphi, \psi \in \mathsf{WFF}$. Each instance of the schema says that the two formulas have the same truth value under every truth assignment.

Theorem 0.6.2 (Logical Equivalence is an Equivalence Relation)**.** *Logical equivalence is reflexive, symmetric, and transitive on* WFF*:*

$$\varphi \equiv \varphi, \qquad \varphi \equiv \psi \Longrightarrow \psi \equiv \varphi,$$

and

$$(\varphi \equiv \psi \wedge \psi \equiv \chi) \Longrightarrow \varphi \equiv \chi.$$

Go to proof.

Theorem 0.6.3 (Basic Boolean Equivalence Laws)**.** *For all formulas* $\varphi, \psi, \chi \in \mathsf{WFF}$*, the following logical equivalences hold.*
Commutativity:

$$\varphi \wedge \psi \equiv \psi \wedge \varphi, \qquad \varphi \vee \psi \equiv \psi \vee \varphi.$$

Associativity:

$$(\varphi \wedge \psi) \wedge \chi \equiv \varphi \wedge (\psi \wedge \chi), \qquad (\varphi \vee \psi) \vee \chi \equiv \varphi \vee (\psi \vee \chi).$$

Idempotence:

$$\varphi \wedge \varphi \equiv \varphi, \qquad \varphi \vee \varphi \equiv \varphi.$$

Double negation:

$$\neg\neg\varphi \equiv \varphi.$$

Go to proof.

Theorem 0.6.4 (Identity and Domination Laws)**.** *Let* $\top$ *be any tautological formula and let* $\bot$ *be any contradictory formula. Then for every* $\varphi \in \mathsf{WFF}$*:*

$$\varphi \wedge \top \equiv \varphi, \qquad \varphi \vee \bot \equiv \varphi,$$

and

$$\varphi \wedge \bot \equiv \bot, \qquad \varphi \vee \top \equiv \top.$$

Go to proof.

Theorem 0.6.5 (De Morgan Laws for Propositional Logic)**.** *For all* $\varphi, \psi \in \mathsf{WFF}$*,*

$$\neg(\varphi \wedge \psi) \equiv \neg\varphi \vee \neg\psi,$$

and

$$\neg(\varphi \vee \psi) \equiv \neg\varphi \wedge \neg\psi.$$

Go to proof.

Theorem 0.6.6 (Distributive Laws for Propositional Logic)**.** *For all* $\varphi, \psi, \chi \in \mathsf{WFF}$*,*

$$\varphi \wedge (\psi \vee \chi) \equiv (\varphi \wedge \psi) \vee (\varphi \wedge \chi),$$

and

$$\varphi \vee (\psi \wedge \chi) \equiv (\varphi \vee \psi) \wedge (\varphi \vee \chi).$$

Go to proof.

Theorem 0.6.7 (Absorption Laws)**.** *For all* $\varphi, \psi \in \mathsf{WFF}$*,*

$$\varphi \wedge (\varphi \vee \psi) \equiv \varphi,$$

and

$$\varphi \vee (\varphi \wedge \psi) \equiv \varphi.$$

Go to proof.

Theorem 0.6.8 (Conditional Equivalence Laws)**.** *For all* $\varphi, \psi \in \mathsf{WFF}$*,*

$$\varphi \to \psi \equiv \neg\varphi \vee \psi,$$

and

$$\neg(\varphi \to \psi) \equiv \varphi \wedge \neg\psi.$$

Go to proof.

Theorem 0.6.9 (Biconditional Equivalence Laws)**.** *For all* $\varphi, \psi \in \mathsf{WFF}$*,*

$$\varphi \leftrightarrow \psi \equiv (\varphi \to \psi) \wedge (\psi \to \varphi),$$

and

$$\varphi \leftrightarrow \psi \equiv (\varphi \wedge \psi) \vee (\neg\varphi \wedge \neg\psi).$$

Go to proof.

Definition 0.6.10 (Formula Context)**.** A formula context $C[-]$ is a well-formed formula with one distinguished placeholder occurrence $[-]$. For each well-formed formula φ, the expression $C[\varphi]$ denotes the formula obtained by replacing the placeholder with φ.

Theorem 0.6.11 (Replacement of Logical Equivalents)**.** *Replacing logically equivalent formulas inside a formula context preserves logical equivalence:*

$$\varphi \equiv \psi \Longrightarrow C[\varphi] \equiv C[\psi].$$

Go to proof.

0.7 Normal Forms

Orientation

Definition 0.7.1 (Literal)**.** A *literal* is either a propositional variable P or the negation $\neg P$ of a propositional variable P.

Definition 0.7.2 (Positive Literal)**.** A *positive literal* is a literal of the form P, where P is a propositional variable.

Definition 0.7.3 (Negative Literal)**.** A *negative literal* is a literal of the form $\neg P$, where P is a propositional variable.

Definition 0.7.4 (Complementary Literals)**.** Two literals are *complementary* if they are P and $\neg P$ for the same propositional variable P.

Definition 0.7.5 (Clause)**.** A *clause* is a finite disjunction of literals.

Definition 0.7.6 (Term)**.** A *term*, also called a *conjunction term*, is a finite conjunction of literals.

Definition 0.7.7 (Empty Clause)**.** The *empty clause* is the empty disjunction of literals. Semantically, it is read as false.

Definition 0.7.8 (Empty Conjunction)**.** The *empty conjunction* is the conjunction over no literals. Semantically, it is read as true.

Definition 0.7.9 (Negation Normal Form)**.** A formula is in *negation normal form* if it uses only propositional variables, negated propositional variables, conjunction, and disjunction, and every negation occurs immediately before a propositional variable.

Definition 0.7.10 (NNF Conversion Procedure)**.** The *NNF conversion procedure* converts a propositional formula to an equivalent formula in negation normal form by the following steps:

1. eliminate biconditionals using biconditional equivalence laws;

2. eliminate conditionals using conditional equivalence laws;

3. eliminate double negations;

4. push negations inward using De Morgan laws.

Theorem 0.7.11 (Existence of Negation Normal Form)**.** *For every formula $\varphi \in \mathsf{WFF}$, there exists a formula $\psi \in \mathsf{WFF}$ such that ψ is in negation normal form and*

$$\varphi \equiv \psi.$$

Go to proof.

Definition 0.7.12 (Elementary Conjunction)**.** An *elementary conjunction* is a finite conjunction of literals. That is, it is a term of the form

$$L_1 \wedge L_2 \wedge \cdots \wedge L_k,$$

where each L_i is a literal. The one-literal case is permitted.

Definition 0.7.13 (Minterm)**.** Let $P_1, \ldots, P_n$ be a finite list of propositional variables. A *minterm* relative to $P_1, \ldots, P_n$ is an elementary conjunction that contains exactly one of P_i or $\neg P_i$ for each $i = 1, \ldots, n$.

Definition 0.7.14 (Disjunctive Normal Form)**.** A formula is in *disjunctive normal form*, abbreviated *DNF*, when it is a finite disjunction of elementary conjunctions. Equivalently, it has the shape

$$C_1 \vee C_2 \vee \cdots \vee C_m,$$

where each C_j is an elementary conjunction.

Theorem 0.7.15 (Existence of Disjunctive Normal Form)**.** *For every formula $\varphi \in \mathsf{WFF}$, there exists a formula $\psi \in \mathsf{WFF}$ such that ψ is in disjunctive normal form and*

$$\varphi \equiv \psi.$$

Go to proof.

Theorem 0.7.16 (DNF Satisfiability Criterion)**.** *Let ψ be a formula in disjunctive normal form. Then ψ is satisfiable if and only if at least one conjunction term in ψ contains no complementary pair of literals.* *Go to proof.*

Definition 0.7.17 (Elementary Disjunction)**.** An *elementary disjunction* is a finite disjunction of literals. That is, it is a clause of the form

$$L_1 \vee L_2 \vee \cdots \vee L_k,$$

where each L_i is a literal. The one-literal case is permitted.

Definition 0.7.18 (Maxterm)**.** Let $P_1, \ldots, P_n$ be a finite list of propositional variables. A *maxterm* relative to $P_1, \ldots, P_n$ is an elementary disjunction that contains exactly one of P_i or $\neg P_i$ for each $i = 1, \ldots, n$.

Definition 0.7.19 (Conjunctive Normal Form)**.** A formula is in *conjunctive normal form*, abbreviated *CNF*, when it is a finite conjunction of clauses. Equivalently, it has the shape

$$C_1 \wedge C_2 \wedge \cdots \wedge C_m,$$

where each C_j is a clause.

Theorem 0.7.20 (Existence of Conjunctive Normal Form)**.** *For every formula $\varphi \in \mathsf{WFF}$, there exists a formula $\psi \in \mathsf{WFF}$ such that ψ is in conjunctive normal form and*

$$\varphi \equiv \psi.$$

Go to proof.

Definition 0.7.21 (Truth-Table Row Conjunction)**.** Fix a finite variable list $P_1, \ldots, P_n$ and a row r of the truth table for those variables. The *truth-table row conjunction* for r is the conjunction

$$L_1 \wedge L_2 \wedge \cdots \wedge L_n,$$

where

$$L_i = \begin{cases} P_i, & \text{if row } r \text{ makes } P_i \text{ true,} \\ \neg P_i, & \text{if row } r \text{ makes } P_i \text{ false.} \end{cases}$$

Definition 0.7.22 (Truth-Table Row Clause)**.** Fix a finite variable list $P_1, \ldots, P_n$ and a row r of the truth table for those variables. The *truth-table row clause* for r is the clause

$$M_1 \vee M_2 \vee \cdots \vee M_n,$$

where

$$M_i = \begin{cases} P_i, & \text{if row } r \text{ makes } P_i \text{ false,} \\ \neg P_i, & \text{if row } r \text{ makes } P_i \text{ true.} \end{cases}$$

Definition 0.7.23 (Canonical Disjunctive Normal Form)**.** Let $\varphi \in \mathsf{WFF}$, and fix an ordering $P_1, \ldots, P_n$ of the finite variable set of φ. The *canonical disjunctive normal form* of φ relative to this ordering is the disjunction of the truth-table row conjunctions corresponding to the rows on which φ is true.

Definition 0.7.24 (Canonical Conjunctive Normal Form)**.** Let $\varphi \in \mathsf{WFF}$, and fix an ordering $P_1, \ldots, P_n$ of the finite variable set of φ. The *canonical conjunctive normal form* of φ relative to this ordering is the conjunction of the truth-table row clauses corresponding to the rows on which φ is false.

Theorem 0.7.25 (Canonical Disjunctive Normal Form Theorem)**.** *Every formula $\varphi \in \mathsf{WFF}$ is logically equivalent to its canonical disjunctive normal form relative to a fixed ordering of its finitely many variables.*

$$\varphi \equiv \mathrm{CDNF}_{P_1,\ldots,P_n}(\varphi).$$

Go to proof.

Theorem 0.7.26 (Canonical Conjunctive Normal Form Theorem)**.** *Every formula $\varphi \in \mathsf{WFF}$ is logically equivalent to its canonical conjunctive normal form relative to a fixed ordering of its finitely many variables.*

$$\varphi \equiv \mathrm{CCNF}_{P_1,\ldots,P_n}(\varphi).$$

Go to proof.

Theorem 0.7.27 (Semantic Classification by Normal Forms)**.** *Normal forms give finite certificates for semantic classification. For any formula $\varphi \in \mathsf{WFF}$:*

1. *φ is satisfiable if and only if some DNF formula equivalent to φ has a satisfiable conjunction term.*
2. *φ is a contradiction if and only if some DNF formula equivalent to φ has no satisfiable conjunction term.*
3. *φ is a tautology if and only if some CNF formula equivalent to φ has no falsifiable clause.*
4. *φ is a contingency if and only if it is satisfiable and not a tautology.*

Go to proof.

Summary

0.8 Functional Completeness

Orientation

Boolean Functions and Representation

Definition 0.8.1 (Representation of a Boolean Function)**.** Let $f : \mathbb{B}^n \to \mathbb{B}$ be a Boolean function. A formula $\varphi \in \mathsf{WFF}$ with variables among $P_1, \ldots, P_n$ *represents* f if for every truth assignment v,

$$\widehat{v}(\varphi) = f(v(P_1), \ldots, v(P_n)).$$

Theorem 0.8.2 (DNF Representation of Boolean Functions)**.** *Every Boolean function $f : \mathbb{B}^n \to \mathbb{B}$ is represented by a formula in disjunctive normal form using variables $P_1, \ldots, P_n$.*
Go to proof.

Connective Bases

Definition 0.8.3 (Connective Basis)**.** A *connective basis* is a specified set B of propositional connectives allowed as primitive connectives for building formulas.

Definition 0.8.4 (Definable Connective)**.** Let B be a connective basis. A connective $\circ$ is *definable from B* if there is a formula built using only connectives from B whose truth function agrees with the truth function of $\circ$.

Definition 0.8.5 (Functionally Complete Connective Set)**.** A connective basis B is *functionally complete* if every Boolean function $f : \mathbb{B}^n \to \mathbb{B}$, for every $n \geq 1$, is represented by some formula using only connectives from B.

Definition 0.8.6 (Expressively Equivalent Connective Sets)**.** Two connective bases B and C are *expressively equivalent* if every connective in B is definable from C, and every connective in C is definable from B.

Completeness of the Standard Connectives

Theorem 0.8.7 (Standard Connectives are Functionally Complete)**.** *The connective basis*

$$\{\neg, \wedge, \vee\}$$

is functionally complete.
Go to proof.

Theorem 0.8.8 (Negation and Conjunction are Functionally Complete)**.** *The connective basis*

$$\{\neg, \wedge\}$$

is functionally complete.
Go to proof.

Theorem 0.8.9 (Negation and Disjunction are Functionally Complete)**.** *The connective basis*

$$\{\neg, \vee\}$$

is functionally complete.
Go to proof.

Reduction of Connective Sets

Definition 0.8.10 (Connective Reduction)**.** A *connective reduction* replaces a formula using one connective basis by a logically equivalent formula using a smaller or different connective basis.

NAND

Definition 0.8.11 (NAND Connective)**.** The *NAND* connective, written $\uparrow$, is the binary connective defined by

$$\varphi \uparrow \psi \equiv \neg(\varphi \wedge \psi).$$

Definition 0.8.12 (NAND-Only Formula)**.** A *NAND-only formula* is a well-formed formula whose only connective is $\uparrow$.

Theorem 0.8.13 (NAND is Functionally Complete)**.** *The one-connective basis* $\{\uparrow\}$ *is functionally complete.*
Go to proof.

NOR

Definition 0.8.14 (NOR Connective)**.** The *NOR* connective, written $\downarrow$, is the binary connective defined by

$$\varphi \downarrow \psi \equiv \neg(\varphi \vee \psi).$$

Definition 0.8.15 (NOR-Only Formula)**.** A *NOR-only formula* is a well-formed formula whose only connective is $\downarrow$.

Theorem 0.8.16 (NOR is Functionally Complete)**.** *The one-connective basis* $\{\downarrow\}$ *is functionally complete.*
Go to proof.

Incomplete Connective Sets

Definition 0.8.17 (Incomplete Connective Set)**.** A connective basis is *incomplete* if it is not functionally complete.

Definition 0.8.18 (Monotone Boolean Function)**.** A Boolean function $f : \mathbb{B}^n \to \mathbb{B}$ is *monotone* if changing input coordinates from False to True never changes the output from True to False.

Theorem 0.8.19 (Conjunction and Disjunction are not Functionally Complete)**.** *The connective basis* $\{\wedge, \vee\}$ *is not functionally complete.*
Go to proof.

Minimal Complete Bases

Definition 0.8.20 (Minimal Complete Basis)**.** A functionally complete connective basis B is *minimal complete* if no proper subset of B is functionally complete.

Theorem 0.8.21 (NAND and NOR are Minimal Complete Bases)**.** *The one-connective bases* $\{\uparrow\}$ *and* $\{\downarrow\}$ *are minimal complete bases.*
Go to proof.

Summary and Transition

0.9 Argument Forms and Informal Proof Patterns

Orientation

Argument Forms

Definition 0.9.1 (Argument Form)**.** An *argument form* is a schematic pattern consisting of a finite list of premise formulas and one conclusion formula.

Definition 0.9.2 (Valid Argument Form)**.** An argument form with premises Γ and conclusion φ is *valid* if every truth assignment that satisfies every formula in Γ also satisfies φ.

Definition 0.9.3 (Counterexample Assignment)**.** A *counterexample assignment* to an argument form (Γ, φ) is a truth assignment that satisfies every premise in Γ and does not satisfy φ.

Theorem 0.9.4 (Truth-Table Validity Test for Argument Forms)**.** *An argument form* (Γ, φ) *is valid if and only if no row of its truth table makes every formula in* Γ *true and* φ *false.*
Go to proof.

Core Valid Inference Patterns

Theorem 0.9.5 (Modus Ponens is Valid)**.** *The argument form*

$$\varphi, \qquad \varphi \to \psi \quad \therefore \quad \psi$$

is valid.
Go to proof.

Theorem 0.9.6 (Modus Tollens is Valid)**.** *The argument form*

$$\varphi \to \psi, \qquad \neg\psi \quad \therefore \quad \neg\varphi$$

is valid.
Go to proof.

Theorem 0.9.7 (Hypothetical Syllogism is Valid)**.** *The argument form*

$$\varphi \to \psi, \qquad \psi \to \chi \quad \therefore \quad \varphi \to \chi$$

is valid.
Go to proof.

Theorem 0.9.8 (Disjunctive Syllogism is Valid)**.** *The argument form*

$$\varphi \vee \psi, \qquad \neg\varphi \quad \therefore \quad \psi$$

is valid.
Go to proof.

Theorem 0.9.9 (Addition is Valid)**.** *The argument form*

$$\varphi \quad \therefore \quad \varphi \vee \psi$$

is valid.
Go to proof.

Theorem 0.9.10 (Simplification is Valid)**.** *The argument form*

$$\varphi \wedge \psi \quad \therefore \quad \varphi$$

is valid.
Go to proof.

Theorem 0.9.11 (Conjunction Introduction is Valid)**.** *The argument form*

$$\varphi, \qquad \psi \quad \therefore \quad \varphi \wedge \psi$$

is valid.
Go to proof.

Theorem 0.9.12 (Constructive Dilemma is Valid). *The argument form*

$$\varphi \to \chi, \qquad \psi \to \theta, \qquad \varphi \vee \psi \quad \therefore \quad \chi \vee \theta$$

is valid.
Go to proof.

Theorem 0.9.13 (Destructive Dilemma is Valid). *The argument form*

$$\varphi \to \chi, \qquad \psi \to \theta, \qquad \neg\chi \vee \neg\theta \quad \therefore \quad \neg\varphi \vee \neg\psi$$

is valid.
Go to proof.

Invalid Argument Forms

Theorem 0.9.14 (Affirming the Consequent is Invalid). *The argument form*

$$\varphi \to \psi, \qquad \psi \quad \therefore \quad \varphi$$

is invalid.
Go to proof.

Theorem 0.9.15 (Denying the Antecedent is Invalid). *The argument form*

$$\varphi \to \psi, \qquad \neg\varphi \quad \therefore \quad \neg\psi$$

is invalid.
Go to proof.

Rules of Replacement

Definition 0.9.16 (Rule of Replacement). A *rule of replacement* is a schematic permission to replace a subformula by a logically equivalent formula inside a formula context.

Theorem 0.9.17 (Replacement Rules Preserve Logical Equivalence). *If a formula ψ is obtained from a formula φ by finitely many applications of rules of replacement, then $\varphi \equiv \psi$.*
Go to proof.

Informal Proof Patterns

Definition 0.9.18 (Informal Proof Pattern). An *informal proof pattern* is a reusable truth-preserving strategy for organizing an ordinary mathematical proof.

Semantic Justification of Argument Forms

Theorem 0.9.19 (Tautological Implication Gives a Valid Argument Form). *Let $\Gamma = \{\gamma_1, \ldots, \gamma_n\} \subseteq \mathsf{WFF}$ and let $\varphi \in \mathsf{WFF}$. If*

$$(\gamma_1 \wedge \cdots \wedge \gamma_n) \to \varphi$$

is a tautology, then the argument form (Γ, φ) is valid.
Go to proof.

Argument Forms as a Bridge to Formal Systems

Summary and Transition

0.10 Compactness and Finite Satisfiability Preview

Orientation

Satisfiable Sets

Definition 0.10.1 (Satisfiable Set of Propositional Formulas)**.** Let $\Gamma \subseteq \mathsf{WFF}$. The set Γ is satisfiable if there exists a truth assignment v such that $v \models \Gamma$.

$$\text{SatisfiableSet}(\Gamma) \iff \exists v \left(\text{TruthAssignment}(v) \wedge v \models \Gamma\right).$$

Definition 0.10.2 (Unsatisfiable Set of Propositional Formulas)**.** Let $\Gamma \subseteq \mathsf{WFF}$. The set Γ is unsatisfiable if it is not satisfiable.

$$\text{UnsatisfiableSet}(\Gamma) \iff \neg\,\text{SatisfiableSet}(\Gamma).$$

Finitely Satisfiable Sets

Definition 0.10.3 (Finite Subset of a Set of Propositional Formulas)**.** Let $\Gamma \subseteq \mathsf{WFF}$. A finite subset of Γ is a set $\Delta \subseteq \Gamma$ containing only finitely many formulas.

Definition 0.10.4 (Finitely Satisfiable Set of Propositional Formulas)**.** Let $\Gamma \subseteq \mathsf{WFF}$. The set Γ is finitely satisfiable if every finite subset of Γ is satisfiable.

$$\text{FinitelySatisfiableSet}(\Gamma) \iff \forall \Delta \Big((\Delta \subseteq \Gamma \wedge \Delta \text{ is finite}) \Rightarrow \text{SatisfiableSet}(\Delta)\Big).$$

Finite Unsatisfiability Witnesses

Definition 0.10.5 (Unsatisfiability Witness)**.** Let $\Gamma \subseteq \mathsf{WFF}$. A subset $\Delta \subseteq \Gamma$ is an unsatisfiability witness for Γ if Δ is unsatisfiable.

Definition 0.10.6 (Finite Unsatisfiability Witness)**.** Let $\Gamma \subseteq \mathsf{WFF}$. A subset $\Delta \subseteq \Gamma$ is a finite unsatisfiability witness for Γ if Δ is finite and unsatisfiable.

$$\text{FiniteUnsatisfiabilityWitness}(\Delta, \Gamma) \iff \Delta \subseteq \Gamma \wedge \Delta \text{ is finite} \wedge \text{UnsatisfiableSet}(\Delta).$$

Theorem 0.10.7 (Finite Unsatisfiability Witness)**.** *Let $\Gamma \subseteq \mathsf{WFF}$. If there exists a finite $\Delta \subseteq \Gamma$ such that Δ is unsatisfiable, then Γ is unsatisfiable.*

Go to proof.

Corollary 0.10.8 (Finite Witness Contrapositive)**.** *Let $\Gamma \subseteq \mathsf{WFF}$. If Γ is satisfiable, then every finite subset of Γ is satisfiable.*

Go to proof.

Propositional Compactness

Theorem 0.10.9 (Propositional Compactness)**.** *Let $\Gamma \subseteq \mathsf{WFF}$. If every finite subset of Γ is satisfiable, then Γ is satisfiable.*

Go to proof.

Finite Truth Table Intuition

Summary

Proofs

Predicate Logic

0.11 Model-Theoretic View

0.12 Axiom Schemas

0.13 Syntax

0.13.1 Syntax of First-Order Logic: Terms and Formulas

Definition 0.13.1 (Variable)**.** For a first-order language $\mathcal{L}$, a *variable* is a symbol drawn from the distinguished variable stock of $\mathcal{L}$. Variables are usually written

$$x, y, z, x_0, x_1, x_2, \ldots.$$

Definition 0.13.2 (Constant Symbol)**.** For a first-order language $\mathcal{L}$, a *constant symbol* is a non-logical symbol of arity 0. A constant symbol may occur as a term without taking input terms.

Definition 0.13.3 (Function Symbol)**.** For a first-order language $\mathcal{L}$, an n-ary *function symbol* is a non-logical symbol of arity $n \geq 1$. If f has arity n, then it forms terms by the pattern

$$f(t_1, \ldots, t_n).$$

Definition 0.13.4 (Predicate Symbol)**.** For a first-order language $\mathcal{L}$, an n-ary *predicate symbol* is a non-logical symbol of arity $n \geq 1$. If P has arity n, then it forms atomic formulas by the pattern

$$P(t_1, \ldots, t_n).$$

Definition 0.13.5 (Term)**.** The set of *terms* of a first-order language $\mathcal{L}$, denoted $\mathsf{Term}_{\mathcal{L}}$, is the smallest set of strings satisfying:

1. every variable is a term;
2. every constant symbol is a term;
3. if f is an n-ary function symbol and $t_1, \ldots, t_n$ are terms, then $f(t_1, \ldots, t_n)$ is a term;
4. no string is a term unless it is generated by finitely many applications of the previous clauses.

Definition 0.13.6 (Atomic Formula)**.** An *atomic formula* of a first-order language $\mathcal{L}$ is a string of the form

$$P(t_1, \ldots, t_n),$$

where P is an n-ary predicate symbol and $t_1, \ldots, t_n$ are terms.

Definition 0.13.7 (Formula)**.** For a first-order language $\mathcal{L}$, a *formula* is an expression formed by the formula grammar of $\mathcal{L}$. The grammar starts from atomic formulas and closes under the logical connectives and quantifiers of the language.

Definition 0.13.8 (Well-Formed Formula)**.** The set of *well-formed formulas* of a first-order language $\mathcal{L}$, denoted $\mathsf{Form}_{\mathcal{L}}$, is the smallest set of strings satisfying:

1. every atomic formula is a well-formed formula;
2. if φ is a well-formed formula, then $\neg\varphi$ is a well-formed formula;
3. if φ and ψ are well-formed formulas and $\circ \in \{\wedge, \vee, \rightarrow, \leftrightarrow\}$, then $(\varphi \circ \psi)$ is a well-formed formula;
4. if φ is a well-formed formula and x is a variable, then $\forall x\, \varphi$ and $\exists x\, \varphi$ are well-formed formulas;
5. no string is a well-formed formula unless it is generated by finitely many applications of the previous clauses.

Definition 0.13.9 (Molecular Formula)**.** A *molecular formula* is a well-formed formula that is not atomic. Equivalently, it is a well-formed formula whose final formation step uses a logical connective or a quantifier.

0.13.2 Syntax: Free Variables, Bound Variables, and Substitution

Definition 0.13.10 (Scope of a Quantifier)**.** Let φ be a formula of a first-order language.
If φ is of the form $\forall x\, \psi$ or $\exists x\, \psi$, then the formula ψ is called the *scope* of the quantifier.
The quantifier is said to *bind* all occurrences of the variable x that appear within its scope.

Definition 0.13.11 (Bound and Free Occurrences)**.** An occurrence of a variable x in a formula φ is *bound* if it lies within the scope of a quantifier $\forall x$ or $\exists x$.
An occurrence of x is *free* if it is not bound by any quantifier in φ.

Definition 0.13.12 (Free Variables of a Formula)**.** The set $\mathrm{FV}(\varphi)$ of *free variables* of a formula φ is defined recursively:

1. If $\varphi = P(t_1, \ldots, t_n)$ is atomic, then $\mathrm{FV}(\varphi) = \bigcup_{i=1}^{n} \mathrm{Var}(t_i)$, where $\mathrm{Var}(t_i)$ is the set of variables in term t_i.
2. $\mathrm{FV}(\neg\varphi) = \mathrm{FV}(\varphi)$.
3. $\mathrm{FV}(\varphi \circ \psi) = \mathrm{FV}(\varphi) \cup \mathrm{FV}(\psi)$ for any binary connective $\circ$.
4. $\mathrm{FV}(\forall x\, \varphi) = \mathrm{FV}(\varphi) \setminus \{x\}$.
5. $\mathrm{FV}(\exists x\, \varphi) = \mathrm{FV}(\varphi) \setminus \{x\}$.

Definition 0.13.13 (Sentence)**.** A *sentence* (or *closed formula*) is a formula with no free variables.
Formally, φ is a sentence if and only if $\mathrm{FV}(\varphi) = \varnothing$.

Definition 0.13.14 (Substitution Notation)**.** Let φ be a formula, x a variable, and t a term.
The expression $\varphi[t/x]$ denotes the formula obtained from φ by replacing every *free* occurrence of x with the term t, leaving bound occurrences of x unchanged.

Definition 0.13.15 (Free for Substitution)**.** A term t is *free for* x in φ if no free occurrence of x in φ lies within the scope of a quantifier $\forall y$ or $\exists y$ where y is a variable occurring in t.
Equivalently, t is free for x in φ if the substitution $\varphi[t/x]$ does not result in any variable in t becoming bound.

Definition 0.13.16 (Alpha-Equivalence)**.** Formulas that differ only by the names of bound variables are logically equivalent. This is called *alpha-equivalence* (or *alphabetic variance*):

$$\forall x\, \varphi \;\equiv\; \forall y\, \varphi[y/x] \quad \text{(provided } y \text{ is not free in } \varphi\text{).}$$

0.13.3 Syntax: Formula Depth

Definition 0.13.17 (Formula Depth)**.** The *depth* (or *complexity*) of a formula φ, denoted $\mathrm{depth}(\varphi)$, is a natural number defined recursively:

1. **Atomic case.** If φ is atomic, then $\mathrm{depth}(\varphi) = 0$.
2. **Negation.** If $\varphi = \neg\psi$, then $\mathrm{depth}(\varphi) = \mathrm{depth}(\psi) + 1$.
3. **Binary connectives.** If $\varphi = (\psi \circ \chi)$ for $\circ \in \{\wedge, \vee, \to, \leftrightarrow\}$, then
$$\mathrm{depth}(\varphi) = \max\{\mathrm{depth}(\psi), \mathrm{depth}(\chi)\} + 1.$$
4. **Quantifiers.** If $\varphi = \forall x\, \psi$ or $\varphi = \exists x\, \psi$, then $\mathrm{depth}(\varphi) = \mathrm{depth}(\psi) + 1$.

0.14 Semantics

0.14.1 Semantics: Structures and Variable Assignments

Definition 0.14.1 (First-Order Language)**.** A *first-order language* $\mathcal{L}$ consists of variables, logical symbols, and a signature of non-logical symbols: constant symbols, function symbols with assigned arities, and predicate symbols with assigned arities. Equality may be included as a distinguished logical predicate.

Definition 0.14.2 (Structure)**.** Let $\mathcal{L}$ be a first-order language. A *structure* for $\mathcal{L}$ is a pair
$$\mathcal{M} = \langle D, I \rangle$$
where D is a nonempty set and I assigns each constant symbol an element of D, each n-ary function symbol a function $D^n \to D$, and each n-ary predicate symbol a relation on D^n.

Definition 0.14.3 (Variable Assignment)**.** Let $\mathcal{M} = \langle D, I \rangle$ be a structure for $\mathcal{L}$. A *variable assignment* for $\mathcal{M}$ is a function
$$s : \mathsf{Var}_{\mathcal{L}} \to D.$$

Definition 0.14.4 (Modified Assignment)**.** Let $s : \mathsf{Var}_{\mathcal{L}} \to D$, let $x \in \mathsf{Var}_{\mathcal{L}}$, and let $d \in D$. The *modified assignment* $s[x \mapsto d]$ is the assignment defined by
$$s[x \mapsto d](y) = \begin{cases} d, & y = x, \\ s(y), & y \neq x. \end{cases}$$

Definition 0.14.5 (Term Evaluation)**.** Let $\mathcal{M} = \langle D, I \rangle$ be a structure and s a variable assignment. The *value* of a term t, denoted $[\![t]\!]_{\mathcal{M},s}$, is defined recursively by
$$[\![x]\!]_{\mathcal{M},s} = s(x), \qquad [\![c]\!]_{\mathcal{M},s} = I(c),$$
and
$$[\![f(t_1, \ldots, t_n)]\!]_{\mathcal{M},s} = I(f)\big([\![t_1]\!]_{\mathcal{M},s}, \ldots, [\![t_n]\!]_{\mathcal{M},s}\big).$$

0.14.2 Semantics: Satisfaction and Truth

Definition 0.14.6 (Satisfaction)**.** Let $\mathcal{M} = \langle D, I \rangle$ be a structure, let s be a variable assignment, and let φ be a well-formed formula. The relation $\mathcal{M}, s \models \varphi$ is defined recursively by the usual clauses: atomic formulas are evaluated by term values and interpreted predicate relations; equality compares term values; connectives follow the truth conditions for $\neg, \wedge, \vee, \to, \leftrightarrow$; and
$$\mathcal{M}, s \models \forall x\, \psi \iff \forall d \in D\ (\mathcal{M}, s[x \mapsto d] \models \psi),$$
$$\mathcal{M}, s \models \exists x\, \psi \iff \exists d \in D\ (\mathcal{M}, s[x \mapsto d] \models \psi).$$

Definition 0.14.7 (Truth in a Structure)**.** Let φ be a sentence and let $\mathcal{M}$ be a structure. The sentence φ is *true in* $\mathcal{M}$, written $\mathcal{M} \models \varphi$, exactly when $\mathcal{M}, s \models \varphi$ for every variable assignment s for $\mathcal{M}$.

Definition 0.14.8 (Validity)**.** A formula φ is *valid*, or *logically true*, exactly when $\mathcal{M}, s \models \varphi$ for every structure $\mathcal{M}$ for the language of φ and every variable assignment s for $\mathcal{M}$.

Definition 0.14.9 (Satisfiability)**.** A formula φ is *satisfiable* exactly when there are a structure $\mathcal{M}$ and a variable assignment s such that $\mathcal{M}, s \models \varphi$.

0.14.3 Semantics: Models, Theories, and Logical Consequence

Definition 0.14.10 (Model of a Sentence)**.** Let φ be a sentence and let $\mathcal{M}$ be a structure for the language of φ. The structure $\mathcal{M}$ is a *model of* φ exactly when $\mathcal{M} \models \varphi$.

Definition 0.14.11 (Model of a Set of Sentences)**.** Let T be a set of sentences and let $\mathcal{M}$ be a structure for their common language. The structure $\mathcal{M}$ is a *model of* T, written $\mathcal{M} \models T$, exactly when $\mathcal{M} \models \theta$ for every $\theta \in T$.

Definition 0.14.12 (Theory)**.** In Volume I, a *theory* is a set of sentences in a fixed first-order language.

Definition 0.14.13 (Logical Consequence)**.** Let T be a set of sentences and let φ be a sentence in the same language. The sentence φ is a *logical consequence* of T, written $T \models \varphi$, exactly when every model of T is a model of φ.

Definition 0.14.14 (Logical Equivalence)**.** Two sentences φ and ψ are *logically equivalent*, written $\varphi \equiv \psi$, exactly when each is a logical consequence of the other:

$$\varphi \models \psi \quad \text{and} \quad \psi \models \varphi.$$

0.14.4 Semantics: Open Formulas, Definable Relations, and Substitution Lemmas

Definition 0.14.15 (Open Formula)**.** An *open formula* is a well-formed formula with at least one free variable. When the free variables are among $x_1, \ldots, x_n$, it may be displayed as $\varphi(x_1, \ldots, x_n)$.

Definition 0.14.16 (Definable Relation)**.** Let $\mathcal{M} = \langle D, I \rangle$ be a structure and let $\varphi(x_1, \ldots, x_n)$ be an open formula. The *relation defined by* φ in $\mathcal{M}$ is the set

$$R^{\mathcal{M}}_{\varphi} = \{(a_1, \ldots, a_n) \in D^n : \mathcal{M}, s[x_1 \mapsto a_1, \ldots, x_n \mapsto a_n] \models \varphi\}.$$

Lemma 0.14.17 (Substitution Lemma for Terms)**.** *Let $\mathcal{M}$ be a structure, let s be a variable assignment, let x be a variable, and let t and u be terms. Then*

$$[\![t[u/x]]\!]_{\mathcal{M},s} = [\![t]\!]_{\mathcal{M},\, s[x \mapsto [\![u]\!]_{\mathcal{M},s}]}.$$

Go to proof.

Lemma 0.14.18 (Substitution Lemma for Formulas)**.** *Let φ be a formula and let u be a term free for x in φ. For every structure $\mathcal{M}$ and assignment s,*

$$\mathcal{M}, s \models \varphi[u/x] \iff \mathcal{M}, s[x \mapsto [\![u]\!]_{\mathcal{M},s}] \models \varphi.$$

Go to proof.

0.15 Quantifiers

Basic Syntax and Semantics

Definition 0.15.1 (Universal Formula). A *universal formula* is a well-formed formula of the form

$$\forall x\, \varphi,$$

where x is a variable and φ is a well-formed formula.

Definition 0.15.2 (Existential Formula). An *existential formula* is a well-formed formula of the form

$$\exists x\, \varphi,$$

where x is a variable and φ is a well-formed formula.

Definition 0.15.3 (Bounded Quantifier). A *bounded quantifier* is an abbreviation that restricts a quantifier to a specified set or predicate:

$$\begin{aligned} \forall x \in A\, \varphi \quad &:= \quad \forall x\, (x \in A \to \varphi), \\ \exists x \in A\, \varphi \quad &:= \quad \exists x\, (x \in A \land \varphi). \end{aligned}$$

Definition 0.15.4 (Sentence). A *sentence* is a well-formed formula with no free variables.

Definition 0.15.5 (Open Formula). An *open formula* is a well-formed formula with at least one free variable.

Definition 0.15.6 (Quantifier Scope). In a formula $Qx\, \varphi$, where $Q \in \{\forall, \exists\}$, the *scope* of the quantifier Qx is the formula φ. Occurrences of x inside that scope are bound by the quantifier unless an inner quantifier for x intervenes.

Theorem 0.15.7 (Negation of Bounded Quantifiers). *Let A be a set and let φ be a formula. Then*

$$\neg(\forall x \in A\, \varphi) \equiv \exists x \in A\, \neg\varphi, \qquad \neg(\exists x \in A\, \varphi) \equiv \forall x \in A\, \neg\varphi.$$

Go to proof.

Quantifier Laws

Theorem 0.15.8 (Quantifier Negation Laws). *For every formula φ,*

$$\neg\forall x\, \varphi \equiv \exists x\, \neg\varphi, \qquad \neg\exists x\, \varphi \equiv \forall x\, \neg\varphi.$$

Go to proof.

Theorem 0.15.9 (Double Negation and Quantifiers). *For every formula φ,*

$$\neg\neg\forall x\, \varphi \equiv \forall x\, \varphi, \qquad \neg\neg\exists x\, \varphi \equiv \exists x\, \varphi.$$

Go to proof.

Theorem 0.15.10 (Same-Type Quantifier Commutation). *Quantifiers of the same type commute:*

$$\forall x\, \forall y\, \varphi \equiv \forall y\, \forall x\, \varphi, \qquad \exists x\, \exists y\, \varphi \equiv \exists y\, \exists x\, \varphi.$$

Go to proof.

Theorem 0.15.11 (Mixed Quantifiers Do Not Commute). *In general,*

$$\forall x\, \exists y\, \varphi \not\equiv \exists y\, \forall x\, \varphi.$$

Go to proof.

Theorem 0.15.12 (Valid Quantifier Distribution). *For all formulas φ, ψ,*

$$\forall x\, (\varphi \wedge \psi) \equiv (\forall x\, \varphi) \wedge (\forall x\, \psi),$$

$$\exists x\, (\varphi \vee \psi) \equiv (\exists x\, \varphi) \vee (\exists x\, \psi).$$

Go to proof.

Theorem 0.15.13 (Invalid Quantifier Distribution). *In general,*

$$\forall x\, (\varphi \vee \psi) \not\equiv (\forall x\, \varphi) \vee (\forall x\, \psi),$$

$$\exists x\, (\varphi \wedge \psi) \not\equiv (\exists x\, \varphi) \wedge (\exists x\, \psi).$$

Go to proof.

Theorem 0.15.14 (Vacuous Quantification). *If x does not occur free in φ, then*

$$\forall x\, \varphi \equiv \varphi, \qquad \exists x\, \varphi \equiv \varphi.$$

Go to proof.

Theorem 0.15.15 (Renaming Bound Variables). *If y is free for x in φ and no capture is introduced, then*

$$\forall x\, \varphi \equiv \forall y\, \varphi[y/x], \qquad \exists x\, \varphi \equiv \exists y\, \varphi[y/x].$$

Go to proof.

Prenex Normal Form

Definition 0.15.16 (Quantifier Prefix). A *quantifier prefix* is a finite string

$$Q_1 x_1\, Q_2 x_2 \cdots Q_n x_n,$$

where each $Q_i \in \{\forall, \exists\}$ and each x_i is a variable.

Definition 0.15.17 (Matrix). In a formula $Q_1 x_1 \cdots Q_n x_n\, \psi$, the *matrix* is the trailing formula ψ. For prenex normal form, the matrix is required to be quantifier-free.

Definition 0.15.18 (Prenex Formula). A *prenex formula* is a formula of the form

$$Q_1 x_1\, Q_2 x_2 \cdots Q_n x_n\, \psi,$$

where $Q_1 x_1 \cdots Q_n x_n$ is a quantifier prefix and ψ is quantifier-free.

Definition 0.15.19 (Prenex Normal Form). A formula ψ is a *prenex normal form* of a formula φ exactly when ψ is prenex and $\varphi \equiv \psi$.

Theorem 0.15.20 (Existence of Prenex Normal Form). *Every first-order formula is logically equivalent to some formula in prenex normal form.* *Go to proof.*

Logical Strength and Quantifier Order

Definition 0.15.21 (Stronger Formula). A sentence A is *stronger than* a sentence B exactly when

$$A \models B \quad \text{and} \quad B \not\models A.$$

Definition 0.15.22 (Weaker Formula). A sentence B is *weaker than* a sentence A exactly when A is stronger than B.

Definition 0.15.23 (Quantifier Order)**.** The *quantifier order* of a prenex formula is the ordered list of quantifiers in its prefix. A change of order is significant when it changes which witnesses may depend on earlier variables.

Theorem 0.15.24 (Universal Implies Existential)**.** *Over nonempty domains,*

$$\forall x\, \varphi(x) \models \exists x\, \varphi(x).$$

Go to proof.

Theorem 0.15.25 (Existential Does Not Imply Universal)**.** *In general,*

$$\exists x\, \varphi(x) \not\models \forall x\, \varphi(x).$$

Go to proof.

Theorem 0.15.26 (Uniform Witness Implies Dependent Witness)**.** *For every formula $\varphi(x, y)$,*

$$\exists y\, \forall x\, \varphi(x, y) \models \forall x\, \exists y\, \varphi(x, y).$$

Go to proof.

Theorem 0.15.27 (Dependent Witness Need Not Be Uniform)**.** *In general,*

$$\forall x\, \exists y\, \varphi(x, y) \not\models \exists y\, \forall x\, \varphi(x, y).$$

Go to proof.

0.16 Proof Theory

0.16.1 Proof Theory: Quantifier Inference Rules

Definition 0.16.1 (Universal Instantiation — UI)**.** From a universally quantified statement, infer any instance obtained by substituting a term for the bound variable:

$$\forall x\, \varphi \;\Rightarrow\; \varphi[t/x].$$

Definition 0.16.2 (Universal Generalization — UG)**.** From a formula, infer a universally quantified statement:

$$\varphi \;\Rightarrow\; \forall x\, \varphi.$$

Definition 0.16.3 (Existential Instantiation — EI)**.** From an existentially quantified statement, infer an instance with a fresh constant (witness):

$$\exists x\, \varphi \;\Rightarrow\; \varphi[c/x].$$

Definition 0.16.4 (Existential Generalization — EG)**.** From a formula containing a term, infer an existential statement:

$$\varphi[t/x] \;\Rightarrow\; \exists x\, \varphi.$$

0.16.2 Proof Strategies for Quantified Statements

Strategy 1: Proving a Universal $\forall x\, \varphi(x)$

Strategy 2: Proving an Existential $\exists x\, \varphi(x)$

Strategy 3: Using a Universal $\forall x\, \varphi(x)$

Strategy 4: Using an Existential $\exists x\, \varphi(x)$

The Central Asymmetry: Universal vs. Existential Goals

Nested Quantifiers: The $\forall\exists$ Pattern in Analysis

A Complete Worked Example

0.16.3 Proof Theory: Equality Rules

Definition 0.16.5 (Equality Introduction — Reflexivity)**.** For any term t, one may infer $t = t$.

Definition 0.16.6 (Equality Elimination — Substitution of Equals)**.** From $t_1 = t_2$ and $\varphi[t_1/x]$, one may infer $\varphi[t_2/x]$.

Theorem 0.16.7 (Symmetry of Equality)**.** *From $t_1 = t_2$, one may infer $t_2 = t_1$.* *Go to proof.*

Theorem 0.16.8 (Transitivity of Equality)**.** *From $t_1 = t_2$ and $t_2 = t_3$, one may infer $t_1 = t_3$.* *Go to proof.*

Theorem 0.16.9 (Term Substitution under Equality)**.** *If $t_1 = t_2$, then for any function symbol f, $f(\dots, t_1, \dots) = f(\dots, t_2, \dots)$.* *Go to proof.*

Theorem 0.16.10 (Predicate Substitution under Equality)**.** *If $t_1 = t_2$ and P is an n-ary predicate symbol, then $P(\dots, t_1, \dots) \Leftrightarrow P(\dots, t_2, \dots)$.* *Go to proof.*

0.16.4 Proof Theory: Soundness and Completeness

Definition 0.16.11 (Soundness)**.** A proof system is *sound* if every provable formula is valid:

$$\Gamma \vdash \varphi \implies \Gamma \models \varphi.$$

Definition 0.16.12 (Completeness)**.** A proof system is *complete* if every valid formula is provable:

$$\Gamma \models \varphi \implies \Gamma \vdash \varphi.$$

Theorem 0.16.13 (Soundness of First-Order Logic)**.** *The standard inference rules for first-order logic (UI, UG, EI, EG, and the propositional rules) are sound: they preserve truth in all structures.*
If $\Gamma \vdash \varphi$, then $\Gamma \models \varphi$. *Go to proof.*

0.17 Translation

English to Logic and Scope Ambiguity

Definition 0.17.1 (Translation Key)**.** A *translation key* is a finite record that fixes a domain of discourse and assigns intended English readings to the constant symbols, predicate symbols, and relation symbols used in a translation.

Definition 0.17.2 (Domain of Discourse)**.** The *domain of discourse* for a translation is the collection of objects over which the variables of the translated formulas range.

Definition 0.17.3 (Predicate and Relation Symbols in a Translation)**.** In a translation key, a unary predicate symbol names a property of one object, and an n-ary relation symbol names a relation among n objects.

Definition 0.17.4 (Constants and Variables in a Translation)**.** A *constant* names a fixed object from the translation key, while a *variable* marks a place bound by a quantifier or left open for later binding.

Proposition 0.17.5 (Universal Translation Pattern)**.** *In a broad domain, an English statement of the form "All A are B" is translated by*

$$\forall x\,(A(x) \to B(x)).$$

Go to proof.

Proposition 0.17.6 (Existential Translation Pattern)**.** *In a broad domain, an English statement of the form "Some A are B" is translated by*

$$\exists x\,(A(x) \wedge B(x)).$$

Go to proof.

Proposition 0.17.7 (Conditional Translation Pattern)**.** *An English statement of the form "If A, then B" is translated by*

$$A \to B,$$

with quantifiers placed according to the variables occurring in A and B.
Go to proof.

Proposition 0.17.8 (Negation Translation Pattern)**.** *The negation of a translated quantified statement is obtained by negating the whole formula and then pushing the negation inward using the quantifier negation laws.*
Go to proof.

Definition 0.17.9 (Scope Ambiguity)**.** A sentence has *scope ambiguity* when the same ordinary-language wording admits two or more translations with different quantifier nesting or different connective scope.

Square of Opposition

Definition 0.17.10 (Universal Affirmative Form)**.** The *universal affirmative* categorical form, called A, translates "All A are B" as

$$\forall x\,(A(x) \to B(x)).$$

Definition 0.17.11 (Universal Negative Form)**.** The *universal negative* categorical form, called E, translates "No A are B" as

$$\forall x\,(A(x) \to \neg B(x)).$$

Definition 0.17.12 (Particular Affirmative Form)**.** The *particular affirmative* categorical form, called I, translates "Some A are B" as

$$\exists x\,(A(x) \wedge B(x)).$$

Definition 0.17.13 (Particular Negative Form)**.** The *particular negative* categorical form, called O, translates "Some A are not B" as

$$\exists x\,(A(x) \wedge \neg B(x)).$$

Proposition 0.17.14 (A-O Contradiction)**.** *The A-form $\forall x\,(A(x) \to B(x))$ and the O-form $\exists x\,(A(x) \wedge \neg B(x))$ are contradictory: exactly one of them is true.*
Go to proof.

Proposition 0.17.15 (E-I Contradiction)**.** *The E-form $\forall x\,(A(x) \to \neg B(x))$ and the I-form $\exists x\,(A(x) \wedge B(x))$ are contradictory: exactly one of them is true.*
Go to proof.

Proposition 0.17.16 (Contrariety Requires Existential Import)**.** *If $\exists x\,A(x)$, then the A-form and E-form cannot both be true.*
Go to proof.

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Languages, Axioms, and Models

Languages, Axioms, and Models Toolkit - Planned

0.32 Signatures

Signatures

Definition 0.32.1 (First-Order Signature)**.** A *first-order signature* is a tuple

$$\Sigma = (\mathcal{C}, \mathcal{F}, \mathcal{R}, \mathrm{ar}),$$

where $\mathcal{C}$ is a set of constant symbols, $\mathcal{F}$ is a set of function symbols, $\mathcal{R}$ is a set of relation symbols, and

$$\mathrm{ar} : \mathcal{F} \cup \mathcal{R} \to \mathbb{Z}_{>0}$$

assigns to each function or relation symbol its number of arguments.

Definition 0.32.2 (Arity)**.** Let $\Sigma = (\mathcal{C}, \mathcal{F}, \mathcal{R}, \mathrm{ar})$ be a first-order signature. The *arity function* is the map

$$\mathrm{ar} : \mathcal{F} \cup \mathcal{R} \to \mathbb{Z}_{>0}$$

that assigns to each function or relation symbol the number of arguments it requires. When constants are treated as 0-ary function symbols, the codomain is enlarged to include 0.

Definition 0.32.3 (Language)**.** A *first-order language* has two standard readings. As vocabulary, it is the same data as a signature,

$$\mathcal{L} = (\mathcal{C}, \mathcal{F}, \mathcal{R}, \mathrm{ar}).$$

As a formal language generated by a signature Σ, it is the set of well-formed strings built from the alphabet

$$A = \Sigma \cup V \cup \Lambda,$$

where V is a countable set of variables and Λ is the fixed stock of logical symbols and punctuation. In this second reading,

$$\mathcal{L}(\Sigma) = \{s \in A^* \mid s \in \mathrm{Terms}(\Sigma) \text{ or } s \in \mathrm{WFF}(\Sigma)\}.$$

Definition 0.32.4 (Constant Symbol)**.** A *constant symbol* is a non-logical symbol that behaves syntactically as a 0-ary function symbol. In the streamlined convention where constants are included among function symbols, this is the condition

$$\mathrm{Constant}(c) \iff c \in \mathcal{F} \text{ and } \mathrm{ar}(c) = 0.$$

Because its arity is 0, the symbol c is already a well-formed term and takes no input terms.

Definition 0.32.5 (Function Symbol)**.** A *function symbol* is a non-logical symbol $f \in \mathcal{F}$ whose assigned arity determines how many terms it must take as inputs. If $\mathrm{ar}(f) = n$, then f is an n-ary term constructor: it takes exactly n terms and produces a new term, not a truth value.

Definition 0.32.6 (Relation Symbol)**.** A *relation symbol*, or predicate symbol, is a non-logical symbol $R \in \mathcal{R}$ whose assigned arity determines how many terms it must take as inputs. If $\mathrm{ar}(R) = n$, then R is an n-ary formula constructor: it takes exactly n terms and produces an atomic formula, hence something with a truth value rather than an object.

0.33 Axioms

Axioms

Definition 0.33.1 (Axiom). Let $\mathcal{L}$ be a first-order language, and let $\mathrm{Sent}(\mathcal{L})$ denote the set of sentences of $\mathcal{L}$. If $T \subseteq \mathrm{Sent}(\mathcal{L})$ is a theory, then an *axiom of* T is an element $\alpha \in T$.

Definition 0.33.2 (Axiom System). An *axiom system* is a collection of axioms expressed in a common language.

Definition 0.33.3 (Assumption). An *assumption* is a statement temporarily accepted for purposes of reasoning.

0.34 Theories

Theories

Definition 0.34.1 (Theory). A *theory* over a first-order language $\mathcal{L}$ is a collection of closed sentences in that language:

$$T \subseteq \mathrm{Sentences}(\mathcal{L}).$$

If Γ is a chosen set of axioms, the theory generated by Γ is the deductive closure

$$T = \{\varphi \in \mathrm{Sentences}(\mathcal{L}) \mid \Gamma \vdash \varphi\}.$$

Definition 0.34.2 (Theorem). A *theorem* of a theory T is a sentence φ for which there is a finite formal derivation from T, written

$$T \vdash \varphi.$$

Equivalently, in proof-theoretic terms, φ is a theorem of T exactly when some derivation has leaves drawn from logical axioms or sentences of T, uses valid inference rules, and has φ as its root.

Definition 0.34.3 (Consequence). A sentence φ is a *semantic consequence* of a set of premises Γ, written $\Gamma \vDash \varphi$, if every model of Γ is also a model of φ. Equivalently, there is no structure in which all sentences of Γ are true while φ is false.

Definition 0.34.4 (Deductive Closure). Let $\Gamma \subseteq \mathrm{Sentences}(\mathcal{L})$ be a set of premises. The *deductive closure* of Γ, denoted $\mathrm{Cn}(\Gamma)$, is the set of all sentences derivable from Γ:

$$\mathrm{Cn}(\Gamma) = \{\varphi \in \mathrm{Sentences}(\mathcal{L}) \mid \Gamma \vdash \varphi\}.$$

0.35 Models

Models

Definition 0.35.1 (Interpretation). Let $\Sigma = (\mathcal{C}, \mathcal{F}, \mathcal{R}, \mathrm{ar})$ be a first-order signature, and let M be a nonempty set. An *interpretation of* Σ *on* M is an assignment $\mathcal{I}$ that maps each symbol of Σ to concrete data over M: constants are assigned elements of M, function symbols are assigned operations on M, and relation symbols are assigned relations on M.

Definition 0.35.2 ($\mathcal{L}$-Structure). Let $\mathcal{L}$ be a first-order signature. An $\mathcal{L}$-*structure* is a pair

$$\mathfrak{M} = (M, \mathcal{I}),$$

where M is a nonempty domain and $\mathcal{I}$ is an interpretation of the symbols of $\mathcal{L}$ on M.

Definition 0.35.3 (Model). Let $\mathcal{L}$ be a first-order signature, let T be a theory over $\mathcal{L}$, and let $\mathfrak{M} = (M, \mathcal{I})$ be an $\mathcal{L}$-structure. The structure $\mathfrak{M}$ is a *model of* T, written $\mathfrak{M} \vDash T$, if every sentence in T is true in $\mathfrak{M}$.

Definition 0.35.4 (Satisfaction)**.** Let $\mathfrak{M} = (M, \mathcal{I})$ be an $\mathcal{L}$-structure. A *variable assignment* is a function $s : \mathrm{Var} \to M$. The assignment extends recursively to a term-evaluation map $\bar{s}$ by

$$\bar{s}(x) = s(x), \qquad \bar{s}(c) = \mathcal{I}(c),$$

and

$$\bar{s}\big(f(t_1, \ldots, t_n)\big) = \mathcal{I}(f)\big(\bar{s}(t_1), \ldots, \bar{s}(t_n)\big).$$

The satisfaction relation $\mathfrak{M} \vDash \varphi[s]$ is then defined by induction on formulas: atomic relations are tested inside the interpreted relation, equality is interpreted as equality in M, Boolean connectives are interpreted by the corresponding truth operations, and quantifiers range over all elements of M.

Definition 0.35.5 (Countermodel)**.** Let $\Gamma \subseteq \mathrm{Sent}(\mathcal{L})$ be a set of premises, let $\varphi \in \mathrm{Sent}(\mathcal{L})$, and let $\mathfrak{M}$ be an $\mathcal{L}$-structure. A *countermodel to the entailment* $\Gamma \vDash \varphi$ is a structure $\mathfrak{M}$ that satisfies every sentence in Γ but does not satisfy φ.

0.36 Consistency

Consistency

Definition 0.36.1 (Contradiction)**.** A *contradiction* is a statement that cannot be true under a given interpretation.

Definition 0.36.2 (Consistent Axiom System)**.** A *consistent axiom system* is an axiom system that does not lead to contradiction.

Definition 0.36.3 (Inconsistent Axiom System)**.** An *inconsistent axiom system* is an axiom system containing or producing contradictory assertions.

Definition 0.36.4 (Model Existence Criterion (Informal))**.** Informally, if an axiom system has a model, then the axioms are mutually compatible.

Definition 0.36.5 (Relative Consistency)**.** Let T_0 and T_1 be theories. The theory T_1 is *consistent relative to* T_0 if

$$\mathrm{Con}(T_0) \implies \mathrm{Con}(T_1).$$

Equivalently, every contradiction in T_1 would imply a contradiction in T_0.

0.37 Independence

Independence Toolkit - Planned

0.38 Comparing Theories

Comparing Theories Toolkit - Planned

0.39 Examples And Previews

Examples And Previews Toolkit - Planned

0.40 Summary

Summary Toolkit - Planned

Proofs

Set Theory

Set Theory

0.41 Model of Set Theory

0.42 Sets

0.42.1 Sets, Membership, and ZFC Axioms

Definition 0.42.1 (Set and Membership)**.** In axiomatic set theory, the notions of *set* and *membership* are *primitive*.

- A *set* is an object.
- *Membership* is a binary relation, denoted by $\in$, between objects.

If x is an object and A is a set, the statement $x \in A$ is read as "x is an element of A." No definition of "set" or "$\in$" is given in more basic terms. Their meaning is determined entirely by the axioms governing them.

0.42.2 Set Constructions and Operations

Definition 0.42.2 (Empty Set)**.** The unique set with no elements is called the *empty set* and is denoted $\varnothing$.

Definition 0.42.3 (Subset)**.** Let A and B be sets. We say A is a *subset* of B, written $A \subseteq B$, if every element of A is also an element of B:

$$A \subseteq B \iff \forall x\,(x \in A \to x \in B).$$

Definition 0.42.4 (Proper Subset)**.** A is a *proper subset* of B, written $A \subsetneq B$, if $A \subseteq B$ and $A \neq B$.

Definition 0.42.5 (Set Equality)**.** Two sets A and B are *equal*, written $A = B$, if they have the same elements:

$$A = B \iff \forall x\,(x \in A \leftrightarrow x \in B).$$

Equivalently, $A = B \iff (A \subseteq B \wedge B \subseteq A)$.

Definition 0.42.6 (Union)**.** The *union* of A and B, denoted $A \cup B$, is the set of all elements belonging to at least one of the two sets:

$$A \cup B \;=\; \{\, x \mid x \in A \vee x \in B \,\}.$$

Definition 0.42.7 (Intersection)**.** The *intersection* of A and B, denoted $A \cap B$, is the set of all elements common to both:

$$A \cap B \;=\; \{\, x \mid x \in A \wedge x \in B \,\}.$$

Definition 0.42.8 (Set Difference)**.** The *set difference* $A \setminus B$ is the set of elements in A but not in B:

$$A \setminus B \;=\; \{\, x \mid x \in A \wedge x \notin B \,\}.$$

Definition 0.42.9 (Symmetric Difference)**.** The *symmetric difference* $A \triangle B$ is the set of elements belonging to exactly one of A and B:

$$A \triangle B \ = \ (A \setminus B) \cup (B \setminus A) \ = \ \{ \, x \mid (x \in A \lor x \in B) \land x \notin A \cap B \, \}.$$

Definition 0.42.10 (Complement)**.** Let U be a fixed universe and $A \subseteq U$. The *complement* of A relative to U, denoted A^c, is:

$$A^c \ = \ U \setminus A.$$

Definition 0.42.11 (Cartesian Product)**.** The *Cartesian product* of sets A and B is:

$$A \times B = \{ \, (a, b) \mid a \in A \text{ and } b \in B \, \}.$$

Definition 0.42.12 (Power Set)**.** The *power set* of A, denoted $\mathcal{P}(A)$, is the set of all subsets of A:

$$\mathcal{P}(A) \ := \ \{ \, S \mid S \subseteq A \, \}.$$

Theorem 0.42.13 (De Morgan's Laws)**.** *Let U be a universe and $A, B \subseteq U$. Then*

$$(A \cup B)^c = A^c \cap B^c \quad \text{and} \quad (A \cap B)^c = A^c \cup B^c.$$

Go to proof.

Definition 0.42.14 (Set Duality)**.** Two set-theoretic expressions over a fixed universe U are *dual* if one is obtained from the other by simultaneously replacing $\cup \leftrightarrow \cap$ and $\varnothing \leftrightarrow U$, with complements unchanged.

Corollary 0.42.15 (Principle of Set Duality)**.** *Any identity involving $\cup$, $\cap$, $\varnothing$, and U that holds for all subsets of a universe remains valid when each operation and constant is replaced by its dual.*

0.42.3 Algebraic Laws of Set Operations

Theorem 0.42.16 (Commutativity of Union and Intersection)**.** *Let A, B be sets. Then*

$$A \cup B = B \cup A \quad \text{and} \quad A \cap B = B \cap A.$$

Go to proof.

Theorem 0.42.17 (Associativity of Union and Intersection)**.** *Let A, B, C be sets. Then*

$$(A \cup B) \cup C = A \cup (B \cup C) \quad \text{and} \quad (A \cap B) \cap C = A \cap (B \cap C).$$

Go to proof.

Theorem 0.42.18 (Distributive Laws)**.** *Let A, B, C be sets. Then*

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C),$$

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C).$$

Go to proof.

Theorem 0.42.19 (Identity and Absorption Laws)**.** *Let A, B be sets and U a universe with $A \subseteq U$. Then*

$$A \cup \varnothing = A, \qquad A \cap U = A,$$

$$A \cup (A \cap B) = A, \qquad A \cap (A \cup B) = A.$$

Go to proof.

Theorem 0.42.20 (Involution of Complement)**.** *For any $A \subseteq U$,*

$$(A^c)^c = A.$$

Go to proof.

0.43 Families

0.43.1 Indexed Families of Sets

Definition 0.43.1 (Indexed Family of Sets)**.** Let I be a set (the *index set*) and U a universe. An *indexed family of sets* is a function

$$F : I \to \mathcal{P}(U),$$

with $F(i)$ typically written A_i. The family is denoted $\{A_i\}_{i\in I}$.

Definition 0.43.2 (Indexed Union)**.** The *union* of the indexed family $\{A_i\}_{i\in I}$ is:

$$\bigcup_{i\in I} A_i \;:=\; \{\, x \mid \exists i \in I \text{ such that } x \in A_i \,\}.$$

Definition 0.43.3 (Indexed Intersection)**.** For $I \neq \varnothing$, the *intersection* of $\{A_i\}_{i\in I}$ is:

$$\bigcap_{i\in I} A_i \;:=\; \{\, x \mid \forall i \in I,\; x \in A_i \,\}.$$

Theorem 0.43.4 (De Morgan's Laws for Indexed Families)**.** *Let $\{A_i\}_{i\in I}$ be an indexed family of subsets of a universe U. Then*

$$U \setminus \bigcup_{i\in I} A_i = \bigcap_{i\in I}(U \setminus A_i).$$

If $I \neq \varnothing$, then

$$U \setminus \bigcap_{i\in I} A_i = \bigcup_{i\in I}(U \setminus A_i).$$

Go to proof.

Definition 0.43.5 (Pairwise Disjoint Family)**.** The family $\{A_i\}_{i\in I}$ is *pairwise disjoint* if

$$\forall i, j \in I,\; i \neq j \to A_i \cap A_j = \varnothing.$$

Definition 0.43.6 (Cover)**.** A collection $\mathcal{C} \subseteq \mathcal{P}(A)$ is a *cover* of A if

$$\bigcup_{C\in\mathcal{C}} C = A.$$

Arbitrary Cartesian Products

Definition 0.43.7 (Arbitrary Cartesian Product)**.** Let $\{A_i\}_{i\in I}$ be an indexed family of sets. The *Cartesian product* is:

$$\prod_{i\in I} A_i \;:=\; \Big\{ f : I \to \bigcup_{i\in I} A_i \;\Big|\; \forall i \in I,\; f(i) \in A_i \Big\}.$$

0.43.2 Covers, Subcovers, and the Finite Intersection Property

Definition 0.43.8 (Cover)**.** Let A be a set and $\mathcal{C}$ a collection of sets. $\mathcal{C}$ is a *cover* of A if every element of A belongs to at least one member of $\mathcal{C}$:

$$A \;\subseteq\; \bigcup_{C\in\mathcal{C}} C.$$

Definition 0.43.9 (Subcover)**.** $\mathcal{C}'$ is a *subcover* of $\mathcal{C}$ if $\mathcal{C}' \subseteq \mathcal{C}$ and $\mathcal{C}'$ is still a cover of A.

Definition 0.43.10 (Finite Cover)**.** A cover is *finite* if it has finitely many members. A *finite subcover* is a subcover $\mathcal{C}' \subseteq \mathcal{C}$ with $|\mathcal{C}'| < \infty$.

Definition 0.43.11 (Open Cover)**.** In a topological space (X, τ), an *open cover* of a subset $A \subseteq X$ is a cover $\mathcal{C}$ of A all of whose members are open sets: $\mathcal{C} \subseteq \tau$.

Definition 0.43.12 (Finite Intersection Property)**.** A collection $\mathcal{F}$ of sets has the *finite intersection property* (FIP) if every finite subcollection has nonempty intersection: for all finite $\mathcal{F}' \subseteq \mathcal{F}$,

$$\bigcap_{F \in \mathcal{F}'} F \neq \varnothing.$$

Proposition 0.43.13 (FIP–Cover Duality)**.** *Let X be a set, $A \subseteq X$, and $\{U_\alpha\}_{\alpha \in I}$ a family of subsets of X. Let $F_\alpha = U_\alpha^c = X \setminus U_\alpha$. Then:*

$$\{U_\alpha\} \text{ covers } A \iff \{F_\alpha \cap A\}_{\alpha \in I} \text{ does not have the FIP.}$$

More precisely: the family $\{U_\alpha\}$ does not *cover A if and only if the family of closed complements $\{F_\alpha\}$ has the FIP relative to A.* *Go to proof.*

Proofs

Relations

0.44 Relations — Quick Reference

0.44.1 Ordered Pairs and Relations

Definition 0.44.1 (Ordered Pair)**.** Let a and b be sets. The *ordered pair* (a, b) is defined (Kuratowski) as

$$(a, b) \;:=\; \{\{a\}, \{a, b\}\}.$$

Theorem 0.44.2 (Uniqueness of Ordered Pairs)**.** *For any sets a, b, c, d,*

$$(a, b) = (c, d) \iff (a = c \wedge b = d).$$

Go to proof.

Definition 0.44.3 (Cartesian Product — as foundation for relations)**.** The *Cartesian product* of sets A and B is:

$$A \times B \;:=\; \{\, (a, b) \mid a \in A \text{ and } b \in B \,\}.$$

Definition 0.44.4 (Relation)**.** A *relation* from A to B is any subset $R \subseteq A \times B$.
If $(a, b) \in R$ we write $a\,R\,b$ and say "a is related to b." When $A = B$ we say R is a *relation on A*.

Definition 0.44.5 (Domain of a Relation)**.** Let $R \subseteq A \times B$ be a relation. The *domain* of R is

$$\mathrm{Dom}(R) \;:=\; \{\, a \in A \mid \exists b \in B,\ (a, b) \in R \,\}.$$

Definition 0.44.6 (Range of a Relation)**.** Let $R \subseteq A \times B$ be a relation. The *range* of R is

$$\mathrm{Ran}(R) \;:=\; \{\, b \in B \mid \exists a \in A,\ (a, b) \in R \,\}.$$

Tarski also calls this class the *counter-domain* or *converse domain*.

Definition 0.44.7 (Converse Relation)**.** If $R \subseteq A \times B$, the *converse* of R is the relation $R^{-1} \subseteq B \times A$ defined by

$$(b, a) \in R^{-1} \iff (a, b) \in R.$$

Equivalently, $b\,R^{-1}\,a$ iff aRb.

Definition 0.44.8 (Identity Relation)**.** For a set A, the *identity relation* on A is

$$\Delta_A \;:=\; \{\, (a, a) \mid a \in A \,\}.$$

Definition 0.44.9 (Composition of Relations)**.** Let $R \subseteq A \times B$ and $S \subseteq B \times C$. The *composition* $S \circ R \subseteq A \times C$ is defined by

$$(a, c) \in S \circ R \iff \exists b \in B,\ (a, b) \in R \text{ and } (b, c) \in S.$$

Theorem 0.44.10 (Associativity of Relation Composition)**.** *Let $R \subseteq A \times B$, $S \subseteq B \times C$, and $T \subseteq C \times D$. Then*

$$T \circ (S \circ R) = (T \circ S) \circ R.$$

Go to proof.

Theorem 0.44.11 (Identity Laws for Relation Composition)**.** *Let* $R \subseteq A \times B$*. Then*

$$R \circ \Delta_A = R \quad \textit{and} \quad \Delta_B \circ R = R.$$

Go to proof.

Definition 0.44.12 (Null Relation)**.** For sets A and B, the *null relation* from A to B is $\varnothing \subseteq A \times B$. No element of A is related to any element of B.

Definition 0.44.13 (Universal Relation)**.** For sets A and B, the *universal relation* from A to B is $A \times B$ itself. Every element of A is related to every element of B.

Definition 0.44.14 (Relation Inclusion)**.** Let $R, S \subseteq A \times B$. Relation inclusion is ordinary set inclusion:

$$R \subseteq S \iff \forall a \in A \, \forall b \in B, \; ((a, b) \in R \Rightarrow (a, b) \in S),$$

Definition 0.44.15 (Relation Equality)**.** Let $R, S \subseteq A \times B$. Relation equality is ordinary set equality:

$$R = S \iff R \subseteq S \text{ and } S \subseteq R.$$

Definition 0.44.16 (Union of Relations)**.** Let $R, S \subseteq A \times B$. The *union* $R \cup S$ is the relation with

$$(a, b) \in R \cup S \iff (a, b) \in R \text{ or } (a, b) \in S.$$

Definition 0.44.17 (Intersection of Relations)**.** Let $R, S \subseteq A \times B$. The *intersection* $R \cap S$ is the relation with

$$(a, b) \in R \cap S \iff (a, b) \in R \text{ and } (a, b) \in S.$$

Definition 0.44.18 (Difference of Relations)**.** Let $R, S \subseteq A \times B$. The *difference* $R \setminus S$ is the relation with

$$(a, b) \in R \setminus S \iff (a, b) \in R \text{ and } (a, b) \notin S.$$

Definition 0.44.19 (Complement of a Relation)**.** Let $R \subseteq A \times B$. The *complement* of R relative to $A \times B$ is

$$(A \times B) \setminus R.$$

Theorem 0.44.20 (Converse Laws for Relation Operations)**.** *Let* $R, S \subseteq A \times B$*, let* $T \subseteq B \times C$*, and take complements relative to the indicated Cartesian products. Then*

$$(R^{-1})^{-1} = R, \qquad (T \circ R)^{-1} = R^{-1} \circ T^{-1},$$
$$(R \cup S)^{-1} = R^{-1} \cup S^{-1}, \qquad (R \cap S)^{-1} = R^{-1} \cap S^{-1},$$
$$(R \setminus S)^{-1} = R^{-1} \setminus S^{-1}, \qquad ((A \times B) \setminus R)^{-1} = (B \times A) \setminus R^{-1}.$$

Go to proof.

Theorem 0.44.21 (Relation Composition and Boolean Operations)**.** *Let* $P, Q \subseteq A \times B$*,* $R \subseteq B \times C$*, and* $S, T \subseteq B \times C$*. Then*

$$R \circ (P \cup Q) = (R \circ P) \cup (R \circ Q), \qquad (S \cup T) \circ P = (S \circ P) \cup (T \circ P).$$

Also,

$$R \circ (P \cap Q) \subseteq (R \circ P) \cap (R \circ Q), \qquad (S \cap T) \circ P \subseteq (S \circ P) \cap (T \circ P).$$

The analogous formulas for difference and complement hold only as inclusions or under extra hypotheses; they are not general distributive laws. *Go to proof.*

Definition 0.44.22 (Transitive Closure)**.** Let R be a relation on A. The *transitive closure* of R is the smallest transitive relation on A containing R.

Definition 0.44.23 (Equivalence Closure)**.** Let R be a relation on A. The *equivalence closure* of R is the smallest equivalence relation on A containing R.

Definition 0.44.24 (One-to-Many Relation)**.** Let $R \subseteq A \times B$. R is *one-to-many* if some input is related to two distinct outputs:

$$\exists a \in A\, \exists b_1, b_2 \in B, \quad (a, b_1) \in R \wedge (a, b_2) \in R \wedge b_1 \neq b_2.$$

Definition 0.44.25 (Many-to-One Relation)**.** Let $R \subseteq A \times B$. R is *many-to-one* if two distinct inputs are related to a common output:

$$\exists a_1, a_2 \in A\, \exists b \in B, \quad (a_1, b) \in R \wedge (a_2, b) \in R \wedge a_1 \neq a_2.$$

Definition 0.44.26 (Many-to-Many Relation)**.** Let $R \subseteq A \times B$. R is *many-to-many* if it is one-to-many and many-to-one.

Definition 0.44.27 (Many-Place Relation)**.** A *ternary relation* among sets A, B, and C is a subset of $A \times B \times C$. More generally, an n-place relation on $A_1, \ldots, A_n$ is a subset of

$$A_1 \times \cdots \times A_n.$$

For example, betweenness in geometry is naturally a three-place relation, and the arithmetic statement $x + y = z$ is naturally a three-place relation among x, y, and z.

0.44.2 Properties of Relations

Definition 0.44.28 (Reflexive Relation)**.** A binary relation $R \subseteq A \times A$ is *reflexive* on A if every element of the domain is related to itself:

$$\forall a \in A,\ (a, a) \in R.$$

In infix notation, this is the condition $\forall a \in A,\ aRa$.

Definition 0.44.29 (Irreflexive Relation)**.** A binary relation $R \subseteq A \times A$ is *irreflexive* on A if no element of the domain is related to itself:

$$\forall a \in A,\ (a, a) \notin R.$$

In infix notation, this is the condition $\forall a \in A,\ \neg(aRa)$.

Definition 0.44.30 (Symmetric Relation)**.** A binary relation $R \subseteq A \times A$ is *symmetric* on A if every related pair also appears in the reverse order:

$$\forall a, b \in A,\ (a, b) \in R \to (b, a) \in R.$$

In infix notation, this is $\forall a, b \in A,\ aRb \to bRa$.

Definition 0.44.31 (Antisymmetric Relation)**.** A binary relation $R \subseteq A \times A$ is *antisymmetric* on A if any two elements related in both directions must be equal:

$$\forall a, b \in A,\ \big((a, b) \in R \wedge (b, a) \in R\big) \to a = b.$$

In infix notation, this is $\forall a, b \in A,\ ((aRb \wedge bRa) \to a = b)$.

Definition 0.44.32 (Asymmetric Relation)**.** A binary relation $R \subseteq A \times A$ is *asymmetric* on A if every related pair excludes the reverse pair:

$$\forall a, b \in A,\ (a, b) \in R \to (b, a) \notin R.$$

In infix notation, this is $\forall a, b \in A,\ (aRb \to \neg(bRa))$.

Definition 0.44.33 (Transitive Relation)**.** A binary relation $R \subseteq A \times A$ is *transitive* on A if any two composable related pairs collapse to a direct related pair:

$$\forall a, b, c \in A,\ \big((a, b) \in R \wedge (b, c) \in R\big) \to (a, c) \in R.$$

In infix notation, this is $\forall a, b, c \in A,\ ((aRb \wedge bRc) \to aRc)$.

Definition 0.44.34 (Total (Connex) Relation)**.** R is *total* (or *connex*) if every pair is comparable:

$$\forall a, b \in A,\ (a, b) \in R \vee (b, a) \in R.$$

Definition 0.44.35 (Equivalence Relation)**.** R is an *equivalence relation* on A if it is reflexive, symmetric, and transitive.

Definition 0.44.36 (Preorder)**.** R is a *preorder* on A if it is reflexive and transitive.

Definition 0.44.37 (Partial Order)**.** R is a *partial order* on A if it is reflexive, antisymmetric, and transitive.

Definition 0.44.38 (Strict Partial Order)**.** R is a *strict partial order* on A if it is irreflexive and transitive.

Definition 0.44.39 (Total Order)**.** R is a *total order* on A if it is a partial order and is total.

0.45 Equivalence

0.45.1 Equivalence Classes and Partitions

Definition 0.45.1 (Equivalence Class)**.** Let R be an equivalence relation on A. For $a \in A$, the *equivalence class* of a is:

$$[a]_R \ := \ \{\, x \in A \mid (a, x) \in R \,\}.$$

When R is clear from context, we write $[a]$.

Definition 0.45.2 (Quotient Set)**.** The *quotient set* of A by R is the set of all equivalence classes:

$$A/R \ := \ \{\, [a]_R \mid a \in A \,\}.$$

Definition 0.45.3 (Index of an Equivalence Relation)**.** The *index* of R on A is the cardinality $|A/R|$, i.e. the number of equivalence classes.

Definition 0.45.4 (Canonical Surjection)**.** The *canonical surjection* (quotient map) is the function

$$\pi : A \to A/R, \qquad \pi(a) := [a].$$

Definition 0.45.5 (Partition)**.** A *partition* of A is a collection $\mathcal{P}$ of subsets of A such that:

(i) every block is nonempty: $\forall P \in \mathcal{P},\ P \neq \varnothing$;

(ii) distinct blocks are disjoint: $\forall P, Q \in \mathcal{P},\ P \neq Q \to P \cap Q = \varnothing$;

(iii) the blocks cover A: $\bigcup_{P \in \mathcal{P}} P = A$.

The sets $P \in \mathcal{P}$ are called the *blocks* of the partition.

Lemma 0.45.6 (Representative Independence Lemma)**.** *Let R be an equivalence relation on A. For any $a, b \in A$,*

$$[a] = [b] \iff (a, b) \in R.$$

Go to proof.

Theorem 0.45.7 (Equivalence Relations and Partitions)**.** *Let A be a set.*

(i) If R is an equivalence relation on A, then A/R is a partition of A.

(ii) If $\mathcal{P}$ is a partition of A, then the relation $R_{\mathcal{P}}$ defined by

$$(a,b) \in R_{\mathcal{P}} \iff \exists P \in \mathcal{P} \text{ with } a \in P \text{ and } b \in P$$

is an equivalence relation on A.

(iii) These constructions are inverse: $R_{A/R} = R$ and $A/R_{\mathcal{P}} = \mathcal{P}$.

Go to proof.

Theorem 0.45.8 (Universal Property of the Quotient Map)**.** *Let R be an equivalence relation on A, let $\pi : A \to A/R$ be the canonical surjection, and let $f : A \to B$ be any function. Then f is* constant on equivalence classes *— meaning $(a,b) \in R$ implies $f(a) = f(b)$ — if and only if there exists a unique function $\bar{f} : A/R \to B$ such that*

$$f = \bar{f} \circ \pi.$$

When it exists, $\bar{f}$ is defined by $\bar{f}([a]) = f(a)$. *Go to proof.*

Proofs

Functions

0.46 Functions — Quick Reference

0.46.1 Functions

Definition 0.46.1 (Functional Relation)**.** Let $R \subseteq A \times B$ be a relation. We say that R is *functional from A to B* if each input has at most one output:

$$\forall a \in A \, \forall b_1, b_2 \in B, \quad \big((a, b_1) \in R \land (a, b_2) \in R\big) \Rightarrow b_1 = b_2.$$

Definition 0.46.2 (Total Relation on a Domain)**.** Let $R \subseteq A \times B$ be a relation. We say that R is *total on A* if each input has at least one output:

$$\forall a \in A \, \exists b \in B, \quad (a, b) \in R.$$

Definition 0.46.3 (Function)**.** Let A and B be sets. A *function* from A to B is a relation $f \subseteq A \times B$, together with the declared domain A and codomain B, such that:

(i) (*Existence / left-total*) for every $a \in A$, there exists $b \in B$ with $(a, b) \in f$;

(ii) (*Uniqueness / right-unique*) if $(a, b_1) \in f$ and $(a, b_2) \in f$ then $b_1 = b_2$.

If $(a, b) \in f$ we write $f(a) = b$.

Theorem 0.46.4 (Functional Relation Characterization of Functions)**.** *Let A and B be sets, and let $R \subseteq A \times B$. Then R is a function from A to B if and only if*

$$\forall a \in A \, \exists! b \in B, \quad (a, b) \in R.$$

Equivalently, R is a function from A to B if and only if R is total on A and functional from A to B. *Go to proof.*

Definition 0.46.5 (Partial Function)**.** Let A and B be sets. A *partial function* from A to B is a functional relation $R \subseteq A \times B$. It need not be total on A: some inputs may have no output.

Proposition 0.46.6 (Partial Functions Are Functional Relations)**.** *For sets A and B and a relation $R \subseteq A \times B$, R is a partial function from A to B if and only if R is functional from A to B.* *Go to proof.*

Definition 0.46.7 (Function Equality)**.** For functions $f : A \to B$ and $g : C \to D$, we write $f = g$ if

$$A = C, \qquad B = D, \qquad \text{and} \qquad \forall a \in A,\ f(a) = g(a).$$

Proposition 0.46.8 (One-to-Many Relations Are Not Functional)**.** *If $R \subseteq A \times B$ is one-to-many, then R is not functional from A to B. Consequently, R is not a function from A to B.* *Go to proof.*

Definition 0.46.9 (Domain of a Function)**.** If f is a function from A to B, we write $f : A \to B$, where A is the *domain* $\operatorname{dom}(f)$.

Definition 0.46.10 (Codomain of a Function)**.** If f is a function from A to B, we write $f : A \to B$, where B is the *codomain* $\mathrm{cod}(f)$.

Definition 0.46.11 (Image of a Function)**.** The *image* (or *range*) of $f : A \to B$ is:

$$\mathrm{im}(f) \;:=\; \{\, b \in B \mid \exists a \in A,\; f(a) = b \,\}.$$

The image may be a proper subset of the codomain.

Definition 0.46.12 (Image of a Set)**.** For $S \subseteq A$, the *image of S under f* is:

$$f(S) \;:=\; \{\, f(a) \mid a \in S \,\}.$$

Note $f(A) = \mathrm{im}(f)$.

Definition 0.46.13 (Preimage)**.** For $T \subseteq B$, the *preimage* (inverse image) of T under f is:

$$f^{-1}(T) \;:=\; \{\, a \in A \mid f(a) \in T \,\}.$$

Definition 0.46.14 (Fiber)**.** The *fiber* of f over $b \in B$ is the preimage of the singleton:

$$f^{-1}(\{b\}) \;=\; \{\, a \in A \mid f(a) = b \,\}.$$

Definition 0.46.15 (Graph of a Function)**.** The *graph* of $f : A \to B$ is:

$$\mathrm{Graph}(f) \;:=\; \{\, (a, b) \in A \times B \mid b = f(a) \,\}.$$

Definition 0.46.16 (Injective Function)**.** $f : A \to B$ is *injective* (one-to-one) if distinct elements of A have distinct images:

$$\forall a_1, a_2 \in A,\; f(a_1) = f(a_2) \implies a_1 = a_2.$$

Definition 0.46.17 (Empty Function)**.** For every set B, the *empty function* from $\varnothing$ to B is the empty relation

$$\varnothing : \varnothing \to B.$$

It is the unique function with domain $\varnothing$ and codomain B.

Definition 0.46.18 (Surjective Function)**.** $f : A \to B$ is *surjective* (onto) if every element of B is achieved:

$$\forall b \in B,\; \exists a \in A \text{ such that } f(a) = b.$$

Equivalently, $\mathrm{im}(f) = B$.

Definition 0.46.19 (Bijective Function)**.** $f : A \to B$ is *bijective* if it is both injective and surjective.

Proposition 0.46.20 (One-One Relations and Converse Functionality)**.** *Let $f : A \to B$ be a function, identified with its graph $G_f \subseteq A \times B$. Then f is injective if and only if the converse relation $G_f^{-1} \subseteq B \times A$ is functional from B to A.* *Go to proof.*

Proposition 0.46.21 (Bijections Have Unique Correspondence Both Ways)**.** *Let $f : A \to B$ be a function. Then f is bijective if and only if*

$$\forall a \in A\, \exists! b \in B, \quad f(a) = b,$$

and

$$\forall b \in B\, \exists! a \in A, \quad f(a) = b.$$

Go to proof.

Definition 0.46.22 (Many-Place Functional Relation)**.** A relation $R \subseteq A_1 \times \cdots \times A_n \times B$ is *functional in the last coordinate* if

$$\forall a_1 \in A_1 \cdots \forall a_n \in A_n\, \forall b_1, b_2 \in B,$$
$$\big((a_1, \ldots, a_n, b_1) \in R \wedge (a_1, \ldots, a_n, b_2) \in R\big) \Rightarrow b_1 = b_2.$$

When such a relation is also total in the first n coordinates, it determines a function of n variables, written $f(a_1, \ldots, a_n) = b$.

Definition 0.46.23 (Identity Function)**.** The *identity function* on A is $\mathrm{id}_A : A \to A$, $\mathrm{id}_A(a) = a$.

Definition 0.46.24 (Inclusion Map)**.** If $A \subseteq B$, the *inclusion map* is $\iota : A \hookrightarrow B$, $\iota(a) = a$.

Definition 0.46.25 (Constant Function)**.** $f : A \to B$ is *constant* if there exists $b_0 \in B$ with $f(a) = b_0$ for all $a \in A$.

0.46.2 Composition, Inverses, and Set-Image Laws

Definition 0.46.26 (Composition)**.** Let $f : A \to B$ and $g : B \to C$. The *composition* $g \circ f : A \to C$ is:

$$(g \circ f)(a) := g\big(f(a)\big) \quad \text{for all } a \in A.$$

Theorem 0.46.27 (Associativity of Composition)**.** *For $f : A \to B$, $g : B \to C$, $h : C \to D$:*

$$h \circ (g \circ f) = (h \circ g) \circ f.$$

Go to proof.

Theorem 0.46.28 (Identity and Composition)**.** *For any $f : A \to B$:*

$$f \circ \mathrm{id}_A = f \quad \textit{and} \quad \mathrm{id}_B \circ f = f.$$

Go to proof.

Theorem 0.46.29 (Injectivity and Surjectivity Under Composition)**.** *Let $f : A \to B$ and $g : B \to C$.*

(i) If f and g are injective, then $g \circ f$ is injective.

(ii) If f and g are surjective, then $g \circ f$ is surjective.

(iii) If $g \circ f$ is injective, then f is injective.

(iv) If $g \circ f$ is surjective, then g is surjective.

Go to proof.

Definition 0.46.30 (Inverse Function)**.** Let $f : A \to B$ be bijective. The *inverse function* is $f^{-1} : B \to A$ defined by: $f^{-1}(b) = a$ iff $f(a) = b$.

Theorem 0.46.31 (Characterization of Inverse Functions)**.** *Let $f : A \to B$ be bijective. Then*

$$f^{-1} \circ f = \mathrm{id}_A \quad \textit{and} \quad f \circ f^{-1} = \mathrm{id}_B\,.$$

Conversely, a function admits an inverse iff it is bijective. *Go to proof.*

Theorem 0.46.32 (Inverse of a Composition)**.** *Let $f : A \to B$ and $g : B \to C$ be bijective. Then $g \circ f$ is bijective and*

$$(g \circ f)^{-1} = f^{-1} \circ g^{-1}.$$

Go to proof.

Definition 0.46.33 (Left Inverse)**.** Let $f : A \to B$. A *left inverse* of f is a function $g : B \to A$ with $g \circ f = \mathrm{id}_A$.

Definition 0.46.34 (Right Inverse)**.** Let $f : A \to B$. A *right inverse* (or *section*) of f is a function $h : B \to A$ with $f \circ h = \mathrm{id}_B$.

Theorem 0.46.35 (One-Sided Inverses and Function Properties)**.** *Let $f : A \to B$.*

(i) f has a left inverse $\iff$ f is injective (and $A \neq \varnothing$ or $A = B = \varnothing$).

(ii) f has a right inverse $\iff$ f is surjective.

(iii) If f has both a left inverse g and a right inverse h, then $g = h$ and f is bijective.

Go to proof.

Definition 0.46.36 (Restriction)**.** Let $f : A \to B$ and $C \subseteq A$. The *restriction* of f to C is $f|_C : C \to B$, $f|_C(a) = f(a)$ for all $a \in C$.

Definition 0.46.37 (Extension)**.** Let $A \subseteq A'$, and let $f : A \to B$. A function $g : A' \to B$ is an *extension* of f if $g|_A = f$.

Theorem 0.46.38 (Preimages Preserve Set Operations)**.** *Let $f : A \to B$ and $S, T \subseteq B$. Then*

(i) $f^{-1}(S \cup T) = f^{-1}(S) \cup f^{-1}(T)$,

(ii) $f^{-1}(S \cap T) = f^{-1}(S) \cap f^{-1}(T)$,

(iii) $f^{-1}(S \setminus T) = f^{-1}(S) \setminus f^{-1}(T)$,

(iv) $f^{-1}(S^c) = (f^{-1}(S))^c$.

Go to proof.

Theorem 0.46.39 (Images and Set Operations)**.** *Let $f : A \to B$ and $S, T \subseteq A$. Then*

(i) $f(S \cup T) = f(S) \cup f(T)$,

(ii) $f(S \cap T) \subseteq f(S) \cap f(T)$,

(iii) $f(S \setminus T) \supseteq f(S) \setminus f(T)$.

Equality holds in (ii) and (iii) for all S, T iff f is injective. *Go to proof.*

Theorem 0.46.40 (Image–Preimage Adjunction)**.** *Let $f : A \to B$, let $S \subseteq A$, and let $T \subseteq B$. Then*

$$S \subseteq f^{-1}(f(S)), \qquad f(f^{-1}(T)) \subseteq T.$$

Moreover, $S = f^{-1}(f(S))$ for every $S \subseteq A$ iff f is injective, and $f(f^{-1}(T)) = T$ for every $T \subseteq B$ iff f is surjective. *Go to proof.*

Proofs

Orderings

0.47 Order

0.47.1 Ordered Sets

Definition 0.47.1 (Ordered Set)**.** An *ordered set* is a set equipped with an order relation. Unless explicitly qualified, "ordered set" means a partially ordered set, and its relation is written $\leq$.

Definition 0.47.2 (Partially Ordered Set)**.** A *partially ordered set* is a pair $(A, \leq)$ where A is a set and $\leq$ is a partial order on A (a relation that is reflexive, antisymmetric, and transitive).

Definition 0.47.3 (Order-Structure Synonyms)**.** The following short forms abbreviate their full names and carry no independent mathematical content; each resolves to the canonical structure it names:

- *poset* := partially ordered set.
- *toset*, *loset*, *linearly ordered set* := totally ordered set (total order and linear order coincide, so these name one structure).
- *woset* := well-ordered set.

Definition 0.47.4 (Strict Order)**.** Let $(A, \leq)$ be an ordered set. The *strict order* $<$ is defined by:

$$a < b \iff (a \leq b \land a \neq b).$$

Definition 0.47.5 (Comparable and Incomparable Elements)**.** In an ordered set $(A, \leq)$, elements $a, b \in A$ are *comparable* if $a \leq b$ or $b \leq a$; they are *incomparable* if neither holds.

Definition 0.47.6 (Totally Ordered Set)**.** A *totally ordered set* (or *linearly ordered set*) is a pair $(A, \leq)$ where $\leq$ is a total order on A — equivalently, a partially ordered set in which every pair of elements is comparable:

$$\forall a, b \in A,\ a \leq b \lor b \leq a.$$

Definition 0.47.7 (Upper and Lower Bounds)**.** Let $(A, \leq)$ be an ordered set and $S \subseteq A$.
An element $u \in A$ is an *upper bound* of S if $\forall s \in S,\ s \leq u$.
An element $\ell \in A$ is a *lower bound* of S if $\forall s \in S,\ \ell \leq s$.

Definition 0.47.8 (Minimal Element)**.** Let $S \subseteq A$ in an ordered set $(A, \leq)$. An element $m \in S$ is a *minimal element* of S if there is no $s \in S$ with $s < m$.

Definition 0.47.9 (Maximal Element)**.** Let $S \subseteq A$ in an ordered set $(A, \leq)$. An element $M \in S$ is a *maximal element* of S if there is no $s \in S$ with $M < s$.

Definition 0.47.10 (Least Element)**.** Let $S \subseteq A$ in an ordered set $(A, \leq)$. An element $\ell \in S$ is the *least element* of S if

$$\forall s \in S,\ \ell \leq s.$$

Definition 0.47.11 (Greatest Element)**.** Let $S \subseteq A$ in an ordered set $(A, \leq)$. An element $g \in S$ is the *greatest element* of S if

$$\forall s \in S,\ s \leq g.$$

Definition 0.47.12 (Order-Preserving Map)**.** Let $(M, \leq)$ and $(M', \leq')$ be partially ordered sets. A function $f : M \to M'$ is *order-preserving* (or *monotone*) if

$$\forall a, b \in M, \quad a \leq b \implies f(a) \leq' f(b).$$

Definition 0.47.13 (Order Isomorphism)**.** Let $(M, \leq)$ and $(M', \leq')$ be partially ordered sets. A function $f : M \to M'$ is an *order isomorphism* if it is bijective and

$$\forall a, b \in M, \quad a \leq b \iff f(a) \leq' f(b).$$

Definition 0.47.14 (Well-Ordered Set)**.** A totally ordered set $(A, \leq)$ is *well-ordered* if every nonempty subset $S \subseteq A$ has a least element:

$$\forall S \subseteq A,\ \big(S \neq \varnothing \ \Rightarrow\ \exists m \in S \text{ s.t. } m \leq s\ \forall s \in S\big).$$

Definition 0.47.15 (Chain and Antichain)**.** A subset $C \subseteq A$ of a poset $(A, \leq)$ is a *chain* if every pair of elements in C is comparable. It is an *antichain* if no two distinct elements of C are comparable.

Definition 0.47.16 (Initial Segment)**.** A subset $I \subseteq A$ is an *initial segment* of $(A, \leq)$ if

$$a \in I \text{ and } b \leq a \ \Rightarrow\ b \in I.$$

Definition 0.47.17 (Directed Set)**.** A preorder $(I, \leq)$ is *directed* (or a *directed set*) if every finite subset of I has an upper bound in I: for all $i, j \in I$ there exists $k \in I$ with $i \leq k$ and $j \leq k$.

Definition 0.47.18 (Net)**.** Let $(I, \leq)$ be a directed set and X a set. A *net* in X is a function $x : I \to X$. The value at $i \in I$ is written x_i, and the net is written $(x_i)_{i \in I}$.

0.47.2 Supremum and Infimum

Definition 0.47.19 (Supremum and Infimum)**.** Let $(A, \leq)$ be a poset and $S \subseteq A$.
The *supremum* (or *least upper bound*) of S, written $\sup S$, is an element $u^* \in A$ satisfying:

(i) u^* is an upper bound of S: $\forall s \in S,\ s \leq u^*$;

(ii) u^* is the *least* upper bound: if u is *any* upper bound of S, then $u^* \leq u$.

The *infimum* (or *greatest lower bound*) of S, written $\inf S$, is an element $\ell^* \in A$ satisfying:

(i) ℓ^* is a lower bound of S: $\forall s \in S,\ \ell^* \leq s$;

(ii) ℓ^* is the *greatest* lower bound: if ℓ is *any* lower bound of S, then $\ell \leq \ell^*$.

Proposition 0.47.20 (Uniqueness of Supremum and Infimum)**.** *Let $(A, \leq)$ be a poset and $S \subseteq A$. If $\sup S$ exists, it is unique. If $\inf S$ exists, it is unique.*

Proposition 0.47.21 (Characterisation of Supremum)**.** *Let $(A, \leq)$ be a poset, $S \subseteq A$, and $u^* \in A$. Then $u^* = \sup S$ if and only if:*

(i) u^* is an upper bound of S; and

(ii) for every $v \in A$ with $v < u^*$, there exists $s \in S$ with $v < s$.

Proposition 0.47.22 (Duality of Supremum and Infimum)**.** *Let $(A, \leq)$ be a poset and $S \subseteq A$. Then $\inf_{(A,\leq)} S = \sup_{(A,\geq)} S$, where $(A, \geq)$ denotes the dual poset.*
In particular: if every nonempty bounded-above subset of $(A, \leq)$ has a supremum, then every nonempty bounded-below subset of $(A, \leq)$ has an infimum.

0.47.3 Lattices

Definition 0.47.23 (Lattice)**.** A *lattice* is a poset $(L, \leq)$ in which every two-element subset $\{a, b\} \subseteq L$ has both a supremum and an infimum in L.

The supremum $\sup\{a, b\}$ is called the *join* of a and b, written $a \vee b$. The infimum $\inf\{a, b\}$ is called the *meet* of a and b, written $a \wedge b$.

Definition 0.47.24 (Complete Lattice)**.** A lattice $(L, \leq)$ is a *complete lattice* if every subset $S \subseteq L$ (including the empty set and L itself) has both a supremum and an infimum in L:

$$\forall S \subseteq L, \quad \sup S \in L \text{ and } \inf S \in L.$$

Definition 0.47.25 (Distributive Lattice)**.** A lattice $(L, \leq)$ is *distributive* if for all $a, b, c \in L$:

$$a \wedge (b \vee c) = (a \wedge b) \vee (a \wedge c).$$

Definition 0.47.26 (Complement and Boolean Lattice)**.** Let $(L, \leq)$ be a bounded distributive lattice with top element $\top$ and bottom element $\bot$. An element $a' \in L$ is a *complement* of $a \in L$ if

$$a \vee a' = \top \quad \text{and} \quad a \wedge a' = \bot.$$

A *Boolean lattice* (or *Boolean algebra*) is a bounded distributive lattice in which every element has a complement.

0.47.4 Hasse Diagrams and the Duality Principle

Definition 0.47.27 (Cover Relation)**.** Let $(A, \leq)$ be a poset and $a, b \in A$. We say b *covers* a, written $a \lessdot b$, if $a < b$ and there is no $c \in A$ with $a < c < b$.

Definition 0.47.28 (Hasse Diagram)**.** The *Hasse diagram* of a finite poset $(A, \leq)$ is the directed graph whose vertices are the elements of A, with an upward edge from a to b whenever $a \lessdot b$. Three graphical conventions apply:

(i) Reflexive loops ($a \leq a$) are suppressed.

(ii) Edges implied by transitivity are suppressed: if $a \lessdot b$ and $b \lessdot c$, no direct edge from a to c is drawn.

(iii) Direction is encoded by height: $a \lessdot b$ is drawn with b strictly above a, so arrowheads are unnecessary.

Definition 0.47.29 (Dual Poset)**.** Let $(A, \leq)$ be a poset. The *dual poset* is $(A, \geq)$, where $a \geq b$ is defined to mean $b \leq a$. The dual is also written $(A, \leq^{\mathrm{op}})$.

Proposition 0.47.30 (Duality Principle for Posets)**.** *Let Φ be any first-order statement about a poset $(A, \leq)$ expressed using only the relation $\leq$. Let Φ^* be the statement obtained from Φ by replacing every occurrence of $\leq$ with $\geq$ (equivalently, working in the dual poset). Then:*

$$(A, \leq) \models \Phi \iff (A, \geq) \models \Phi.$$

In particular, if Φ is a theorem about all posets, then so is Φ^*.

0.47.5 Order Extensions

Definition 0.47.31 (Symmetric and Asymmetric Parts)**.** For any binary relation R on X, define:

- the *symmetric part*: $xIy \iff xRy$ and yRx
- the *asymmetric part*: $xPy \iff xRy$ but not yRx

Note that $R = P \cup I$.

Definition 0.47.32 (Maximal Elements and Upper Bounds)**.** Let $\succsim$ be a binary relation on X.

- The set of *maximal elements* of X is

$$\mathrm{Max}(X, \succsim) = \{x \in X \mid y \succsim x \text{ for no } y \in X \text{ with } y \neq x\}.$$

- The set of *upper bounds* of X is

$$M(X, \succsim) = \{x \in X \mid x \succsim y \text{ for all } y \in X\}.$$

Definition 0.47.33 (Extension of a Preorder)**.** Let $\succsim$ be a preorder on X. A preorder $\succsim'$ is an *extension* of $\succsim$ if

$$x \succsim y \;\Rightarrow\; x \succsim' y, \qquad x \succ y \;\Rightarrow\; x \succ' y.$$

A *complete extension* is an extension that is also complete.

Corollary 0.47.34 (Complete preorder extension)**.** *For any nonempty set X and preorder $\succsim$ on X, there exists a complete preorder that is an extension of $\succsim$.*

Definition 0.47.35 (and Proposition (Only Weak Cycles))**.** Let $\succsim$ be a binary relation on X with asymmetric part $\succ$ and transitive closure $T(\succsim)$. We say $\succsim$ satisfies *only weak cycles* (OWC) if

$$xT(\succsim)y \;\Rightarrow\; \neg(y \succ x).$$

Proposition. A binary relation $\succsim$ on a nonempty set X can be extended to a complete preorder if and only if it satisfies OWC.

0.47.6 Induced Orders and Order Embeddings

Definition 0.47.36 (Induced Order)**.** Let $(B, \leq')$ be a partially ordered set and let $f : A \to B$ be a function. The *order induced by f* (or *pullback order along f*) is the relation $\leq_f$ on A defined by:

$$x \;\leq_f\; y \iff f(x) \;\leq'\; f(y).$$

Proposition 0.47.37 ($\leq_f$ is always a preorder)**.** *For any function $f : A \to B$ and partial order $(B, \leq')$, the relation $\leq_f$ is a preorder on A: it is reflexive and transitive.*

Definition 0.47.38 (f-Indistinguishable Elements)**.** Let $f : A \to B$ and $\leq_f$ be the induced order. Two elements $x, y \in A$ are *f-indistinguishable*, written $x \sim_f y$, if both $x \leq_f y$ and $y \leq_f x$, i.e. if

$$f(x) \leq' f(y) \quad \text{and} \quad f(y) \leq' f(x).$$

Proposition 0.47.39 ($\leq_f$ is a partial order iff f is injective)**.** *Let $(B, \leq')$ be a partially ordered set and $f : A \to B$. The induced order $\leq_f$ is a partial order on A if and only if f is injective.*

Definition 0.47.40 (Suborder / Restriction)**.** Let $(B, \leq')$ be a partially ordered set and $S \subseteq B$. The *suborder* on S (or *induced order on S*) is the relation $\leq'_S$ defined by:

$$x \;\leq'_S\; y \iff x \leq' y, \quad \text{for } x, y \in S.$$

Definition 0.47.41 (Order Embedding)**.** Let $(A, \leq)$ and $(B, \leq')$ be partially ordered sets. A function $f : A \to B$ is an *order embedding* if for all $x, y \in A$:

$$x \;\leq\; y \iff f(x) \;\leq'\; f(y).$$

Proposition 0.47.42 (Order embeddings are injective)**.** *Every order embedding $f : (A, \leq) \to (B, \leq')$ is injective. Go to proof.*

Proposition 0.47.43 (Order embedding is isomorphism onto image)**.** *If $f : (A, \leq) \to (B, \leq')$ is an order embedding, then f is an order isomorphism from $(A, \leq)$ to the suborder $(f(A), \leq'_{f(A)})$. Go to proof.*

0.48 Zorn

0.48.1 Zorn's Lemma and the Axiom of Choice

Proofs

Cardinality

0.49 Cardinality

0.49.1 Cardinality and Infinite Sets

Definition 0.49.1 (Same Cardinality)**.** Two sets A and B have the *same cardinality*, written $A \sim B$, if there exists a bijection $f : A \to B$.

Definition 0.49.2 (Finite and Infinite Sets)**.** A set A is *finite* if $A = \varnothing$ or $A \sim \{1, 2, \ldots, n\}$ for some $n \in \mathbb{N}$. A set is *infinite* if it is not finite.

Definition 0.49.3 (Countable and Uncountable Sets)**.** A set A is *countably infinite* if $A \sim \mathbb{N}$. A set is *countable* if it is finite or countably infinite. A set is *uncountable* if it is infinite but not countable.

Definition 0.49.4 (Comparing Cardinalities)**.** Write $|A| \leq |B|$ if there exists an injection $f : A \hookrightarrow B$. Write $|A| < |B|$ if $|A| \leq |B|$ but $A \not\sim B$.

Theorem 0.49.5 ($\mathbb{Q}$ is countable)**.** *The set $\mathbb{Q}$ of rational numbers is countable.* *Go to proof.*

Theorem 0.49.6 (Countable union of countable sets is countable)**.** *Let $\{A_n\}_{n \in \mathbb{N}}$ be a countable family of countable sets. Then $\bigcup_{n=1}^{\infty} A_n$ is countable.* *Go to proof.*

Theorem 0.49.7 ($\mathbb{R}$ is uncountable)**.** *The set $\mathbb{R}$ of real numbers is not countable.* *Go to proof.*

Theorem 0.49.8 (Cantor's Theorem)**.** *For any set A, there is no surjection $A \to \mathcal{P}(A)$. In particular, $|A| < |\mathcal{P}(A)|$.* *Go to proof.*

Definition 0.49.9 (Function Space B^A)**.** For sets A and B, the *function space* B^A denotes the set of all functions from A to B:

$$B^A := \{ f : A \to B \}.$$

Theorem 0.49.10 (Schröder–Bernstein Theorem)**.** *If $|A| \leq |B|$ and $|B| \leq |A|$, then $A \sim B$.*

Equivalently: if there exist injections $f : A \hookrightarrow B$ and $g : B \hookrightarrow A$, then there exists a bijection $h : A \to B$. *Go to proof.*

Proofs

Foundational Geometry

Euclidean Geometry

Definitions

Definition 0.49.11 (Point)**.** A point is that which has no part.

Definition 0.49.12 (Line)**.** A line is breadthless length.

Definition 0.49.13 (Ends of a line)**.** The ends of a line are points.

Definition 0.49.14 (Straight line)**.** A straight line is a line which lies evenly with the points on itself.

Definition 0.49.15 (Surface)**.** A surface is that which has length and breadth only.

Definition 0.49.16 (Edges of a surface)**.** The edges of a surface are lines.

Definition 0.49.17 (Plane surface)**.** A plane surface is a surface which lies evenly with the straight lines on itself.

Definition 0.49.18 (Plane angle)**.** A plane angle is the inclination to one another of two lines in a plane which meet one another and do not lie in a straight line.

Definition 0.49.19 (Rectilinear angle)**.** And when the lines containing the angle are straight, the angle is called rectilinear.

Definition 0.49.20 (Right angle and perpendicular)**.** When a straight line standing on a straight line makes the adjacent angles equal to one another, each of the equal angles is right, and the straight line standing on the other is called a perpendicular to that on which it stands.

Definition 0.49.21 (Obtuse angle)**.** An obtuse angle is an angle greater than a right angle.

Definition 0.49.22 (Acute angle)**.** An acute angle is an angle less than a right angle.

Definition 0.49.23 (Boundary)**.** A boundary is that which is an extremity of anything.

Definition 0.49.24 (Figure)**.** A figure is that which is contained by any boundary or boundaries.

Definition 0.49.25 (Circle)**.** A circle is a plane figure contained by one line such that all the straight lines falling upon it from one point among those lying within the figure equal one another.

Definition 0.49.26 (Center of a circle)**.** And the point is called the center of the circle.

Definition 0.49.27 (Diameter)**.** A diameter of the circle is any straight line drawn through the center and terminated in both directions by the circumference of the circle, and such a straight line also bisects the circle.

Definition 0.49.28 (Semicircle)**.** A semicircle is the figure contained by the diameter and the circumference cut off by it. And the center of the semicircle is the same as that of the circle.

Definition 0.49.29 (Rectilinear figures)**.** Rectilinear figures are those which are contained by straight lines, trilateral figures being those contained by three, quadrilateral those contained by four, and multilateral those contained by more than four straight lines.

Definition 0.49.30 (Equilateral, isosceles, and scalene triangles)**.** Of trilateral figures, an equilateral triangle is that which has its three sides equal, an isosceles triangle that which has two of its sides alone equal, and a scalene triangle that which has its three sides unequal.

Definition 0.49.31 (Right-angled, obtuse-angled, and acute-angled triangles)**.** Further, of trilateral figures, a right-angled triangle is that which has a right angle, an obtuse-angled triangle that which has an obtuse angle, and an acute-angled triangle that which has its three angles acute.

Definition 0.49.32 (Quadrilateral classes)**.** Of quadrilateral figures, a square is that which is both equilateral and right-angled; an oblong that which is right-angled but not equilateral; a rhombus that which is equilateral but not right-angled; and a rhomboid that which has its opposite sides and angles equal to one another but is neither equilateral nor right-angled. And let quadrilaterals other than these be called trapezia.

Definition 0.49.33 (Parallel straight lines)**.** Parallel straight lines are straight lines which, being in the same plane and being produced indefinitely in both directions, do not meet one another in either direction.

Postulates

0.50 Foundations

Axioms

Propositions

Theorem 0.50.1 (Euclid I.1: Constructing an equilateral triangle)**.** *Given a finite straight line segment AB, there exists an equilateral triangle ABC constructed on AB.*

Theorem 0.50.2 (Euclid I.2: Copying a segment from a given point)**.** *Given a point A and a finite straight line segment BC, there exists a straight line segment beginning at A and equal to BC.*

Theorem 0.50.3 (Euclid I.3: Cutting off an equal segment)**.** *Given two unequal finite straight line segments, there exists a construction which cuts off from the greater a straight line segment equal to the less.*

Theorem 0.50.4 (Euclid I.4: Side-angle-side congruence)**.** *If two triangles have two sides equal to two sides respectively, and have the angles contained by the equal straight lines equal, then they also have the base equal to the base, the triangle equal to the triangle, and the remaining angles equal to the remaining angles respectively.*

Theorem 0.50.5 (Euclid I.31: Drawing a parallel through a point)**.** *Through a given point, to draw a straight line parallel to a given straight line.*

Theorem 0.50.6 (Euclid I.32: Triangle angle sum)**.** *In any triangle, if one of the sides is produced, then the exterior angle is equal to the two interior and opposite angles, and the three interior angles of the triangle are equal to two right angles.*

Theorem 0.50.7 (Euclid I.41: Triangle and parallelogram on the same base)**.** *If a parallelogram has the same base with a triangle and is in the same parallels, then the parallelogram is double the triangle.*

Theorem 0.50.8 (Euclid I.46: Constructing a square on a segment)**.** *On a given straight line, to describe a square.*

Theorem 0.50.9 (Euclid I.47: Pythagorean theorem)**.** *In right-angled triangles, the square on the side subtending the right angle is equal to the squares on the sides containing the right angle.*

0.51 Compass-and-Straightedge Constructions

Proofs

Trigonometry

0.51.1 Trigonometric Functions from Euclidean Geometry

Definition 0.51.1 (Right-triangle trigonometric ratios)**.** Let T be a right triangle and let θ be one of its acute angles. Write

$$\operatorname{opp}(T,\theta), \qquad \operatorname{adj}(T,\theta), \qquad \operatorname{hyp}(T)$$

for the side lengths opposite θ, adjacent to θ, and opposite the right angle, respectively. Define

$$\sin\theta = \frac{\operatorname{opp}(T,\theta)}{\operatorname{hyp}(T)}, \quad \cos\theta = \frac{\operatorname{adj}(T,\theta)}{\operatorname{hyp}(T)}, \quad \tan\theta = \frac{\operatorname{opp}(T,\theta)}{\operatorname{adj}(T,\theta)},$$

$$\cot\theta = \frac{\operatorname{adj}(T,\theta)}{\operatorname{opp}(T,\theta)}, \quad \sec\theta = \frac{\operatorname{hyp}(T)}{\operatorname{adj}(T,\theta)}, \quad \csc\theta = \frac{\operatorname{hyp}(T)}{\operatorname{opp}(T,\theta)}.$$

Definition 0.51.2 (Sine and cosine)**.** For an acute Euclidean angle θ, choose any right triangle T having θ as an acute angle and set

$$\sin\theta = \frac{\operatorname{opp}(T,\theta)}{\operatorname{hyp}(T)}, \qquad \cos\theta = \frac{\operatorname{adj}(T,\theta)}{\operatorname{hyp}(T)}.$$

Theorem 0.51.3 (Similarity invariance of trigonometric ratios)**.** *If two right triangles have the same acute angle θ, then their corresponding opposite-to-hypotenuse and adjacent-to-hypotenuse ratios are equal.*

Theorem 0.51.4 (Well-definedness of sine and cosine)**.** *For each acute Euclidean angle θ, the values $\sin\theta$ and $\cos\theta$ are independent of the right triangle chosen to compute them.*

Theorem 0.51.5 (Six trigonometric functions)**.** *The six right-triangle trigonometric ratios define functions of an acute Euclidean angle, subject to the denominator restrictions in their defining quotients.*

Definition 0.51.6 (Unit-circle sine and cosine)**.** Let θ be an angle in standard position, and let $P(\theta) = (x_\theta, y_\theta)$ be the point where its terminal ray meets the unit circle. Define

$$\cos\theta = x_\theta, \qquad \sin\theta = y_\theta.$$

Theorem 0.51.7 (Unit-circle agreement with right-triangle trigonometry)**.** *For $0 < \theta < 90°$, the unit-circle definitions of sine and cosine agree with the right-triangle definitions.*

Definition 0.51.8 (Extended tangent, cotangent, secant, and cosecant)**.** Where the displayed denominators are nonzero, define

$$\tan\theta = \frac{\sin\theta}{\cos\theta}, \quad \cot\theta = \frac{\cos\theta}{\sin\theta}, \quad \sec\theta = \frac{1}{\cos\theta}, \quad \csc\theta = \frac{1}{\sin\theta}.$$

Definition 0.51.9 (Radian measure)**.** One radian is the angle that cuts off arc length 1 on the unit circle. A full turn has measure 2π radians.

Theorem 0.51.10 (Symmetry and periodicity of sine and cosine)**.** *For every angle in the unit-circle domain,*

$$\cos(-\theta) = \cos\theta, \qquad \sin(-\theta) = -\sin\theta,$$

and one full turn returns the same sine and cosine values.

Theorem 0.51.11 (Pythagorean identity)**.** *For every angle in the unit-circle domain,*

$$\cos^2\theta + \sin^2\theta = 1.$$

Theorem 0.51.12 (Angle-sum and difference identities)**.** *The sine and cosine of a sum or difference of angles are determined by the standard angle-sum and angle-difference formulas.*

Proofs

Analytical Geometry

0.52 Analytical Geometry

0.53 Lines

Definition 0.53.1 (Point-slope form)**.** Let $P_1 = (x_1, y_1)$ be a point in the Euclidean plane and let $m \in \mathbb{R}$. The point-slope form of the line through P_1 with slope m is

$$y - y_1 = m(x - x_1).$$

Definition 0.53.2 (Two-point form)**.** Let $P_1 = (x_1, y_1)$ and $P_2 = (x_2, y_2)$ be distinct points with $x_1 \neq x_2$. The two-point form of the line through P_1 and P_2 is

$$y - y_1 = \left(\frac{y_2 - y_1}{x_2 - x_1}\right)(x - x_1).$$

Definition 0.53.3 (General linear form)**.** The general linear form of a line in $\mathbb{R}^2$ is

$$Ax + By + C = 0,$$

where $A, B, C \in \mathbb{R}$ and $A^2 + B^2 \neq 0$.

Definition 0.53.4 (Vector parametric form)**.** A line in $\mathbb{R}^2$ may be represented as the image of $\mathbf{r} : \mathbb{R} \to \mathbb{R}^2$ given by

$$\mathbf{r}(t) = \mathbf{r}_0 + t\mathbf{v},$$

where $\mathbf{r}_0 \in \mathbb{R}^2$ is a point on the line and $\mathbf{v} \in \mathbb{R}^2$ is a nonzero direction vector.

Definition 0.53.5 (Intercept form)**.** If a line meets the x-axis at $(a, 0)$ and the y-axis at $(0, b)$ with $a, b \neq 0$, its intercept form is

$$\frac{x}{a} + \frac{y}{b} = 1.$$

Definition 0.53.6 (Affine parametric line in $\mathbb{R}^n$)**.** Given distinct points $\mathbf{x}_1, \mathbf{x}_2 \in \mathbb{R}^n$, the line through them is

$$\mathbf{x}(t) = \mathbf{x}_1 + t(\mathbf{x}_2 - \mathbf{x}_1), \qquad t \in \mathbb{R}.$$

Definition 0.53.7 (Normal form)**.** A line in $\mathbb{R}^2$ may be written in normal form as

$$x \cos\alpha + y \sin\alpha - p = 0,$$

where $p \geq 0$ is the distance from the origin to the line.

Definition 0.53.8 (Symmetric form)**.** Given a point (x_1, y_1) and direction cosines $(\cos\theta, \sin\theta)$, the symmetric form of a line is

$$\frac{x - x_1}{\cos\theta} = \frac{y - y_1}{\sin\theta} = r.$$

Proofs

Part II

Volume II: Origins of Numbers

Discrete Number Systems

Formalizing the Number Systems

Proofs

Capstone

Identity, Equality, and Equivalence

0.54 Identity, Equality, and Equivalence

0.55 Equality

Definition 0.55.1 (Identity)**.** Two expressions x and y are *identical* when they denote the same object.

Definition 0.55.2 (Equality on a Domain)**.** Let A be a domain. *Equality on A* is the relation, written $=$, that holds exactly when two terms or constructions denote the same element of A.

Definition 0.55.3 (Definitional and Propositional Equality)**.** *Definitional equality* records an introduced meaning, written with $:=$. *Propositional equality* is a statement of sameness inside the theory, written with $=$.

0.56 Properties of Equality

Theorem 0.56.1 (Symmetry of Equality)**.**

$$\forall x\, \forall y,\ (x = y \implies y = x).$$

Theorem 0.56.2 (Transitivity of Equality)**.**

$$\forall x\, \forall y\, \forall z,\ \big(x = y \land y = z \implies x = z\big).$$

Corollary 0.56.3 (Equality Is an Equivalence Relation)**.** *The equality relation is reflexive, symmetric, and transitive.*

0.57 Substitution

Proposition 0.57.1 (Substitution Preserves Predicates)**.**

$$x = y \implies \big(P(x) \iff P(y)\big).$$

Proposition 0.57.2 (Substitution Preserves Relations on the Left)**.**

$$x = y \implies \big(R(x, z) \iff R(y, z)\big).$$

Proposition 0.57.3 (Substitution Preserves Relations on the Right)**.**

$$x = y \implies \big(R(z, x) \iff R(z, y)\big).$$

Proposition 0.57.4 (Substitution Preserves Relations)**.**

$$x = x' \land y = y' \implies \big(R(x, y) \iff R(x', y')\big).$$

Proposition 0.57.5 (Substitution Preserves Functions)**.**

$$x = y \Longrightarrow f(x) = f(y).$$

Proposition 0.57.6 (Left Congruence for Operations)**.**

$$x = x' \Longrightarrow x \, O \, y = x' \, O \, y.$$

Proposition 0.57.7 (Right Congruence for Operations)**.**

$$y = y' \Longrightarrow x \, O \, y = x \, O \, y'.$$

Proposition 0.57.8 (Full Congruence for Operations)**.**

$$x = x' \wedge y = y' \Longrightarrow x \, O \, y = x' \, O \, y'.$$

Proposition 0.57.9 (Congruence with Respect to Equality Is Automatic)**.** *Every function, predicate, relation, and operation respects equality in its displayed arguments.*

0.58 Equivalence

Definition 0.58.1 (Equivalence Relation)**.** Let A be a set. A relation $\sim$ on A is an *equivalence relation* if it is reflexive, symmetric, and transitive.

Corollary 0.58.2 (Equality Is the Finest Equivalence)**.** *Let $\sim$ be an equivalence relation on A. For all $a, b \in A$,*

$$a = b \Longrightarrow a \sim b.$$

Definition 0.58.3 (Equivalence Class)**.** Let $\sim$ be an equivalence relation on A. The *equivalence class* of $a \in A$ is

$$[a] := \{x \in A \mid x \sim a\}.$$

Theorem 0.58.4 (Equivalence Classes Partition the Domain)**.** *The equivalence classes of an equivalence relation on A are nonempty, cover A, and are pairwise disjoint.*

Proposition 0.58.5 (Representatives Related iff Classes Equal)**.**

$$a \sim b \Longleftrightarrow [a] = [b].$$

Canonical Projection

Definition 0.58.6 (Canonical Projection)**.** Let $\sim$ be an equivalence relation on A. The *canonical projection* is the map

$$\pi : A \to A/{\sim}, \qquad \pi(a) := [a].$$

Proposition 0.58.7 (Projection Is Surjective)**.** *The canonical projection $\pi : A \to A/{\sim}$ is surjective.*

Proposition 0.58.8 (Projection Identifies Exactly the Equivalent Elements)**.** *For all $a, b \in A$,*

$$\pi(a) = \pi(b) \Longleftrightarrow a \sim b.$$

Theorem 0.58.9 (Existence of a Quotient)**.** *For any equivalence relation $\sim$ on A, the quotient set $A/{\sim}$ exists as the set of equivalence classes, and the canonical projection*

$$\pi : A \to A/{\sim}$$

is well-defined.

Induced Structure on a Quotient

Definition 0.58.10 (Operation Respects an Equivalence)**.** Let $O : A \times A \to A$ be a binary operation and let $\sim$ be an equivalence relation on A. The operation O *respects* $\sim$ if

$$\forall a, a', b, b' \in A,\ \big(a \sim a' \wedge b \sim b'\big) \Longrightarrow \big(a\, O\, b\big) \sim \big(a'\, O\, b'\big).$$

Definition 0.58.11 (Left and Right Respect)**.** The operation O *respects* $\sim$ on the left if

$$a \sim a' \Longrightarrow (a\, O\, b) \sim (a'\, O\, b),$$

and respects $\sim$ on the right if

$$b \sim b' \Longrightarrow (a\, O\, b) \sim (a\, O\, b').$$

Lemma 0.58.12 (Respect Splits)**.** *A binary operation O respects $\sim$ if and only if it respects $\sim$ on the left and on the right.*

Theorem 0.58.13 (Induced Operation on a Quotient)**.** *If $O : A \times A \to A$ respects $\sim$, then*

$$[a]\, \overline{O}\, [b] := [a\, O\, b]$$

defines a well-defined binary operation on $A/{\sim}$.

Theorem 0.58.14 (Induced Unary Operation)**.** *Let $f : A \to A$. If*

$$a \sim a' \Longrightarrow f(a) \sim f(a'),$$

then

$$\overline{f}([a]) := [f(a)]$$

defines a well-defined unary operation on $A/{\sim}$.

Definition 0.58.15 (Predicate Respects an Equivalence)**.** Let P be a predicate on A. The predicate P *respects* $\sim$ if

$$\forall a, a' \in A,\ a \sim a' \Longrightarrow \big(P(a) \Longleftrightarrow P(a')\big).$$

Theorem 0.58.16 (Induced Predicate on a Quotient)**.** *If P respects $\sim$, then*

$$\overline{P}([a]) :\Longleftrightarrow P(a)$$

defines a well-defined predicate on $A/{\sim}$.

Definition 0.58.17 (Relation Respects an Equivalence)**.** Let R be a binary relation on A. The relation R *respects* $\sim$ if

$$\forall a, a', b, b' \in A,\ \big(a \sim a' \wedge b \sim b'\big) \Longrightarrow \big(R(a, b) \Longleftrightarrow R(a', b')\big).$$

Theorem 0.58.18 (Induced Relation on a Quotient)**.** *If R respects $\sim$, then*

$$\overline{R}([a], [b]) :\Longleftrightarrow R(a, b)$$

defines a well-defined binary relation on $A/{\sim}$.

Theorem 0.58.19 (Induced Map out of a Quotient)**.** *Let $g : A \to B$. If*

$$a \sim a' \Longrightarrow g(a) = g(a'),$$

then there is a unique well-defined map

$$\overline{g} : A/{\sim} \to B$$

such that

$$\overline{g}([a]) = g(a) \qquad \text{and} \qquad \overline{g} \circ \pi = g,$$

where $\pi : A \to A/{\sim}$ is the canonical projection.

Theorem 0.58.20 (Induced Partial Operation on a Quotient)**.** *Let $D \subseteq A$ be saturated under $\sim$, meaning*

$$a \in D \wedge a \sim a' \implies a' \in D.$$

If $f : D \to A$ respects $\sim$ on D, then

$$\overline{f}([a]) := [f(a)]$$

is a well-defined partial operation on $A/\sim$ with domain

$$D/\sim := \{[a] \in A/\sim : a \in D\}.$$

Proposition 0.58.21 (Representative Independence Is Necessary)**.** *If a rule on classes*

$$[a]\, \overline{O}\, [b] := [a \, O \, b]$$

is well-defined on $A/\sim$, then O respects $\sim$.

Lemma 0.58.22 (Distinct Equivalence Classes Are Disjoint)**.**

$$[a] \neq [b] \implies [a] \cap [b] = \emptyset.$$

Proofs

Capstone

Arithmetic Operations and Relations

0.59 Arithmetic Operations and Relations

0.60 Closure

Definition 0.60.1 (Unary Arithmetic Operation)**.** Let K be a domain. A *unary arithmetic operation* on K is a function

$$O : K \to K.$$

Definition 0.60.2 (Binary Arithmetic Operation)**.** Let K be a domain. A *binary arithmetic operation* on K is a function

$$O : K \times K \to K.$$

Definition 0.60.3 (Unary Closure)**.** A unary operation O is *closed on K* if applying it to an element of K produces an element of K.

Definition 0.60.4 (Binary Closure)**.** A binary operation O is *closed on K* if combining two elements of K produces an element of K.

Definition 0.60.5 (Operation Well-Definedness)**.** Let $\sim$ be an equivalence relation on A. A binary operation O respects $\sim$ if

$$a \sim a' \wedge b \sim b' \implies a\,O\,b \sim a'\,O\,b'.$$

Lemma 0.60.6 (Left Respect)**.** *If a binary operation respects $\sim$, then*

$$a \sim a' \implies a\,O\,b \sim a'\,O\,b.$$

Lemma 0.60.7 (Right Respect)**.** *If a binary operation respects $\sim$, then*

$$b \sim b' \implies a\,O\,b \sim a\,O\,b'.$$

Theorem 0.60.8 (Induced Operation)**.** *The operation*

$$[a]\,\overline{O}\,[b] := [a\,O\,b]$$

on $A/{\sim}$ is well-defined if and only if O respects $\sim$.

0.61 Identity Elements

Definition 0.61.1 (Left Identity)**.**

$$\forall x,\ e_L\,O\,x = x.$$

Definition 0.61.2 (Right Identity)**.**

$$\forall x,\ x\,O\,e_R = x.$$

Definition 0.61.3 (Two-Sided Identity)**.** An element e is a *two-sided identity* for O if it is both a left identity and a right identity.

Definition 0.61.4 (Absorbing Element)**.** An element z is an *absorbing element* for a binary operation O if

$$z \, O \, x = z \quad \text{and} \quad x \, O \, z = z$$

for every element x in the domain of the operation.

Lemma 0.61.5 (Left and Right Identities Coincide)**.** *If e_L is a left identity and e_R is a right identity for the same operation, then $e_L = e_R$.*

Theorem 0.61.6 (Uniqueness of Identity)**.** *A two-sided identity for a binary operation is unique.*

0.62 Inverse Operations

Definition 0.62.1 (Left Inverse)**.** An element x' is a left inverse of x with respect to identity e if

$$x' \, O \, x = e.$$

Definition 0.62.2 (Right Inverse)**.** An element x'' is a right inverse of x with respect to identity e if

$$x \, O \, x'' = e.$$

Definition 0.62.3 (Two-Sided Inverse)**.** An element y is a two-sided inverse of x if

$$y \, O \, x = e \quad \text{and} \quad x \, O \, y = e.$$

Lemma 0.62.4 (Left and Right Inverses Coincide)**.** *In an associative operation with identity, a left inverse and a right inverse of the same element are equal.*

Theorem 0.62.5 (Uniqueness of Inverse)**.** *In an associative operation with identity, inverses are unique.*

Definition 0.62.6 (Right Solvability)**.** Right solvability means

$$\forall x, y \, \exists z, \; x = y \, O \, z,$$

Definition 0.62.7 (Left Solvability)**.** Left solvability means

$$\forall x, y \, \exists z, \; x = z \, O \, y.$$

Definition 0.62.8 (Two-Sided Solvability)**.** Two-sided solvability means both left solvability and right solvability hold.

Proposition 0.62.9 (Solvability and Inverses)**.** *For an associative operation with a two-sided identity, solvability is equivalent to the existence of inverses.*

0.63 Algebraic Laws

Definition 0.63.1 (Associativity)**.**

$$\forall x, y, z, \; x \, O \, (y \, O \, z) = (x \, O \, y) \, O \, z.$$

Theorem 0.63.2 (General Associativity)**.** *If O is associative, then every bracketing of $x_1 \, O \cdots O \, x_n$ has the same value.*

Definition 0.63.3 (Commutativity)**.**

$$\forall x, y, \; x \, O \, y = y \, O \, x.$$

Theorem 0.63.4 (General Commutativity)**.** *If O is associative and commutative, then any finite reordering of $x_1 \, O \cdots O \, x_n$ has the same value.*

Definition 0.63.5 (Left Distributivity). The operation $\otimes$ is left distributive over $\oplus$ if

$$x \otimes (y \oplus z) = (x \otimes y) \oplus (x \otimes z),$$

Definition 0.63.6 (Right Distributivity). The operation $\otimes$ is right distributive over $\oplus$ if

$$(y \oplus z) \otimes x = (y \otimes x) \oplus (z \otimes x),$$

Definition 0.63.7 (Two-Sided Distributivity). The operation $\otimes$ is two-sided distributive over $\oplus$ if it is both left distributive and right distributive.

Definition 0.63.8 (Left Cancellation). An operation has left cancellation if

$$x\, O\, y = x\, O\, z \Longrightarrow y = z,$$

Definition 0.63.9 (Right Cancellation). An operation has right cancellation if

$$y\, O\, x = z\, O\, x \Longrightarrow y = z.$$

Definition 0.63.10 (Two-Sided Cancellation). An operation has two-sided cancellation if it has both left cancellation and right cancellation.

Proposition 0.63.11 (Invertibility Implies Left Cancellation). *In an associative operation with identity, invertible elements satisfy left cancellation.*

Proposition 0.63.12 (Invertibility Implies Cancellation). *In an associative operation with identity, invertible elements satisfy two-sided cancellation.*

Proposition 0.63.13 (Invertibility Implies Right Cancellation). *In an associative operation with identity, invertible elements satisfy right cancellation.*

0.64 Derived Laws

Proposition 0.64.1 (Left Zero Absorption). *When multiplication distributes over addition and* 0 *is the additive identity,*

$$0 \cdot a = 0.$$

Proposition 0.64.2 (Right Zero Absorption). *When multiplication distributes over addition and* 0 *is the additive identity,*

$$a \cdot 0 = 0.$$

Proposition 0.64.3 (Zero Absorption). *When multiplication distributes over addition and* 0 *is the additive identity,*

$$0 \cdot a = 0 \qquad \textit{and} \qquad a \cdot 0 = 0.$$

Proposition 0.64.4 (Left Sign Rule). *When multiplication distributes over addition and additive inverses exist,*

$$(-a) \cdot b = -(a \cdot b).$$

Proposition 0.64.5 (Right Sign Rule). *When multiplication distributes over addition and additive inverses exist,*

$$a \cdot (-b) = -(a \cdot b).$$

Proposition 0.64.6 (Sign Rule). *When multiplication distributes over addition and additive inverses exist,*

$$(-a) \cdot b = -(a \cdot b) \qquad \textit{and} \qquad a \cdot (-b) = -(a \cdot b).$$

Corollary 0.64.7 (Negative One Rule). *When multiplication distributes over addition and additive inverses exist,*

$$(-1) \cdot a = -a.$$

Proposition 0.64.8 (Product of Negatives)**.** *When multiplication distributes over addition and additive inverses exist,*

$$(-a) \cdot (-b) = a \cdot b.$$

Definition 0.64.9 (Zero Divisor)**.** Let K be an arithmetic system with multiplication and zero. A nonzero element $a \in K$ is a *zero divisor* if there exists a nonzero element $b \in K$ such that $ab = 0$.

Theorem 0.64.10 (No Zero Divisors)**.** *In an arithmetic system with no nonzero zero divisors,*

$$a \cdot b = 0 \implies a = 0 \vee b = 0.$$

0.65 Order Relations

Definition 0.65.1 (Reflexive Relation)**.**

$$\forall x,\ xRx.$$

Definition 0.65.2 (Irreflexive Relation)**.**

$$\forall x,\ \neg(xRx).$$

Definition 0.65.3 (Symmetric Relation)**.**

$$\forall x, y,\ xRy \implies yRx.$$

Definition 0.65.4 (Antisymmetric Relation)**.**

$$\forall x, y,\ (xRy \wedge yRx) \implies x = y.$$

Definition 0.65.5 (Asymmetric Relation)**.**

$$\forall x, y,\ xRy \implies \neg(yRx).$$

Definition 0.65.6 (Transitive Relation)**.**

$$\forall x, y, z,\ (xRy \wedge yRz) \implies xRz.$$

Definition 0.65.7 (Total Relation)**.**

$$\forall x, y,\ xRy \vee yRx.$$

Definition 0.65.8 (Trichotomy)**.** The relation $<$ satisfies *trichotomy* if exactly one of

$$x < y, \qquad x = y, \qquad y < x$$

holds for every pair x, y.

0.66 Order Compatibility

Definition 0.66.1 (Right Monotony)**.** Right monotony means

$$yRz \implies (x\,O\,y)R(x\,O\,z),$$

Definition 0.66.2 (Left Monotony)**.** Left monotony means

$$yRz \implies (y\,O\,x)R(z\,O\,x).$$

Definition 0.66.3 (Two-Sided Monotony)**.** Two-sided monotony means both left monotony and right monotony hold.

Definition 0.66.4 (Right Order Reflection)**.** Right reflection means

$$(x\,O\,y)R(x\,O\,z) \Longrightarrow yRz,$$

Definition 0.66.5 (Left Order Reflection)**.** Left reflection means

$$(y\,O\,x)R(z\,O\,x) \Longrightarrow yRz.$$

Definition 0.66.6 (Two-Sided Order Reflection)**.** Two-sided order reflection means both left order reflection and right order reflection hold.

Definition 0.66.7 (Positive cone)**.** Let K be an arithmetic system with addition, multiplication, and zero. A subset $P \subset K$ is called a *positive cone* if it satisfies:

1. if $a, b \in P$, then $a + b \in P$;
2. if $a, b \in P$, then $ab \in P$;
3. for every $a \in K$, exactly one of the following holds:

$$a \in P, \qquad a = 0, \qquad -a \in P.$$

The strict order associated to P is

$$a < b \iff b - a \in P.$$

Definition 0.66.8 (Monotone Map)**.**

$$x \leq y \Longrightarrow f(x) \leq f(y).$$

Definition 0.66.9 (Strictly Monotone Map)**.**

$$x < y \Longrightarrow f(x) < f(y).$$

Proposition 0.66.10 (Addition Preserves Order on the Left)**.**

$$y < z \Longrightarrow x + y < x + z$$

in ordered arithmetic systems where addition is strictly order-preserving.

Proposition 0.66.11 (Addition Preserves Order on the Right)**.**

$$y < z \Longrightarrow y + x < z + x$$

in ordered arithmetic systems where addition is strictly order-preserving.

Proposition 0.66.12 (Addition Preserves Order)**.** *In ordered arithmetic systems where addition is strictly order-preserving, addition preserves order on both sides.*

Proposition 0.66.13 (Multiplication Preserves Order on the Left)**.**

$$0 < x \wedge y < z \Longrightarrow x \cdot y < x \cdot z$$

in ordered arithmetic systems where multiplication by positive elements is strictly order-preserving.

Proposition 0.66.14 (Multiplication Preserves Order on the Right)**.**

$$0 < x \wedge y < z \Longrightarrow y \cdot x < z \cdot x$$

in ordered arithmetic systems where multiplication by positive elements is strictly order-preserving.

Proposition 0.66.15 (Multiplication Preserves Order)**.** *In ordered arithmetic systems where multiplication by positive elements is strictly order-preserving, multiplication preserves order on both sides.*

0.67 Composed Relations

Definition 0.67.1 (Strict Partial Order)**.** A *strict partial order* is an irreflexive and transitive relation.

Definition 0.67.2 (Strict Total Order)**.** A *strict total order* is a strict partial order satisfying trichotomy.

0.68 Structure-Preserving Maps

Definition 0.68.1 (Homomorphism)**.** A map f is a *homomorphism* for an operation O when

$$f(x \, O \, y) = f(x) \, O' \, f(y).$$

For ring-like structures it preserves both addition and multiplication, and preserves 1 when the language includes a distinguished multiplicative identity.

Definition 0.68.2 (Injective Homomorphism)**.** An *injective homomorphism* is a homomorphism whose underlying function is injective.

Definition 0.68.3 (Order Embedding)**.**

$$xRy \iff f(x)R'f(y).$$

Definition 0.68.4 (Structure Embedding)**.** A *structure embedding* is an injective homomorphism that also embeds the relevant order relation.

Proposition 0.68.5 (Composition of Homomorphisms)**.** *The composition of two homomorphisms is a homomorphism.*

Corollary 0.68.6 (Identity Map Is a Homomorphism)**.** *The identity map on a structure is a homomorphism from that structure to itself.*

0.69 Algebraic Structures

0.70 Field and Ordered-Field Laws

Proofs

Capstone

Constructing a Number System

0.71 Ground Rules

Construction Obligations

Definition 0.71.1 (Number-System Construction Target)**.** A *number-system construction target* is the specified algebraic and order structure that a proposed system is meant to satisfy. The target fixes the required carrier, operations, relations, distinguished elements, and laws.

Definition 0.71.2 (Construction Data)**.** The *construction data* for a number system consists of the raw carrier objects, the intended equality or equivalence relation, the distinguished elements, and the operation formulas from which the system is built.

Definition 0.71.3 (Construction Verification)**.** *Construction verification* is the proof that the construction data actually satisfies the construction target. It includes, as applicable, closure of operations, equivalence-relation proofs, well-definedness of operations on classes, algebraic laws, order laws, and compatibility of embeddings with earlier systems.

0.72 Finite Modular Number Systems

Congruence and Residue Classes

Definition 0.72.1 (Congruence Modulo an Integer)**.** Let $n \in \mathbb{Z}$ be positive. For $a, b \in \mathbb{Z}$, define

$$a \equiv b \pmod{n} \quad :\Longleftrightarrow \quad n \mid (a - b).$$

Theorem 0.72.2 (Congruence Modulo an Integer Is an Equivalence Relation)**.** *For each positive integer n, congruence modulo n is an equivalence relation on $\mathbb{Z}$.*

Definition 0.72.3 (Integers Modulo n)**.** Let $n \in \mathbb{Z}$ be positive. The *integers modulo n* are the quotient set

$$\mathbb{Z}/n\mathbb{Z} := \mathbb{Z}/\equiv \pmod{n}.$$

Definition 0.72.4 (Residue Class Modulo n)**.** For $a \in \mathbb{Z}$, the *residue class of a modulo n* is

$$[a]_n := \{b \in \mathbb{Z} : b \equiv a \pmod{n}\}.$$

Operations

Definition 0.72.5 (Addition Modulo n)**.** For residue classes modulo a positive integer n, define

$$[a]_n + [b]_n := [a + b]_n.$$

Definition 0.72.6 (Multiplication Modulo n)**.** For residue classes modulo a positive integer n, define

$$[a]_n \cdot [b]_n := [ab]_n.$$

Theorem 0.72.7 (Modular Addition and Multiplication Are Well-Defined)**.** *For each positive integer n, addition and multiplication modulo n are well-defined operations on $\mathbb{Z}/n\mathbb{Z}$.*

Definition 0.72.8 (Finite Modular Number System)**.** For a positive integer n, the *finite modular number system modulo n* is

$$\mathbb{Z}/n\mathbb{Z}$$

with addition and multiplication modulo n, additive identity $[0]_n$, and multiplicative identity $[1]_n$.

Basic Example

Theorem 0.72.9 ($\mathbb{Z}/4\mathbb{Z}$ Has Zero Divisors)**.** *In $\mathbb{Z}/4\mathbb{Z}$, the nonzero class $[2]_4$ satisfies*

$$[2]_4 \cdot [2]_4 = [0]_4.$$

Therefore $\mathbb{Z}/4\mathbb{Z}$ is not a field.

0.73 The $\{0, 1\}$ Number System

The Carrier and Operations

Definition 0.73.1 (Two-Element Carrier)**.** The *two-element carrier* is the set

$$\mathbb{B}_2 := \{0, 1\}.$$

Definition 0.73.2 (Addition on the Two-Element System)**.** Addition on $\mathbb{B}_2$ is the binary operation $+_2$ determined by

$$\begin{array}{c|cc} +_2 & 0 & 1 \\ \hline 0 & 0 & 1 \\ 1 & 1 & 0 \end{array}$$

Definition 0.73.3 (Multiplication on the Two-Element System)**.** Multiplication on $\mathbb{B}_2$ is the binary operation $\cdot_2$ determined by

$$\begin{array}{c|cc} \cdot_2 & 0 & 1 \\ \hline 0 & 0 & 0 \\ 1 & 0 & 1 \end{array}$$

Definition 0.73.4 (Two-Element Number System)**.** The *two-element number system* is

$$\mathbb{F}_2 := (\mathbb{B}_2, +_2, \cdot_2, 0, 1).$$

Verification

Theorem 0.73.5 (The Two-Element System Is a Field)**.** *The two-element number system $\mathbb{F}_2$ is a field.*

Theorem 0.73.6 (The Two-Element System Is Not an Ordered Field)**.** *There is no order on $\mathbb{B}_2$ that makes $\mathbb{F}_2$ an ordered field.*

As Modular Arithmetic

Theorem 0.73.7 ($\mathbb{Z}/2\mathbb{Z}$ Is the Two-Element System)**.** *The finite modular number system $\mathbb{Z}/2\mathbb{Z}$ has exactly two classes, $[0]_2$ and $[1]_2$, and its addition and multiplication tables are the tables of $\mathbb{F}_2$.*

Proofs

Capstone

Peano Systems

0.74 Presburger Arithmetic

Definition 0.74.1 (Presburger Signature)**.** The *Presburger signature* is the first-order signature

$$\mathcal{L}_{\text{Pres}} = \{0, S, +, <, =\},$$

where 0 is a constant symbol, S is a unary function symbol, $+$ is a binary function symbol, and $<$ and $=$ are binary relation symbols.

Definition 0.74.2 (Presburger Arithmetic)**.** *Presburger arithmetic* is the first-order theory of the natural numbers in the Presburger signature. Its intended structure is

$$(\mathbb{N}, 0, S, +, <, =),$$

and its axioms describe the behavior of zero, successor, addition, order, and induction for formulas written in the language $\mathcal{L}_{\text{Pres}}$.

0.75 Defining Peano Systems

Definition 0.75.1 (Peano System)**.** A triple $(P, S, 1)$ is called a *Peano system* exactly when P is a set, $1 \in P$, $S : P \to P$ is a function, and the following axioms hold:

P1. $1 \in P$.

P2. For every $n \in P$, $S(n) \in P$.

P3. For every $n \in P$, $S(n) \neq 1$.

P4. For all $m, n \in P$, if $S(m) = S(n)$ then $m = n$.

P5. If $A \subseteq P$, $1 \in A$, and

$$\forall n \in P \ (n \in A \Longrightarrow S(n) \in A),$$

then $A = P$.

First Theorems on Peano Systems

Theorem 0.75.2 (Induction Principle for a Peano System)**.** *Let $(P, S, 1)$ be a Peano system, and let φ be a predicate on P. If*

$$\varphi(1) \qquad \textit{and} \qquad \forall n \in P \ (\varphi(n) \Longrightarrow \varphi(S(n))),$$

then

$$\forall n \in P \ \varphi(n).$$

Go to proof.

Theorem 0.75.3 (Strong Induction on a Peano System). *Let $(P, S, 1)$ be a Peano system, and let φ be a predicate on P. If*

$$\forall n \in P \ \big(\forall k \in P \ (k < n \Longrightarrow \varphi(k))\big) \Longrightarrow \varphi(n),$$

then

$$\forall n \in P \ \varphi(n).$$

Go to proof.

Definition 0.75.4 (Successor-Closed Subset). Let $(P, S, 1)$ be a Peano system, and let $A \subseteq P$. The subset A is called *successor-closed* exactly when

$$\forall n \in P \ (n \in A \Longrightarrow S(n) \in A).$$

Definition 0.75.5 (Inductive Subset of a Peano System). Let $(P, S, 1)$ be a Peano system, and let $A \subseteq P$. The subset A is called *inductive* exactly when $1 \in A$ and A is successor-closed.

Theorem 0.75.6 (Minimality of a Peano System). *Let $(P, S, 1)$ be a Peano system. If $A \subseteq P$ contains 1 and is closed under successor, then $A = P$.*

$$\big(1 \in A \wedge \forall n \in P \ (n \in A \Longrightarrow S(n) \in A)\big) \Longrightarrow A = P.$$

Go to proof.

Definition 0.75.7 (Predecessor in a Peano System). Let $(P, S, 1)$ be a Peano system, and let $x, u \in P$. The element u is called a *predecessor* of x exactly when

$$S(u) = x.$$

Theorem 0.75.8 (Every Element Is Either 1 or a Successor). *Let $(P, S, 1)$ be a Peano system. For every $x \in P$, either $x = 1$ or there exists $u \in P$ such that $S(u) = x$. Go to proof.*

Theorem 0.75.9 (Successor Inequality Reflection). *Let $(P, S, 1)$ be a Peano system. For all $a, b \in P$, if $a \neq b$, then $S(a) \neq S(b)$. Go to proof.*

Corollary 0.75.10 (Non-One Elements Have a Predecessor). *Let $(P, S, 1)$ be a Peano system. For every $x \in P$, if $x \neq 1$, then there exists $u \in P$ such that $S(u) = x$. Go to proof.*

Corollary 0.75.11 (Predecessor Existence and Uniqueness Away from 1). *Let $(P, S, 1)$ be a Peano system, and let $x \in P$ with $x \neq 1$. Then there exists a unique $u \in P$ such that $S(u) = x$. Go to proof.*

Corollary 0.75.12 (Unique Predecessor Characterization Away from 1). *Let $(P, S, 1)$ be a Peano system. For every $x \in P$, $x \neq 1$ if and only if there exists a unique $u \in P$ such that $S(u) = x$. Go to proof.*

Corollary 0.75.13 (The Distinguished Element Is the Unique Non-Successor). *Let $(P, S, 1)$ be a Peano system. An element $x \in P$ is not a successor of any element of P if and only if $x = 1$. Go to proof.*

Theorem 0.75.14 (No Object Is Its Own Successor). *Let $(P, S, 1)$ be a Peano system. For every $n \in P$,*

$$S(n) \neq n.$$

Go to proof.

Peano System Figure Plates

0.76 Examples of Peano Systems

The Standard One-Based System

Von Neumann Ordinals

Nonnumeric Peano Systems

A Finite Alphabet Is Not a Peano System

Summary

0.77 Recursion on Peano Systems

Arity Examples Before the Theorem

Formal Development

Definition 0.77.1 (Iterator Data)**.** Let $(P, S, 1)$ be a Peano system, let W be a set, let $c \in W$, and let $g : W \to W$ be a function. The triple (W, c, g) is called *iterator data* for $(P, S, 1)$.

Definition 0.77.2 (Iterator Relation)**.** Let $(P, S, 1)$ be a Peano system, and let (W, c, g) be iterator data for it. A relation $R \subseteq P \times W$ is called an *iterator relation* for (W, c, g) exactly when

$$(1, c) \in R$$

and

$$\forall n \in P \, \forall w \in W \, \big((n, w) \in R \Longrightarrow (S(n), g(w)) \in R\big).$$

Definition 0.77.3 (Minimal Iterator Relation)**.** Let $(P, S, 1)$ be a Peano system, and let (W, c, g) be iterator data for it. The *minimal iterator relation* for (W, c, g) is

$$R^* = \bigcap \{R \subseteq P \times W : R \text{ is an iterator relation for } (W, c, g)\}.$$

Lemma 0.77.4 (Consistency of the Minimal Iterator Relation)**.** *Let $(P, S, 1)$ be a Peano system, let (W, c, g) be iterator data for it, and let R^* be the minimal iterator relation for (W, c, g). Then R^* is an iterator relation for (W, c, g).* *Go to proof.*

Lemma 0.77.5 (Forced Values Are Unique)**.** *Let $(P, S, 1)$ be a Peano system, let (W, c, g) be iterator data for it, and let R^* be the minimal iterator relation. If $(n, w_0) \in R^*$ and $(n, w_1) \in R^*$, then $w_0 = w_1$.* *Go to proof.*

Lemma 0.77.6 (Determinism of the Minimal Iterator Relation)**.** *Let $(P, S, 1)$ be a Peano system, let (W, c, g) be iterator data for it, and let R^* be the minimal iterator relation. For each $n \in P$, there is at most one $w \in W$ such that $(n, w) \in R^*$.* *Go to proof.*

Lemma 0.77.7 (Completeness of the Minimal Iterator Relation)**.** *Let $(P, S, 1)$ be a Peano system, let (W, c, g) be iterator data for it, and let R^* be the minimal iterator relation. For each $n \in P$, there exists $w \in W$ such that $(n, w) \in R^*$.* *Go to proof.*

Lemma 0.77.8 (Uniqueness of Iterator Functions)**.** *Let $(P, S, 1)$ be a Peano system, and let (W, c, g) be iterator data for it. If $f, h : P \to W$ satisfy*

$$f(1) = c, \qquad h(1) = c,$$

and

$$\forall n \in P \; (f(S(n)) = g(f(n))) \qquad \textit{and} \qquad \forall n \in P \; (h(S(n)) = g(h(n))),$$

then $f = h$. *Go to proof.*

Lemma 0.77.9 (Existence of an Iterator Function)**.** *Let $(P, S, 1)$ be a Peano system, and let (W, c, g) be iterator data for it. Then there exists a function $f : P \to W$ such that*

$$f(1) = c \qquad \textit{and} \qquad \forall n \in P \; (f(S(n)) = g(f(n))).$$

Go to proof.

Theorem 0.77.10 (Peano Iterator Theorem)**.** *Let $(P, S, 1)$ be a Peano system, and let (W, c, g) be iterator data for it. Then there exists a unique function $f : P \to W$ such that*

$$f(1) = c \qquad \textit{and} \qquad \forall n \in P \; (f(S(n)) = g(f(n))).$$

Go to proof.

Definition 0.77.11 (Iterator-Generated Function)**.** Let $(P, S, 1)$ be a Peano system, and let (W, c, g) be iterator data for it. The unique function $f : P \to W$ satisfying

$$f(1) = c \qquad \text{and} \qquad \forall n \in P \; (f(S(n)) = g(f(n)))$$

is called the *iterator-generated function determined by* (W, c, g).

Theorem 0.77.12 (Iterator Base Clause)**.** *Let $(P, S, 1)$ be a Peano system, let (W, c, g) be iterator data for it, and let f be the iterator-generated function determined by (W, c, g). Then*

$$f(1) = c.$$

Go to proof.

Theorem 0.77.13 (Iterator Successor Clause)**.** *Let $(P, S, 1)$ be a Peano system, let (W, c, g) be iterator data for it, and let f be the iterator-generated function determined by (W, c, g). Then*

$$\forall n \in P \; (f(S(n)) = g(f(n))).$$

Go to proof.

Definition 0.77.14 (Iteration of a Self-Map)**.** Let $(P, S, 1)$ be a Peano system, let W be a set, let $c \in W$, and let $g : W \to W$. The iterator-generated function determined by (W, c, g) is called the *iteration of g from c along P*.

Corollary 0.77.15 (Uniqueness of Binary Iterator Operations)**.** *Let $(P, S, 1)$ be a Peano system, let A be a set, let W be a set, and suppose each $a \in A$ determines iterator data (W, c_a, g_a) for $(P, S, 1)$. If $F, G : A \times P \to W$ satisfy*

$$F(a, 1) = c_a, \qquad G(a, 1) = c_a$$

and

$$F(a, S(n)) = g_a(F(a, n)), \qquad G(a, S(n)) = g_a(G(a, n))$$

for all $a \in A$ and all $n \in P$, then $F = G$. Go to proof.

Definition 0.77.16 (Stage-Dependent Step Rule)**.** Let $(P, S, 1)$ be a Peano system and let W be a set. A *stage-dependent step rule* on W over P is a function

$$\Phi : P \times W \to W.$$

Definition 0.77.17 (General Recursive Function)**.** Let $(P, S, 1)$ be a Peano system, let W be a set, let $c \in W$, and let $\Phi : P \times W \to W$ be a stage-dependent step rule. A function $f : P \to W$ is called the *general recursive function determined by* (W, c, Φ) exactly when

$$f(1) = c \qquad \text{and} \qquad \forall n \in P \; (f(S(n)) = \Phi(n, f(n))).$$

Lemma 0.77.18 (Uniqueness of General Recursive Functions)**.** *Let $(P, S, 1)$ be a Peano system, let W be a set, let $c \in W$, and let $\Phi : P \times W \to W$ be a stage-dependent step rule. If $f, h : P \to W$ satisfy*

$$f(1) = c, \qquad h(1) = c,$$

and

$$\forall n \in P\ (f(S(n)) = \Phi(n, f(n))) \qquad \text{and} \qquad \forall n \in P\ (h(S(n)) = \Phi(n, h(n))),$$

then $f = h$. *Go to proof.*

Theorem 0.77.19 (General Recursion by State Encoding)**.** *Let $(P, S, 1)$ be a Peano system, let W be a set, let $c \in W$, and let $\Phi : P \times W \to W$ be a stage-dependent step rule. There exists a function $H : P \to P \times W$ such that*

$$H(1) = (1, c)$$

and

$$\forall n \in P\ \big(H(S(n)) = (S(\pi_1(H(n))), \Phi(\pi_1(H(n)), \pi_2(H(n))))\big).$$

Go to proof.

Theorem 0.77.20 (General Recursion Theorem)**.** *Let $(P, S, 1)$ be a Peano system, let W be a set, let $c \in W$, and let $\Phi : P \times W \to W$ be a stage-dependent step rule. Then there exists a unique function $f : P \to W$ such that*

$$f(1) = c \qquad \text{and} \qquad \forall n \in P\ (f(S(n)) = \Phi(n, f(n))).$$

Go to proof.

Theorem 0.77.21 (Uniqueness of Peano Systems up to Isomorphism)**.** *Let $(P, S_P, 1_P)$ and $(Q, S_Q, 1_Q)$ be Peano systems. Then there exists a unique bijection $\theta : P \to Q$ such that*

$$\theta(1_P) = 1_Q$$

and

$$\forall n \in P\ (\theta(S_P(n)) = S_Q(\theta(n))).$$

Thus any two Peano systems are isomorphic. *Go to proof.*

Proofs

Capstone

Natural Numbers ($\mathbb{N}$)

0.78 Axioms for $\mathbb{N}$

0.78.1 Axioms for $\mathbb{N}$

Definition 0.78.1 (The Natural Numbers)**.** The natural numbers $\mathbb{N}$ are the canonical Peano system

$$(\mathbb{N}, 1, S),$$

where $1 \in \mathbb{N}$ is the distinguished base element and $S : \mathbb{N} \to \mathbb{N}$ is the successor operation.

0.79 Canonical Natural Number System

0.80 Addition on N

Definition 0.80.1 (Addition on a Peano System)**.** Let $(P, S, 1)$ be a Peano system. For each $m \in P$, define the function

$$f_m : P \to P$$

to be the unique function satisfying

$$f_m(1) = S(m) \qquad \text{and} \qquad \forall n \in P\ (f_m(S(n)) = S(f_m(n))).$$

For $m, n \in P$, define

$$m + n \coloneqq f_m(n).$$

Theorem 0.80.2 (Addition Is Well-Defined on a Peano System)**.** *Let $(P, S, 1)$ be a Peano system. For every $m \in P$, there exists a unique function*

$$f_m : P \to P$$

such that

$$f_m(1) = S(m) \qquad \textit{and} \qquad \forall n \in P\ (f_m(S(n)) = S(f_m(n))).$$

Consequently the rule

$$(m, n) \longmapsto m + n \coloneqq f_m(n)$$

defines a unique binary operation on P. *Go to proof.*

Theorem 0.80.3 (Addition With 1)**.** *Let $(P, S, 1)$ be a Peano system. For every $m \in P$,*

$$m + 1 = S(m).$$

Go to proof.

Theorem 0.80.4 (Addition Successor on the Right)**.** *Let $(P, S, 1)$ be a Peano system. For every $m, n \in P$,*

$$m + S(n) = S(m + n).$$

Go to proof.

Theorem 0.80.5 (One Plus n)**.** *Let $(P, S, 1)$ be a Peano system. For every $n \in P$,*

$$1 + n = S(n).$$

Go to proof.

Theorem 0.80.6 (Addition Is Associative)**.** *Let $(P, S, 1)$ be a Peano system. For every $a, b, c \in P$,*

$$(a + b) + c = a + (b + c).$$

Go to proof.

Theorem 0.80.7 (Addition Is Commutative)**.** *Let $(P, S, 1)$ be a Peano system. For every $a, b \in P$,*

$$a + b = b + a.$$

Go to proof.

Theorem 0.80.8 (Left Cancellation for Addition)**.** *Let $(P, S, 1)$ be a Peano system. For every $a, b, c \in P$,*

$$a + b = a + c \Longrightarrow b = c.$$

Go to proof.

Theorem 0.80.9 (Right Cancellation for Addition)**.** *Let $(P, S, 1)$ be a Peano system. For every $a, b, c \in P$,*

$$b + a = c + a \Longrightarrow b = c.$$

Go to proof.

Theorem 0.80.10 (Cancellation for Addition)**.** *Let $(P, S, 1)$ be a Peano system. Addition on P satisfies cancellation on both sides:*

$$a + b = a + c \Longrightarrow b = c \qquad \textit{and} \qquad b + a = c + a \Longrightarrow b = c.$$

Go to proof.

Theorem 0.80.11 (No Natural Number Equals Itself Plus a Natural Number)**.** *Let $(P, S, 1)$ be a Peano system. For every $m, n \in P$,*

$$n \neq m + n.$$

Go to proof.

0.81 Multiplication on N

Definition 0.81.1 (Multiplication on a Peano System)**.** Let $(P, S, 1)$ be a Peano system. For each $m \in P$, define the function

$$g_m : P \to P$$

to be the unique function satisfying

$$g_m(1) = m \qquad \text{and} \qquad \forall n \in P\ (g_m(S(n)) = g_m(n) + m).$$

For $m, n \in P$, define

$$m \cdot n \coloneqq g_m(n).$$

Theorem 0.81.2 (Multiplication Is Well-Defined on a Peano System)**.** *Let $(P, S, 1)$ be a Peano system. For every $m \in P$, there exists a unique function*

$$g_m : P \to P$$

such that

$$g_m(1) = m \qquad \textit{and} \qquad \forall n \in P\ (g_m(S(n)) = g_m(n) + m).$$

Consequently the rule

$$(m, n) \longmapsto m \cdot n \coloneqq g_m(n)$$

defines a unique binary operation on P. Go to proof.

Theorem 0.81.3 (Multiplication With 1). *Let $(P, S, 1)$ be a Peano system. For every $m \in P$,*

$$m \cdot 1 = m.$$

Go to proof.

Theorem 0.81.4 (Multiplication Successor on the Right). *Let $(P, S, 1)$ be a Peano system. For every $m, n \in P$,*

$$m \cdot S(n) = (m \cdot n) + m.$$

Go to proof.

Theorem 0.81.5 (One Times n). *Let $(P, S, 1)$ be a Peano system. For every $n \in P$,*

$$1 \cdot n = n.$$

Go to proof.

Theorem 0.81.6 (Multiplication Distributes Over Addition). *Let $(P, S, 1)$ be a Peano system. For every $a, b, c \in P$,*

$$a \cdot (b + c) = (a \cdot b) + (a \cdot c).$$

Go to proof.

Theorem 0.81.7 (Left Distributivity of Multiplication Over Addition). *Let $(P, S, 1)$ be a Peano system. For every $a, b, c \in P$,*

$$(a + b) \cdot c = (a \cdot c) + (b \cdot c).$$

Go to proof.

Theorem 0.81.8 (Multiplication Distributes over Addition on Both Sides). *Let $(P, S, 1)$ be a Peano system. For every $a, b, c \in P$,*

$$a \cdot (b + c) = (a \cdot b) + (a \cdot c) \qquad \textit{and} \qquad (a + b) \cdot c = (a \cdot c) + (b \cdot c).$$

Go to proof.

Theorem 0.81.9 (Multiplication Is Associative). *Let $(P, S, 1)$ be a Peano system. For every $a, b, c \in P$,*

$$(a \cdot b) \cdot c = a \cdot (b \cdot c).$$

Go to proof.

Theorem 0.81.10 (Multiplication Is Commutative). *Let $(P, S, 1)$ be a Peano system. For every $a, b \in P$,*

$$a \cdot b = b \cdot a.$$

Go to proof.

Theorem 0.81.11 (Left Cancellation for Multiplication). *Let $(P, S, 1)$ be a Peano system. For every $a, b, c \in P$,*

$$a \cdot b = a \cdot c \Longrightarrow b = c.$$

Go to proof.

Theorem 0.81.12 (Right Cancellation for Multiplication). *Let $(P, S, 1)$ be a Peano system. For every $a, b, c \in P$,*

$$b \cdot a = c \cdot a \Longrightarrow b = c.$$

Go to proof.

Theorem 0.81.13 (Cancellation for Multiplication). *Let $(P, S, 1)$ be a Peano system. Multiplication on P satisfies cancellation on both sides:*

$$a \cdot b = a \cdot c \Longrightarrow b = c \qquad \textit{and} \qquad b \cdot a = c \cdot a \Longrightarrow b = c.$$

Go to proof.

0.82 Order on N

Definition 0.82.1 (Strict Less Than on a Peano System)**.** Let $(P, S, 1)$ be a Peano system, and let $a, b \in P$. One writes

$$a < b$$

exactly when there exists $c \in P$ such that

$$b = a + c.$$

Definition 0.82.2 (Less Than or Equal on a Peano System)**.** Let $(P, S, 1)$ be a Peano system, and let $a, b \in P$. One writes

$$a \leq b$$

exactly when

$$a < b \qquad \text{or} \qquad a = b.$$

Theorem 0.82.3 (Order Is Reflexive on a Peano System)**.** *Let $(P, S, 1)$ be a Peano system. For every $a \in P$,*

$$a \leq a.$$

Go to proof.

Theorem 0.82.4 (Order Is Transitive on a Peano System)**.** *Let $(P, S, 1)$ be a Peano system. For every $a, b, c \in P$,*

$$\big(a \leq b \wedge b \leq c\big) \implies a \leq c.$$

Go to proof.

Theorem 0.82.5 (Order Is Antisymmetric on a Peano System)**.** *Let $(P, S, 1)$ be a Peano system. For every $a, b \in P$,*

$$\big(a \leq b \wedge b \leq a\big) \implies a = b.$$

Go to proof.

Theorem 0.82.6 (Addition Preserves Order on the Right)**.** *Let $(P, S, 1)$ be a Peano system. For every $a, b, c \in P$,*

$$a \leq b \implies a + c \leq b + c.$$

Go to proof.

Theorem 0.82.7 (Addition Preserves Order on the Left)**.** *Let $(P, S, 1)$ be a Peano system. For every $a, b, c \in P$,*

$$a \leq b \implies c + a \leq c + b.$$

Go to proof.

Theorem 0.82.8 (Addition Preserves Order)**.** *Let $(P, S, 1)$ be a Peano system. For every $a, b, c \in P$,*

$$a \leq b \implies a + c \leq b + c \qquad \textit{and} \qquad a \leq b \implies c + a \leq c + b.$$

Go to proof.

Theorem 0.82.9 (Multiplication Preserves and Reflects Strict Order)**.** *Let $(P, S, 1)$ be a Peano system. For every $k, m, n \in P$,*

$$m < n \iff k \cdot m < k \cdot n.$$

Go to proof.

Theorem 0.82.10 (Strict Order Successor Characterization)**.** *Let $(P, S, 1)$ be a Peano system. For every $a, b \in P$,*

$$a < b \iff S(a) \leq b.$$

Go to proof.

Theorem 0.82.11 (Trichotomy on a Peano System). *Let $(P, S, 1)$ be a Peano system. For every $a, b \in P$, exactly one of the following holds:*

$$a < b, \qquad a = b, \qquad b < a.$$

Go to proof.

Theorem 0.82.12 (Natural-Number Order Is Total). *Let $(P, S, 1)$ be a Peano system. For every $a, b \in P$,*

$$a \leq b \qquad \text{or} \qquad b \leq a.$$

Consequently $\leq$ is a total order on P. Go to proof.

0.83 Powers and Growth

Definition 0.83.1 (Exponentiation on a Peano System). Let $(P, S, 1)$ be a Peano system. For each $a \in P$, define the function

$$h_a : P \to P$$

to be the unique function satisfying

$$h_a(1) = a \qquad \text{and} \qquad \forall n \in P\ (h_a(S(n)) = h_a(n) \cdot a).$$

For $a, n \in P$, define

$$a^n := h_a(n).$$

Theorem 0.83.2 (Exponentiation Is Well-Defined on a Peano System). *Let $(P, S, 1)$ be a Peano system. For every $a \in P$, there exists a unique function*

$$h_a : P \to P$$

such that

$$h_a(1) = a \qquad \text{and} \qquad \forall n \in P\ (h_a(S(n)) = h_a(n) \cdot a).$$

Consequently the rule

$$(a, n) \longmapsto a^n := h_a(n)$$

defines a unique binary operation on P. Go to proof.

Theorem 0.83.3 (Exponent Sum Law). *Let $(P, S, 1)$ be a Peano system. For all $a, m, n \in P$,*

$$a^{m+n} = a^m \cdot a^n.$$

Go to proof.

Theorem 0.83.4 (Exponent Product Law). *Let $(P, S, 1)$ be a Peano system. For all $a, b, n \in P$,*

$$(a \cdot b)^n = a^n \cdot b^n.$$

Go to proof.

Theorem 0.83.5 (Exponent Power Law). *Let $(P, S, 1)$ be a Peano system. For all $a, m, n \in P$,*

$$(a^m)^n = a^{m \cdot n}.$$

Go to proof.

Theorem 0.83.6 (Powers with Common Exponent Preserve Strict Order). *Let $(P, S, 1)$ be a Peano system. For every $a, b, n \in P$,*

$$a < b \iff a^n < b^n.$$

Go to proof.

Theorem 0.83.7 (Powers with Common Base Preserve Strict Order). *Let $(P, S, 1)$ be a Peano system. For every $a, m, n \in P$,*

$$a^m < a^n \iff 1 < a \land m < n.$$

Go to proof.

Theorem 0.83.8 ($n < 2^n$ for All $n \in \mathbb{N}$). *Let $(P, S, 1)$ be a Peano system, and let $2 \coloneqq S(1)$. For every $n \in P$,*

$$n < 2^n.$$

Go to proof.

0.84 Well Ordering Principle

Theorem 0.84.1 (Well-Ordering Principle). *Let $(P, S, 1)$ be a Peano system. Every nonempty subset of P has a least element with respect to the order $\leq$ on P.* *Go to proof.*

Theorem 0.84.2 (Induction, Strong Induction, and Well-Ordering Are Equivalent). *Let $(P, S, 1)$ be a Peano system equipped with the order $\leq$. The following are equivalent:*

1. *The induction principle: for every predicate φ on P, if $\varphi(1)$ holds and the successor step holds, then $\varphi(n)$ holds for all $n \in P$.*

2. *The strong induction principle: for every predicate φ on P, if $\varphi(k)$ holds for every $k < n$ implies $\varphi(n)$, then $\varphi(n)$ holds for every $n \in P$.*

3. *The well-ordering principle: every nonempty subset of P has a least element with respect to $\leq$.*

Go to proof.

0.85 Induction

Theorem 0.85.1 (Induction from an Arbitrary Base). *Let $(P, S, 1)$ be a Peano system, let $m \in P$, and let φ be a predicate on P. If*

$$\varphi(m) \qquad \text{and} \qquad \forall n \in P \ \big(n \geq m \land \varphi(n) \Longrightarrow \varphi(S(n))\big),$$

then

$$\forall n \in P \ \big(n \geq m \Longrightarrow \varphi(n)\big).$$

Go to proof.

0.86 Algebraic Structure of N

Theorem 0.86.1 ($\mathbb{N}$ Is Closed, Associative, and Commutative Under Addition). *Let $(P, S, 1)$ be a Peano system. The binary operation $+$ on P satisfies:*

1. ***Closure.*** *For all $a, b \in P$, $a + b \in P$.*

2. ***Associativity.*** *For all $a, b, c \in P$, $(a + b) + c = a + (b + c)$.*

3. ***Commutativity.*** *For all $a, b \in P$, $a + b = b + a$.*

4. ***Left cancellation.*** *For all $a, b, c \in P$, if $a + b = a + c$ then $b = c$.*

5. ***Right cancellation.*** *For all $a, b, c \in P$, if $b + a = c + a$ then $b = c$.*

Go to proof.

Theorem 0.86.2 ($\mathbb{N}$ Is a Commutative Monoid Under Multiplication)**.** *Let $(P, S, 1)$ be a Peano system. The binary operation $\cdot$ on P satisfies:*

1. ***Closure.*** *For all $a, b \in P$, $a \cdot b \in P$.*
2. ***Associativity.*** *For all $a, b, c \in P$, $(a \cdot b) \cdot c = a \cdot (b \cdot c)$.*
3. ***Commutativity.*** *For all $a, b \in P$, $a \cdot b = b \cdot a$.*
4. ***Identity.*** *For all $a \in P$, $1 \cdot a = a = a \cdot 1$.*

Consequently $(P, \cdot, 1)$ is a commutative monoid. Go to proof.

Theorem 0.86.3 ($\mathbb{N}$ Is a Semiring)**.** *Let $(P, S, 1)$ be a Peano system. Then $(P, +, \cdot)$ satisfies:*

1. *$(P, +)$ is a commutative semigroup.*
2. *$(P, \cdot, 1)$ is a commutative monoid.*
3. *Multiplication distributes over addition: for all $a, b, c \in P$,*

$$a \cdot (b + c) = (a \cdot b) + (a \cdot c).$$

Go to proof.

Proofs

Capstone

Constructing the Whole Numbers

0.87 Why Whole Numbers Are Introduced

The Missing Identity

0.88 Definition of the Whole Numbers

The Underlying Set

Definition 0.88.1 (Whole Numbers)**.** Let $\mathbb{N}$ be the one-based natural-number system developed in the preceding chapter. Let 0 be a new element with $0 \notin \mathbb{N}$. Define

$$\mathbb{W} := \mathbb{N} \cup \{0\}.$$

Theorem 0.88.2 ($0 \notin \mathbb{N}$)**.** *The adjoined element* 0 *is not an element of the one-based natural-number system:*

$$0 \notin \mathbb{N}.$$

Go to proof.

0.89 Extending Successor

The Extended Successor

Definition 0.89.1 (Successor on $\mathbb{W}$)**.** Define $S_{\mathbb{W}} : \mathbb{W} \to \mathbb{W}$ by

$$S_{\mathbb{W}}(0) = 1 \qquad \text{and} \qquad S_{\mathbb{W}}(n) = S_{\mathbb{N}}(n) \quad (n \in \mathbb{N}).$$

Theorem 0.89.2 (Successor on $\mathbb{W}$ Is Well-Defined)**.** *The rule defining $S_{\mathbb{W}}$ determines a function $S_{\mathbb{W}} : \mathbb{W} \to \mathbb{W}$. Go to proof.*

Theorem 0.89.3 (0 Is Not a Successor in $\mathbb{W}$)**.** *There is no $a \in \mathbb{W}$ such that $S_{\mathbb{W}}(a) = 0$. Go to proof.*

Theorem 0.89.4 (Every Nonzero Element of $\mathbb{W}$ Is a Successor)**.** *If $a \in \mathbb{W}$ and $a \neq 0$, then there exists $b \in \mathbb{W}$ such that $a = S_{\mathbb{W}}(b)$. Go to proof.*

Theorem 0.89.5 (Successor on $\mathbb{W}$ Is Injective)**.** *For all $a, b \in \mathbb{W}$,*

$$S_{\mathbb{W}}(a) = S_{\mathbb{W}}(b) \implies a = b.$$

Go to proof.

Theorem 0.89.6 (($\mathbb{W}, 0, S_{\mathbb{W}}$) Satisfies the Successor Properties)**.** *The triple $(\mathbb{W}, 0, S_{\mathbb{W}})$ satisfies the successor-side properties:*

$$0 \in \mathbb{W}, \qquad S_{\mathbb{W}} : \mathbb{W} \to \mathbb{W},$$
$$\neg \exists a \in \mathbb{W}\ (S_{\mathbb{W}}(a) = 0), \qquad \forall a, b \in \mathbb{W}\ (S_{\mathbb{W}}(a) = S_{\mathbb{W}}(b) \implies a = b).$$

Go to proof.

0.90 Extending Addition to $\mathbb{W}$

0.90.1 Extending Addition to $\mathbb{W}$

The Extended Addition Operation

Definition 0.90.1 (Addition on $\mathbb{W}$)**.** Define $+_{\mathbb{W}} : \mathbb{W} \times \mathbb{W} \to \mathbb{W}$ by

$$0 +_{\mathbb{W}} a = a, \qquad a +_{\mathbb{W}} 0 = a,$$

and, for $a, b \in \mathbb{N}$,

$$a +_{\mathbb{W}} b = a +_{\mathbb{N}} b.$$

Theorem 0.90.2 (Addition on $\mathbb{W}$ Is Well-Defined)**.** *The case definition of $+_{\mathbb{W}}$ determines a binary operation on $\mathbb{W}$. Go to proof.*

Theorem 0.90.3 (Zero Is an Additive Identity on $\mathbb{W}$)**.** *For every $a \in \mathbb{W}$,*

$$0 +_{\mathbb{W}} a = a \qquad \text{and} \qquad a +_{\mathbb{W}} 0 = a.$$

Go to proof.

Theorem 0.90.4 (Left Additive Identity on $\mathbb{W}$)**.** *For every $a \in \mathbb{W}$,*

$$0 +_{\mathbb{W}} a = a.$$

Go to proof.

Theorem 0.90.5 (Right Additive Identity on $\mathbb{W}$)**.** *For every $a \in \mathbb{W}$,*

$$a +_{\mathbb{W}} 0 = a.$$

Go to proof.

Theorem 0.90.6 (Addition on $\mathbb{W}$ Extends Addition on $\mathbb{N}$)**.** *For all $a, b \in \mathbb{N}$,*

$$a +_{\mathbb{W}} b = a +_{\mathbb{N}} b.$$

Go to proof.

Theorem 0.90.7 (Addition on $\mathbb{W}$ Is Associative)**.** *For all $a, b, c \in \mathbb{W}$,*

$$(a +_{\mathbb{W}} b) +_{\mathbb{W}} c = a +_{\mathbb{W}} (b +_{\mathbb{W}} c).$$

Go to proof.

Theorem 0.90.8 (Addition on $\mathbb{W}$ Is Commutative)**.** *For all $a, b \in \mathbb{W}$,*

$$a +_{\mathbb{W}} b = b +_{\mathbb{W}} a.$$

Go to proof.

Theorem 0.90.9 (Addition on $\mathbb{W}$ Satisfies Cancellation)**.** *For all $a, b, c \in \mathbb{W}$,*

$$a +_{\mathbb{W}} b = a +_{\mathbb{W}} c \Longrightarrow b = c.$$

Go to proof.

0.91 Extending Multiplication to $\mathbb{W}$

0.91.1 Extending Multiplication to $\mathbb{W}$

The Extended Multiplication Operation

Definition 0.91.1 (Multiplication on $\mathbb{W}$)**.** Define $\cdot_{\mathbb{W}} : \mathbb{W} \times \mathbb{W} \to \mathbb{W}$ by

$$0 \cdot_{\mathbb{W}} a = 0, \qquad a \cdot_{\mathbb{W}} 0 = 0,$$

and, for $a, b \in \mathbb{N}$,

$$a \cdot_{\mathbb{W}} b = a \cdot_{\mathbb{N}} b.$$

Theorem 0.91.2 (Multiplication on $\mathbb{W}$ Is Well-Defined)**.** *The case definition of $\cdot_{\mathbb{W}}$ determines a binary operation on $\mathbb{W}$. Go to proof.*

Theorem 0.91.3 (Zero Is Absorbing for Multiplication on $\mathbb{W}$)**.** *For every $a \in \mathbb{W}$,*

$$0 \cdot_{\mathbb{W}} a = 0 \qquad \textit{and} \qquad a \cdot_{\mathbb{W}} 0 = 0.$$

Go to proof.

Theorem 0.91.4 (Left Zero Absorption on $\mathbb{W}$)**.** *For every $a \in \mathbb{W}$,*

$$0 \cdot_{\mathbb{W}} a = 0.$$

Go to proof.

Theorem 0.91.5 (Right Zero Absorption on $\mathbb{W}$)**.** *For every $a \in \mathbb{W}$,*

$$a \cdot_{\mathbb{W}} 0 = 0.$$

Go to proof.

Theorem 0.91.6 (One Is a Multiplicative Identity on $\mathbb{W}$)**.** *For every $a \in \mathbb{W}$,*

$$1 \cdot_{\mathbb{W}} a = a \qquad \textit{and} \qquad a \cdot_{\mathbb{W}} 1 = a.$$

Go to proof.

Theorem 0.91.7 (Multiplication on $\mathbb{W}$ Extends Multiplication on $\mathbb{N}$)**.** *For all $a, b \in \mathbb{N}$,*

$$a \cdot_{\mathbb{W}} b = a \cdot_{\mathbb{N}} b.$$

Go to proof.

Theorem 0.91.8 (Multiplication on $\mathbb{W}$ Is Associative)**.** *For all $a, b, c \in \mathbb{W}$,*

$$(a \cdot_{\mathbb{W}} b) \cdot_{\mathbb{W}} c = a \cdot_{\mathbb{W}} (b \cdot_{\mathbb{W}} c).$$

Go to proof.

Theorem 0.91.9 (Multiplication on $\mathbb{W}$ Is Commutative)**.** *For all $a, b \in \mathbb{W}$,*

$$a \cdot_{\mathbb{W}} b = b \cdot_{\mathbb{W}} a.$$

Go to proof.

Theorem 0.91.10 (Multiplication Distributes over Addition on $\mathbb{W}$)**.** *For all $a, b, c \in \mathbb{W}$,*

$$a \cdot_{\mathbb{W}} (b +_{\mathbb{W}} c) = (a \cdot_{\mathbb{W}} b) +_{\mathbb{W}} (a \cdot_{\mathbb{W}} c).$$

Go to proof.

0.92 Order on $\mathbb{W}$

0.92.1 Order on $\mathbb{W}$

The Extended Order

Definition 0.92.1 (Order on $\mathbb{W}$)**.** Define $\leq_{\mathbb{W}}$ on $\mathbb{W}$ by declaring

$$0 \leq_{\mathbb{W}} a \quad (a \in \mathbb{W}),$$

and, for $a, b \in \mathbb{N}$,

$$a \leq_{\mathbb{W}} b \quad \Longleftrightarrow \quad a \leq_{\mathbb{N}} b.$$

Definition 0.92.2 (Strict Order on $\mathbb{W}$)**.** Define $<_{\mathbb{W}}$ on $\mathbb{W}$ by declaring

$$0 <_{\mathbb{W}} n \quad (n \in \mathbb{N}),$$

and, for $a, b \in \mathbb{N}$,

$$a <_{\mathbb{W}} b \quad \Longleftrightarrow \quad a <_{\mathbb{N}} b.$$

No element of $\mathbb{W}$ is strictly less than 0.

Theorem 0.92.3 (Order on $\mathbb{W}$ Extends Order on $\mathbb{N}$)**.** *For all $a, b \in \mathbb{N}$,*

$$a \leq_{\mathbb{W}} b \Longleftrightarrow a \leq_{\mathbb{N}} b, \qquad a <_{\mathbb{W}} b \Longleftrightarrow a <_{\mathbb{N}} b.$$

Go to proof.

Theorem 0.92.4 (Zero Is the Least Element of $\mathbb{W}$)**.** *For every $a \in \mathbb{W}$,*

$$0 \leq_{\mathbb{W}} a.$$

Go to proof.

Theorem 0.92.5 (Order on $\mathbb{W}$ Is Reflexive)**.** *For every $a \in \mathbb{W}$, $a \leq_{\mathbb{W}} a$.* *Go to proof.*

Theorem 0.92.6 (Order on $\mathbb{W}$ Is Antisymmetric)**.** *For all $a, b \in \mathbb{W}$,*

$$a \leq_{\mathbb{W}} b \text{ and } b \leq_{\mathbb{W}} a \Longrightarrow a = b.$$

Go to proof.

Theorem 0.92.7 (Order on $\mathbb{W}$ Is Transitive)**.** *For all $a, b, c \in \mathbb{W}$,*

$$a \leq_{\mathbb{W}} b \text{ and } b \leq_{\mathbb{W}} c \Longrightarrow a \leq_{\mathbb{W}} c.$$

Go to proof.

Theorem 0.92.8 (Order on $\mathbb{W}$ Is Total)**.** *For all $a, b \in \mathbb{W}$,*

$$a \leq_{\mathbb{W}} b \quad \text{or} \quad b \leq_{\mathbb{W}} a.$$

Go to proof.

Theorem 0.92.9 (Well-Ordering Principle for $\mathbb{W}$)**.** *Every nonempty subset of $\mathbb{W}$ has a least element with respect to $\leq_{\mathbb{W}}$.* *Go to proof.*

0.93 Algebraic Structure of $\mathbb{W}$

0.93.1 Algebraic Structure of $\mathbb{W}$

Algebraic Packages

Theorem 0.93.1 ($(\mathbb{W}, +, 0)$ Is a Commutative Monoid)**.** *The structure* $(\mathbb{W}, +_{\mathbb{W}}, 0)$ *is a commutative monoid. Go to proof.*

Theorem 0.93.2 ($(\mathbb{W}, \cdot, 1)$ Is a Commutative Monoid)**.** *The structure* $(\mathbb{W}, \cdot_{\mathbb{W}}, 1)$ *is a commutative monoid. Go to proof.*

Theorem 0.93.3 ($(\mathbb{W}, +, \cdot, 0, 1)$ Is a Commutative Semiring)**.** *The structure* $(\mathbb{W}, +_{\mathbb{W}}, \cdot_{\mathbb{W}}, 0, 1)$ *is a commutative semiring. Go to proof.*

Theorem 0.93.4 ($\mathbb{W}$ Is a Commutative Ordered Semiring)**.** *With the operations* $+_{\mathbb{W}}$ *and* $\cdot_{\mathbb{W}}$ *and the order* $\leq_{\mathbb{W}}$*, the whole numbers form a commutative ordered semiring. Go to proof.*

Theorem 0.93.5 ($\mathbb{N}$ Embeds into $\mathbb{W}$ as the Positive Part)**.** *The inclusion map identifies* $\mathbb{N}$ *with the positive part* $\mathbb{W} \setminus \{0\}$ *and preserves successor,* 1*, addition, multiplication, and order. Go to proof.*

Theorem 0.93.6 ($\mathbb{W}$ Has No Additive Inverses Except 0)**.** *If* $a, b \in \mathbb{W}$ *and*

$$a +_{\mathbb{W}} b = 0,$$

then $a = 0$ *and* $b = 0$*. Go to proof.*

Theorem 0.93.7 ($\mathbb{W}$ Is Not a Ring)**.** *With the operations* $+_{\mathbb{W}}$ *and* $\cdot_{\mathbb{W}}$*, the whole numbers do not form a ring. Go to proof.*

0.94 Transition to $\mathbb{Z}$

0.94.1 Transition to $\mathbb{Z}$

The Next Construction

Proofs

Capstone

Integers ($\mathbb{Z}$)

0.95 Construction of $\mathbb{Z}$

Integer Prepairs

Definition 0.95.1 (Integer Prepair)**.** An *integer prepair* is an ordered pair $(a, b) \in \mathbb{W} \times \mathbb{W}$, read as the formal difference $a - b$.

Definition 0.95.2 (Integer Prepair Set)**.** The integer prepair set is

$$\operatorname{Pre}\mathbb{Z} := \mathbb{W} \times \mathbb{W}.$$

Formal-Difference Equivalence

Definition 0.95.3 (Formal-Difference Equivalence)**.** For $(a, b), (c, d) \in \operatorname{Pre}\mathbb{Z}$, define

$$(a, b) \sim_{\mathbb{Z}} (c, d) \quad :\Longleftrightarrow \quad a + d = c + b.$$

Lemma 0.95.4 (Formal-Difference Equivalence Is Reflexive)**.** *For every* $(a, b) \in \operatorname{Pre}\mathbb{Z}$*,*

$$(a, b) \sim_{\mathbb{Z}} (a, b).$$

Lemma 0.95.5 (Formal-Difference Equivalence Is Symmetric)**.** *For all prepairs* $(a, b), (c, d)$*,*

$$(a, b) \sim_{\mathbb{Z}} (c, d) \Longrightarrow (c, d) \sim_{\mathbb{Z}} (a, b).$$

Lemma 0.95.6 (Formal-Difference Equivalence Is Transitive)**.** *For all prepairs* $(a, b), (c, d), (e, f)$*,*

$$(a, b) \sim_{\mathbb{Z}} (c, d) \wedge (c, d) \sim_{\mathbb{Z}} (e, f) \Longrightarrow (a, b) \sim_{\mathbb{Z}} (e, f).$$

Theorem 0.95.7 (Formal-Difference Equivalence Is an Equivalence Relation)**.** *The relation* $\sim_{\mathbb{Z}}$ *is an equivalence relation on* $\operatorname{Pre}\mathbb{Z}$*.*

Definition of $\mathbb{Z}$

Definition 0.95.8 (Integers)**.** The integers are the quotient

$$\mathbb{Z} := \operatorname{Pre}\mathbb{Z}/{\sim_{\mathbb{Z}}}.$$

Definition 0.95.9 (Integer Class Notation)**.** The notation $[a, b]$ denotes the $\sim_{\mathbb{Z}}$-equivalence class of the prepair (a, b).

Zero and One in $\mathbb{Z}$

Definition 0.95.10 (Zero in $\mathbb{Z}$)**.** Define

$$0_{\mathbb{Z}} := [0, 0].$$

Definition 0.95.11 (One in $\mathbb{Z}$)**.** Define

$$1_{\mathbb{Z}} := [1, 0].$$

Theorem 0.95.12 (Zero Is an Integer)**.** *The class $0_{\mathbb{Z}}$ is an element of $\mathbb{Z}$.*

Theorem 0.95.13 (One Is an Integer)**.** *The class $1_{\mathbb{Z}}$ is an element of $\mathbb{Z}$.*

Theorem 0.95.14 (Zero Is Not One in $\mathbb{Z}$)**.** *In $\mathbb{Z}$,*

$$0_{\mathbb{Z}} \neq 1_{\mathbb{Z}}.$$

0.96 Operations on $\mathbb{Z}$-Classes

Addition on $\mathbb{Z}$-Classes

Definition 0.96.1 (Addition on $\mathbb{Z}$-Classes)**.** Define addition on representatives by

$$[a, b] + [c, d] := [a + c, b + d].$$

Lemma 0.96.2 (Addition Respects Formal-Difference Equivalence on the Left)**.** *If $[a, b] = [a', b']$, then*

$$[a, b] + [c, d] = [a', b'] + [c, d].$$

Lemma 0.96.3 (Addition Respects Formal-Difference Equivalence on the Right)**.** *If $[c, d] = [c', d']$, then*

$$[a, b] + [c, d] = [a, b] + [c', d'].$$

Lemma 0.96.4 (Addition Respects Formal-Difference Equivalence)**.** *If $[a, b] = [a', b']$ and $[c, d] = [c', d']$, then*

$$[a, b] + [c, d] = [a', b'] + [c', d'].$$

Theorem 0.96.5 (Addition on $\mathbb{Z}$ Is Well-Defined)**.** *Addition determines a well-defined binary operation $\mathbb{Z} \times \mathbb{Z} \to \mathbb{Z}$.*

Theorem 0.96.6 (Addition on $\mathbb{Z}$ Is Associative)**.** *For all $x, y, z \in \mathbb{Z}$,*

$$(x + y) + z = x + (y + z).$$

Theorem 0.96.7 (Addition on $\mathbb{Z}$ Is Commutative)**.** *For all $x, y \in \mathbb{Z}$,*

$$x + y = y + x.$$

Theorem 0.96.8 (Zero Is a Left Additive Identity on $\mathbb{Z}$)**.** *For every $x \in \mathbb{Z}$,*

$$0_{\mathbb{Z}} + x = x.$$

Corollary 0.96.9 (Zero Is a Right Additive Identity on $\mathbb{Z}$)**.** *For every $x \in \mathbb{Z}$,*

$$x + 0_{\mathbb{Z}} = x.$$

Theorem 0.96.10 (Zero Is an Additive Identity on $\mathbb{Z}$)**.** *For every $x \in \mathbb{Z}$,*

$$0_{\mathbb{Z}} + x = x \qquad \textit{and} \qquad x + 0_{\mathbb{Z}} = x.$$

Negation on $\mathbb{Z}$-Classes

Definition 0.96.11 (Negation on $\mathbb{Z}$-Classes)**.** Define negation on representatives by

$$-[a, b] := [b, a].$$

Lemma 0.96.12 (Negation Respects Formal-Difference Equivalence)**.** *If* $[a, b] = [a', b']$*, then*

$$-[a, b] = -[a', b'].$$

Theorem 0.96.13 (Negation on $\mathbb{Z}$ Is Well-Defined)**.** *Negation determines a well-defined unary operation* $\mathbb{Z} \to \mathbb{Z}$*.*

Theorem 0.96.14 (Negation Gives a Left Additive Inverse on $\mathbb{Z}$)**.** *For every* $x \in \mathbb{Z}$*,*

$$(-x) + x = 0_{\mathbb{Z}}.$$

Corollary 0.96.15 (Negation Gives a Right Additive Inverse on $\mathbb{Z}$)**.** *For every* $x \in \mathbb{Z}$*,*

$$x + (-x) = 0_{\mathbb{Z}}.$$

Theorem 0.96.16 (Every Integer Has an Additive Inverse)**.** *For every* $x \in \mathbb{Z}$ *there exists* $y \in \mathbb{Z}$ *such that*

$$x + y = 0_{\mathbb{Z}} \qquad \text{and} \qquad y + x = 0_{\mathbb{Z}}.$$

Subtraction on $\mathbb{Z}$-Classes

Definition 0.96.17 (Subtraction on $\mathbb{Z}$)**.** For $x, y \in \mathbb{Z}$, define

$$x - y := x + (-y).$$

Theorem 0.96.18 (Subtraction on $\mathbb{Z}$ Is Well-Defined)**.** *Subtraction determines a well-defined binary operation* $\mathbb{Z} \times \mathbb{Z} \to \mathbb{Z}$*.*

Multiplication on $\mathbb{Z}$-Classes

Definition 0.96.19 (Multiplication on $\mathbb{Z}$-Classes)**.** Define multiplication on representatives by

$$[a, b] \cdot [c, d] := [ac + bd, ad + bc].$$

Lemma 0.96.20 (Multiplication Respects Formal-Difference Equivalence on the Left)**.** *If* $[a, b] = [a', b']$*, then*

$$[a, b] \cdot [c, d] = [a', b'] \cdot [c, d].$$

Lemma 0.96.21 (Multiplication Respects Formal-Difference Equivalence on the Right)**.** *If* $[c, d] = [c', d']$*, then*

$$[a, b] \cdot [c, d] = [a, b] \cdot [c', d'].$$

Lemma 0.96.22 (Multiplication Respects Formal-Difference Equivalence)**.** *If* $[a, b] = [a', b']$ *and* $[c, d] = [c', d']$*, then*

$$[a, b] \cdot [c, d] = [a', b'] \cdot [c', d'].$$

Theorem 0.96.23 (Multiplication on $\mathbb{Z}$ Is Well-Defined)**.** *Multiplication determines a well-defined binary operation* $\mathbb{Z} \times \mathbb{Z} \to \mathbb{Z}$*.*

Theorem 0.96.24 (Multiplication on $\mathbb{Z}$ Is Associative)**.** *For all* $x, y, z \in \mathbb{Z}$*,*

$$(x \cdot y) \cdot z = x \cdot (y \cdot z).$$

Theorem 0.96.25 (Multiplication on $\mathbb{Z}$ Is Commutative)**.** *For all $x, y \in \mathbb{Z}$,*

$$x \cdot y = y \cdot x.$$

Theorem 0.96.26 (One Is a Left Multiplicative Identity on $\mathbb{Z}$)**.** *For every $x \in \mathbb{Z}$,*

$$1_{\mathbb{Z}} \cdot x = x.$$

Corollary 0.96.27 (One Is a Right Multiplicative Identity on $\mathbb{Z}$)**.** *For every $x \in \mathbb{Z}$,*

$$x \cdot 1_{\mathbb{Z}} = x.$$

Theorem 0.96.28 (One Is a Multiplicative Identity on $\mathbb{Z}$)**.** *For every $x \in \mathbb{Z}$,*

$$1_{\mathbb{Z}} \cdot x = x \qquad \text{and} \qquad x \cdot 1_{\mathbb{Z}} = x.$$

Distributivity on $\mathbb{Z}$-Classes

Theorem 0.96.29 (Left Distributivity on $\mathbb{Z}$)**.** *For all $x, y, z \in \mathbb{Z}$,*

$$x \cdot (y + z) = x \cdot y + x \cdot z.$$

Corollary 0.96.30 (Right Distributivity on $\mathbb{Z}$)**.** *For all $x, y, z \in \mathbb{Z}$,*

$$(y + z) \cdot x = y \cdot x + z \cdot x.$$

Theorem 0.96.31 (Multiplication Distributes over Addition on $\mathbb{Z}$)**.** *Multiplication distributes over addition on both sides in $\mathbb{Z}$.*

Class-Level Ring Summary

Theorem 0.96.32 ($\mathbb{Z}$ Is an Additive Abelian Group)**.** *The structure $(\mathbb{Z}, +, 0_{\mathbb{Z}})$ is an abelian group.*

Theorem 0.96.33 ($\mathbb{Z}$ Is a Multiplicative Commutative Monoid)**.** *The structure $(\mathbb{Z}, \cdot, 1_{\mathbb{Z}})$ is a commutative monoid.*

Theorem 0.96.34 ($\mathbb{Z}$ Is a Commutative Ring)**.** *The structure $(\mathbb{Z}, +, \cdot, 0_{\mathbb{Z}}, 1_{\mathbb{Z}})$ is a commutative ring.*

0.97 Algebraic Structure of $\mathbb{Z}$

Integral Domain Structure

Theorem 0.97.1 ($\mathbb{Z}$ Has No Zero Divisors)**.** *For all $x, y \in \mathbb{Z}$,*

$$x \cdot y = 0 \Longrightarrow x = 0 \text{ or } y = 0.$$

Theorem 0.97.2 ($\mathbb{Z}$ Is an Integral Domain)**.** *The integers form an integral domain: $\mathbb{Z}$ is a commutative ring, $0 \neq 1$, and $\mathbb{Z}$ has no zero divisors.*

Corollary 0.97.3 (Multiplicative Cancellation on $\mathbb{Z}$)**.** *For all $x, y, z \in \mathbb{Z}$,*

$$z \neq 0 \wedge z \cdot x = z \cdot y \Longrightarrow x = y.$$

Derived Laws on $\mathbb{Z}$

Corollary 0.97.4 (Zero Absorption on $\mathbb{Z}$)**.** *For every $x \in \mathbb{Z}$,*

$$0 \cdot x = 0 \qquad \textit{and} \qquad x \cdot 0 = 0.$$

Corollary 0.97.5 (Sign Rule on $\mathbb{Z}$)**.** *For all $x, y \in \mathbb{Z}$,*

$$(-x) \cdot y = -(x \cdot y) \qquad \textit{and} \qquad x \cdot (-y) = -(x \cdot y).$$

Corollary 0.97.6 (Negative One Rule on $\mathbb{Z}$)**.** *For every $x \in \mathbb{Z}$,*

$$(-1) \cdot x = -x.$$

Corollary 0.97.7 (Product of Negatives on $\mathbb{Z}$)**.** *For all $x, y \in \mathbb{Z}$,*

$$(-x) \cdot (-y) = x \cdot y.$$

0.98 Order on $\mathbb{Z}$

Positivity on $\mathbb{Z}$

Definition 0.98.1 (Positive Integer)**.** An integer $x \in \mathbb{Z}$ is *positive* if $x = [a, b]$ for some $a, b \in \mathbb{W}$ with $b < a$.

Lemma 0.98.2 (Positivity Respects Formal-Difference Equivalence)**.** *If $[a, b] = [a', b']$, then*

$$b < a \Longleftrightarrow b' < a'.$$

Theorem 0.98.3 (Positivity on $\mathbb{Z}$ Is Well-Defined)**.** *The predicate "x is positive" is well-defined on equivalence classes in $\mathbb{Z}$.*

Definition 0.98.4 (Strict Order on $\mathbb{Z}$)**.** For $x, y \in \mathbb{Z}$, define

$$x < y \quad :\Longleftrightarrow \quad y - x \text{ is positive.}$$

Definition 0.98.5 (Order on $\mathbb{Z}$)**.** For $x, y \in \mathbb{Z}$, define

$$x \leq y \quad :\Longleftrightarrow \quad x < y \text{ or } x = y.$$

Proposition 0.98.6 (Product of Positives Is Positive)**.** *For all $x, y \in \mathbb{Z}$,*

$$0 < x \wedge 0 < y \Longrightarrow 0 < x \cdot y.$$

Canonical Form and Sign

Theorem 0.98.7 (Canonical Form of Integers)**.** *For every $x \in \mathbb{Z}$, exactly one of the following holds:*

$$x = [n, 0] \textit{ for some nonzero } n \in \mathbb{W}, \qquad x = [0, 0], \qquad x = [0, n] \textit{ for some nonzero } n \in \mathbb{W}.$$

Corollary 0.98.8 (Integers Split by Sign)**.** *Every integer is nonnegative or the negative of a nonnegative integer, and the two descriptions overlap only at $0_{\mathbb{Z}}$.*

Corollary 0.98.9 (Image of the Embedding Is the Nonnegative Integers)**.** *The image of the canonical embedding $\mathbb{W} \to \mathbb{Z}$ is exactly the set of nonnegative integers.*

Theorem 0.98.10 (Sign Trichotomy on $\mathbb{Z}$)**.** *For every $x \in \mathbb{Z}$, exactly one of*

$$x < 0, \qquad x = 0, \qquad 0 < x$$

holds.

Theorem 0.98.11 (Trichotomy on $\mathbb{Z}$)**.** *For all $x, y \in \mathbb{Z}$, exactly one of*

$$x < y, \qquad x = y, \qquad y < x$$

holds.

Total Order on $\mathbb{Z}$

Theorem 0.98.12 (Strict Order on $\mathbb{Z}$ Is Irreflexive). *For every $x \in \mathbb{Z}$, not $x < x$.*

Theorem 0.98.13 (Strict Order on $\mathbb{Z}$ Is Asymmetric). *For all $x, y \in \mathbb{Z}$,*

$$x < y \implies \neg(y < x).$$

Theorem 0.98.14 (Strict Order on $\mathbb{Z}$ Is Transitive). *For all $x, y, z \in \mathbb{Z}$,*

$$x < y \wedge y < z \implies x < z.$$

Theorem 0.98.15 (Order on $\mathbb{Z}$ Is Reflexive). *For every $x \in \mathbb{Z}$, $x \leq x$.*

Theorem 0.98.16 (Order on $\mathbb{Z}$ Is Antisymmetric). *For all $x, y \in \mathbb{Z}$,*

$$x \leq y \wedge y \leq x \implies x = y.$$

Theorem 0.98.17 (Order on $\mathbb{Z}$ Is Transitive). *For all $x, y, z \in \mathbb{Z}$,*

$$x \leq y \wedge y \leq z \implies x \leq z.$$

Theorem 0.98.18 (Order on $\mathbb{Z}$ Is Total). *For all $x, y \in \mathbb{Z}$,*

$$x \leq y \text{ or } y \leq x.$$

Theorem 0.98.19 ($\mathbb{Z}$ Is a Totally Ordered Set). *The ordered pair $(\mathbb{Z}, \leq)$ is a totally ordered set.*

Addition Order Compatibility on $\mathbb{Z}$

Theorem 0.98.20 (Left Addition Preserves Strict Order on $\mathbb{Z}$). *For all $x, y, z \in \mathbb{Z}$,*

$$x < y \implies z + x < z + y.$$

Corollary 0.98.21 (Right Addition Preserves Strict Order on $\mathbb{Z}$). *For all $x, y, z \in \mathbb{Z}$,*

$$x < y \implies x + z < y + z.$$

Theorem 0.98.22 (Addition Preserves Strict Order on $\mathbb{Z}$). *Addition preserves strict order on both sides in $\mathbb{Z}$.*

Theorem 0.98.23 (Left Addition Preserves Order on $\mathbb{Z}$). *For all $x, y, z \in \mathbb{Z}$,*

$$x \leq y \implies z + x \leq z + y.$$

Corollary 0.98.24 (Right Addition Preserves Order on $\mathbb{Z}$). *For all $x, y, z \in \mathbb{Z}$,*

$$x \leq y \implies x + z \leq y + z.$$

Theorem 0.98.25 (Addition Preserves Order on $\mathbb{Z}$). *Addition preserves non-strict order on both sides in $\mathbb{Z}$.*

Corollary 0.98.26 (Addition Reflects Order on $\mathbb{Z}$). *Addition by a fixed integer reflects both strict and non-strict order.*

Multiplication Order Compatibility on $\mathbb{Z}$

Theorem 0.98.27 (Left Multiplication by Positives Preserves Strict Order on $\mathbb{Z}$). *For all $x, y, z \in \mathbb{Z}$,*

$$0 < z \wedge x < y \implies z \cdot x < z \cdot y.$$

Corollary 0.98.28 (Right Multiplication by Positives Preserves Strict Order on $\mathbb{Z}$)**.** *For all $x, y, z \in \mathbb{Z}$,*

$$0 < z \wedge x < y \Longrightarrow x \cdot z < y \cdot z.$$

Theorem 0.98.29 (Multiplication by Positives Preserves Strict Order on $\mathbb{Z}$)**.** *Multiplication by a positive integer preserves strict order on both sides.*

Theorem 0.98.30 (Left Multiplication by Positives Preserves Order on $\mathbb{Z}$)**.** *For all $x, y, z \in \mathbb{Z}$,*

$$0 < z \wedge x \leq y \Longrightarrow z \cdot x \leq z \cdot y.$$

Corollary 0.98.31 (Right Multiplication by Positives Preserves Order on $\mathbb{Z}$)**.** *For all $x, y, z \in \mathbb{Z}$,*

$$0 < z \wedge x \leq y \Longrightarrow x \cdot z \leq y \cdot z.$$

Theorem 0.98.32 (Multiplication by Positives Preserves Order on $\mathbb{Z}$)**.** *Multiplication by a positive integer preserves non-strict order on both sides.*

Theorem 0.98.33 (Multiplication by Negatives Reverses Strict Order on $\mathbb{Z}$)**.** *For all $x, y, z \in \mathbb{Z}$,*

$$z < 0 \wedge x < y \Longrightarrow z \cdot y < z \cdot x.$$

Corollary 0.98.34 (Multiplication by Negatives Reverses Order on $\mathbb{Z}$)**.** *For all $x, y, z \in \mathbb{Z}$,*

$$z < 0 \wedge x \leq y \Longrightarrow z \cdot y \leq z \cdot x.$$

Theorem 0.98.35 (Multiplication by Zero Collapses Order on $\mathbb{Z}$)**.** *For all $x, y \in \mathbb{Z}$,*

$$0 \cdot x = 0 \cdot y.$$

Absolute Value on $\mathbb{Z}$

Definition 0.98.36 (Absolute Value on $\mathbb{Z}$)**.** Define

$$|x| := \begin{cases} x, & 0 \leq x, \\ -x, & x < 0. \end{cases}$$

Theorem 0.98.37 (Absolute Value Is Nonnegative)**.** *For every $x \in \mathbb{Z}$,*

$$0 \leq |x|.$$

Theorem 0.98.38 (Absolute Value Vanishes Only at Zero)**.** *For every $x \in \mathbb{Z}$,*

$$|x| = 0 \Longleftrightarrow x = 0.$$

Theorem 0.98.39 (Absolute Value Is Symmetric under Negation)**.** *For every $x \in \mathbb{Z}$,*

$$|-x| = |x|.$$

Theorem 0.98.40 (Absolute Value Is Multiplicative)**.** *For all $x, y \in \mathbb{Z}$,*

$$|x \cdot y| = |x| \cdot |y|.$$

Theorem 0.98.41 (Bounding by Absolute Value)**.** *For every $x \in \mathbb{Z}$,*

$$-|x| \leq x \leq |x|.$$

Theorem 0.98.42 (Triangle Inequality on $\mathbb{Z}$)**.** *For all $x, y \in \mathbb{Z}$,*

$$|x + y| \leq |x| + |y|.$$

Well-Ordering Fails on $\mathbb{Z}$

Theorem 0.98.43 ($\mathbb{Z}$ Is Not Well-Ordered)**.** *The ordered set* $(\mathbb{Z}, \leq)$ *is not a well-order.*

0.99 Embedding $\mathbb{W}$ into $\mathbb{Z}$

Embedding $\mathbb{W}$ into $\mathbb{Z}$

Definition 0.99.1 (Embedding of $\mathbb{W}$ into $\mathbb{Z}$)**.** Define

$$\iota : \mathbb{W} \to \mathbb{Z}, \qquad \iota(n) := [n, 0].$$

Theorem 0.99.2 (Embedding Preserves Zero)**.** *The canonical embedding satisfies*

$$\iota(0_{\mathbb{W}}) = 0_{\mathbb{Z}}.$$

Theorem 0.99.3 (Embedding Preserves One)**.** *The canonical embedding satisfies*

$$\iota(1_{\mathbb{W}}) = 1_{\mathbb{Z}}.$$

Theorem 0.99.4 (Embedding $\mathbb{W}$ into $\mathbb{Z}$ Is Injective)**.** *For all* $a, b \in \mathbb{W}$*,*

$$\iota(a) = \iota(b) \Longrightarrow a = b.$$

Theorem 0.99.5 (Embedding Preserves Addition)**.** *For all* $a, b \in \mathbb{W}$*,*

$$\iota(a + b) = \iota(a) + \iota(b).$$

Theorem 0.99.6 (Embedding Preserves Multiplication)**.** *For all* $a, b \in \mathbb{W}$*,*

$$\iota(a \cdot b) = \iota(a) \cdot \iota(b).$$

Theorem 0.99.7 (Embedding Preserves Order)**.** *For all* $a, b \in \mathbb{W}$*,*

$$a \leq b \Longleftrightarrow \iota(a) \leq \iota(b).$$

Theorem 0.99.8 ($\mathbb{W}$ Embeds into $\mathbb{Z}$)**.** *The map* $\iota : \mathbb{W} \to \mathbb{Z}$ *is an injective order-preserving semiring homomorphism.*

0.100 Divisibility in $\mathbb{Z}$

Divisibility in $\mathbb{Z}$

Definition 0.100.1 (Divisibility in $\mathbb{Z}$)**.** For $a, b \in \mathbb{Z}$, define

$$a \mid b \quad :\Longleftrightarrow \quad \exists c \in \mathbb{Z}\ (b = a \cdot c).$$

Basic Divisibility Laws

Proposition 0.100.2 (Divisibility Is Reflexive on $\mathbb{Z}$)**.** *For every* $a \in \mathbb{Z}$*,*

$$a \mid a.$$

Proposition 0.100.3 (Divisibility Is Transitive on $\mathbb{Z}$)**.** *For all* $a, b, c \in \mathbb{Z}$*,*

$$a \mid b \wedge b \mid c \Longrightarrow a \mid c.$$

Proposition 0.100.4 (One and Negative One Divide Every Integer)**.** *For every $a \in \mathbb{Z}$,*

$$1 \mid a \qquad \textit{and} \qquad (-1) \mid a.$$

Proposition 0.100.5 (Every Integer Divides Zero)**.** *For every $a \in \mathbb{Z}$,*

$$a \mid 0.$$

Proposition 0.100.6 (Divisibility Is Compatible with Negation on the Divisor)**.** *For all $a, b \in \mathbb{Z}$,*

$$a \mid b \iff (-a) \mid b.$$

Proposition 0.100.7 (Divisibility Is Compatible with Negation on the Dividend)**.** *For all $a, b \in \mathbb{Z}$,*

$$a \mid b \iff a \mid (-b).$$

Proposition 0.100.8 (Divisibility Is Compatible with Negation on $\mathbb{Z}$)**.** *For all $a, b \in \mathbb{Z}$,*

$$a \mid b \iff (-a) \mid b \iff a \mid (-b).$$

Proposition 0.100.9 (Divisibility Is Compatible with Addition on $\mathbb{Z}$)**.** *For all $a, b, c \in \mathbb{Z}$,*

$$a \mid b \wedge a \mid c \Longrightarrow a \mid (b + c).$$

Proposition 0.100.10 (Divisibility Respects Linear Combinations on $\mathbb{Z}$)**.** *For all $a, b, c, x, y \in \mathbb{Z}$,*

$$a \mid b \wedge a \mid c \Longrightarrow a \mid (b \cdot x + c \cdot y).$$

Units in $\mathbb{Z}$

Definition 0.100.11 (Unit in $\mathbb{Z}$)**.** An integer $u \in \mathbb{Z}$ is a *unit* if

$$\exists v \in \mathbb{Z}\ (u \cdot v = 1).$$

Theorem 0.100.12 (The Units of $\mathbb{Z}$ Are Exactly ± 1)**.** *An integer u is a unit if and only if*

$$u = 1 \quad \textit{or} \quad u = -1.$$

Definition 0.100.13 (Associates in $\mathbb{Z}$)**.** Integers $a, b \in \mathbb{Z}$ are *associates* if $a = u \cdot b$ for some unit $u \in \mathbb{Z}$.

Division Algorithm

Theorem 0.100.14 (Division Algorithm on $\mathbb{Z}$)**.** *If $a, b \in \mathbb{Z}$ and $b \neq 0$, then there exist unique $q, r \in \mathbb{Z}$ such that*

$$a = bq + r \qquad \textit{and} \qquad 0 \leq r < |b|.$$

Greatest Common Divisor

Definition 0.100.15 (Greatest Common Divisor in $\mathbb{Z}$)**.** A greatest common divisor $\gcd(a, b)$ is the positive common divisor of a and b divisible by every other common divisor.

Bezout Identity

Theorem 0.100.16 (Bezout Identity on $\mathbb{Z}$)**.** *For $a, b \in \mathbb{Z}$, there exist $x, y \in \mathbb{Z}$ such that*

$$\gcd(a, b) = a \cdot x + b \cdot y.$$

0.101 Transition to $\mathbb{Q}$

Proofs

Capstone

The Continuum

Rational Numbers ($\mathbb{Q}$)

0.102 Construction of $\mathbb{Q}$

Prefractions

Definition 0.102.1 (Nonzero Integers)**.** The set of nonzero integers is

$$\mathbb{Z}^* := \mathbb{Z} \setminus \{0_{\mathbb{Z}}\}.$$

Definition 0.102.2 (Prefraction)**.** A *prefraction* is an ordered pair $(a, b) \in \mathbb{Z} \times \mathbb{Z}^*$, read as the formal quotient a/b.

Definition 0.102.3 (Prefraction Set)**.** The prefraction set is

$$\operatorname{Pre}\mathbb{Q} := \mathbb{Z} \times \mathbb{Z}^*.$$

Fraction Equivalence

Definition 0.102.4 (Fraction Equivalence)**.** For $(a, b), (c, d) \in \operatorname{Pre}\mathbb{Q}$, define

$$(a, b) \sim_{\mathbb{Q}} (c, d) \quad :\iff \quad a \cdot d = b \cdot c.$$

Lemma 0.102.5 (Fraction Equivalence Is Reflexive)**.** *For every* $(a, b) \in \operatorname{Pre}\mathbb{Q}$*,*

$$(a, b) \sim_{\mathbb{Q}} (a, b).$$

Lemma 0.102.6 (Fraction Equivalence Is Symmetric)**.** *For all prefractions* $(a, b), (c, d)$*,*

$$(a, b) \sim_{\mathbb{Q}} (c, d) \implies (c, d) \sim_{\mathbb{Q}} (a, b).$$

Lemma 0.102.7 (Fraction Equivalence Is Transitive)**.** *For all prefractions* $(a, b), (c, d), (e, f)$*,*

$$(a, b) \sim_{\mathbb{Q}} (c, d) \wedge (c, d) \sim_{\mathbb{Q}} (e, f) \implies (a, b) \sim_{\mathbb{Q}} (e, f).$$

Theorem 0.102.8 (Fraction Equivalence Is an Equivalence Relation)**.** *The relation* $\sim_{\mathbb{Q}}$ *is an equivalence relation on* $\operatorname{Pre}\mathbb{Q}$*.*

Definition of $\mathbb{Q}$

Definition 0.102.9 (Rational Numbers)**.** The rational numbers are the quotient

$$\mathbb{Q} := \operatorname{Pre}\mathbb{Q}/{\sim_{\mathbb{Q}}}.$$

Definition 0.102.10 (Rational Class Notation)**.** The notation $[a, b]$ denotes the $\sim_{\mathbb{Q}}$-equivalence class of the prefraction (a, b), where $b \neq 0_{\mathbb{Z}}$.

Proposition 0.102.11 (Equality Criterion for Rational Classes)**.** *For prefractions* $(a, b), (c, d) \in \operatorname{Pre}\mathbb{Q}$*,*

$$[a, b] = [c, d] \quad \iff \quad a \cdot d = b \cdot c.$$

Zero and One in $\mathbb{Q}$

Definition 0.102.12 (Zero in $\mathbb{Q}$)**.** Define
$$0_\mathbb{Q} := [0_\mathbb{Z}, 1_\mathbb{Z}].$$

Definition 0.102.13 (One in $\mathbb{Q}$)**.** Define
$$1_\mathbb{Q} := [1_\mathbb{Z}, 1_\mathbb{Z}].$$

Lemma 0.102.14 (Vanishing Criterion in $\mathbb{Q}$)**.** *For every prefraction* $(a, b) \in \operatorname{Pre}\mathbb{Q}$*,*
$$[a, b] = 0_\mathbb{Q} \iff a = 0_\mathbb{Z}.$$

Theorem 0.102.15 (Zero Is a Rational)**.** *The class* $0_\mathbb{Q}$ *is an element of* $\mathbb{Q}$*.*

Theorem 0.102.16 (One Is a Rational)**.** *The class* $1_\mathbb{Q}$ *is an element of* $\mathbb{Q}$*.*

Theorem 0.102.17 (Zero Is Not One in $\mathbb{Q}$)**.** *In* $\mathbb{Q}$*,*
$$0_\mathbb{Q} \neq 1_\mathbb{Q}.$$

0.103 Operations on $\mathbb{Q}$-Classes

Addition on $\mathbb{Q}$-Classes

Definition 0.103.1 (Addition on $\mathbb{Q}$-Classes)**.** Define addition on representatives by
$$[a, b] + [c, d] := [a \cdot d + b \cdot c,\ b \cdot d].$$

Lemma 0.103.2 (Addition Lands in $\operatorname{Pre}\mathbb{Q}$)**.** *If* $b \neq 0_\mathbb{Z}$ *and* $d \neq 0_\mathbb{Z}$*, then*
$$b \cdot d \neq 0_\mathbb{Z}.$$

Lemma 0.103.3 (Addition Respects Fraction Equivalence on the Left)**.** *If* $[a, b] = [a', b']$*, then*
$$[a, b] + [c, d] = [a', b'] + [c, d].$$

Lemma 0.103.4 (Addition Respects Fraction Equivalence on the Right)**.** *If* $[c, d] = [c', d']$*, then*
$$[a, b] + [c, d] = [a, b] + [c', d'].$$

Lemma 0.103.5 (Addition Respects Fraction Equivalence)**.** *If* $[a, b] = [a', b']$ *and* $[c, d] = [c', d']$*, then*
$$[a, b] + [c, d] = [a', b'] + [c', d'].$$

Theorem 0.103.6 (Addition on $\mathbb{Q}$ Is Well-Defined)**.** *Addition determines a well-defined binary operation* $\mathbb{Q} \times \mathbb{Q} \to \mathbb{Q}$*.*

Theorem 0.103.7 (Addition on $\mathbb{Q}$ Is Associative)**.** *For all* $x, y, z \in \mathbb{Q}$*,*
$$(x + y) + z = x + (y + z).$$

Theorem 0.103.8 (Addition on $\mathbb{Q}$ Is Commutative)**.** *For all* $x, y \in \mathbb{Q}$*,*
$$x + y = y + x.$$

Theorem 0.103.9 (Zero Is a Left Additive Identity on $\mathbb{Q}$)**.** *For every* $x \in \mathbb{Q}$*,*
$$0_\mathbb{Q} + x = x.$$

Corollary 0.103.10 (Zero Is a Right Additive Identity on $\mathbb{Q}$)**.** *For every* $x \in \mathbb{Q}$*,*
$$x + 0_\mathbb{Q} = x.$$

Theorem 0.103.11 (Zero Is an Additive Identity on $\mathbb{Q}$)**.** *For every* $x \in \mathbb{Q}$*,*
$$0_\mathbb{Q} + x = x \qquad \textit{and} \qquad x + 0_\mathbb{Q} = x.$$

Negation on $\mathbb{Q}$-Classes

Definition 0.103.12 (Negation on $\mathbb{Q}$-Classes)**.** Define negation on representatives by

$$-[a, b] := [-a, b].$$

Lemma 0.103.13 (Negation Respects Fraction Equivalence)**.** *If* $[a, b] = [a', b']$*, then*

$$-[a, b] = -[a', b'].$$

Theorem 0.103.14 (Negation on $\mathbb{Q}$ Is Well-Defined)**.** *Negation determines a well-defined unary operation* $\mathbb{Q} \to \mathbb{Q}$.

Theorem 0.103.15 (Negation Gives a Left Additive Inverse on $\mathbb{Q}$)**.** *For every* $x \in \mathbb{Q}$*,*

$$(-x) + x = 0_{\mathbb{Q}}.$$

Corollary 0.103.16 (Negation Gives a Right Additive Inverse on $\mathbb{Q}$)**.** *For every* $x \in \mathbb{Q}$*,*

$$x + (-x) = 0_{\mathbb{Q}}.$$

Theorem 0.103.17 (Every Rational Has an Additive Inverse)**.** *For every* $x \in \mathbb{Q}$ *there exists* $y \in \mathbb{Q}$ *such that*

$$x + y = 0_{\mathbb{Q}} \qquad \textit{and} \qquad y + x = 0_{\mathbb{Q}}.$$

Subtraction on $\mathbb{Q}$

Definition 0.103.18 (Subtraction on $\mathbb{Q}$)**.** For $x, y \in \mathbb{Q}$, define

$$x - y := x + (-y).$$

Theorem 0.103.19 (Subtraction on $\mathbb{Q}$ Is Well-Defined)**.** *Subtraction determines a well-defined binary operation* $\mathbb{Q} \times \mathbb{Q} \to \mathbb{Q}$.

Multiplication on $\mathbb{Q}$-Classes

Definition 0.103.20 (Multiplication on $\mathbb{Q}$-Classes)**.** Define multiplication on representatives by

$$[a, b] \cdot [c, d] := [a \cdot c,\ b \cdot d].$$

Lemma 0.103.21 (Multiplication Lands in $\operatorname{Pre}\mathbb{Q}$)**.** *If* $b \neq 0_{\mathbb{Z}}$ *and* $d \neq 0_{\mathbb{Z}}$*, then*

$$b \cdot d \neq 0_{\mathbb{Z}}.$$

Lemma 0.103.22 (Multiplication Respects Fraction Equivalence on the Left)**.** *If* $[a, b] = [a', b']$*, then*

$$[a, b] \cdot [c, d] = [a', b'] \cdot [c, d].$$

Lemma 0.103.23 (Multiplication Respects Fraction Equivalence on the Right)**.** *If* $[c, d] = [c', d']$*, then*

$$[a, b] \cdot [c, d] = [a, b] \cdot [c', d'].$$

Lemma 0.103.24 (Multiplication Respects Fraction Equivalence)**.** *If* $[a, b] = [a', b']$ *and* $[c, d] = [c', d']$*, then*

$$[a, b] \cdot [c, d] = [a', b'] \cdot [c', d'].$$

Theorem 0.103.25 (Multiplication on $\mathbb{Q}$ Is Well-Defined)**.** *Multiplication determines a well-defined binary operation* $\mathbb{Q} \times \mathbb{Q} \to \mathbb{Q}$.

Theorem 0.103.26 (Multiplication on $\mathbb{Q}$ Is Associative)**.** *For all $x, y, z \in \mathbb{Q}$,*

$$(x \cdot y) \cdot z = x \cdot (y \cdot z).$$

Theorem 0.103.27 (Multiplication on $\mathbb{Q}$ Is Commutative)**.** *For all $x, y \in \mathbb{Q}$,*

$$x \cdot y = y \cdot x.$$

Theorem 0.103.28 (One Is a Left Multiplicative Identity on $\mathbb{Q}$)**.** *For every $x \in \mathbb{Q}$,*

$$1_{\mathbb{Q}} \cdot x = x.$$

Corollary 0.103.29 (One Is a Right Multiplicative Identity on $\mathbb{Q}$)**.** *For every $x \in \mathbb{Q}$,*

$$x \cdot 1_{\mathbb{Q}} = x.$$

Theorem 0.103.30 (One Is a Multiplicative Identity on $\mathbb{Q}$)**.** *For every $x \in \mathbb{Q}$,*

$$1_{\mathbb{Q}} \cdot x = x \qquad \textit{and} \qquad x \cdot 1_{\mathbb{Q}} = x.$$

Reciprocal on Nonzero $\mathbb{Q}$-Classes

Lemma 0.103.31 (Nonzero Predicate Is Well-Defined on $\mathbb{Q}$)**.** *For every prefraction $(a, b) \in \operatorname{Pre}\mathbb{Q}$,*

$$[a, b] \neq 0_{\mathbb{Q}} \iff a \neq 0_{\mathbb{Z}}.$$

Thus nonzeroness is independent of the representative.

Definition 0.103.32 (Reciprocal on Nonzero $\mathbb{Q}$-Classes)**.** For $[a, b] \neq 0_{\mathbb{Q}}$, define

$$[a, b]^{-1} := [b, a].$$

Lemma 0.103.33 (Reciprocal Lands in $\operatorname{Pre}\mathbb{Q}$)**.** *If $[a, b] \neq 0_{\mathbb{Q}}$, then $a \neq 0_{\mathbb{Z}}$, so (b, a) is a prefraction.*

Lemma 0.103.34 (Reciprocal Respects Fraction Equivalence)**.** *If $[a, b] = [a', b']$ and $[a, b] \neq 0_{\mathbb{Q}}$, then*

$$[a, b]^{-1} = [a', b']^{-1}.$$

Theorem 0.103.35 (Reciprocal on $\mathbb{Q}$ Is Well-Defined)**.** *Reciprocal determines a well-defined map*

$$\mathbb{Q} \setminus \{0_{\mathbb{Q}}\} \to \mathbb{Q} \setminus \{0_{\mathbb{Q}}\}.$$

Theorem 0.103.36 (Reciprocal Gives a Left Multiplicative Inverse on $\mathbb{Q}$)**.** *For every nonzero $x \in \mathbb{Q}$,*

$$x^{-1} \cdot x = 1_{\mathbb{Q}}.$$

Corollary 0.103.37 (Reciprocal Gives a Right Multiplicative Inverse on $\mathbb{Q}$)**.** *For every nonzero $x \in \mathbb{Q}$,*

$$x \cdot x^{-1} = 1_{\mathbb{Q}}.$$

Theorem 0.103.38 (Every Nonzero Rational Has a Multiplicative Inverse)**.** *For every $x \in \mathbb{Q}$ with $x \neq 0_{\mathbb{Q}}$, there exists $y \in \mathbb{Q}$ such that*

$$x \cdot y = 1_{\mathbb{Q}} \qquad \textit{and} \qquad y \cdot x = 1_{\mathbb{Q}}.$$

Division on $\mathbb{Q}$

Definition 0.103.39 (Division on $\mathbb{Q}$)**.** For $x, y \in \mathbb{Q}$ with $y \neq 0_{\mathbb{Q}}$, define

$$x/y := x \cdot y^{-1}.$$

Theorem 0.103.40 (Division on $\mathbb{Q}$ Is Well-Defined)**.** *Division determines a well-defined map*

$$\mathbb{Q} \times (\mathbb{Q} \setminus \{0_{\mathbb{Q}}\}) \to \mathbb{Q}.$$

Distributivity on $\mathbb{Q}$-Classes

Theorem 0.103.41 (Left Distributivity on $\mathbb{Q}$). *For all $x, y, z \in \mathbb{Q}$,*

$$x \cdot (y + z) = x \cdot y + x \cdot z.$$

Corollary 0.103.42 (Right Distributivity on $\mathbb{Q}$). *For all $x, y, z \in \mathbb{Q}$,*

$$(y + z) \cdot x = y \cdot x + z \cdot x.$$

Theorem 0.103.43 (Multiplication Distributes over Addition on $\mathbb{Q}$). *Multiplication distributes over addition on both sides in $\mathbb{Q}$.*

Class-Level Field Summary

Theorem 0.103.44 ($\mathbb{Q}$ Is an Additive Abelian Group). *The structure $(\mathbb{Q}, +, 0_\mathbb{Q})$ is an abelian group.*

Theorem 0.103.45 (Nonzero $\mathbb{Q}$ Is a Multiplicative Abelian Group). *The structure $(\mathbb{Q} \setminus \{0_\mathbb{Q}\}, \cdot, 1_\mathbb{Q})$ is an abelian group.*

Theorem 0.103.46 ($\mathbb{Q}$ Is a Field). *The structure $(\mathbb{Q}, +, \cdot, 0_\mathbb{Q}, 1_\mathbb{Q})$ is a field.*

0.104 Field Structure of Q

Derived Field Laws on $\mathbb{Q}$

Corollary 0.104.1 (Zero Product on $\mathbb{Q}$). *For all $x, y \in \mathbb{Q}$,*

$$x \cdot y = 0_\mathbb{Q} \iff x = 0_\mathbb{Q} \text{ or } y = 0_\mathbb{Q}.$$

Corollary 0.104.2 (Double Negation on $\mathbb{Q}$). *For every $x \in \mathbb{Q}$,*

$$-(-x) = x.$$

Corollary 0.104.3 (Negation of a Product on $\mathbb{Q}$). *For all $x, y \in \mathbb{Q}$,*

$$(-x) \cdot y = -(x \cdot y) \qquad \text{and} \qquad x \cdot (-y) = -(x \cdot y).$$

Corollary 0.104.4 (Product of Negatives on $\mathbb{Q}$). *For all $x, y \in \mathbb{Q}$,*

$$(-x) \cdot (-y) = x \cdot y.$$

Corollary 0.104.5 (Additive Cancellation on $\mathbb{Q}$). *For all $x, y, z \in \mathbb{Q}$,*

$$x + z = y + z \Longrightarrow x = y.$$

Corollary 0.104.6 (Multiplicative Cancellation on $\mathbb{Q}$). *For all $x, y, z \in \mathbb{Q}$,*

$$z \neq 0_\mathbb{Q} \wedge x \cdot z = y \cdot z \Longrightarrow x = y.$$

Corollary 0.104.7 (Uniqueness of Additive Inverses on $\mathbb{Q}$). *Every additive inverse in $\mathbb{Q}$ is unique.*

Corollary 0.104.8 (Uniqueness of Multiplicative Inverses on $\mathbb{Q}$). *Every multiplicative inverse of a nonzero rational number is unique.*

Corollary 0.104.9 (The Inverse of One Is One on $\mathbb{Q}$). *In $\mathbb{Q}$,*

$$1_\mathbb{Q}^{-1} = 1_\mathbb{Q}.$$

Corollary 0.104.10 (Inverse of a Product on $\mathbb{Q}$). *For all nonzero $x, y \in \mathbb{Q}$,*

$$(x \cdot y)^{-1} = x^{-1} \cdot y^{-1}.$$

Corollary 0.104.11 (Inverse of an Inverse on $\mathbb{Q}$). *For every nonzero $x \in \mathbb{Q}$,*

$$(x^{-1})^{-1} = x.$$

Integer Powers on $\mathbb{Q}$

Definition 0.104.12 (Integer Power on $\mathbb{Q}$)**.** Let $x \in \mathbb{Q}$. For $n \in \mathbb{W}$, define x^n by the usual recursive power operation. If $x \neq 0_{\mathbb{Q}}$ and $n \in \mathbb{Z}$ is negative, define

$$x^n := (x^{-1})^{-n}.$$

Corollary 0.104.13 (Exponent Addition Law on $\mathbb{Q}$)**.** *For every nonzero $x \in \mathbb{Q}$ and all $m, n \in \mathbb{Z}$,*

$$x^{m+n} = x^m \cdot x^n.$$

Corollary 0.104.14 (Power of a Power on $\mathbb{Q}$)**.** *For every nonzero $x \in \mathbb{Q}$ and all $m, n \in \mathbb{Z}$,*

$$(x^m)^n = x^{m \cdot n}.$$

Corollary 0.104.15 (Power of a Product on $\mathbb{Q}$)**.** *For all nonzero $x, y \in \mathbb{Q}$ and every $n \in \mathbb{Z}$,*

$$(x \cdot y)^n = x^n \cdot y^n.$$

0.105 Order on $\mathbb{Q}$

Positivity on $\mathbb{Q}$

Definition 0.105.1 (Positive Rational)**.** A rational number $[a, b] \in \mathbb{Q}$ is *positive* if

$$a \cdot b > 0_{\mathbb{Z}}.$$

Lemma 0.105.2 (Positivity Respects Fraction Equivalence on $\mathbb{Q}$)**.** *If $[a, b] = [c, d]$, then*

$$a \cdot b > 0_{\mathbb{Z}} \iff c \cdot d > 0_{\mathbb{Z}}.$$

Theorem 0.105.3 (Positivity on $\mathbb{Q}$ Is Well-Defined)**.** *The predicate "x is positive" is well-defined on equivalence classes in $\mathbb{Q}$.*

Definition 0.105.4 (Strict Order on $\mathbb{Q}$)**.** For $x, y \in \mathbb{Q}$, define

$$x < y \quad :\iff \quad y - x \text{ is positive.}$$

Definition 0.105.5 (Order on $\mathbb{Q}$)**.** For $x, y \in \mathbb{Q}$, define

$$x \leq y \quad :\iff \quad x < y \text{ or } x = y.$$

Proposition 0.105.6 (Product of Positives Is Positive on $\mathbb{Q}$)**.** *For all $x, y \in \mathbb{Q}$,*

$$0_{\mathbb{Q}} < x \wedge 0_{\mathbb{Q}} < y \implies 0_{\mathbb{Q}} < x \cdot y.$$

Trichotomy on $\mathbb{Q}$

Theorem 0.105.7 (Sign Trichotomy on $\mathbb{Q}$)**.** *For every $x \in \mathbb{Q}$, exactly one of*

$$x < 0_{\mathbb{Q}}, \qquad x = 0_{\mathbb{Q}}, \qquad 0_{\mathbb{Q}} < x$$

holds.

Theorem 0.105.8 (Trichotomy on $\mathbb{Q}$)**.** *For all $x, y \in \mathbb{Q}$, exactly one of*

$$x < y, \qquad x = y, \qquad y < x$$

holds.

Total Order on $\mathbb{Q}$

Theorem 0.105.9 (Strict Order on $\mathbb{Q}$ Is Irreflexive). *For every $x \in \mathbb{Q}$, not $x < x$.*

Theorem 0.105.10 (Strict Order on $\mathbb{Q}$ Is Asymmetric). *For all $x, y \in \mathbb{Q}$,*

$$x < y \implies \neg(y < x).$$

Theorem 0.105.11 (Strict Order on $\mathbb{Q}$ Is Transitive). *For all $x, y, z \in \mathbb{Q}$,*

$$x < y \wedge y < z \implies x < z.$$

Theorem 0.105.12 (Order on $\mathbb{Q}$ Is Reflexive). *For every $x \in \mathbb{Q}$, $x \leq x$.*

Theorem 0.105.13 (Order on $\mathbb{Q}$ Is Antisymmetric). *For all $x, y \in \mathbb{Q}$,*

$$x \leq y \wedge y \leq x \implies x = y.$$

Theorem 0.105.14 (Order on $\mathbb{Q}$ Is Transitive). *For all $x, y, z \in \mathbb{Q}$,*

$$x \leq y \wedge y \leq z \implies x \leq z.$$

Theorem 0.105.15 (Order on $\mathbb{Q}$ Is Total). *For all $x, y \in \mathbb{Q}$,*

$$x \leq y \text{ or } y \leq x.$$

Theorem 0.105.16 ($\mathbb{Q}$ Is a Totally Ordered Set). *The ordered pair $(\mathbb{Q}, \leq)$ is a totally ordered set.*

Addition Order Compatibility on $\mathbb{Q}$

Theorem 0.105.17 (Left Addition Preserves Strict Order on $\mathbb{Q}$). *For all $x, y, z \in \mathbb{Q}$,*

$$x < y \implies z + x < z + y.$$

Corollary 0.105.18 (Right Addition Preserves Strict Order on $\mathbb{Q}$). *For all $x, y, z \in \mathbb{Q}$,*

$$x < y \implies x + z < y + z.$$

Theorem 0.105.19 (Addition Preserves Strict Order on $\mathbb{Q}$). *Addition preserves strict order on both sides in $\mathbb{Q}$.*

Theorem 0.105.20 (Left Addition Preserves Order on $\mathbb{Q}$). *For all $x, y, z \in \mathbb{Q}$,*

$$x \leq y \implies z + x \leq z + y.$$

Corollary 0.105.21 (Right Addition Preserves Order on $\mathbb{Q}$). *For all $x, y, z \in \mathbb{Q}$,*

$$x \leq y \implies x + z \leq y + z.$$

Theorem 0.105.22 (Addition Preserves Order on $\mathbb{Q}$). *Addition preserves non-strict order on both sides in $\mathbb{Q}$.*

Corollary 0.105.23 (Addition Reflects Order on $\mathbb{Q}$). *Addition by a fixed rational reflects both strict and non-strict order.*

Multiplication Order Compatibility on $\mathbb{Q}$

Theorem 0.105.24 (Left Multiplication by Positives Preserves Strict Order on $\mathbb{Q}$)**.** *For all $x, y, z \in \mathbb{Q}$,*

$$0_{\mathbb{Q}} < z \wedge x < y \Longrightarrow z \cdot x < z \cdot y.$$

Corollary 0.105.25 (Right Multiplication by Positives Preserves Strict Order on $\mathbb{Q}$)**.** *For all $x, y, z \in \mathbb{Q}$,*

$$0_{\mathbb{Q}} < z \wedge x < y \Longrightarrow x \cdot z < y \cdot z.$$

Theorem 0.105.26 (Multiplication by Positives Preserves Strict Order on $\mathbb{Q}$)**.** *Multiplication by a positive rational preserves strict order on both sides.*

Theorem 0.105.27 (Left Multiplication by Positives Preserves Order on $\mathbb{Q}$)**.** *For all $x, y, z \in \mathbb{Q}$,*

$$0_{\mathbb{Q}} < z \wedge x \leq y \Longrightarrow z \cdot x \leq z \cdot y.$$

Corollary 0.105.28 (Right Multiplication by Positives Preserves Order on $\mathbb{Q}$)**.** *For all $x, y, z \in \mathbb{Q}$,*

$$0_{\mathbb{Q}} < z \wedge x \leq y \Longrightarrow x \cdot z \leq y \cdot z.$$

Theorem 0.105.29 (Multiplication by Positives Preserves Order on $\mathbb{Q}$)**.** *Multiplication by a positive rational preserves non-strict order on both sides.*

Theorem 0.105.30 (Multiplication by Negatives Reverses Strict Order on $\mathbb{Q}$)**.** *For all $x, y, z \in \mathbb{Q}$,*

$$z < 0_{\mathbb{Q}} \wedge x < y \Longrightarrow z \cdot y < z \cdot x.$$

Corollary 0.105.31 (Multiplication by Negatives Reverses Order on $\mathbb{Q}$)**.** *For all $x, y, z \in \mathbb{Q}$,*

$$z < 0_{\mathbb{Q}} \wedge x \leq y \Longrightarrow z \cdot y \leq z \cdot x.$$

Reciprocal Order Laws on $\mathbb{Q}$

Theorem 0.105.32 (Sign of a Reciprocal on $\mathbb{Q}$)**.** *For every nonzero $x \in \mathbb{Q}$,*

$$0_{\mathbb{Q}} < x \Longrightarrow 0_{\mathbb{Q}} < x^{-1}, \qquad x < 0_{\mathbb{Q}} \Longrightarrow x^{-1} < 0_{\mathbb{Q}}.$$

Theorem 0.105.33 (Reciprocal Reverses Order on Positives in $\mathbb{Q}$)**.** *For all $x, y \in \mathbb{Q}$,*

$$0_{\mathbb{Q}} < x < y \Longrightarrow 0_{\mathbb{Q}} < y^{-1} < x^{-1}.$$

Absolute Value on $\mathbb{Q}$

Definition 0.105.34 (Absolute Value on $\mathbb{Q}$)**.** For $x \in \mathbb{Q}$, define

$$|x| := \begin{cases} x, & 0_{\mathbb{Q}} \leq x, \\ -x, & x < 0_{\mathbb{Q}}. \end{cases}$$

Theorem 0.105.35 (Absolute Value Is Nonnegative on $\mathbb{Q}$)**.** *For every $x \in \mathbb{Q}$,*

$$0_{\mathbb{Q}} \leq |x|.$$

Theorem 0.105.36 (Absolute Value Vanishes Only at Zero on $\mathbb{Q}$)**.** *For every $x \in \mathbb{Q}$,*

$$|x| = 0_{\mathbb{Q}} \iff x = 0_{\mathbb{Q}}.$$

Theorem 0.105.37 (Absolute Value Is Multiplicative on $\mathbb{Q}$)**.** *For all $x, y \in \mathbb{Q}$,*

$$|x \cdot y| = |x| \cdot |y|.$$

Theorem 0.105.38 (Triangle Inequality on $\mathbb{Q}$)**.** *For all $x, y \in \mathbb{Q}$,*

$$|x + y| \leq |x| + |y|.$$

Theorem 0.105.39 (Absolute Value of a Reciprocal on $\mathbb{Q}$)**.** *For every nonzero $x \in \mathbb{Q}$,*

$$|x^{-1}| = |x|^{-1}.$$

0.106 Ordered Field Structure of Q

Ordered Field Instantiation

Theorem 0.106.1 (ℚ Is an Ordered Field). *The structure*

$$(\mathbb{Q}, +, \cdot, 0_{\mathbb{Q}}, 1_{\mathbb{Q}}, \leq)$$

is an ordered field.

Corollary 0.106.2 (Ordered Field Laws Hold on ℚ). *Every theorem that depends only on the ordered-field axioms applies to* $\mathbb{Q}$ *with its constructed operations and order.*

Standard Ordered-Field Consequences

Corollary 0.106.3 (Squares Are Nonnegative on ℚ). *For every* $x \in \mathbb{Q}$*,*

$$0_{\mathbb{Q}} \leq x^2.$$

Corollary 0.106.4 (One Is Positive in ℚ). *In* $\mathbb{Q}$*,*

$$0_{\mathbb{Q}} < 1_{\mathbb{Q}}.$$

Corollary 0.106.5 (Natural Numbers Are Positive in ℚ). *Every natural number embedded in* $\mathbb{Q}$ *is positive.*

Corollary 0.106.6 (Positive Rationals Have Positive Inverses). *For every* $x \in \mathbb{Q}$*,*

$$0_{\mathbb{Q}} < x \Longrightarrow 0_{\mathbb{Q}} < x^{-1}.$$

Corollary 0.106.7 (Fraction Comparison in ℚ). *For rational classes with positive denominator product,*

$$[a, b] < [c, d] \quad \Longleftrightarrow \quad a \cdot d < c \cdot b.$$

0.107 Arbitrary Closeness in ℚ

Density and the Archimedean Property

Definition 0.107.1 (Two in ℚ). Define

$$2_{\mathbb{Q}} := 1_{\mathbb{Q}} + 1_{\mathbb{Q}}.$$

Lemma 0.107.2 (Two Is Nonzero in ℚ). *In* $\mathbb{Q}$*,*

$$2_{\mathbb{Q}} \neq 0_{\mathbb{Q}}.$$

Lemma 0.107.3 (Midpoint Lies Strictly Between Rationals). *For all* $x, y \in \mathbb{Q}$*,*

$$x < y \Longrightarrow x < (x + y) \cdot 2_{\mathbb{Q}}^{-1} < y.$$

Theorem 0.107.4 (Density of ℚ in Itself). *For all* $x, y \in \mathbb{Q}$*,*

$$x < y \Longrightarrow \exists r \in \mathbb{Q}\ (x < r < y).$$

Theorem 0.107.5 (Archimedean Property of ℚ). *For all* $x, y \in \mathbb{Q}$*,*

$$0_{\mathbb{Q}} < x \Longrightarrow \exists n \in \mathbb{N}\ (y < n \cdot x).$$

From Discreteness to Arbitrary Closeness

Definition 0.107.6 (Arbitrarily Close Distinct Points)**.** A number system has arbitrarily close distinct points if for every point x and every positive tolerance ε, there is a point $y \neq x$ such that

$$|x - y| < \varepsilon.$$

Theorem 0.107.7 ($\mathbb{Q}$ Has Arbitrarily Close Distinct Points)**.** *For every $x \in \mathbb{Q}$ and every $\varepsilon \in \mathbb{Q}$,*

$$0_{\mathbb{Q}} < \varepsilon \Longrightarrow \exists y \in \mathbb{Q}\ (y \neq x \text{ and } |x - y| < \varepsilon).$$

Corollary 0.107.8 ($\mathbb{Q}$ Has No Adjacent Points)**.** *There do not exist rational numbers $x < y$ with no rational number strictly between them.*

Proposition 0.107.9 ($\mathbb{Q}$ Has No Least Positive Rational)**.** *For every $x \in \mathbb{Q}$,*

$$0_{\mathbb{Q}} < x \Longrightarrow \exists y \in \mathbb{Q}\ (0_{\mathbb{Q}} < y < x).$$

Corollary 0.107.10 (Reciprocals Become Arbitrarily Small in $\mathbb{Q}$)**.** *For every $\varepsilon \in \mathbb{Q}$,*

$$0_{\mathbb{Q}} < \varepsilon \Longrightarrow \exists n \in \mathbb{N}\ \left(0_{\mathbb{Q}} < n^{-1} < \varepsilon\right).$$

Limit-Like Behavior Before Limits

Proposition 0.107.11 (One-Sided Rational Approximation)**.** *For every $x \in \mathbb{Q}$ and every $\varepsilon \in \mathbb{Q}$,*

$$0_{\mathbb{Q}} < \varepsilon \Longrightarrow \exists y, z \in \mathbb{Q}\ (x - \varepsilon < y < x < z < x + \varepsilon).$$

0.108 Rational Gaps

No Rational Square Root of Two

Theorem 0.108.1 (No Rational Square Root of Two)**.** *There is no rational number $q \in \mathbb{Q}$ such that*

$$q \cdot q = 2_{\mathbb{Q}}.$$

The Square-Two Gap

Definition 0.108.2 (The Gap Set in $\mathbb{Q}$)**.** Define

$$S := \{q \in \mathbb{Q} : 0_{\mathbb{Q}} < q \text{ and } q \cdot q < 2_{\mathbb{Q}}\}.$$

Proposition 0.108.3 (The Gap Set Is Bounded Above in $\mathbb{Q}$)**.** *The set S is nonempty and bounded above in $\mathbb{Q}$.*

Theorem 0.108.4 (The Square-Two Gap Has No Rational Supremum)**.** *The set S has no least upper bound in $\mathbb{Q}$.*

Corollary 0.108.5 ($\mathbb{Q}$ Is Order-Incomplete)**.** *The ordered field $\mathbb{Q}$ is not order-complete: there exists a nonempty bounded-above subset of $\mathbb{Q}$ with no supremum in $\mathbb{Q}$.*

Cauchy Sequences in $\mathbb{Q}$

Definition 0.108.6 (Cauchy Sequence in $\mathbb{Q}$)**.** A sequence (x_n) in $\mathbb{Q}$ is Cauchy if

$$\forall \varepsilon \in \mathbb{Q}\ (0_{\mathbb{Q}} < \varepsilon \Longrightarrow \exists N \in \mathbb{N}\, \forall m, n \geq N\ |x_m - x_n| < \varepsilon).$$

Definition 0.108.7 (Null Sequence in $\mathbb{Q}$)**.** A sequence (x_n) in $\mathbb{Q}$ is null if

$$\forall \varepsilon \in \mathbb{Q}\ (0_{\mathbb{Q}} < \varepsilon \Longrightarrow \exists N \in \mathbb{N}\, \forall n \geq N\ |x_n| < \varepsilon).$$

Lemma 0.108.8 (Rational Limits Are Unique)**.** *A convergent sequence in $\mathbb{Q}$ has at most one rational limit.*

0.109 Embedding $\mathbb{Z}$ into $\mathbb{Q}$

The Integer Embedding

Definition 0.109.1 (Embedding of $\mathbb{Z}$ into $\mathbb{Q}$)**.** Define

$$\iota : \mathbb{Z} \to \mathbb{Q}, \qquad \iota(n) := [n, 1_{\mathbb{Z}}].$$

Theorem 0.109.2 (Embedding Preserves Zero)**.** *The embedding sends integer zero to rational zero:*

$$\iota(0_{\mathbb{Z}}) = 0_{\mathbb{Q}}.$$

Theorem 0.109.3 (Embedding Preserves One)**.** *The embedding sends integer one to rational one:*

$$\iota(1_{\mathbb{Z}}) = 1_{\mathbb{Q}}.$$

Theorem 0.109.4 (Embedding $\mathbb{Z}$ into $\mathbb{Q}$ Is Injective)**.** *For all $m, n \in \mathbb{Z}$,*

$$\iota(m) = \iota(n) \implies m = n.$$

Preservation of Algebra and Order

Theorem 0.109.5 (Embedding Preserves Addition)**.** *For all $m, n \in \mathbb{Z}$,*

$$\iota(m + n) = \iota(m) + \iota(n).$$

Theorem 0.109.6 (Embedding Preserves Multiplication)**.** *For all $m, n \in \mathbb{Z}$,*

$$\iota(m \cdot n) = \iota(m) \cdot \iota(n).$$

Theorem 0.109.7 (Embedding Preserves Order)**.** *For all $m, n \in \mathbb{Z}$,*

$$m \leq n \iff \iota(m) \leq \iota(n).$$

Theorem 0.109.8 ($\mathbb{Z}$ Embeds into $\mathbb{Q}$)**.** *The map $\iota : \mathbb{Z} \to \mathbb{Q}$ is an injective order-preserving ring homomorphism.*

Field of Fractions

Theorem 0.109.9 ($\mathbb{Q}$ Is the Field of Fractions of $\mathbb{Z}$)**.** *For every $x \in \mathbb{Q}$, there exist $m, n \in \mathbb{Z}$ with $n \neq 0_{\mathbb{Z}}$ such that*

$$x = \iota(m) \cdot \iota(n)^{-1}.$$

Corollary 0.109.10 (The Embedding of $\mathbb{Z}$ into $\mathbb{Q}$ Is Not Surjective)**.** *The image of $\iota : \mathbb{Z} \to \mathbb{Q}$ is a proper subset of $\mathbb{Q}$.*

0.110 Transition to $\mathbb{R}$

Why the Real Numbers Are Introduced

0.111 Canonical Rational Number Statement Plan

Proofs

Real Numbers ($\mathbb{R}$)

0.112 Construction of $\mathbb{R}$

Dedekind Cuts

Definition 0.112.1 (Dedekind Cut)**.** A *Dedekind cut* is a set $\alpha \subseteq \mathbb{Q}$ satisfying:

$$\alpha \neq \emptyset, \qquad \alpha \neq \mathbb{Q},$$

$$p \in \alpha \wedge q < p \Longrightarrow q \in \alpha,$$

and

$$p \in \alpha \Longrightarrow \exists r \in \alpha \, (p < r).$$

Definition 0.112.2 (The Real Numbers)**.** The real numbers are the set

$$\mathbb{R} := \{\alpha : \alpha \text{ is a Dedekind cut in } \mathbb{Q}\}.$$

Rational Cuts

Definition 0.112.3 (Rational Cut)**.** For $q \in \mathbb{Q}$, define the rational cut

$$q^* := \{r \in \mathbb{Q} : r < q\}.$$

Lemma 0.112.4 (Rational Cuts Are Cuts)**.** *For every $q \in \mathbb{Q}$, the set q^* is a Dedekind cut.*

Zero and One in $\mathbb{R}$

Definition 0.112.5 (Zero in $\mathbb{R}$)**.** Define

$$0_{\mathbb{R}} := (0_{\mathbb{Q}})^* = \{r \in \mathbb{Q} : r < 0_{\mathbb{Q}}\}.$$

Definition 0.112.6 (One in $\mathbb{R}$)**.** Define

$$1_{\mathbb{R}} := (1_{\mathbb{Q}})^* = \{r \in \mathbb{Q} : r < 1_{\mathbb{Q}}\}.$$

Theorem 0.112.7 (Zero and One Are Reals)**.** *The cuts $0_{\mathbb{R}}$ and $1_{\mathbb{R}}$ are elements of $\mathbb{R}$.*

Theorem 0.112.8 (Zero Is Not One in $\mathbb{R}$)**.** *In $\mathbb{R}$,*

$$0_{\mathbb{R}} \neq 1_{\mathbb{R}}.$$

0.113 Order on $\mathbb{R}$

Order by Inclusion

Definition 0.113.1 (Order on $\mathbb{R}$)**.** For cuts $\alpha, \beta \in \mathbb{R}$, define

$$\alpha \leq \beta \quad :\Longleftrightarrow \quad \alpha \subseteq \beta.$$

Definition 0.113.2 (Strict Order on $\mathbb{R}$)**.** For cuts $\alpha, \beta \in \mathbb{R}$, define

$$\alpha < \beta \quad :\Longleftrightarrow \quad \alpha \subsetneq \beta.$$

Theorem 0.113.3 (Order on $\mathbb{R}$ Is Reflexive)**.** *For every $\alpha \in \mathbb{R}$, $\alpha \leq \alpha$.*

Theorem 0.113.4 (Order on $\mathbb{R}$ Is Antisymmetric)**.** *For all $\alpha, \beta \in \mathbb{R}$,*

$$\alpha \leq \beta \wedge \beta \leq \alpha \Longrightarrow \alpha = \beta.$$

Theorem 0.113.5 (Order on $\mathbb{R}$ Is Transitive)**.** *For all $\alpha, \beta, \gamma \in \mathbb{R}$,*

$$\alpha \leq \beta \wedge \beta \leq \gamma \Longrightarrow \alpha \leq \gamma.$$

Theorem 0.113.6 (Comparability of Cuts)**.** *For all cuts $\alpha, \beta \in \mathbb{R}$,*

$$\alpha \subseteq \beta \vee \beta \subseteq \alpha.$$

Theorem 0.113.7 (Order on $\mathbb{R}$ Is Total)**.** *For all $\alpha, \beta \in \mathbb{R}$,*

$$\alpha \leq \beta \vee \beta \leq \alpha.$$

Theorem 0.113.8 (Trichotomy on $\mathbb{R}$)**.** *For all $\alpha, \beta \in \mathbb{R}$, exactly one of*

$$\alpha < \beta, \qquad \alpha = \beta, \qquad \beta < \alpha$$

holds.

Theorem 0.113.9 ($\mathbb{R}$ Is a Totally Ordered Set)**.** *The ordered pair $(\mathbb{R}, \leq)$ is a total order.*

0.114 Completeness of $\mathbb{R}$

Definition 0.114.1 (Upper Bound in $\mathbb{R}$)**.** Let $S \subseteq \mathbb{R}$. A cut $\gamma \in \mathbb{R}$ is an upper bound of S if

$$\forall \alpha \in S \, (\alpha \leq \gamma).$$

The set S is bounded above if it has an upper bound in $\mathbb{R}$.

Definition 0.114.2 (Supremum in $\mathbb{R}$)**.** For $S \subseteq \mathbb{R}$, a cut σ is the supremum of S if σ is an upper bound of S and every upper bound γ of S satisfies $\sigma \leq \gamma$.

Definition 0.114.3 (Least-Upper-Bound Property on $\mathbb{R}$)**.** The ordered set $\mathbb{R}$ has the *least-upper-bound property* if every nonempty subset of $\mathbb{R}$ that is bounded above has a supremum in $\mathbb{R}$.

Lemma 0.114.4 (Union of a Bounded Nonempty Family of Cuts Is a Cut)**.** *If $S \subseteq \mathbb{R}$ is nonempty and bounded above, then*

$$\bigcup_{\alpha \in S} \alpha$$

is a Dedekind cut.

Theorem 0.114.5 (Least-Upper-Bound Property)**.** *Every nonempty subset $S \subseteq \mathbb{R}$ that is bounded above has a least upper bound, namely*

$$\sup S = \bigcup_{\alpha \in S} \alpha.$$

Corollary 0.114.6 (Greatest-Lower-Bound Property)**.** *Every nonempty subset $S \subseteq \mathbb{R}$ that is bounded below has a greatest lower bound.*

0.115 Addition on $\mathbb{R}$

Definition 0.115.1 (Addition of Cuts). For cuts $\alpha, \beta \in \mathbb{R}$, define

$$\alpha + \beta := \{p + q : p \in \alpha,\ q \in \beta\}.$$

Lemma 0.115.2 (Sum of Cuts Is a Cut). *If $\alpha, \beta \in \mathbb{R}$, then $\alpha + \beta$ is a Dedekind cut.*

Theorem 0.115.3 (Addition on $\mathbb{R}$ Is Associative). *For all $\alpha, \beta, \gamma \in \mathbb{R}$,*

$$(\alpha + \beta) + \gamma = \alpha + (\beta + \gamma).$$

Theorem 0.115.4 (Addition on $\mathbb{R}$ Is Commutative). *For all $\alpha, \beta \in \mathbb{R}$,*

$$\alpha + \beta = \beta + \alpha.$$

Theorem 0.115.5 (Zero Is an Additive Identity on $\mathbb{R}$). *For every $\alpha \in \mathbb{R}$,*

$$\alpha + 0_{\mathbb{R}} = \alpha.$$

Definition 0.115.6 (Negation of a Cut). For $\alpha \in \mathbb{R}$, define

$$-\alpha := \{p \in \mathbb{Q} : \exists r \in \mathbb{Q}\,(r > 0 \wedge -p - r \notin \alpha)\}.$$

Lemma 0.115.7 (Negation of a Cut Is a Cut). *If $\alpha \in \mathbb{R}$, then $-\alpha$ is a Dedekind cut.*

Theorem 0.115.8 (Negation Is an Additive Inverse on $\mathbb{R}$). *For every $\alpha \in \mathbb{R}$,*

$$\alpha + (-\alpha) = 0_{\mathbb{R}}.$$

Theorem 0.115.9 ($\mathbb{R}$ Is an Additive Abelian Group). *The structure $(\mathbb{R}, +, 0_{\mathbb{R}})$ is an abelian group.*

Definition 0.115.10 (Subtraction on $\mathbb{R}$). For cuts $\alpha, \beta \in \mathbb{R}$, define

$$\alpha - \beta := \alpha + (-\beta).$$

0.116 Multiplication on $\mathbb{R}$

Definition 0.116.1 (Nonnegative Cut). A cut $\alpha \in \mathbb{R}$ is nonnegative if $0_{\mathbb{R}} \leq \alpha$ and positive if $0_{\mathbb{R}} < \alpha$.

Definition 0.116.2 (Product of Nonnegative Cuts). For nonnegative cuts $\alpha, \beta \in \mathbb{R}$, define

$$\alpha \cdot \beta := \{r \in \mathbb{Q} : r < 0_{\mathbb{Q}}\} \cup \{p \cdot q : p \in \alpha,\ q \in \beta,\ p \geq 0_{\mathbb{Q}},\ q \geq 0_{\mathbb{Q}}\}.$$

Lemma 0.116.3 (Product of Nonnegative Cuts Is a Cut). *If $\alpha, \beta \geq 0_{\mathbb{R}}$, then $\alpha \cdot \beta$ is a Dedekind cut.*

Definition 0.116.4 (Absolute Value on $\mathbb{R}$). For $\alpha \in \mathbb{R}$, define

$$|\alpha| := \begin{cases} \alpha, & 0_{\mathbb{R}} \leq \alpha, \\ -\alpha, & \alpha < 0_{\mathbb{R}}. \end{cases}$$

Theorem 0.116.5 (Triangle Inequality). *For all $\alpha, \beta \in \mathbb{R}$,*

$$|\alpha + \beta| \leq |\alpha| + |\beta|.$$

Go to proof.

Definition 0.116.6 (Multiplication of Cuts). For cuts $\alpha, \beta \in \mathbb{R}$, define $\alpha \cdot \beta$ by sign cases: use $|\alpha| \cdot |\beta|$ when α and β have the same sign, use $-(|\alpha| \cdot |\beta|)$ when they have opposite signs, and use $0_{\mathbb{R}}$ when either factor is $0_{\mathbb{R}}$.

Lemma 0.116.7 (Product of Cuts Is a Cut). *If $\alpha, \beta \in \mathbb{R}$, then $\alpha \cdot \beta \in \mathbb{R}$.*

Theorem 0.116.8 (Multiplication on $\mathbb{R}$ Is Associative). *For all $\alpha, \beta, \gamma \in \mathbb{R}$,*

$$(\alpha \cdot \beta) \cdot \gamma = \alpha \cdot (\beta \cdot \gamma).$$

Theorem 0.116.9 (Multiplication on $\mathbb{R}$ Is Commutative). *For all $\alpha, \beta \in \mathbb{R}$,*

$$\alpha \cdot \beta = \beta \cdot \alpha.$$

Theorem 0.116.10 (One Is a Multiplicative Identity on $\mathbb{R}$). *For every $\alpha \in \mathbb{R}$,*

$$\alpha \cdot 1_{\mathbb{R}} = \alpha.$$

Definition 0.116.11 (Reciprocal of a Positive Cut). For $\alpha > 0_{\mathbb{R}}$, define

$$\alpha^{-1} := \{r \in \mathbb{Q} : r \leq 0_{\mathbb{Q}}\} \cup \{r \in \mathbb{Q} : r > 0_{\mathbb{Q}} \wedge \exists s \in \mathbb{Q}\, (s > r \wedge s^{-1} \notin \alpha)\}.$$

Definition 0.116.12 (Reciprocal of a Nonzero Cut). For $\alpha \neq 0_{\mathbb{R}}$, define α^{-1} to be $|\alpha|^{-1}$ if $\alpha > 0_{\mathbb{R}}$ and $-(|\alpha|^{-1})$ if $\alpha < 0_{\mathbb{R}}$.

Lemma 0.116.13 (Reciprocal of a Nonzero Cut Is a Cut). *If $\alpha \neq 0_{\mathbb{R}}$, then $\alpha^{-1} \in \mathbb{R}$.*

Theorem 0.116.14 (Reciprocal Is a Multiplicative Inverse on $\mathbb{R}$). *If $\alpha \neq 0_{\mathbb{R}}$, then*

$$\alpha \cdot \alpha^{-1} = 1_{\mathbb{R}}.$$

Definition 0.116.15 (Division on $\mathbb{R}$). For $\beta \neq 0_{\mathbb{R}}$, define

$$\alpha/\beta := \alpha \cdot \beta^{-1}.$$

Theorem 0.116.16 (Distributivity on $\mathbb{R}$). *For all $\alpha, \beta, \gamma \in \mathbb{R}$,*

$$\alpha \cdot (\beta + \gamma) = \alpha \cdot \beta + \alpha \cdot \gamma.$$

0.117 Field and Ordered-Field Structure of $\mathbb{R}$

Theorem 0.117.1 (Nonzero $\mathbb{R}$ Is a Multiplicative Abelian Group). *The structure $(\mathbb{R} \setminus \{0_{\mathbb{R}}\}, \cdot, 1_{\mathbb{R}})$ is an abelian group.*

Theorem 0.117.2 ($\mathbb{R}$ Is a Field). *The structure $(\mathbb{R}, +, \cdot, 0_{\mathbb{R}}, 1_{\mathbb{R}})$ is a field.*

Corollary 0.117.3 (Field Laws Hold on $\mathbb{R}$). *Every theorem proved generically for fields holds in $\mathbb{R}$, including zero product, double negation, sign rules, cancellation, uniqueness of inverses, inverse laws, and exponent laws.*

Order Compatibility

Theorem 0.117.4 (Addition Preserves Order on $\mathbb{R}$). *For all $\alpha, \beta, \gamma \in \mathbb{R}$,*

$$\alpha < \beta \Longrightarrow \alpha + \gamma < \beta + \gamma,$$

and the corresponding non-strict statement also holds.

Theorem 0.117.5 (Multiplication by Positives Preserves Order on $\mathbb{R}$). *For all $\alpha, \beta, \gamma \in \mathbb{R}$,*

$$0_{\mathbb{R}} < \gamma \wedge \alpha < \beta \Longrightarrow \gamma \cdot \alpha < \gamma \cdot \beta.$$

Theorem 0.117.6 ($\mathbb{R}$ Is an Ordered Field). *The structure $(\mathbb{R}, +, \cdot, 0_{\mathbb{R}}, 1_{\mathbb{R}}, \leq)$ is an ordered field.*

Corollary 0.117.7 (Ordered-Field Laws Hold on $\mathbb{R}$). *Every theorem proved generically for ordered fields holds in $\mathbb{R}$.*

Theorem 0.117.8 ($\mathbb{R}$ Is a Complete Ordered Field). *The ordered field $\mathbb{R}$ satisfies the least-upper-bound property.*

0.118 Embedding $\mathbb{Q}$ into $\mathbb{R}$

Definition 0.118.1 (Embedding of $\mathbb{Q}$ into $\mathbb{R}$)**.** Define $\iota : \mathbb{Q} \to \mathbb{R}$ by

$$\iota(q) := q^*.$$

Theorem 0.118.2 (Embedding Preserves Zero and One)**.** *The embedding satisfies*

$$\iota(0_{\mathbb{Q}}) = 0_{\mathbb{R}} \qquad \textit{and} \qquad \iota(1_{\mathbb{Q}}) = 1_{\mathbb{R}}.$$

Theorem 0.118.3 (Embedding $\mathbb{Q}$ into $\mathbb{R}$ Is Injective)**.** *For all $p, q \in \mathbb{Q}$,*

$$\iota(p) = \iota(q) \implies p = q.$$

Theorem 0.118.4 (Embedding Preserves Addition and Multiplication)**.** *For all $p, q \in \mathbb{Q}$,*

$$\iota(p+q) = \iota(p) + \iota(q) \qquad \textit{and} \qquad \iota(p \cdot q) = \iota(p) \cdot \iota(q).$$

Theorem 0.118.5 (Embedding Preserves Order)**.** *For all $p, q \in \mathbb{Q}$,*

$$p \leq q \iff \iota(p) \leq \iota(q).$$

Theorem 0.118.6 ($\mathbb{Q}$ Embeds into $\mathbb{R}$ as an Ordered Field)**.** *The map ι is an injective order-preserving field homomorphism.*

Theorem 0.118.7 (Density of $\mathbb{Q}$ in $\mathbb{R}$)**.** *For all $\alpha, \beta \in \mathbb{R}$,*

$$\alpha < \beta \implies \exists q \in \mathbb{Q}\, (\alpha < \iota(q) < \beta).$$

Theorem 0.118.8 (Archimedean Property of $\mathbb{R}$)**.** *For $\alpha, \beta \in \mathbb{R}$,*

$$0_{\mathbb{R}} < \alpha \implies \exists n \in \mathbb{N}\, (\beta < \iota(n) \cdot \alpha).$$

0.119 The Gap Filled

Definition 0.119.1 (The Square-Root-of-Two Cut)**.** Define

$$\sqrt{2} := \{r \in \mathbb{Q} : r < 0_{\mathbb{Q}} \vee r \cdot r < 2_{\mathbb{Q}}\}.$$

Lemma 0.119.2 (The Square-Root-of-Two Cut Is a Cut)**.** *The set $\sqrt{2}$ is a Dedekind cut.*

Theorem 0.119.3 (Its Square Is Two)**.** *In $\mathbb{R}$,*

$$\sqrt{2} \cdot \sqrt{2} = 2_{\mathbb{R}},$$

where $2_{\mathbb{R}} := \iota(2_{\mathbb{Q}})$.

Corollary 0.119.4 ($\mathbb{R}$ Contains the Supremum $\mathbb{Q}$ Was Missing)**.** *Let*

$$S = \{q \in \mathbb{Q} : 0_{\mathbb{Q}} < q \wedge q \cdot q < 2_{\mathbb{Q}}\}.$$

Then

$$\sqrt{2} = \sup \iota(S).$$

Thus the bounded-above rational set with no rational supremum has a supremum in $\mathbb{R}$.

0.120 Characterization of $\mathbb{R}$

Definition 0.120.1 (Complete Ordered Field)**.** A *complete ordered field* is an ordered field satisfying the least-upper-bound property.

Theorem 0.120.2 (Uniqueness of the Complete Ordered Field)**.** *Any two complete ordered fields are isomorphic by a unique order-isomorphism.*

Corollary 0.120.3 ($\mathbb{R}$ Is the Real Numbers)**.** *The field $\mathbb{R}$ is determined up to unique order-isomorphism by being a complete ordered field.*

0.121 Transition and Forward

Proofs

Capstone

Complex Numbers

0.122 Construction of $\mathbb{C}$

Complex Prepairs

Definition 0.122.1 (Complex Prepair)**.** A *complex prepair* is an ordered pair $(a, b) \in \mathbb{R} \times \mathbb{R}$, read informally as the expression $a + bi$.

Definition 0.122.2 (Complex Prepair Set)**.** The complex prepair set is

$$\operatorname{Pre}\mathbb{C} := \mathbb{R} \times \mathbb{R}.$$

Coordinate Equivalence

Definition 0.122.3 (Complex Coordinate Equivalence)**.** For $(a, b), (c, d) \in \operatorname{Pre}\mathbb{C}$, define

$$(a, b) \sim_{\mathbb{C}} (c, d) \quad :\Longleftrightarrow \quad a = c \text{ and } b = d.$$

Lemma 0.122.4 (Complex Coordinate Equivalence Is Reflexive)**.** *For every* $(a, b) \in \operatorname{Pre}\mathbb{C}$*,*

$$(a, b) \sim_{\mathbb{C}} (a, b).$$

Lemma 0.122.5 (Complex Coordinate Equivalence Is Symmetric)**.** *For all prepairs* $(a, b), (c, d)$*,*

$$(a, b) \sim_{\mathbb{C}} (c, d) \Longrightarrow (c, d) \sim_{\mathbb{C}} (a, b).$$

Lemma 0.122.6 (Complex Coordinate Equivalence Is Transitive)**.** *For all prepairs* $(a, b), (c, d), (e, f)$*,*

$$(a, b) \sim_{\mathbb{C}} (c, d) \wedge (c, d) \sim_{\mathbb{C}} (e, f) \Longrightarrow (a, b) \sim_{\mathbb{C}} (e, f).$$

Theorem 0.122.7 (Complex Coordinate Equivalence Is an Equivalence Relation)**.** *The relation* $\sim_{\mathbb{C}}$ *is an equivalence relation on* $\operatorname{Pre}\mathbb{C}$*.*

Definition of $\mathbb{C}$

Definition 0.122.8 (Complex Numbers)**.** The complex numbers are the quotient

$$\mathbb{C} := \operatorname{Pre}\mathbb{C}/{\sim_{\mathbb{C}}}.$$

Definition 0.122.9 (Complex Class Notation)**.** The notation $[a, b]_{\mathbb{C}}$ denotes the $\sim_{\mathbb{C}}$-equivalence class of the prepair (a, b).

Definition 0.122.10 (Zero, One, and the Imaginary Unit in $\mathbb{C}$)**.** Define

$$0_{\mathbb{C}} := [0, 0]_{\mathbb{C}}, \qquad 1_{\mathbb{C}} := [1, 0]_{\mathbb{C}}, \qquad i := [0, 1]_{\mathbb{C}}.$$

0.123 Operations on $\mathbb{C}$-Classes

Addition on $\mathbb{C}$-Classes

Definition 0.123.1 (Addition on $\mathbb{C}$-Classes)**.** Define addition on representatives by

$$[a, b]_{\mathbb{C}} + [c, d]_{\mathbb{C}} := [a + c, b + d]_{\mathbb{C}}.$$

Lemma 0.123.2 (Addition Respects Complex Coordinate Equivalence)**.** *If $[a, b]_{\mathbb{C}} = [a', b']_{\mathbb{C}}$ and $[c, d]_{\mathbb{C}} = [c', d']_{\mathbb{C}}$, then*

$$[a, b]_{\mathbb{C}} + [c, d]_{\mathbb{C}} = [a', b']_{\mathbb{C}} + [c', d']_{\mathbb{C}}.$$

Theorem 0.123.3 (Addition on $\mathbb{C}$ Is Well-Defined)**.** *Addition determines a well-defined binary operation*

$$\mathbb{C} \times \mathbb{C} \to \mathbb{C}.$$

Negation and Subtraction

Definition 0.123.4 (Negation on $\mathbb{C}$-Classes)**.** Define negation by

$$-[a, b]_{\mathbb{C}} := [-a, -b]_{\mathbb{C}}.$$

Lemma 0.123.5 (Negation Respects Complex Coordinate Equivalence)**.** *If $[a, b]_{\mathbb{C}} = [a', b']_{\mathbb{C}}$, then*

$$-[a, b]_{\mathbb{C}} = -[a', b']_{\mathbb{C}}.$$

Theorem 0.123.6 (Negation on $\mathbb{C}$ Is Well-Defined)**.** *Negation determines a well-defined unary operation $\mathbb{C} \to \mathbb{C}$.*

Definition 0.123.7 (Subtraction on $\mathbb{C}$)**.** For $z, w \in \mathbb{C}$, define

$$z - w := z + (-w).$$

Theorem 0.123.8 (Subtraction on $\mathbb{C}$ Is Well-Defined)**.** *Subtraction determines a well-defined binary operation $\mathbb{C} \times \mathbb{C} \to \mathbb{C}$.*

Multiplication on $\mathbb{C}$-Classes

Definition 0.123.9 (Multiplication on $\mathbb{C}$-Classes)**.** Define multiplication on representatives by

$$[a, b]_{\mathbb{C}} \cdot [c, d]_{\mathbb{C}} := [ac - bd, ad + bc]_{\mathbb{C}}.$$

Lemma 0.123.10 (Multiplication Respects Complex Coordinate Equivalence)**.** *If $[a, b]_{\mathbb{C}} = [a', b']_{\mathbb{C}}$ and $[c, d]_{\mathbb{C}} = [c', d']_{\mathbb{C}}$, then*

$$[a, b]_{\mathbb{C}} \cdot [c, d]_{\mathbb{C}} = [a', b']_{\mathbb{C}} \cdot [c', d']_{\mathbb{C}}.$$

Theorem 0.123.11 (Multiplication on $\mathbb{C}$ Is Well-Defined)**.** *Multiplication determines a well-defined binary operation*

$$\mathbb{C} \times \mathbb{C} \to \mathbb{C}.$$

Conjugation and Inverses

Definition 0.123.12 (Complex Conjugation)**.** For $z = [a, b]_{\mathbb{C}}$, define its *complex conjugate* by

$$\overline{z} := [a, -b]_{\mathbb{C}}.$$

Theorem 0.123.13 (Complex Conjugation Is Well-Defined)**.** *Complex conjugation determines a well-defined unary operation $\mathbb{C} \to \mathbb{C}$.*

Lemma 0.123.14 (Product with the Conjugate Is Real and Nonnegative)**.** *For* $z = [a, b]_{\mathbb{C}}$*,*

$$z\overline{z} = [a^2 + b^2, 0]_{\mathbb{C}}.$$

Moreover, if $z \neq 0_{\mathbb{C}}$*, then* $a^2 + b^2 \neq 0$*.*

Definition 0.123.15 (Multiplicative Inverse in $\mathbb{C}$)**.** For $z = [a, b]_{\mathbb{C}} \neq 0_{\mathbb{C}}$, define

$$z^{-1} := \left[\frac{a}{a^2 + b^2}, \frac{-b}{a^2 + b^2}\right]_{\mathbb{C}}.$$

Theorem 0.123.16 (Inversion on $\mathbb{C}^{\times}$ Is Well-Defined)**.** *Inversion determines a well-defined operation on the nonzero complex numbers.*

0.124 Field Structure of $\mathbb{C}$

Additive Structure

Theorem 0.124.1 (Addition on $\mathbb{C}$ Is Associative)**.** *For all* $z, w, u \in \mathbb{C}$*,*

$$(z + w) + u = z + (w + u).$$

Theorem 0.124.2 (Addition on $\mathbb{C}$ Is Commutative)**.** *For all* $z, w \in \mathbb{C}$*,*

$$z + w = w + z.$$

Theorem 0.124.3 (Zero Is an Additive Identity on $\mathbb{C}$)**.** *For every* $z \in \mathbb{C}$*,*

$$0_{\mathbb{C}} + z = z \qquad \textit{and} \qquad z + 0_{\mathbb{C}} = z.$$

Theorem 0.124.4 (Every Complex Number Has an Additive Inverse)**.** *For every* $z \in \mathbb{C}$*,*

$$z + (-z) = 0_{\mathbb{C}} \qquad \textit{and} \qquad (-z) + z = 0_{\mathbb{C}}.$$

Multiplicative Structure

Theorem 0.124.5 (Multiplication on $\mathbb{C}$ Is Associative)**.** *For all* $z, w, u \in \mathbb{C}$*,*

$$(zw)u = z(wu).$$

Theorem 0.124.6 (Multiplication on $\mathbb{C}$ Is Commutative)**.** *For all* $z, w \in \mathbb{C}$*,*

$$zw = wz.$$

Theorem 0.124.7 (One Is a Multiplicative Identity on $\mathbb{C}$)**.** *For every* $z \in \mathbb{C}$*,*

$$1_{\mathbb{C}}z = z \qquad \textit{and} \qquad z1_{\mathbb{C}} = z.$$

Theorem 0.124.8 (Every Nonzero Complex Number Has a Multiplicative Inverse)**.** *For every* $z \in \mathbb{C}$ *with* $z \neq 0_{\mathbb{C}}$*,*

$$zz^{-1} = 1_{\mathbb{C}} \qquad \textit{and} \qquad z^{-1}z = 1_{\mathbb{C}}.$$

Distributivity and the Field Theorem

Theorem 0.124.9 (Multiplication Distributes over Addition on $\mathbb{C}$)**.** *For all* $z, w, u \in \mathbb{C}$*,*

$$z(w + u) = zw + zu \qquad \textit{and} \qquad (w + u)z = wz + uz.$$

Theorem 0.124.10 ($\mathbb{C}$ Is a Field)**.** *With the operations defined above,*

$$(\mathbb{C}, +, \cdot, 0_{\mathbb{C}}, 1_{\mathbb{C}})$$

is a field.

The Imaginary Unit

Theorem 0.124.11 (The Imaginary Unit Squares to Negative One)**.** *In* $\mathbb{C}$*,*

$$i^2 = -1_{\mathbb{C}}.$$

Theorem 0.124.12 ($\mathbb{C}$ Is Not an Ordered Field)**.** *There is no order relation on* $\mathbb{C}$ *making* $\mathbb{C}$ *an ordered field and extending the usual order on* $\mathbb{R}$*.*

0.125 Embedding $\mathbb{R}$ into $\mathbb{C}$

The Real Axis

Definition 0.125.1 (Canonical Embedding of $\mathbb{R}$ into $\mathbb{C}$)**.** Define

$$\iota : \mathbb{R} \to \mathbb{C}, \qquad \iota(a) := [a, 0]_{\mathbb{C}}.$$

Theorem 0.125.2 (The Real Embedding into $\mathbb{C}$ Is Injective)**.** *If* $\iota(a) = \iota(b)$*, then* $a = b$*.*

Theorem 0.125.3 (The Real Embedding Preserves Addition and Multiplication)**.** *For all* $a, b \in \mathbb{R}$*,*

$$\iota(a + b) = \iota(a) + \iota(b), \qquad \iota(ab) = \iota(a)\iota(b).$$

Theorem 0.125.4 (The Real Embedding Preserves Zero and One)**.**

$$\iota(0) = 0_{\mathbb{C}}, \qquad \iota(1) = 1_{\mathbb{C}}.$$

Real and Imaginary Parts

Definition 0.125.5 (Real and Imaginary Parts)**.** For $z = [a, b]_{\mathbb{C}}$, define

$$\mathrm{Re}(z) := a, \qquad \mathrm{Im}(z) := b.$$

Theorem 0.125.6 (Real and Imaginary Parts Are Well-Defined)**.** *The functions*

$$\mathrm{Re}, \mathrm{Im} : \mathbb{C} \to \mathbb{R}$$

are well-defined.

Proofs

Capstone

Number Lines and Intervals

0.126 Number Lines

Definition 0.126.1 (Number Line)**.** A *number line* for a linearly ordered system $(A, <_A)$ is a linearly ordered set of points L together with an order isomorphism $\lambda : A \to L$, so that for all $a, a' \in A$,

$$a <_A a' \iff \lambda(a) \text{ lies to the left of } \lambda(a').$$

0.127 Intervals on a Number Line

Definition 0.127.1 (Open Interval)**.** Let $a, b \in \mathbb{R}$ with $a \leq b$. The *open interval* with endpoints a and b is

$$(a, b) = \{\, x \in \mathbb{R} : a < x < b \,\}.$$

Definition 0.127.2 (Closed Interval)**.** Let $a, b \in \mathbb{R}$ with $a \leq b$. The *closed interval* with endpoints a and b is

$$[a, b] = \{\, x \in \mathbb{R} : a \leq x \leq b \,\}.$$

Definition 0.127.3 (Left-Half-Open Interval)**.** Let $a, b \in \mathbb{R}$ with $a \leq b$. The *left-half-open interval* with endpoints a and b is

$$(a, b] = \{\, x \in \mathbb{R} : a < x \leq b \,\}.$$

Definition 0.127.4 (Right-Half-Open Interval)**.** Let $a, b \in \mathbb{R}$ with $a \leq b$. The *right-half-open interval* with endpoints a and b is

$$[a, b) = \{\, x \in \mathbb{R} : a \leq x < b \,\}.$$

Definition 0.127.5 (Ray)**.** Let $a \in \mathbb{R}$. The four *rays* determined by a are

$$(a, \infty) = \{x : x > a\},\ [a, \infty) = \{x : x \geq a\},\ (-\infty, a) = \{x : x < a\},\ (-\infty, a] = \{x : x \leq a\}.$$

Proofs

Capstone

Embedding the Number Systems

0.128 Embeddings as Structure Preserving Maps

Definition 0.128.1 (Structure-Preserving Map)**.** Let A and B be ordered arithmetic systems with a specified common source structure: selected constants, selected operations, and a strict order. A map $\varphi : A \to B$ is *structure-preserving* exactly when it preserves each selected constant and, for every selected binary operation $\star$ and all $a, a' \in A$,

$$\varphi(a \star_A a') = \varphi(a) \star_B \varphi(a'),$$

and $a <_A a' \Longrightarrow \varphi(a) <_B \varphi(a')$.

Definition 0.128.2 (Embedding of Number Systems)**.** Let A and B be ordered arithmetic systems. A map $\varphi : A \to B$ is an *embedding* exactly when φ is injective and structure-preserving and is an order embedding.

Definition 0.128.3 (Image of an Embedding)**.** Let $\varphi : A \to B$ be an embedding. The *image* of φ is the set $\varphi(A) = \{ \varphi(a) : a \in A \} \subseteq B$, equipped with the selected constants, operations, and order of B restricted to $\varphi(A)$.

Theorem 0.128.4 (Embeddings Identify a System with Its Image)**.** *Let $\varphi : A \to B$ be an embedding of ordered arithmetic systems relative to a selected source structure. Then the corestriction $\varphi : A \to \varphi(A)$ is an isomorphism onto its image: it is a bijection that preserves and reflects the selected constants, operations, and order. Consequently A may be identified with the corresponding substructure $\varphi(A) \subseteq B$.* *Go to proof.*

0.129 Embedding N Into W

Definition 0.129.1 (Embedding $\mathbb{N}$ into $\mathbb{W}$)**.** The *canonical embedding* $\iota : \mathbb{N} \to \mathbb{W}$ is defined by

$$\iota(n) = n \quad \text{(the copy of each natural number inside the whole numbers).}$$

Theorem 0.129.2 ($\mathbb{N}$ Embeds in $\mathbb{W}$)**.** *The canonical map $\iota : \mathbb{N} \to \mathbb{W}$ identifies $\mathbb{N}$ with the positive part $\mathbb{W} \setminus \{0\}$ and preserves successor, 1, addition, multiplication, and order.* *Go to proof.*

0.130 Embedding W Into Z

Definition 0.130.1 (Embedding $\mathbb{W}$ into $\mathbb{Z}$)**.** The *canonical embedding* $\iota : \mathbb{W} \to \mathbb{Z}$ is defined by

$$\iota(w) = [(w, 0)] \quad \text{(the integer represented by } w \text{ and } 0).$$

Theorem 0.130.2 ($\mathbb{W}$ Embeds in $\mathbb{Z}$)**.** *The canonical map $\iota : \mathbb{W} \to \mathbb{Z}$ is an embedding of ordered arithmetic systems: it is injective, preserves 0, 1, addition, and multiplication, and reflects the order.* *Go to proof.*

0.131 Embedding Z Into Q

Definition 0.131.1 (Embedding $\mathbb{Z}$ into $\mathbb{Q}$)**.** The *canonical embedding* $\iota : \mathbb{Z} \to \mathbb{Q}$ is defined by

$$\iota(z) = [(z, 1)] \quad \text{(the rational represented by } z/1).$$

Theorem 0.131.2 ($\mathbb{Z}$ Embeds in $\mathbb{Q}$)**.** *The canonical map $\iota : \mathbb{Z} \to \mathbb{Q}$ is an embedding of ordered arithmetic systems: it is injective, preserves* 0, 1, *addition, and multiplication, and reflects the order.* *Go to proof.*

0.132 Embedding Q Into R

Definition 0.132.1 (Embedding $\mathbb{Q}$ into $\mathbb{R}$)**.** The *canonical embedding* $\iota : \mathbb{Q} \to \mathbb{R}$ is defined by

$$\iota(q) = q^* \quad \text{(the real number determined by } q\text{)}.$$

Theorem 0.132.2 ($\mathbb{Q}$ Embeds in $\mathbb{R}$)**.** *The canonical map $\iota : \mathbb{Q} \to \mathbb{R}$ is an embedding of ordered arithmetic systems: it is injective, preserves* 0, 1, *addition, and multiplication, and reflects the order.* *Go to proof.*

0.133 Compatibility of Arithmetic and Order

Theorem 0.133.1 (Compatibility of the Embedding Chain)**.** *Let $\iota_1 : \mathbb{N} \to \mathbb{W}$, $\iota_2 : \mathbb{W} \to \mathbb{Z}$, $\iota_3 : \mathbb{Z} \to \mathbb{Q}$, and $\iota_4 : \mathbb{Q} \to \mathbb{R}$ be the canonical embeddings. For every $n \in \mathbb{N}$ the composite $(\iota_4 \circ \iota_3 \circ \iota_2 \circ \iota_1)(n)$ is the real number identified with n, and the composite preserves the arithmetic and order carried by the source system. Hence arithmetic computed in any system agrees with arithmetic on its image in $\mathbb{R}$.* *Go to proof.*

0.134 One Chain Many Structures

Proofs

Capstone

Summary: The Number Systems

0.135 Axiom Systems for $\mathbb{N}$, $\mathbb{W}$, $\mathbb{Z}$, $\mathbb{Q}$, and $\mathbb{R}$

0.136 Number Systems Summary

The Natural Numbers $\mathbb{N}$

The Whole Numbers $\mathbb{W}$

The Integers $\mathbb{Z}$

Derived Properties of $\mathbb{Z}$ (from the Ring Axioms)

Structural Comparison: $\mathbb{N}$, $\mathbb{W}$, $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$

Proofs

Capstone

Part III

Volume III: Classical Analysis

Bounds, Sequences, and Series

Structure of the Real Line

0.137 Real Line One Dimensional

0.138 Order Distance Length

Definition 0.138.1 (Distance on the Real Line)**.** The *distance* on $\mathbb{R}$ is the map $d : \mathbb{R} \times \mathbb{R} \to \mathbb{R}$ given by

$$d(x, y) = |x - y|.$$

Definition 0.138.2 (Length of a Bounded Interval)**.** Let $a, b \in \mathbb{R}$ with $a \le b$. The *length* of any bounded interval with endpoints a and b is

$$\ell = b - a.$$

Theorem 0.138.3 (The Distance Function Is a Metric)**.** *The map $d(x, y) = |x-y|$ satisfies, for all $x, y, z \in \mathbb{R}$: (i) $d(x, y) \ge 0$ with $d(x, y) = 0$ iff $x = y$; (ii) $d(x, y) = d(y, x)$; and (iii) $d(x, z) \le d(x, y) + d(y, z)$. Hence $(\mathbb{R}, d)$ is a metric space.* *Go to proof.*

0.139 Absolute Value as Distance

Proposition 0.139.1 (Absolute Value Is Distance to the Origin)**.** *For every $a \in \mathbb{R}$, $|a| = d(a, 0)$.* *Go to proof.*

0.140 Intervals as Subsets

Definition 0.140.1 (Bounded Subset of the Real Line)**.** A set $E \subseteq \mathbb{R}$ is *bounded* exactly when there exists $M \ge 0$ such that $|x| \le M$ for all $x \in E$; equivalently, E is contained in some closed interval $[-M, M]$.

Set Operations on Intervals

Definition 0.140.2 (Union of Intervals)**.** Let $I, J \subseteq \mathbb{R}$ be intervals. Their *union* is the set

$$I \cup J = \{x \in \mathbb{R} : x \in I \text{ or } x \in J\}.$$

Definition 0.140.3 (Intersection of Intervals)**.** Let $I, J \subseteq \mathbb{R}$ be intervals. Their *intersection* is the set

$$I \cap J = \{x \in \mathbb{R} : x \in I \text{ and } x \in J\}.$$

Definition 0.140.4 (Complement of an Interval)**.** Let $I \subseteq \mathbb{R}$ be an interval. Its *complement* in $\mathbb{R}$ is the set

$$I^c = \mathbb{R} \setminus I = \{x \in \mathbb{R} : x \notin I\}.$$

Definition 0.140.5 (Difference of Intervals)**.** Let $I, J \subseteq \mathbb{R}$ be intervals. The *difference* of I by J is the set

$$I \setminus J = \{x \in \mathbb{R} : x \in I \text{ and } x \notin J\} = I \cap J^c.$$

0.141 Neighborhoods and Balls

Definition 0.141.1 (Open Ball on the Real Line). Let $x \in \mathbb{R}$ and $r > 0$. The *open ball* of radius r about x is

$$B_r(x) = \{\, y \in \mathbb{R} : |y - x| < r \,\} = (x - r,\, x + r).$$

Definition 0.141.2 (Neighborhood of a Point). A set $N \subseteq \mathbb{R}$ is a *neighborhood* of $x \in \mathbb{R}$ exactly when there exists $r > 0$ with $B_r(x) \subseteq N$.

0.142 Open Sets

Definition 0.142.1 (Open Set in $\mathbb{R}$). A set $U \subseteq \mathbb{R}$ is *open* exactly when

$$\forall x \in U\ \exists r > 0\ (B_r(x) \subseteq U).$$

Theorem 0.142.2 (Open Intervals Are Open). *Every open interval (a, b), every open ray (a, ∞) and $(-\infty, b)$, and $\mathbb{R}$ itself are open sets.* *Go to proof.*

Theorem 0.142.3 (Unions and Finite Intersections of Open Sets). *An arbitrary union of open subsets of $\mathbb{R}$ is open, and a finite intersection of open subsets of $\mathbb{R}$ is open.* *Go to proof.*

0.143 Closed Sets

Definition 0.143.1 (Closed Set in $\mathbb{R}$). A set $F \subseteq \mathbb{R}$ is *closed* exactly when its complement $\mathbb{R} \setminus F$ is open.

Theorem 0.143.2 (Closed Sets Contain Their Limit Points). *A set $F \subseteq \mathbb{R}$ is closed if and only if it contains every one of its limit points.* *Go to proof.*

0.144 Interior Exterior Boundary

Definition 0.144.1 (Interior Point). Let $E \subseteq \mathbb{R}$ and $x \in \mathbb{R}$. Then x is an interior point of E exactly when

$$\exists r > 0\ (B_r(x) \subseteq E).$$

Definition 0.144.2 (Exterior Point). Let $E \subseteq \mathbb{R}$ and $x \in \mathbb{R}$. Then x is an exterior point of E exactly when

$$\exists r > 0\ (B_r(x) \subseteq \mathbb{R} \setminus E).$$

Definition 0.144.3 (Boundary Point). Let $E \subseteq \mathbb{R}$ and $x \in \mathbb{R}$. Then x is a boundary point of E exactly when

$$\forall r > 0\ (B_r(x) \cap E \neq \emptyset\ \wedge\ B_r(x) \cap (\mathbb{R} \setminus E) \neq \emptyset).$$

Definition 0.144.4 (Interior of a Set). The *interior* of $E \subseteq \mathbb{R}$, written $\operatorname{int}(E)$, is the set of all interior points of E.

0.145 Limit and Isolated Points

Definition 0.145.1 (Limit Point). Let $E \subseteq \mathbb{R}$ and $x \in \mathbb{R}$. Then x is a *limit point* of E exactly when

$$\forall r > 0\ \exists y \in E\ \big(0 < |y - x| < r\big).$$

Definition 0.145.2 (Isolated Point). Let $E \subseteq \mathbb{R}$ and $x \in E$. Then x is an *isolated point* of E exactly when

$$\exists r > 0\ \forall y \in E\ \big(|y - x| < r \Rightarrow y = x\big).$$

0.146 Unions and Intersections

Theorem 0.146.1 (Closure Laws for Closed Sets)**.** *An arbitrary intersection of closed subsets of* $\mathbb{R}$ *is closed, and a finite union of closed subsets of* $\mathbb{R}$ *is closed.* *Go to proof.*

0.147 Covers and Subcovers

Definition 0.147.1 (Open Cover)**.** Let $E \subseteq \mathbb{R}$. An *open cover* of E is a family $\{U_i\}_{i\in I}$ of open subsets of $\mathbb{R}$ with

$$E \subseteq \bigcup_{i\in I} U_i.$$

Definition 0.147.2 (Subcover and Finite Subcover)**.** Let $\{U_i\}_{i\in I}$ be an open cover of E. A *subcover* is a subfamily $\{U_i\}_{i\in J}$ with $J \subseteq I$ that still covers E. It is a *finite subcover* when J is finite.

0.148 Compactness on R

Definition 0.148.1 (Compact Set in $\mathbb{R}$)**.** A set $K \subseteq \mathbb{R}$ is *compact* exactly when every open cover of K admits a finite subcover:

$$\forall \{U_i\}_{i\in I} \text{ open cover of } K\ \exists \text{finite } J \subseteq I \left(K \subseteq \bigcup_{i\in J} U_i\right).$$

Theorem 0.148.2 (Compact Subsets of $\mathbb{R}$ Are Closed and Bounded)**.** *If* $K \subseteq \mathbb{R}$ *is compact, then* K *is closed and bounded.* *Go to proof.*

0.149 Heine Borel

Theorem 0.149.1 (Closed Bounded Intervals Are Compact)**.** *Every closed bounded interval* $[a, b] \subseteq \mathbb{R}$ *is compact.* *Go to proof.*

Theorem 0.149.2 (Heine–Borel Theorem for $\mathbb{R}$)**.** *A set* $K \subseteq \mathbb{R}$ *is compact if and only if* K *is closed and bounded.* *Go to proof.*

Proofs

Bounds

0.150 Completeness Axiom

Axiom of Completeness

Definition 0.150.1 (Least Upper Bound Property)**.** Let $(S, \leq)$ be an ordered set. The set S has the *Least Upper Bound Property* if and only if every nonempty subset of S that is bounded above in S has a supremum in S.

0.150.1 Nested Intervals

Nested Interval Property

Theorem 0.150.2 (Nested Interval Property)**.** *Let (I_n) be a sequence of nonempty closed intervals in $\mathbb{R}$. If $I_{n+1} \subseteq I_n$ for every $n \in \mathbb{N}$, then*

$$\bigcap_{n=1}^{\infty} I_n \neq \varnothing.$$

Go to proof.

0.150.2 Archimedean Property

Archimedean Property

Theorem 0.150.3 (Archimedean Property)**.** *For every real number x, there exists a natural number n such that $n > x$.*

Go to proof.

Corollary 0.150.4 (Archimedean Reciprocal Corollary)**.** *For every $x > 0$ and every $y \in \mathbb{R}$, there exists $n \in \mathbb{N}$ such that $nx > y$.*

Go to proof.

0.150.3 Density

Density

Definition 0.150.5 (Dense Subset)**.** Let $D \subseteq S \subseteq \mathbb{R}$. The set D is *dense in S* if and only if for every $x \in S$ and every $\varepsilon > 0$ there exists $d \in D$ such that $|x - d| < \varepsilon$.

Definition 0.150.6 (Order Dense Subset)**.** Let X be a linearly ordered set and let $A \subseteq X$. The subset A is *order dense in X* if and only if for every $a, b \in X$ with $a < b$ there exists $c \in A$ such that $a < c < b$.

Theorem 0.150.7 (Density of the Rational Numbers in $\mathbb{R}$)**.** *For every $a, b \in \mathbb{R}$ with $a < b$, there exists $q \in \mathbb{Q}$ such that $a < q < b$.* *Go to proof.*

Theorem 0.150.8 (Density of the Irrational Numbers in $\mathbb{R}$)**.** *For every $a, b \in \mathbb{R}$ with $a < b$, there exists $s \in \mathbb{R} \setminus \mathbb{Q}$ such that $a < s < b$.* *Go to proof.*

Corollary 0.150.9 (Rational and Irrational Points in Every Open Interval)**.** *Every open interval in $\mathbb{R}$ contains both a rational number and an irrational number.* *Go to proof.*

0.150.4 Completeness Summary

0.151 Bounds and Extremal Values

0.151.1 Upper and Lower Bounds

Definition 0.151.1 (Bound)**.** Let $(S, \leq)$ be an ordered set, let $A \subseteq S$ be nonempty, and let $b \in S$. A *bound* of A means a one-sided comparison between b and every element of A: either $x \leq b$ for all $x \in A$, or $b \leq x$ for all $x \in A$.

Definition 0.151.2 (Upper Bound)**.** Let $(S, \leq)$ be an ordered set, let $A \subseteq S$ be nonempty, and let $u \in S$. The element u is an *upper bound* of A if every element of A is less than or equal to u.

Definition 0.151.3 (Lower Bound)**.** Let $(S, \leq)$ be an ordered set, let $A \subseteq S$ be nonempty, and let $\ell \in S$. The element ℓ is a *lower bound* of A if every element of A is greater than or equal to ℓ.

Definition 0.151.4 (Bounded Above)**.** Let $(S, \leq)$ be an ordered set and let $A \subseteq S$ be nonempty. The set A is *bounded above* if it has at least one upper bound in S.

Definition 0.151.5 (Bounded Below)**.** Let $(S, \leq)$ be an ordered set and let $A \subseteq S$ be nonempty. The set A is *bounded below* if it has at least one lower bound in S.

Definition 0.151.6 (Bounded Set)**.** Let $(S, \leq)$ be an ordered set and let $A \subseteq S$ be nonempty. The set A is *bounded* if it is bounded above and bounded below.

0.151.2 Suprema and Infima

Definition 0.151.7 (Supremum)**.** Let $(S, \leq)$ be an ordered set, let $A \subseteq S$ be nonempty, and let $s \in S$. The element s is the *supremum* of A if s is an upper bound of A and no smaller element of S is an upper bound of A.

Definition 0.151.8 (Infimum)**.** Let $(S, \leq)$ be an ordered set, let $A \subseteq S$ be nonempty, and let $i \in S$. The element i is the *infimum* of A if i is a lower bound of A and no larger element of S is a lower bound of A.

Proposition 0.151.9 (Uniqueness of the Supremum)**.** *Let $(S, \leq)$ be an ordered set and let $A \subseteq S$ be nonempty. If $s, t \in S$ are both suprema of A, then $s = t$.*
Go to proof.

Proposition 0.151.10 (Uniqueness of the Infimum)**.** *Let $(S, \leq)$ be an ordered set and let $A \subseteq S$ be nonempty. If $i, j \in S$ are both infima of A, then $i = j$.*

Proposition 0.151.11 (Upper Bound Property of the Supremum)**.** *Let $(S, \leq)$ be an ordered set, let $A \subseteq S$ be nonempty, and suppose that $s = \sup A$ exists in S. Then every element of A lies below s:*

$$(\forall x \in A)(x \leq s).$$

Go to proof.

Proposition 0.151.12 (Lower Bound Property of the Infimum)**.** *Let $(S, \leq)$ be an ordered set, let $A \subseteq S$ be nonempty, and suppose that $i = \inf A$ exists in S. Then every element of A lies above i:*

$$(\forall x \in A)(i \leq x).$$

Lemma 0.151.13 (Subset Inclusion Preserves Upper Bounds)**.** *Let $A \subseteq B \subseteq S$ be nonempty subsets of an ordered set $(S, \leq)$. If u is an upper bound of B, then u is an upper bound of A.*
Go to proof.

Lemma 0.151.14 (Subset Inclusion Preserves Lower Bounds)**.** *Let $A \subseteq B \subseteq S$ be nonempty subsets of an ordered set $(S, \leq)$. If ℓ is a lower bound of B, then ℓ is a lower bound of A.*

Theorem 0.151.15 (Supremum Is Monotone Under Inclusion)**.** *Let $A, B \subseteq \mathbb{R}$ be nonempty and bounded above. If $A \subseteq B$, then $\sup A \leq \sup B$.*
Go to proof.

Theorem 0.151.16 (Infimum Is Monotone Under Inclusion)**.** *Let $A, B \subseteq \mathbb{R}$ be nonempty and bounded below. If $A \subseteq B$, then $\inf B \leq \inf A$.*

Proposition 0.151.17 (Upper Bound Comparison with the Supremum)**.** *Let $A \subseteq \mathbb{R}$ be nonempty and bounded above, and let $s = \sup A$. For $u \in \mathbb{R}$, the number u is an upper bound of A if and only if $s \leq u$.*
Go to proof.

Proposition 0.151.18 (Lower Bound Comparison with the Infimum)**.** *Let $A \subseteq \mathbb{R}$ be nonempty and bounded below, and let $i = \inf A$. For $\ell \in \mathbb{R}$, the number ℓ is a lower bound of A if and only if $\ell \leq i$.*

Proposition 0.151.19 (Every Element Lies Below the Supremum)**.** *Let $A \subseteq \mathbb{R}$ be nonempty and bounded above, and let $s = \sup A$. Then $x \leq s$ for every $x \in A$.*
Go to proof.

Proposition 0.151.20 (Every Element Lies Above the Infimum)**.** *Let $A \subseteq \mathbb{R}$ be nonempty and bounded below, and let $i = \inf A$. Then $i \leq x$ for every $x \in A$.*

Theorem 0.151.21 (Infimum Is Less Than or Equal to the Supremum)**.** *Let $A \subseteq \mathbb{R}$ be nonempty, bounded below, and bounded above. Then $\inf A \leq \sup A$.*
Go to proof.

Proposition 0.151.22 (Order Comparison of Suprema)**.** *Let $A, B \subseteq \mathbb{R}$ be nonempty and bounded above. If every $x \in A$ is below some $y \in B$, then $\sup A \leq \sup B$.*
Go to proof.

Theorem 0.151.23 (Least Upper Bound Property Implies Existence of Suprema)**.** *Let $S \subseteq \mathbb{R}$ be nonempty and bounded above. By the least upper bound property of the real numbers, there exists a real number s such that $s = \sup S$.*

Theorem 0.151.24 (Greatest Lower Bound Property Implies Existence of Infima)**.** *Let $S \subseteq \mathbb{R}$ be nonempty and bounded below. Then there exists a real number i such that $i = \inf S$.*

0.151.3 Maxima and Minima

Definition 0.151.25 (Maximum)**.** Let $(S, \leq)$ be an ordered set and let $A \subseteq S$ be nonempty. An element $m \in S$ is the *maximum* of A if $m \in A$ and every element of A is less than or equal to m.

Definition 0.151.26 (Minimum)**.** Let $(S, \leq)$ be an ordered set and let $A \subseteq S$ be nonempty. An element $m \in S$ is the *minimum* of A if $m \in A$ and every element of A is greater than or equal to m.

Proposition 0.151.27 (Uniqueness of the Maximum)**.** *Let $(S, \leq)$ be an ordered set and let $A \subseteq S$ be nonempty. If $m, n \in S$ are both maxima of A, then $m = n$.*
Go to proof.

Proposition 0.151.28 (Uniqueness of the Minimum)**.** *Let $(S, \leq)$ be an ordered set and let $A \subseteq S$ be nonempty. If $m, n \in S$ are both minima of A, then $m = n$.*

Proposition 0.151.29 (Maximum Implies Supremum)**.** *Let $(S, \leq)$ be an ordered set and let $A \subseteq S$ be nonempty. If $m = \max A$, then $m = \sup A$.*

Go to proof.

Proposition 0.151.30 (Minimum Implies Infimum)**.** *Let $(S, \leq)$ be an ordered set and let $A \subseteq S$ be nonempty. If $m = \min A$, then $m = \inf A$.*

Proposition 0.151.31 (Supremum in the Set is the Maximum)**.** *Let $(S, \leq)$ be an ordered set and let $A \subseteq S$ be nonempty. If $s = \sup A$ and $s \in A$, then $s = \max A$.*

Go to proof.

Proposition 0.151.32 (Infimum in the Set is the Minimum)**.** *Let $(S, \leq)$ be an ordered set and let $A \subseteq S$ be nonempty. If $i = \inf A$ and $i \in A$, then $i = \min A$.*

0.151.4 Epsilon Characterization

ε-Characterization of Supremum

Definition 0.151.33 (Epsilon Characterization of Supremum)**.** Let $A \subseteq \mathbb{R}$ and let $s \in \mathbb{R}$. The element s satisfies the *epsilon characterization of the supremum* of A if and only if s is an upper bound of A and for every $\varepsilon > 0$ there exists $a \in A$ such that $s - \varepsilon < a$.

Definition 0.151.34 (Epsilon Characterization of Infimum)**.** Let $A \subseteq \mathbb{R}$ and let $t \in \mathbb{R}$. The element t satisfies the *epsilon characterization of the infimum* of A if and only if t is a lower bound of A and for every $\varepsilon > 0$ there exists $a \in A$ such that $a < t + \varepsilon$.

0.151.5 Extremal Bounds Summary

0.152 Algebra of Bounds

0.152.1 Set Operations and Bounds

Lemma 0.152.1 (Union Preserves Upper Bounds)**.** *Let $A, B \subseteq \mathbb{R}$. If u_A is an upper bound of A and u_B is an upper bound of B, then $\max\{u_A, u_B\}$ is an upper bound of $A \cup B$. Go to proof.*

Lemma 0.152.2 (Union Preserves Lower Bounds)**.** *Let $A, B \subseteq \mathbb{R}$. If ℓ_A is a lower bound of A and ℓ_B is a lower bound of B, then $\min\{\ell_A, \ell_B\}$ is a lower bound of $A \cup B$. Go to proof.*

Proposition 0.152.3 (Union Is Bounded Above Exactly When Both Pieces Are)**.** *Let $A, B \subseteq \mathbb{R}$ be nonempty. Then $A \cup B$ is bounded above if and only if A is bounded above and B is bounded above. Go to proof.*

Proposition 0.152.4 (Union Is Bounded Below Exactly When Both Pieces Are)**.** *Let $A, B \subseteq \mathbb{R}$ be nonempty. Then $A \cup B$ is bounded below if and only if A is bounded below and B is bounded below. Go to proof.*

Corollary 0.152.5 (Union Is Bounded Exactly When Both Pieces Are)**.** *Let $A, B \subseteq \mathbb{R}$ be nonempty. Then $A \cup B$ is bounded if and only if A is bounded and B is bounded. Go to proof.*

Lemma 0.152.6 (Subsets Preserve Upper Bounds)**.** *Let $C \subseteq A \subseteq \mathbb{R}$. If u is an upper bound of A, then u is an upper bound of C. Go to proof.*

Lemma 0.152.7 (Subsets Preserve Lower Bounds)**.** *Let $C \subseteq A \subseteq \mathbb{R}$. If ℓ is a lower bound of A, then ℓ is a lower bound of C. Go to proof.*

Corollary 0.152.8 (Intersections Inherit Bounds)**.** *Let $A, B \subseteq \mathbb{R}$. Every upper bound of A is an upper bound of $A \cap B$, and every lower bound of A is a lower bound of $A \cap B$. Go to proof.*

Corollary 0.152.9 (Differences Inherit Bounds)**.** *Let $A, B \subseteq \mathbb{R}$. Every upper bound of A is an upper bound of $A \setminus B$, and every lower bound of A is a lower bound of $A \setminus B$. Go to proof.*

Corollary 0.152.10 (Complements Inherit Ambient Bounds)**.** *Let $A \subseteq E \subseteq \mathbb{R}$, and take complements relative to E. Every upper bound of E is an upper bound of $E \setminus A$, and every lower bound of E is a lower bound of $E \setminus A$. Go to proof.*

0.152.2 Lattice Operations and Bounds

Definition 0.152.11 (Pointwise Maximum and Minimum Sets)**.** Let $A, B \subseteq \mathbb{R}$ be nonempty. Define

$$\max(A, B) := \{\max\{a, b\} : a \in A,\ b \in B\},$$

and

$$\min(A, B) := \{\min\{a, b\} : a \in A,\ b \in B\}.$$

Theorem 0.152.12 (Supremum of the Pointwise Maximum Set)**.** *Let $A, B \subseteq \mathbb{R}$ be nonempty and bounded above. Then*

$$\sup \max(A, B) = \max\{\sup A, \sup B\}.$$

Go to proof.

Theorem 0.152.13 (Infimum of the Pointwise Maximum Set)**.** *Let $A, B \subseteq \mathbb{R}$ be nonempty and bounded below. Then*

$$\inf \max(A, B) = \max\{\inf A, \inf B\}.$$

Go to proof.

Theorem 0.152.14 (Supremum of the Pointwise Minimum Set)**.** *Let $A, B \subseteq \mathbb{R}$ be nonempty and bounded above. Then*

$$\sup \min(A, B) = \min\{\sup A, \sup B\}.$$

Go to proof.

Theorem 0.152.15 (Infimum of the Pointwise Minimum Set)**.** *Let $A, B \subseteq \mathbb{R}$ be nonempty and bounded below. Then*

$$\inf \min(A, B) = \min\{\inf A, \inf B\}.$$

Go to proof.

0.152.3 Images, Inverse Images, and Extrema

Theorem 0.152.16 (Increasing Image Preserves Suprema)**.** *Let $I \subseteq \mathbb{R}$, let $A \subseteq I$ be nonempty and bounded above, and let $f : I \to \mathbb{R}$ be increasing. Suppose $s = \sup A$ belongs to I and f is continuous at s. Then*

$$\sup f(A) = f(\sup A), \qquad f(A) = \{f(a) : a \in A\}.$$

Go to proof.

Theorem 0.152.17 (Increasing Image Preserves Infima)**.** *Let $I \subseteq \mathbb{R}$, let $A \subseteq I$ be nonempty and bounded below, and let $f : I \to \mathbb{R}$ be increasing. Suppose $t = \inf A$ belongs to I and f is continuous at t. Then*

$$\inf f(A) = f(\inf A).$$

Go to proof.

Theorem 0.152.18 (Decreasing Image Sends Infimum to Supremum)**.** *Let $I \subseteq \mathbb{R}$, let $A \subseteq I$ be nonempty and bounded below, and let $f : I \to \mathbb{R}$ be decreasing. Suppose $t = \inf A$ belongs to I and f is continuous at t. Then*

$$\sup f(A) = f(\inf A).$$

Go to proof.

Theorem 0.152.19 (Decreasing Image Sends Supremum to Infimum)**.** *Let $I \subseteq \mathbb{R}$, let $A \subseteq I$ be nonempty and bounded above, and let $f : I \to \mathbb{R}$ be decreasing. Suppose $s = \sup A$ belongs to I and f is continuous at s. Then*

$$\inf f(A) = f(\sup A).$$

Go to proof.

Theorem 0.152.20 (Increasing Inverse Preserves Suprema). *Let $I, J \subseteq \mathbb{R}$, and let $f : I \to J$ be bijective. Suppose $f^{-1} : J \to I$ is increasing and continuous. If $B \subseteq J$ is nonempty and bounded above, and if $\sup B \in J$, then*

$$\sup f^{-1}(B) = f^{-1}(\sup B).$$

Go to proof.

Theorem 0.152.21 (Increasing Inverse Preserves Infima). *Let $I, J \subseteq \mathbb{R}$, and let $f : I \to J$ be bijective. Suppose $f^{-1} : J \to I$ is increasing and continuous. If $B \subseteq J$ is nonempty and bounded below, and if $\inf B \in J$, then*

$$\inf f^{-1}(B) = f^{-1}(\inf B).$$

Go to proof.

Theorem 0.152.22 (Decreasing Inverse Sends Infimum to Supremum). *Let $I, J \subseteq \mathbb{R}$, and let $f : I \to J$ be bijective. Suppose $f^{-1} : J \to I$ is decreasing and continuous. If $B \subseteq J$ is nonempty and bounded below, and if $\inf B \in J$, then*

$$\sup f^{-1}(B) = f^{-1}(\inf B).$$

Go to proof.

Theorem 0.152.23 (Decreasing Inverse Sends Supremum to Infimum). *Let $I, J \subseteq \mathbb{R}$, and let $f : I \to J$ be bijective. Suppose $f^{-1} : J \to I$ is decreasing and continuous. If $B \subseteq J$ is nonempty and bounded above, and if $\sup B \in J$, then*

$$\inf f^{-1}(B) = f^{-1}(\sup B).$$

Go to proof.

0.152.4 Dilation and Displacement

Definition 0.152.24 (Displacement, Dilation, and Reflection). Let $A \subseteq \mathbb{R}$ and let $c, \lambda \in \mathbb{R}$. Define

$$A + c := \{a + c : a \in A\}, \qquad A - c := \{a - c : a \in A\}, \qquad \lambda A := \{\lambda a : a \in A\}.$$

The set $-A := (-1)A$ is the *reflection* of A through the origin.

Proposition 0.152.25 (Translation Preserves Upper Bounds). *Let $A \subseteq \mathbb{R}$ be nonempty. If u is an upper bound of A, then $u + c$ is an upper bound of $A + c$. Go to proof.*

Proposition 0.152.26 (Translation Preserves Lower Bounds). *Let $A \subseteq \mathbb{R}$ be nonempty. If ℓ is a lower bound of A, then $\ell + c$ is a lower bound of $A + c$. Go to proof.*

Proposition 0.152.27 (Positive Dilation Preserves Upper Bounds). *Let $A \subseteq \mathbb{R}$ be nonempty and let $\lambda > 0$. If u is an upper bound of A, then λu is an upper bound of λA. Go to proof.*

Proposition 0.152.28 (Positive Dilation Preserves Lower Bounds). *Let $A \subseteq \mathbb{R}$ be nonempty and let $\lambda > 0$. If ℓ is a lower bound of A, then $\lambda \ell$ is a lower bound of λA. Go to proof.*

Proposition 0.152.29 (Negative Dilation Sends Lower Bounds to Upper Bounds). *Let $A \subseteq \mathbb{R}$ be nonempty and let $\lambda < 0$. If ℓ is a lower bound of A, then $\lambda \ell$ is an upper bound of λA. Go to proof.*

Proposition 0.152.30 (Negative Dilation Sends Upper Bounds to Lower Bounds). *Let $A \subseteq \mathbb{R}$ be nonempty and let $\lambda < 0$. If u is an upper bound of A, then λu is a lower bound of λA. Go to proof.*

Corollary 0.152.31 (Reflection Swaps Upper and Lower Bounds). *Let $A \subseteq \mathbb{R}$ be nonempty. If ℓ is a lower bound of A, then $-\ell$ is an upper bound of $-A$. If u is an upper bound of A, then $-u$ is a lower bound of $-A$. Go to proof.*

0.152.5 Algebra of Suprema and Infima

Current Algebraic Properties of Supremum and Infimum

Definition 0.152.32 (Set Arithmetic Images)**.** Let $A, B \subseteq \mathbb{R}$. Define

$$A + B := \{a + b : a \in A,\ b \in B\}, \qquad A - B := \{a - b : a \in A,\ b \in B\},$$

$$AB := \{ab : a \in A,\ b \in B\}, \qquad A/B := \{a/b : a \in A,\ b \in B,\ b \neq 0\}.$$

Definition 0.152.33 (Reciprocal Image)**.** Let $A \subseteq \mathbb{R}$ and suppose $0 \notin A$. The reciprocal image of A is

$$A^{-1} := \{1/a : a \in A\}.$$

Theorem 0.152.34 (Translation Invariance of the Supremum)**.** *Let $A \subseteq \mathbb{R}$ be nonempty and bounded above, and let $c \in \mathbb{R}$. Then*

$$\sup(A + c) = \sup A + c, \qquad A + c = \{a + c : a \in A\}.$$

Go to proof.

Theorem 0.152.35 (Translation Invariance of the Infimum)**.** *Let $A \subseteq \mathbb{R}$ be nonempty and bounded below, and let $c \in \mathbb{R}$. Then*

$$\inf(A + c) = \inf A + c.$$

Go to proof.

Theorem 0.152.36 (Scalar Multiplication by a Positive Scalar)**.** *Let $A \subseteq \mathbb{R}$ be nonempty and bounded above, and let $k > 0$. Then*

$$\sup(kA) = k \sup A, \qquad kA = \{ka : a \in A\}.$$

Go to proof.

Theorem 0.152.37 (Positive Scalar Multiplication of the Infimum)**.** *Let $A \subseteq \mathbb{R}$ be nonempty and bounded below, and let $k > 0$. Then*

$$\inf(kA) = k \inf A.$$

Go to proof.

Theorem 0.152.38 (Scalar Multiplication by a Negative Scalar)**.** *Let $A \subseteq \mathbb{R}$ be nonempty and bounded below, and let $k < 0$. Then*

$$\sup(kA) = k \inf A, \qquad kA = \{ka : a \in A\}.$$

Go to proof.

Theorem 0.152.39 (Negative Scalar Multiplication of the Infimum)**.** *Let $A \subseteq \mathbb{R}$ be nonempty and bounded above, and let $k < 0$. Then*

$$\inf(kA) = k \sup A.$$

Go to proof.

Theorem 0.152.40 (Negation Exchanges Supremum and Infimum)**.** *Let $A \subseteq \mathbb{R}$ be nonempty and bounded below. Then $-A = \{-a : a \in A\}$ is bounded above and*

$$\sup(-A) = -\inf A.$$

Go to proof.

Theorem 0.152.41 (Supremum of a Sum Set)**.** *Let $A, B \subseteq \mathbb{R}$ be nonempty and bounded above. Then*

$$\sup(A + B) = \sup A + \sup B, \qquad A + B = \{a + b : a \in A,\ b \in B\}.$$

Go to proof.

Theorem 0.152.42 (Infimum of a Sum Set). *Let $A, B \subseteq \mathbb{R}$ be nonempty and bounded below. Then*

$$\inf(A+B) = \inf A + \inf B.$$

Go to proof.

Theorem 0.152.43 (Supremum of a Difference Set). *Let $A, B \subseteq \mathbb{R}$ be nonempty, with A bounded above and B bounded below. Then*

$$\sup(A-B) = \sup A - \inf B, \qquad A - B = \{a - b : a \in A,\ b \in B\}.$$

Go to proof.

Theorem 0.152.44 (Infimum of a Difference Set). *Let $A, B \subseteq \mathbb{R}$ be nonempty, with A bounded below and B bounded above. Then*

$$\inf(A-B) = \inf A - \sup B.$$

Go to proof.

Theorem 0.152.45 (Supremum of a Dilation). *Let $A \subseteq \mathbb{R}$ be nonempty and bounded, and let $\lambda \in \mathbb{R}$. Then*

$$\sup(\lambda A) = \max\{\lambda \sup A, \lambda \inf A\}, \qquad \lambda A = \{\lambda a : a \in A\}.$$

Go to proof.

Theorem 0.152.46 (Supremum of the Absolute-Value Image). *Let $A \subseteq \mathbb{R}$ be nonempty and bounded. Then*

$$\sup |A| = \max\{|\sup A|, |\inf A|\}, \qquad |A| = \{|a| : a \in A\}.$$

Go to proof.

Theorem 0.152.47 (Supremum of a Reciprocal Set). *Let $A \subseteq \mathbb{R}$ be nonempty and bounded, and suppose that $0 < \inf A$ or $\sup A < 0$. Define*

$$A^{-1} := \{1/a : a \in A\}.$$

Then

$$\sup(A^{-1}) = \frac{1}{\inf A}.$$

Go to proof.

Theorem 0.152.48 (Infimum of a Reciprocal Set). *Let $A \subseteq \mathbb{R}$ be nonempty and bounded, and suppose that $0 < \inf A$ or $\sup A < 0$. Define*

$$A^{-1} := \{1/a : a \in A\}.$$

Then

$$\inf(A^{-1}) = \frac{1}{\sup A}.$$

Go to proof.

Theorem 0.152.49 (Supremum of a Product Set). *Let $A, B \subseteq \mathbb{R}$ be nonempty and bounded. Then*

$$\sup(AB) = \max\{(\inf A)(\inf B), (\inf A)(\sup B), (\sup A)(\inf B), (\sup A)(\sup B)\}.$$

Go to proof.

Theorem 0.152.50 (Infimum of a Product Set). *Let $A, B \subseteq \mathbb{R}$ be nonempty and bounded. Then*

$$\inf(AB) = \min\{(\inf A)(\inf B), (\inf A)(\sup B), (\sup A)(\inf B), (\sup A)(\sup B)\}.$$

Go to proof.

Theorem 0.152.51 (Supremum of a Quotient Set). *Let $A, B \subseteq \mathbb{R}$ be nonempty and bounded, and suppose $0 < \inf B$ or $\sup B < 0$. Then*

$$\sup(A/B) = \max\left\{\frac{\inf A}{\inf B}, \frac{\inf A}{\sup B}, \frac{\sup A}{\inf B}, \frac{\sup A}{\sup B}\right\}.$$

Go to proof.

Theorem 0.152.52 (Infimum of a Quotient Set). *Let $A, B \subseteq \mathbb{R}$ be nonempty and bounded, and suppose $0 < \inf B$ or $\sup B < 0$. Then*

$$\inf(A/B) = \min\left\{\frac{\inf A}{\inf B}, \frac{\inf A}{\sup B}, \frac{\sup A}{\inf B}, \frac{\sup A}{\sup B}\right\}.$$

Go to proof.

Proposition 0.152.53 (Bounds for Suprema and Infima Under Set Addition). *Let $A, B \subseteq \mathbb{R}$ be nonempty and bounded. Then*

$$\inf A + \inf B \leq \inf(A + B) \qquad \text{and} \qquad \sup(A + B) \leq \sup A + \sup B.$$

Go to proof.

Proposition 0.152.54 (Upper Bounds Depend on the Ambient Order). *Let $(S, \leq_S)$ and $(T, \leq_T)$ be ordered sets, and let $A \subseteq S \cap T$. Whether u is an upper bound of A must be interpreted relative to the chosen ambient order.* *Go to proof.*

Proposition 0.152.55 (Suprema Depend on the Ambient Set). *Let $A \subseteq S \subseteq S'$ be subsets of an ordered set. The supremum of A relative to S, if it exists, need not agree with the supremum of A relative to S'.* *Go to proof.*

Theorem 0.152.56 (Ambient Existence of Supremum). *There are ordered sets $S \subseteq S'$ and a subset $A \subseteq S$ such that A has a supremum relative to S' but has no supremum relative to S.* *Go to proof.*

Lemma 0.152.57 (Rational Gap Example for Suprema). *Let*

$$A = \{q \in \mathbb{Q} : q^2 < 2\}.$$

Then A is nonempty and bounded above in $\mathbb{Q}$, but A has no supremum relative to $\mathbb{Q}$. *Go to proof.*

Proofs

Capstone

Functions

0.153 Algebra of Functions

Algebra of Functions

Proposition 0.153.1 (Composition of Injections Is Injective). *Let $f : A \to B$ and let $g : B \to C$. If f is injective and g is injective, then $g \circ f : A \to C$ is injective.*

Go to proof.

Proposition 0.153.2 (Composition of Surjections Is Surjective). $\text{Surjective}(f) \wedge \text{Surjective}(g) \Rightarrow \text{Surjective}(g \circ f)$.

Go to proof.

Corollary 0.153.3 (Composition of Bijections Is Bijective). *Let $f : A \to B$ and let $g : B \to C$. If f is bijective and g is bijective, then $g \circ f : A \to C$ is bijective.*

Go to proof.

Proposition 0.153.4 (Inverse of a Bijection Is a Bijection). $\text{Bijective}(f) \Rightarrow \text{Bijective}(f^{-1})$.

Go to proof.

Proposition 0.153.5 (Preimage Distributes over Union and Intersection). *Let $f : A \to B$, $S, T \subseteq B$. Then:*

(i) $f^{-1}(S \cup T) = f^{-1}(S) \cup f^{-1}(T)$.

(ii) $f^{-1}(S \cap T) = f^{-1}(S) \cap f^{-1}(T)$.

(iii) $f^{-1}(S^c) = (f^{-1}(S))^c$.

Go to proof.

0.154 Real-Valued Functions

Real-Valued Functions

Definition 0.154.1 (Pointwise Sum of Functions). Let $X \subseteq \mathbb{R}$ and let $f, g : X \to \mathbb{R}$. The pointwise sum $f + g : X \to \mathbb{R}$ is the function defined by

$$(f + g)(x) = f(x) + g(x) \qquad (x \in X).$$

Definition 0.154.2 (Pointwise Difference of Functions). Let $X \subseteq \mathbb{R}$ and let $f, g : X \to \mathbb{R}$. The pointwise difference $f - g : X \to \mathbb{R}$ is the function defined by

$$(f - g)(x) = f(x) - g(x) \qquad (x \in X).$$

Definition 0.154.3 (Pointwise Product of Functions)**.** Let $X \subseteq \mathbb{R}$ and let $f, g : X \to \mathbb{R}$. The pointwise product $fg : X \to \mathbb{R}$ is the function defined by

$$(fg)(x) = f(x)g(x) \qquad (x \in X).$$

Definition 0.154.4 (Pointwise Scalar Multiple of a Function)**.** Let $X \subseteq \mathbb{R}$, let $f : X \to \mathbb{R}$, and let $c \in \mathbb{R}$. The pointwise scalar multiple $cf : X \to \mathbb{R}$ is the function defined by

$$(cf)(x) = cf(x) \qquad (x \in X).$$

Definition 0.154.5 (Pointwise Absolute Value of a Function)**.** Let $X \subseteq \mathbb{R}$ and let $f : X \to \mathbb{R}$. The pointwise absolute value $|f| : X \to \mathbb{R}$ is the function defined by

$$|f|(x) = |f(x)| \qquad (x \in X).$$

Definition 0.154.6 (Pointwise Maximum of Two Functions)**.** Let $X \subseteq \mathbb{R}$ and let $f, g : X \to \mathbb{R}$. The pointwise maximum $\max(f, g) : X \to \mathbb{R}$ is the function defined by

$$\max(f, g)(x) = \max\{f(x), g(x)\} \qquad (x \in X).$$

Definition 0.154.7 (Pointwise Quotient of Functions)**.** Let $X \subseteq \mathbb{R}$ and let $f, g : X \to \mathbb{R}$. Suppose $g(x) \neq 0$ for every $x \in X$. The pointwise quotient $f/g : X \to \mathbb{R}$ is the function defined by

$$(f/g)(x) = \frac{f(x)}{g(x)} \qquad (x \in X).$$

Definition 0.154.8 (Linear Combination of Real-Valued Functions)**.** Let $X \subseteq \mathbb{R}$, let $f, g : X \to \mathbb{R}$, and let $\alpha, \beta \in \mathbb{R}$. The real linear combination $\alpha f + \beta g : X \to \mathbb{R}$ is the function defined by

$$(\alpha f + \beta g)(x) = \alpha f(x) + \beta g(x) \qquad (x \in X).$$

Definition 0.154.9 (Real-Linear Rule)**.** Let $\mathcal{C}$ be a class of real-valued functions closed under real linear combinations. A rule T on $\mathcal{C}$ is real-linear if, whenever $f, g \in \mathcal{C}$ and $\alpha, \beta \in \mathbb{R}$,

$$T(\alpha f + \beta g) = \alpha T(f) + \beta T(g)$$

whenever both sides are defined.

Definition 0.154.10 (Bounded Above Function)**.** Let $A \subseteq \mathbb{R}$ and let $f : A \to \mathbb{R}$. The function f is bounded above on A if there exists $M \in \mathbb{R}$ such that

$$f(x) \leq M \qquad (x \in A).$$

Definition 0.154.11 (Bounded Below Function)**.** Let $A \subseteq \mathbb{R}$ and let $f : A \to \mathbb{R}$. The function f is bounded below on A if there exists $m \in \mathbb{R}$ such that

$$m \leq f(x) \qquad (x \in A).$$

Definition 0.154.12 (Bounded Function)**.** Let $A \subseteq \mathbb{R}$ and let $f : A \to \mathbb{R}$. The function f is bounded on A if there exists $B > 0$ such that

$$|f(x)| \leq B \qquad (x \in A).$$

Definition 0.154.13 (Supremum of a Function on a Set)**.** Let $A \subseteq \mathbb{R}$ and let $f : A \to \mathbb{R}$. The supremum of f on A is

$$\sup_{x \in A} f(x) = \sup f(A),$$

provided the supremum of $f(A)$ exists.

Definition 0.154.14 (Infimum of a Function on a Set)**.** Let $A \subseteq \mathbb{R}$ and let $f : A \to \mathbb{R}$. The infimum of f on A is

$$\inf_{x \in A} f(x) = \inf f(A),$$

provided the infimum of $f(A)$ exists.

Definition 0.154.15 (Maximum Point of a Function)**.** Let $A \subseteq \mathbb{R}$ and let $f : A \to \mathbb{R}$. A point $x^* \in A$ is a maximum point of f on A if

$$f(x) \leq f(x^*) \qquad (x \in A).$$

Definition 0.154.16 (Minimum Point of a Function)**.** Let $A \subseteq \mathbb{R}$ and let $f : A \to \mathbb{R}$. A point $x_* \in A$ is a minimum point of f on A if

$$f(x_*) \leq f(x) \qquad (x \in A).$$

Definition 0.154.17 (Increasing Function)**.** Let $A \subseteq \mathbb{R}$ and let $f : A \to \mathbb{R}$. The function f is increasing on A if

$$x_1 < x_2 \implies f(x_1) \leq f(x_2) \qquad (x_1, x_2 \in A).$$

Definition 0.154.18 (Strictly Increasing Function)**.** Let $A \subseteq \mathbb{R}$ and let $f : A \to \mathbb{R}$. The function f is strictly increasing on A if

$$x_1 < x_2 \implies f(x_1) < f(x_2) \qquad (x_1, x_2 \in A).$$

Definition 0.154.19 (Decreasing Function)**.** Let $A \subseteq \mathbb{R}$ and let $f : A \to \mathbb{R}$. The function f is decreasing on A if

$$x_1 < x_2 \implies f(x_1) \geq f(x_2) \qquad (x_1, x_2 \in A).$$

Definition 0.154.20 (Strictly Decreasing Function)**.** Let $A \subseteq \mathbb{R}$ and let $f : A \to \mathbb{R}$. The function f is strictly decreasing on A if

$$x_1 < x_2 \implies f(x_1) > f(x_2) \qquad (x_1, x_2 \in A).$$

Definition 0.154.21 (Monotone Function)**.** Let $A \subseteq \mathbb{R}$ and let $f : A \to \mathbb{R}$. The function f is monotone on A if f is increasing on A or decreasing on A.

Definition 0.154.22 (Constant Function)**.** Let $A \subseteq \mathbb{R}$ and let $f : A \to \mathbb{R}$. The function f is constant on A if

$$f(x_1) = f(x_2) \qquad (x_1, x_2 \in A).$$

0.155 Subsets of $\mathbb{R}$

Subsets of $\mathbb{R}$

Definition 0.155.1 (ε-Neighbourhood)**.** Let $x \in \mathbb{R}$ and let $\varepsilon > 0$. The **ε-neighbourhood** of x is

$$V_\varepsilon(x) := (x - \varepsilon, x + \varepsilon) = \{y \in \mathbb{R} : |y - x| < \varepsilon\}.$$

Definition 0.155.2 (Deleted ε-Neighbourhood)**.** Let $x \in \mathbb{R}$ and let $\varepsilon > 0$. The **deleted ε-neighbourhood** of x is

$$V_\varepsilon^*(x) := V_\varepsilon(x) \setminus \{x\} = \{y \in \mathbb{R} : 0 < |y - x| < \varepsilon\}.$$

Definition 0.155.3 (Cluster Point)**.** Let $X \subseteq \mathbb{R}$ and $x \in \mathbb{R}$. The point x is a **cluster point** of X if

$$\forall \varepsilon > 0,\ \exists\, y \in X \setminus \{x\} :\ |y - x| < \varepsilon.$$

Theorem 0.155.4 (Sequential Characterization of Cluster Points)**.** *c is a cluster point of A if and only if there exists $(a_n) \subseteq A \setminus \{c\}$ with $a_n \to c$.*

Go to proof.

Definition 0.155.5 (Adherent Point)**.** Let $X \subseteq \mathbb{R}$ and let $x \in \mathbb{R}$. The point x is an **adherent point** of X if

$$\forall \varepsilon > 0\ \exists y \in X\ (|y - x| < \varepsilon).$$

Definition 0.155.6 (Isolated Point)**.** Let $X \subseteq \mathbb{R}$ and let $x \in X$. The point x is an **isolated point** of X if

$$\exists \varepsilon > 0\ \big(V_\varepsilon(x) \cap X = \{x\}\big).$$

Definition 0.155.7 (Interior Point)**.** Let $X \subseteq \mathbb{R}$ and let $x \in \mathbb{R}$. The point x is an **interior point** of X if

$$\exists \varepsilon > 0\ \big(V_\varepsilon(x) \subseteq X\big).$$

Definition 0.155.8 (Boundary Point)**.** Let $X \subseteq \mathbb{R}$ and let $x \in \mathbb{R}$. The point x is a **boundary point** of X if

$$\forall \varepsilon > 0\ \big(V_\varepsilon(x) \cap X \neq \emptyset\big) \wedge \big(V_\varepsilon(x) \cap X^c \neq \emptyset\big).$$

Definition 0.155.9 (Interior)**.** Let $X \subseteq \mathbb{R}$. The **interior** of X is

$$X^\circ := \{x \in \mathbb{R} : \exists \varepsilon > 0\ V_\varepsilon(x) \subseteq X\}.$$

Definition 0.155.10 (Boundary)**.** Let $X \subseteq \mathbb{R}$. The **boundary** of X is

$$\partial X := \{x \in \mathbb{R} : \forall \varepsilon > 0\ (V_\varepsilon(x) \cap X \neq \emptyset \wedge V_\varepsilon(x) \cap X^c \neq \emptyset)\}.$$

Definition 0.155.11 (Closure)**.**

$$\overline{X} := \{x \in \mathbb{R} : \forall \varepsilon > 0\ V_\varepsilon(x) \cap X \neq \emptyset\}.$$

Lemma 0.155.12 (Elementary Properties of Closures)**.** *Let $X, Y \subseteq \mathbb{R}$. Then: (i) $X \subseteq \overline{X}$; (ii) $\overline{X \cup Y} = \overline{X} \cup \overline{Y}$; (iii) $\overline{X \cap Y} \subseteq \overline{X} \cap \overline{Y}$; (iv) $X \subseteq Y \Rightarrow \overline{X} \subseteq \overline{Y}$.*

Go to proof.

Definition 0.155.13 (Closed Subset of $\mathbb{R}$)**.** E is closed $\iff$ $\overline{E} = E$.

Corollary 0.155.14 (Closed iff Sequentially Closed)**.** *X is closed $\iff$ $\forall (a_n) \subseteq X,\ a_n \to x \Rightarrow x \in X$.*

Go to proof.

Corollary 0.155.15 (Every Point of an Interval Is a Cluster Point)**.** *If I is an interval, then every $x \in I$ is a cluster point of I.*

Go to proof.

Definition 0.155.16 (Bounded Subset of $\mathbb{R}$)**.** X is bounded $\iff$ $\exists M > 0,\ \forall x \in X : |x| \leq M$.

Theorem 0.155.17 (Heine–Borel Theorem for $\mathbb{R}$)**.** *X is closed $\wedge$ X is bounded $\iff$ every $(a_n) \subseteq X$ has a subsequence converging to a limit in X.*

Go to proof.

Definition 0.155.18 (True Near a Point)**.** Property $P(f(x))$ is **true near** x_0 if $\exists \delta > 0,\ \forall x : 0 < |x - x_0| < \delta) \Rightarrow P(f(x))$.

Proofs

Discrete Calculus

0.156 Motivation

Motivation

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A Structural View

0.157 Foundations

0.157.1 Foundations

0.157.2 Summation Notation

Basic Properties of Summation

Splitting Sums

Changing the Index

Empty Sums

Nested Summations

Common Examples

0.157.3 The Binomial Theorem

Theorem 0.157.1 (Binomial Theorem)**.** *For any integer $n \geq 0$ and real numbers x, y,*

$$(x+y)^n = \sum_{k=0}^{n} \binom{n}{k} x^{n-k} y^k,$$

where the binomial coefficients *are*

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}.$$

Basic Properties of Binomial Coefficients

Connection with Discrete Calculus

Proposition 0.157.2 (Higher difference formula)**.** *For any function f,*

$$\Delta^n f(x) = \sum_{k=0}^{n} (-1)^{n-k} \binom{n}{k} f(x+k).$$

Go to proof.

Explicit Examples

0.158 Algebra of Summations

Linearity

Splitting a Sum

Index Renaming

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Common Summation Identities

0.159 Difference Operators

0.159.1 Difference Operators

Motivation

Forward Difference

Definition 0.159.1 (Forward Difference Operator)**.** Let $f : \mathbb{Z} \to \mathbb{R}$. The *forward difference operator* Δ is defined by

$$\Delta f(n) = f(n+1) - f(n).$$

The function Δf measures the change in f when the input increases by one unit.

Example

0.159.2 Higher Differences

Definition 0.159.2 (Higher Differences)**.** The *second difference* of f is defined by

$$\Delta^2 f(n) = \Delta(\Delta f(n)).$$

More generally, the k-th difference $\Delta^k f(n)$ denotes the k-fold application of the difference operator, defined recursively by

$$\Delta^0 f = f, \qquad \Delta^k f = \Delta(\Delta^{k-1} f).$$

Example

0.159.3 Properties of the Difference Operator

Linearity

Proposition 0.159.3 (Linearity of Δ)**.** *For functions $f, g : \mathbb{Z} \to \mathbb{R}$ and constants $a, b \in \mathbb{R}$,*

$$\Delta(af + bg) = a\,\Delta f + b\,\Delta g.$$

Constant Rule

Proposition 0.159.4 (Constant rule)**.** *If c is a constant function, then $\Delta c = 0$.*

The Shift Operator and Operator Algebra

Definition 0.159.5 (Shift Operator)**.** The *shift operator* E and the *identity operator* I are defined by

$$Ef(n) = f(n+1), \qquad If(n) = f(n).$$

Theorem 0.159.6 ($\Delta = E - I$)**.**

$$\Delta = E - I.$$

0.159.4 Difference Tables and Polynomial Detection

Definition 0.159.7 (Difference Table)**.** Given a sequence $f(0), f(1), f(2), \ldots$, the *difference table* is constructed by listing the successive values of the function and then computing their forward differences $\Delta f(n) = f(n+1) - f(n)$, then $\Delta^2 f$, $\Delta^3 f$, and so on.

Example

Theorem 0.159.8 (Polynomial detection)**.** *Let $f(n)$ be a sequence generated by a polynomial of degree d. Then $\Delta^d f(n)$ is constant, and $\Delta^{d+1} f(n) = 0$.*

0.160 Telescoping Sums

0.160.1 Telescoping Sums

A Simple Example

General Form

The Telescoping Identity

Cancellation Pattern

Operator Form

0.161 Discrete Antiderivatives

0.161.1 Discrete Antiderivatives

Definition 0.161.1 (Discrete Antiderivative)**.** Let $f : \mathbb{Z} \to \mathbb{R}$ be a discrete function. A function $F : \mathbb{Z} \to \mathbb{R}$ is called a *discrete antiderivative* (or *discrete indefinite sum*) of f if

$$\Delta F(n) = f(n),$$

that is, $F(n+1) - F(n) = f(n)$ for all $n \in \mathbb{Z}$.

Examples

Constant of Integration

Proposition 0.161.2 (Constant of integration)**.** *If F is a discrete antiderivative of f, then so is $F + C$ for any constant $C \in \mathbb{R}$.*

Fundamental Theorem of Finite Calculus

0.162 Polynomial Basis

0.162.1 Polynomial Basis: Falling Factorials

The Difference Rule for Falling Factorials

Summary Table

Polynomial Representation in the Falling Factorial Basis

Consequences of the Difference Rule

Higher Differences of Falling Factorials

Connection to Binomial Coefficients

Worked Example: Expanding n^2

0.163 Calculus Rules

0.163.1 Calculus Rules

Product Rule

Summation by Parts

0.164 Expansions

0.164.1 Newton's Forward Difference Expansion

Structural Parallel with the Taylor Series

Proof Sketch

Reading Coefficients from the Difference Table

Worked Example: Expanding $f(n) = n^3$

Application: Computing $\sum_{k=0}^{n-1} k^2$

0.165 Special Functions

0.165.1 Special Functions in Discrete Calculus

The Discrete Exponential

Definition 0.165.1 (Discrete Exponential)**.** For a constant $a \neq -1$ and $n \in \mathbb{Z}$, the *discrete exponential* with base $(1 + a)$ is the function

$$E_a(n) = (1 + a)^n.$$

Proposition 0.165.2 (Discrete exponential rule)**.**

$$\Delta\,(1 + a)^n = a\,(1 + a)^n.$$

0.165.2 The Function 2^n in Discrete Calculus

Differencing 2^n

Antiderivative and Geometric Sum

Higher Differences of 2^n

Proposition 0.165.3 (Higher differences of 2^n)**.** *For all $k \geq 0$,*

$$\Delta^k 2^n = 2^n.$$

2^n in Combinatorics and the Newton Expansion

Discrete Differential Equation

Proposition 0.165.4 (Discrete IVP)**.** *The unique function $F : \mathbb{N} \to \mathbb{R}$ satisfying*

$$F(n+1) = 2\,F(n), \qquad F(0) = 1$$

is $F(n) = 2^n$.

Difference Formulas for Special Functions

Antiderivative Table

Proofs

Sequences

0.166 Sequence Definitions

0.166.1 Definitions

Definition 0.166.1 (Sequence)**.** Let X be a nonempty set. A **sequence** in X is a function

$$x : \mathbb{N} \to X.$$

The value $x(n)$ is written x_n and called the ***n*-th term** of the sequence. The sequence itself is denoted $(x_n)_{n\in\mathbb{N}}$, or simply (x_n) when the index set is clear.

Definition 0.166.2 (Real Sequence)**.** A **real sequence** is a sequence whose image is contained in $\mathbb{R}$. Equivalently, a real sequence is a function

$$x : \mathbb{N} \to \mathbb{R}.$$

0.166.2 Null and Constant Sequences

Null and Constant Sequences

Definition 0.166.3 (Constant Sequence)**.** Let (x_n) be a real sequence. The sequence (x_n) is **constant** exactly when there exists $c \in \mathbb{R}$ such that $x_n = c$ for every $n \in \mathbb{N}$.

Definition 0.166.4 (Null Sequence)**.** Let (x_n) be a real sequence. The sequence (x_n) is a **null sequence** exactly when for every $\varepsilon > 0$ there exists $N \in \mathbb{N}$ such that $|x_n| < \varepsilon$ for every $n \geq N$.

Theorem 0.166.5 (Constant Sequence Convergence)**.** *Let (x_n) be a real sequence, and let $c \in \mathbb{R}$. If $x_n = c$ for every $n \in \mathbb{N}$, then (x_n) converges to c.*

Go to proof.

Theorem 0.166.6 (Zero Sequence is Null)**.** *Let (x_n) be a real sequence. If $x_n = 0$ for every $n \in \mathbb{N}$, then (x_n) is a null sequence.*

Go to proof.

Theorem 0.166.7 (Constant Null Sequence)**.** *Let (x_n) be a real sequence, and let $c \in \mathbb{R}$. If $x_n = c$ for every $n \in \mathbb{N}$, then (x_n) is a null sequence if and only if $c = 0$.*

Go to proof.

Definition 0.166.8 (Ultimately Constant Sequence)**.** Let (x_n) be a real sequence. The sequence (x_n) is **ultimately constant** exactly when there exist $N \in \mathbb{N}$ and $c \in \mathbb{R}$ such that $x_n = c$ for every $n \geq N$.

Theorem 0.166.9 (Difference from the Limit is Null)**.** *Let (x_n) be a real sequence, and let $L \in \mathbb{R}$. The sequence (x_n) converges to L if and only if the sequence $(x_n - L)$ is a null sequence.*

Go to proof.

Theorem 0.166.10 (Ultimately Constant Sequence Convergence)**.** *Let (x_n) be a real sequence. If there exist $N_0 \in \mathbb{N}$ and $c \in \mathbb{R}$ such that $x_n = c$ for every $n \geq N_0$, then (x_n) converges to c.*

Go to proof.

Theorem 0.166.11 (Constant Implies Ultimately Constant)**.** *Every constant real sequence is ultimately constant.*

Go to proof.

Theorem 0.166.12 (Ultimately Zero Sequence is Null)**.** *Let (x_n) be a real sequence. If there exists $N_0 \in \mathbb{N}$ such that $x_n = 0$ for every $n \geq N_0$, then (x_n) is a null sequence.*

Go to proof.

Theorem 0.166.13 (Ultimately Constant Null Sequence)**.** *Let (x_n) be a real sequence. Suppose there exist $N_0 \in \mathbb{N}$ and $c \in \mathbb{R}$ such that $x_n = c$ for every $n \geq N_0$. Then (x_n) is a null sequence if and only if $c = 0$.*

Go to proof.

Theorem 0.166.14 (Tail Equality Preserves Convergence)**.** *Let (x_n) and (y_n) be real sequences. If there exists $N_0 \in \mathbb{N}$ such that $x_n = y_n$ for every $n \geq N_0$, then for every $L \in \mathbb{R}$, (x_n) converges to L if and only if (y_n) converges to L.*

Go to proof.

Definition 0.166.15 (Sequence Bounded Above)**.** Let (x_n) be a real sequence. The sequence (x_n) is **bounded above** exactly when there exists $M \in \mathbb{R}$ such that $x_n \leq M$ for every $n \in \mathbb{N}$.

Definition 0.166.16 (Sequence Bounded Below)**.** Let (x_n) be a real sequence. The sequence (x_n) is **bounded below** exactly when there exists $m \in \mathbb{R}$ such that $m \leq x_n$ for every $n \in \mathbb{N}$.

Definition 0.166.17 (Bounded Sequence)**.** Let (x_n) be a real sequence. The sequence (x_n) is **bounded** exactly when there exists $M > 0$ such that $|x_n| \leq M$ for every $n \in \mathbb{N}$.

Theorem 0.166.18 (Eventually Bounded Above Tail Formulation)**.** *Let (x_n) be a real sequence. The sequence (x_n) is bounded above if and only if there exist $N_0 \in \mathbb{N}$ and $M \in \mathbb{R}$ such that $x_n \leq M$ for every $n \geq N_0$.*

Go to proof.

Theorem 0.166.19 (Eventually Bounded Below Tail Formulation)**.** *Let (x_n) be a real sequence. The sequence (x_n) is bounded below if and only if there exist $N_0 \in \mathbb{N}$ and $m \in \mathbb{R}$ such that $m \leq x_n$ for every $n \geq N_0$.*

Go to proof.

Theorem 0.166.20 (Eventually Bounded Tail Formulation)**.** *Let (x_n) be a real sequence. The sequence (x_n) is bounded if and only if there exist $N_0 \in \mathbb{N}$ and $M > 0$ such that $|x_n| \leq M$ for every $n \geq N_0$.*

Go to proof.

Theorem 0.166.21 (Bounded Sequence If and Only If Bounded Above and Below)**.** *Let (x_n) be a real sequence. The sequence (x_n) is bounded if and only if it is both bounded above and bounded below.*

Go to proof.

Theorem 0.166.22 (Absolute Bound Gives Upper and Lower Bounds)**.** *Let (x_n) be a real sequence, and let $K > 0$. If $|x_n| \leq K$ for every $n \in \mathbb{N}$, then $-K \leq x_n \leq K$ for every $n \in \mathbb{N}$.*

Go to proof.

Theorem 0.166.23 (Upper and Lower Bounds Give an Absolute Bound)**.** *Let (x_n) be a real sequence. If there exist $m, M \in \mathbb{R}$ such that $m \leq x_n \leq M$ for every $n \in \mathbb{N}$, then there exists $K > 0$ such that $|x_n| \leq K$ for every $n \in \mathbb{N}$.*

Go to proof.

0.167 Examples and Counterexamples

Examples and Counterexamples

0.168 Convergence

Definition 0.168.1 (Convergence of a Sequence)**.** Let $(x_n) \subseteq \mathbb{R}$. The sequence (x_n) **converges** to $L \in \mathbb{R}$ if

$$\forall \varepsilon > 0\ \exists N \in \mathbb{N}\ \forall n \geq N\ \big(|x_n - L| < \varepsilon\big).$$

The value L is called the **limit** of (x_n), written $x_n \to L$ or $\lim_{n\to\infty} x_n = L$. A sequence that converges to some $L \in \mathbb{R}$ is called **convergent**; otherwise it is called **divergent**.

Definition 0.168.2 (Convergence — Neighborhood Formulation)**.** Let $(x_n) \subseteq \mathbb{R}$ and let $L \in \mathbb{R}$. The sequence (x_n) **converges** to L if for every $\varepsilon > 0$ the ε-neighborhood

$$V_\varepsilon(L)\ :=\ (L - \varepsilon,\ L + \varepsilon)$$

contains all but finitely many terms of (x_n); that is,

$$\forall \varepsilon > 0\ \exists N \in \mathbb{N}\ \forall n \geq N\ \big(x_n \in V_\varepsilon(L)\big).$$

Theorem 0.168.3 (Equivalence of Convergence Formulations)**.** *Let $(x_n) \subseteq \mathbb{R}$ and let $L \in \mathbb{R}$. The following statements are equivalent.*

(a) $x_n \to L$.

(b) $\forall \varepsilon > 0\ \exists K \in \mathbb{N}\ \forall n \geq K\ \big(|x_n - L| < \varepsilon\big)$.

(c) $\forall \varepsilon > 0\ \exists K \in \mathbb{N}\ \forall n \geq K\ \big(L - \varepsilon < x_n < L + \varepsilon\big)$.

(d) $\forall \varepsilon > 0\ \exists K \in \mathbb{N}\ \forall n \geq K\ \big(x_n \in V_\varepsilon(L)\big)$.

Go to proof.

0.168.1 Limits

Theorem 0.168.4 (Uniqueness of Limits)**.** *Let $(x_n) \subseteq \mathbb{R}$. If $x_n \to L$ and $x_n \to K$, then $L = K$.*

Go to proof.

Theorem 0.168.5 (Limit Preserves Eventual Order)**.** *Let (x_n) and (y_n) be real sequences, and let $L, M \in \mathbb{R}$. Suppose that $x_n \to L$ and $y_n \to M$. If there exists $N_0 \in \mathbb{N}$ such that $x_n \leq y_n$ for every $n \geq N_0$, then $L \leq M$.*

Go to proof.

Theorem 0.168.6 (Strict Limit Separation Gives Eventual Order)**.** *Let (x_n) and (y_n) be real sequences, and let $A, B \in \mathbb{R}$. Suppose that $x_n \to A$ and $y_n \to B$. If $A < B$, then there exists $N \in \mathbb{N}$ such that $x_n < y_n$ for every $n \geq N$.*

Go to proof.

Theorem 0.168.7 (Eventual Strict Comparison Preserves Weak Limit Order)**.** *Let (x_n) and (y_n) be real sequences, and let $A, B \in \mathbb{R}$. Suppose that $x_n \to A$ and $y_n \to B$.*

(a) If there exists $N \in \mathbb{N}$ such that $x_n < y_n$ for every $n \geq N$, then $A \leq B$.

(b) If there exists $N \in \mathbb{N}$ such that $x_n > y_n$ for every $n \geq N$, then $A \geq B$.

Go to proof.

Theorem 0.168.8 (Constant Comparison for Sequence Limits)**.** *Let (x_n) be a real sequence, and let $A, B \in \mathbb{R}$. Suppose that $x_n \to A$.*

(a) If there exists $N \in \mathbb{N}$ such that $x_n \le B$ for every $n \ge N$, then $A \le B$.

(b) If there exists $N \in \mathbb{N}$ such that $x_n < B$ for every $n \ge N$, then $A \le B$.

(c) If there exists $N \in \mathbb{N}$ such that $x_n \ge B$ for every $n \ge N$, then $A \ge B$.

(d) If there exists $N \in \mathbb{N}$ such that $x_n > B$ for every $n \ge N$, then $A \ge B$.

Go to proof.

Theorem 0.168.9 (Constant Squeeze Theorem)**.** *Let (x_n) be a real sequence, and let $L \in \mathbb{R}$. If there exists $N_0 \in \mathbb{N}$ such that $L \le x_n \le L$ for every $n \ge N_0$, then $x_n \to L$.*

Go to proof.

Theorem 0.168.10 (Sequence Squeeze Theorem)**.** *Let (a_n), (x_n), and (b_n) be real sequences, and let $L \in \mathbb{R}$. Suppose that $a_n \to L$ and $b_n \to L$. If there exists $N_0 \in \mathbb{N}$ such that $a_n \le x_n \le b_n$ for every $n \ge N_0$, then $x_n \to L$.*

Go to proof.

Theorem 0.168.11 (Absolute-Value Squeeze Theorem)**.** *Let (x_n) and (u_n) be real sequences, and let $L \in \mathbb{R}$. Suppose that $u_n \to 0$. If there exists $N_0 \in \mathbb{N}$ such that $|x_n - L| \le u_n$ for every $n \ge N_0$, then $x_n \to L$.*

Go to proof.

0.168.2 Algebra of Limits

Algebra of Convergent Sequences

Definition 0.168.12 (Pointwise Sum of Sequences)**.** Let (x_n) and (y_n) be real sequences. The **pointwise sum** of (x_n) and (y_n) is the real sequence $(x_n + y_n)$ defined by

$$(x + y)_n := x_n + y_n \qquad \text{for every } n \in \mathbb{N}.$$

Definition 0.168.13 (Pointwise Difference of Sequences)**.** Let (x_n) and (y_n) be real sequences. The **pointwise difference** of (x_n) and (y_n) is the real sequence $(x_n - y_n)$ defined by

$$(x - y)_n := x_n - y_n \qquad \text{for every } n \in \mathbb{N}.$$

Definition 0.168.14 (Scalar Multiple of a Sequence)**.** Let (x_n) be a real sequence, and let $\alpha \in \mathbb{R}$. The **scalar multiple** $\alpha(x_n)$ is the real sequence (αx_n) defined by

$$(\alpha x)_n := \alpha x_n \qquad \text{for every } n \in \mathbb{N}.$$

Definition 0.168.15 (Linear Combination of Real Sequences)**.** Let (x_n) and (y_n) be real sequences, and let $\alpha, \beta \in \mathbb{R}$. The real linear combination $\alpha(x_n) + \beta(y_n)$ is the real sequence $(\alpha x_n + \beta y_n)$ defined by

$$(\alpha x + \beta y)_n := \alpha x_n + \beta y_n \qquad \text{for every } n \in \mathbb{N}.$$

Definition 0.168.16 (Pointwise Negation of a Sequence)**.** Let (x_n) be a real sequence. The **pointwise negation** of (x_n) is the real sequence $(-x_n)$ defined by

$$(-x)_n := -x_n \qquad \text{for every } n \in \mathbb{N}.$$

Definition 0.168.17 (Pointwise Product of Sequences)**.** Let (x_n) and (y_n) be real sequences. The **pointwise product** of (x_n) and (y_n) is the real sequence $(x_n y_n)$ defined by

$$(xy)_n := x_n y_n \qquad \text{for every } n \in \mathbb{N}.$$

Definition 0.168.18 (Reciprocal Sequence). Let (x_n) be a real sequence such that $x_n \neq 0$ for every $n \in \mathbb{N}$. The **reciprocal sequence** of (x_n) is the real sequence $(1/x_n)$ defined by

$$\left(\frac{1}{x}\right)_n := \frac{1}{x_n} \qquad \text{for every } n \in \mathbb{N}.$$

Definition 0.168.19 (Pointwise Quotient of Sequences). Let (x_n) and (y_n) be real sequences such that $y_n \neq 0$ for every $n \in \mathbb{N}$. The **pointwise quotient** of (x_n) by (y_n) is the real sequence (x_n/y_n) defined by

$$\left(\frac{x}{y}\right)_n := \frac{x_n}{y_n} \qquad \text{for every } n \in \mathbb{N}.$$

Definition 0.168.20 (Square Sequence). Let (x_n) be a real sequence. The **square sequence** of (x_n) is the real sequence (x_n^2) defined by

$$(x^2)_n := x_n^2 \qquad \text{for every } n \in \mathbb{N}.$$

Definition 0.168.21 (Absolute Value Sequence). Let (x_n) be a real sequence. The **absolute value sequence** of (x_n) is the real sequence $(|x_n|)$ defined by

$$(|x|)_n := |x_n| \qquad \text{for every } n \in \mathbb{N}.$$

Definition 0.168.22 (Square Root Sequence). Let (x_n) be a real sequence such that $0 \leq x_n$ for every $n \in \mathbb{N}$. The **square root sequence** of (x_n) is the real sequence $(\sqrt{x_n})$ defined by

$$(\sqrt{x})_n := \sqrt{x_n} \qquad \text{for every } n \in \mathbb{N}.$$

Theorem 0.168.23 (Limit of a Scalar Multiple). *Let (x_n) be a real sequence, let $L \in \mathbb{R}$, and let $\alpha \in \mathbb{R}$. If $x_n \to L$, then $\alpha x_n \to \alpha L$.*

Go to proof.

Theorem 0.168.24 (Limit of a Sum). *Let (x_n) and (y_n) be real sequences, and let $L, M \in \mathbb{R}$. If $x_n \to L$ and $y_n \to M$, then $x_n + y_n \to L + M$.*

Go to proof.

Theorem 0.168.25 (Limit of a Negation). *Let (x_n) be a real sequence, and let $L \in \mathbb{R}$. If $x_n \to L$, then $-x_n \to -L$.*

Go to proof.

Theorem 0.168.26 (Limit of a Difference). *Let (x_n) and (y_n) be real sequences, and let $L, M \in \mathbb{R}$. If $x_n \to L$ and $y_n \to M$, then $x_n - y_n \to L - M$.*

Go to proof.

Theorem 0.168.27 (Limit of a Product). *Let (x_n) and (y_n) be real sequences, and let $L, M \in \mathbb{R}$. If $x_n \to L$ and $y_n \to M$, then $x_n y_n \to LM$.*

Go to proof.

Theorem 0.168.28 (Nonzero Limit is Eventually Nonzero). *Let (x_n) be a real sequence, and let $L \in \mathbb{R}$ with $L \neq 0$. If $x_n \to L$, then there exists $N \in \mathbb{N}$ such that $x_n \neq 0$ for every $n \geq N$.*

Go to proof.

Theorem 0.168.29 (Limit of a Reciprocal). *Let (x_n) be a real sequence such that $x_n \neq 0$ for every $n \in \mathbb{N}$, and let $L \in \mathbb{R}$ with $L \neq 0$. If $x_n \to L$, then $1/x_n \to 1/L$.*

Go to proof.

Theorem 0.168.30 (Limit of a Quotient). *Let (x_n) and (y_n) be real sequences such that $y_n \neq 0$ for every $n \in \mathbb{N}$, and let $L, M \in \mathbb{R}$ with $M \neq 0$. If $x_n \to L$ and $y_n \to M$, then $x_n/y_n \to L/M$.*

Go to proof.

Theorem 0.168.31 (Limit of a Square)**.** *Let (x_n) be a real sequence, and let $L \in \mathbb{R}$. If $x_n \to L$, then $x_n^2 \to L^2$.*

Go to proof.

Theorem 0.168.32 (Limit of an Absolute Value)**.** *Let (x_n) be a real sequence, and let $L \in \mathbb{R}$. If $x_n \to L$, then $|x_n| \to |L|$.*

Go to proof.

Theorem 0.168.33 (Positive Limit is Eventually Positive)**.** *Let (x_n) be a real sequence, and let $L > 0$. If $x_n \to L$, then there exists $N \in \mathbb{N}$ such that $0 < x_n$ for every $n \geq N$.*

Go to proof.

Theorem 0.168.34 (Limit of a Square Root)**.** *Let (x_n) be a real sequence such that $0 \leq x_n$ for every $n \in \mathbb{N}$, and let $L \in \mathbb{R}$. If $x_n \to L$, then $0 \leq L$ and $\sqrt{x_n} \to \sqrt{L}$.*

Go to proof.

Theorem 0.168.35 (Polynomial Sequence Limit)**.** *Let (x_n) be a real sequence, let $L \in \mathbb{R}$, and let $p \in \mathbb{R}[t]$ be a polynomial. If $x_n \to L$, then $p(x_n) \to p(L)$.*

Go to proof.

Theorem 0.168.36 (Rational Sequence Limit)**.** *Let (x_n) be a real sequence, let $L \in \mathbb{R}$, and let $p, q \in \mathbb{R}[t]$ be polynomials. Suppose $q(L) \neq 0$ and $q(x_n) \neq 0$ for every $n \in \mathbb{N}$. If $x_n \to L$, then*

$$\frac{p(x_n)}{q(x_n)} \to \frac{p(L)}{q(L)}.$$

Go to proof.

0.169 Tails

Definition 0.169.1 (M-Tail of a Sequence)**.** Let $(x_n) \subseteq \mathbb{R}$ and let $M \in \mathbb{N}$. The M**-tail** of (x_n) is the sequence

$$(x_n)_{n \geq M} \;:=\; \big(x_M,\, x_{M+1},\, x_{M+2},\, \ldots\big).$$

Theorem 0.169.2 (Convergence of a Tail)**.** *Let $(x_n) \subseteq \mathbb{R}$ and let $m \in \mathbb{N}$. The m-tail $(x_{m+n})_{n \in \mathbb{N}}$ converges if and only if (x_n) converges. Moreover, in this case,*

$$\lim_{n \to \infty} x_{m+n} \;=\; \lim_{n \to \infty} x_n.$$

Go to proof.

Theorem 0.169.3 (Convergence by Domination)**.** *Let $(x_n) \subseteq \mathbb{R}$ and let $L \in \mathbb{R}$. Let (a_n) be a sequence of positive real numbers with $\lim_{n \to \infty} a_n = 0$. If there exist a constant $c > 0$ and an index $m \in \mathbb{N}$ such that*

$$|x_n - L| \;\leq\; c\, a_n \qquad \forall n \geq m,$$

then $\lim_{n \to \infty} x_n = L$.

Go to proof.

Theorem 0.169.4 (Ratio Limit Less Than One Implies Null)**.** *Let (x_n) be a sequence of positive real numbers. Suppose that the limit*

$$L := \lim_{n \to \infty} \frac{x_{n+1}}{x_n}$$

exists. If $L < 1$, then (x_n) converges and

$$\lim_{n \to \infty} x_n = 0.$$

Go to proof.

0.170 Monotonicity

Monotonicity of Sequences

Definition 0.170.1 (Increasing Sequence). Let (x_n) be a real sequence. The sequence (x_n) is **increasing** if

$$x_n \leq x_{n+1} \qquad \text{for every } n \in \mathbb{N}.$$

Definition 0.170.2 (Decreasing Sequence). Let (x_n) be a real sequence. The sequence (x_n) is **decreasing** if

$$x_{n+1} \leq x_n \qquad \text{for every } n \in \mathbb{N}.$$

Definition 0.170.3 (Strictly Increasing Sequence). Let (x_n) be a real sequence. The sequence (x_n) is **strictly increasing** if

$$x_n < x_{n+1} \qquad \text{for every } n \in \mathbb{N}.$$

Definition 0.170.4 (Strictly Decreasing Sequence). Let (x_n) be a real sequence. The sequence (x_n) is **strictly decreasing** if

$$x_{n+1} < x_n \qquad \text{for every } n \in \mathbb{N}.$$

Definition 0.170.5 (Monotone Sequence). Let (x_n) be a real sequence. The sequence (x_n) is **monotone** if it is increasing or decreasing.

Definition 0.170.6 (Eventually Increasing Sequence). Let (x_n) be a real sequence. The sequence (x_n) is **eventually increasing** if there exists $N \in \mathbb{N}$ such that

$$x_n \leq x_{n+1} \qquad \text{for every } n \geq N.$$

Definition 0.170.7 (Eventually Decreasing Sequence). Let (x_n) be a real sequence. The sequence (x_n) is **eventually decreasing** if there exists $N \in \mathbb{N}$ such that

$$x_{n+1} \leq x_n \qquad \text{for every } n \geq N.$$

Definition 0.170.8 (Eventually Monotone Sequence). Let (x_n) be a real sequence. The sequence (x_n) is **eventually monotone** if it is eventually increasing or eventually decreasing.

Theorem 0.170.9 (Monotone Convergence Theorem). *Let (x_n) be a real sequence.*

(a) If (x_n) is increasing and bounded above, then (x_n) converges and

$$\lim_{n\to\infty} x_n = \sup\{x_n : n \in \mathbb{N}\}.$$

(b) If (x_n) is decreasing and bounded below, then (x_n) converges and

$$\lim_{n\to\infty} x_n = \inf\{x_n : n \in \mathbb{N}\}.$$

Go to proof.

Theorem 0.170.10 (Strict Monotonicity Implies Monotonicity). *Let (x_n) be a real sequence.*

(a) If (x_n) is strictly increasing, then (x_n) is increasing.

(b) If (x_n) is strictly decreasing, then (x_n) is decreasing.

Go to proof.

Theorem 0.170.11 (Bounded Monotone Sequence Equivalences). *Let (x_n) be a real sequence.*

(a) If (x_n) is increasing, then (x_n) converges if and only if (x_n) is bounded above.

(b) If (x_n) is decreasing, then (x_n) converges if and only if (x_n) is bounded below.

(c) If (x_n) is monotone, then (x_n) converges if and only if (x_n) is bounded.

Go to proof.

Theorem 0.170.12 (Eventually Monotone Convergence Theorem)**.** *Let (x_n) be a real sequence.*

(a) If (x_n) is eventually increasing and bounded above, then (x_n) converges.

(b) If (x_n) is eventually decreasing and bounded below, then (x_n) converges.

(c) If (x_n) is eventually monotone and bounded, then (x_n) converges.

Go to proof.

Theorem 0.170.13 (Unbounded Monotone Divergence)**.** *Let (x_n) be a real sequence.*

(a) If (x_n) is increasing and is not bounded above, then $x_n \to +\infty$.

(b) If (x_n) is decreasing and is not bounded below, then $x_n \to -\infty$.

Go to proof.

Theorem 0.170.14 (Algebraic Transformations Preserve Monotonicity)**.** *Let (x_n) be a real sequence and let $c, \alpha \in \mathbb{R}$.*

(a) If (x_n) is increasing, then $(x_n + c)$ is increasing.

(b) If (x_n) is decreasing, then $(x_n + c)$ is decreasing.

(c) If (x_n) is increasing and $\alpha > 0$, then (αx_n) is increasing.

(d) If (x_n) is increasing and $\alpha < 0$, then (αx_n) is decreasing.

(e) If (x_n) is increasing, then $(-x_n)$ is decreasing.

(f) If (x_n) is decreasing, then $(-x_n)$ is increasing.

Go to proof.

0.171 Subsequences

Subsequences

Definition 0.171.1 (Strictly Increasing Index Map)**.** Let $\sigma : \mathbb{N} \to \mathbb{N}$ be a function. The function σ is a **strictly increasing index map** if

$$k < \ell \Longrightarrow \sigma(k) < \sigma(\ell) \qquad \text{for all } k, \ell \in \mathbb{N}.$$

Definition 0.171.2 (Subsequence)**.** Let (x_n) be a real sequence. A **subsequence** of (x_n) is a sequence $(x_{\sigma(k)})_{k \in \mathbb{N}}$, where $\sigma : \mathbb{N} \to \mathbb{N}$ is a strictly increasing index map.

Definition 0.171.3 (Subsequential Limit)**.** Let (x_n) be a real sequence and let $L \in \mathbb{R}$. The real number L is a **subsequential limit** of (x_n) if there exists a subsequence $(x_{\sigma(k)})$ such that

$$\lim_{k \to \infty} x_{\sigma(k)} = L.$$

Definition 0.171.4 (Has Convergent Subsequence). Let (x_n) be a real sequence. The sequence (x_n) **has a convergent subsequence** if there exist a strictly increasing index map $\sigma : \mathbb{N} \to \mathbb{N}$ and a real number L such that

$$\lim_{k\to\infty} x_{\sigma(k)} = L.$$

Theorem 0.171.5 (Subsequence Indices Dominate the Identity). *Let $\sigma : \mathbb{N} \to \mathbb{N}$ be a strictly increasing index map. Then*

$$k \leq \sigma(k) \qquad \text{for every } k \in \mathbb{N}.$$

Go to proof.

Theorem 0.171.6 (Subsequences Preserve Limits). *Let (x_n) be a real sequence and let $L \in \mathbb{R}$. If $x_n \to L$, then every subsequence of (x_n) converges to L.*

Go to proof.

Theorem 0.171.7 (Subsequential Limit of a Convergent Sequence). *Let (x_n) be a real sequence and let $L \in \mathbb{R}$. If $x_n \to L$, then L is the only possible subsequential limit of (x_n).*

Go to proof.

Theorem 0.171.8 (Divergence by Two Subsequential Limits). *Let (x_n) be a real sequence. If (x_n) has two distinct subsequential limits, then (x_n) does not converge.*

Go to proof.

Theorem 0.171.9 (Boundedness Passes to Subsequences). *Every subsequence of a bounded real sequence is bounded.*

Go to proof.

Theorem 0.171.10 (Monotonicity Passes to Subsequences). *Every subsequence of an increasing real sequence is increasing, and every subsequence of a decreasing real sequence is decreasing.*

Go to proof.

Theorem 0.171.11 (Subsequence of a Subsequence). *Every subsequence of a subsequence of (x_n) is a subsequence of (x_n).*

Go to proof.

Theorem 0.171.12 (Eventual Properties Pass to Subsequences). *Let $P(n)$ be a property of indices. If there exists $N \in \mathbb{N}$ such that $P(n)$ holds for every $n \geq N$, then for every strictly increasing index map σ there exists $K \in \mathbb{N}$ such that $P(\sigma(k))$ holds for every $k \geq K$.*

Go to proof.

Theorem 0.171.13 (Frequent Properties Yield Subsequences). *Let $P(n)$ be a property of indices. If for every $N \in \mathbb{N}$ there exists $n \geq N$ such that $P(n)$ holds, then there exists a strictly increasing index map $\sigma : \mathbb{N} \to \mathbb{N}$ such that $P(\sigma(k))$ holds for every $k \in \mathbb{N}$.*

Go to proof.

Theorem 0.171.14 (Subsequential Limits Respect Bounds). *Let (x_n) be a real sequence and let L be a subsequential limit of (x_n). If $m \leq x_n \leq M$ for every $n \in \mathbb{N}$, then $m \leq L \leq M$.*

Go to proof.

Theorem 0.171.15 (Squeeze Passes to Subsequences). *Let (a_n), (x_n), and (b_n) be real sequences. If $a_n \leq x_n \leq b_n$ eventually, then for every subsequence $(x_{\sigma(k)})$ there exists $K \in \mathbb{N}$ such that*

$$a_{\sigma(k)} \leq x_{\sigma(k)} \leq b_{\sigma(k)} \qquad \text{for every } k \geq K.$$

Go to proof.

Theorem 0.171.16 (Monotone Subsequence Theorem)**.** *Every real sequence has a monotone subsequence.*

Go to proof.

Theorem 0.171.17 (Bolzano-Weierstrass Theorem for Sequences)**.** *Every bounded real sequence has a convergent subsequence.*

Go to proof.

Theorem 0.171.18 (Sequential Compactness of Closed Bounded Intervals)**.** *Let $a, b \in \mathbb{R}$ with $a \leq b$. Every sequence in $[a, b]$ has a convergent subsequence whose limit belongs to $[a, b]$.*

Go to proof.

0.172 Divergence

Divergence

Definition 0.172.1 (Divergent Sequence)**.** Let (x_n) be a real sequence. The sequence (x_n) is **divergent** if there is no $L \in \mathbb{R}$ such that $x_n \to L$.

Definition 0.172.2 (Divergence to Positive Infinity)**.** Let (x_n) be a real sequence. The sequence (x_n) **diverges to** $+\infty$ if

$$\forall M \in \mathbb{R}\, \exists N \in \mathbb{N}\, \forall n \geq N\, \big(M < x_n\big).$$

Definition 0.172.3 (Divergence to Negative Infinity)**.** Let (x_n) be a real sequence. The sequence (x_n) **diverges to** $-\infty$ if

$$\forall M \in \mathbb{R}\, \exists N \in \mathbb{N}\, \forall n \geq N\, \big(x_n < M\big).$$

Definition 0.172.4 (Oscillatory Sequence)**.** Let (x_n) be a real sequence. The sequence (x_n) is **oscillatory** if it is divergent, does not diverge to $+\infty$, and does not diverge to $-\infty$.

Theorem 0.172.5 (Divergence to Infinity Implies Real Divergence)**.** *Let (x_n) be a real sequence. If $x_n \to +\infty$ or $x_n \to -\infty$, then (x_n) is divergent.*

Go to proof.

Theorem 0.172.6 (Two Subsequential Limits Force Divergence)**.** *Let (x_n) be a real sequence. If (x_n) has two subsequential limits $L, K \in \mathbb{R}$ with $L \neq K$, then (x_n) is divergent.*

Go to proof.

Theorem 0.172.7 (Unbounded Above Sequence Has a Positive-Infinity Subsequence)**.** *Let (x_n) be a real sequence. If (x_n) is not bounded above, then there exists a subsequence $(x_{\sigma(k)})$ such that $x_{\sigma(k)} \to +\infty$.*

Go to proof.

Theorem 0.172.8 (Unbounded Below Sequence Has a Negative-Infinity Subsequence)**.** *Let (x_n) be a real sequence. If (x_n) is not bounded below, then there exists a subsequence $(x_{\sigma(k)})$ such that $x_{\sigma(k)} \to -\infty$.*

Go to proof.

Theorem 0.172.9 (Bounded Divergence Produces Two Subsequential Limits)**.** *Let (x_n) be a bounded real sequence. If (x_n) is divergent, then there exist $L, K \in \mathbb{R}$ with $L \neq K$ such that L and K are subsequential limits of (x_n).*

Go to proof.

0.173 Liminf and Limsup

Liminf and Limsup

Definition 0.173.1 (Tail Supremum Sequence)**.** Let (x_n) be a bounded real sequence. The **tail supremum sequence** of (x_n) is the real sequence (s_n) defined by

$$s_n := \sup\{x_k : k \geq n\} \qquad \text{for every } n \in \mathbb{N}.$$

Definition 0.173.2 (Tail Infimum Sequence)**.** Let (x_n) be a bounded real sequence. The **tail infimum sequence** of (x_n) is the real sequence (i_n) defined by

$$i_n := \inf\{x_k : k \geq n\} \qquad \text{for every } n \in \mathbb{N}.$$

Definition 0.173.3 (Limit Superior of a Sequence)**.** Let (x_n) be a bounded real sequence, and let (s_n) be its tail supremum sequence. The **limit superior** of (x_n) is

$$\limsup_{n\to\infty} x_n := \lim_{n\to\infty} s_n.$$

Definition 0.173.4 (Limit Inferior of a Sequence)**.** Let (x_n) be a bounded real sequence, and let (i_n) be its tail infimum sequence. The **limit inferior** of (x_n) is

$$\liminf_{n\to\infty} x_n := \lim_{n\to\infty} i_n.$$

Theorem 0.173.5 (Tail Suprema Are Decreasing)**.** *Let (x_n) be a bounded real sequence, and let (s_n) be its tail supremum sequence. Then (s_n) is decreasing.*

Go to proof.

Theorem 0.173.6 (Tail Infima Are Increasing)**.** *Let (x_n) be a bounded real sequence, and let (i_n) be its tail infimum sequence. Then (i_n) is increasing.*

Go to proof.

Theorem 0.173.7 (Liminf Is Below Limsup)**.** *Let (x_n) be a bounded real sequence. Then*

$$\liminf_{n\to\infty} x_n \leq \limsup_{n\to\infty} x_n.$$

Go to proof.

Theorem 0.173.8 (Convergence iff Liminf Equals Limsup)**.** *Let (x_n) be a bounded real sequence, and let $L \in \mathbb{R}$. Then $x_n \to L$ if and only if*

$$\liminf_{n\to\infty} x_n = \limsup_{n\to\infty} x_n = L.$$

Go to proof.

Theorem 0.173.9 (Limsup Is the Largest Subsequential Limit)**.** *Let (x_n) be a bounded real sequence, and let $S = \limsup_{n\to\infty} x_n$. Then S is a subsequential limit of (x_n), and every subsequential limit of (x_n) is less than or equal to S.*

Go to proof.

Theorem 0.173.10 (Liminf Is the Smallest Subsequential Limit)**.** *Let (x_n) be a bounded real sequence, and let $I = \liminf_{n\to\infty} x_n$. Then I is a subsequential limit of (x_n), and every subsequential limit of (x_n) is greater than or equal to I.*

Go to proof.

Theorem 0.173.11 (Oscillation Criterion via Liminf and Limsup). *Let (x_n) be a bounded real sequence. Then*

$$\liminf_{n\to\infty} x_n < \limsup_{n\to\infty} x_n$$

if and only if (x_n) has two distinct subsequential limits.

Go to proof.

Theorem 0.173.12 (Limsup Comparison Under Eventual Order). *Let (x_n) and (y_n) be bounded real sequences. If there exists $N \in \mathbb{N}$ such that $x_n \le y_n$ for every $n \ge N$, then*

$$\limsup_{n\to\infty} x_n \le \limsup_{n\to\infty} y_n.$$

Go to proof.

Theorem 0.173.13 (Liminf Comparison Under Eventual Order). *Let (x_n) and (y_n) be bounded real sequences. If there exists $N \in \mathbb{N}$ such that $x_n \le y_n$ for every $n \ge N$, then*

$$\liminf_{n\to\infty} x_n \le \liminf_{n\to\infty} y_n.$$

Go to proof.

Theorem 0.173.14 (Limsup Squeeze Under Eventual Order). *Let (a_n), (x_n), and (b_n) be bounded real sequences. If there exists $N \in \mathbb{N}$ such that*

$$a_n \le x_n \le b_n \qquad \text{for every } n \ge N,$$

then

$$\limsup_{n\to\infty} a_n \le \limsup_{n\to\infty} x_n \le \limsup_{n\to\infty} b_n.$$

Go to proof.

Theorem 0.173.15 (Liminf Squeeze Under Eventual Order). *Let (a_n), (x_n), and (b_n) be bounded real sequences. If there exists $N \in \mathbb{N}$ such that*

$$a_n \le x_n \le b_n \qquad \text{for every } n \ge N,$$

then

$$\liminf_{n\to\infty} a_n \le \liminf_{n\to\infty} x_n \le \liminf_{n\to\infty} b_n.$$

Go to proof.

0.174 Order-Theoretic Limit Constructions

Order-Theoretic Limit Constructions

Theorem 0.174.1 (Increasing Sequence Limit as Supremum). *Let (x_n) be an increasing real sequence that is bounded above. Then*

$$\lim_{n\to\infty} x_n = \sup\{x_n : n \in \mathbb{N}\}.$$

Go to proof.

Theorem 0.174.2 (Decreasing Sequence Limit as Infimum). *Let (x_n) be a decreasing real sequence that is bounded below. Then*

$$\lim_{n\to\infty} x_n = \inf\{x_n : n \in \mathbb{N}\}.$$

Go to proof.

Theorem 0.174.3 (Tail Suprema and Infima Converge)**.** *Let (x_n) be a bounded real sequence. Define*

$$s_n = \sup\{x_k : k \geq n\}, \qquad i_n = \inf\{x_k : k \geq n\}.$$

Then (s_n) and (i_n) converge.

Go to proof.

Theorem 0.174.4 (Bounded Sequence Has Limsup and Liminf)**.** *Every bounded real sequence has a limit superior and a limit inferior.*

Go to proof.

0.175 Cluster Values of Sequences

Cluster Values of Sequences

Definition 0.175.1 (Cluster Value of a Sequence)**.** Let (x_n) be a real sequence, and let $L \in \mathbb{R}$. The number L is a **cluster value** of (x_n) if for every $\varepsilon > 0$ and every $N \in \mathbb{N}$ there exists $n \geq N$ such that

$$|x_n - L| < \varepsilon.$$

Theorem 0.175.2 (Cluster Values Are Subsequential Limits)**.** *Let (x_n) be a real sequence, and let $L \in \mathbb{R}$. Then L is a cluster value of (x_n) if and only if L is a subsequential limit of (x_n).*

Go to proof.

Theorem 0.175.3 (Bounded Sequences Have Cluster Values)**.** *Every bounded real sequence has at least one cluster value.*

Go to proof.

Theorem 0.175.4 (Limsup and Liminf Are Extremal Cluster Values)**.** *Let (x_n) be a bounded real sequence. Then $\limsup_{n\to\infty} x_n$ is the largest cluster value of (x_n), and $\liminf_{n\to\infty} x_n$ is the smallest cluster value of (x_n).*

Go to proof.

0.176 Cauchy

Cauchy Sequences

Definition 0.176.1 (Cauchy Sequence)**.** Let (x_n) be a real sequence. The sequence (x_n) is a **Cauchy sequence** if

$$\forall \varepsilon > 0 \; \exists N \in \mathbb{N} \; \forall m, n \geq N \; \big(|x_m - x_n| < \varepsilon\big).$$

Theorem 0.176.2 (Convergent Sequences Are Cauchy)**.** *Let (x_n) be a real sequence. If (x_n) converges, then (x_n) is Cauchy.*

Go to proof.

Theorem 0.176.3 (Cauchy Sequences Are Bounded)**.** *Every Cauchy real sequence is bounded.*

Go to proof.

Theorem 0.176.4 (Cauchy Sequence with a Convergent Subsequence)**.** *Let (x_n) be a Cauchy real sequence. If a subsequence of (x_n) converges to $L \in \mathbb{R}$, then (x_n) converges to L.*

Go to proof.

Theorem 0.176.5 (Cauchy Criterion for Real Sequences). *Let (x_n) be a real sequence. Then (x_n) converges if and only if (x_n) is Cauchy.*

Go to proof.

Theorem 0.176.6 (Cauchy Criterion via Tails). *Let (x_n) be a real sequence. Then (x_n) is Cauchy if and only if for every $\varepsilon > 0$ there exists $N \in \mathbb{N}$ such that the N-tail has pairwise term distances less than ε.*

Go to proof.

Theorem 0.176.7 (Cauchy Tail Diameter Criterion). *Let (x_n) be a real sequence. Then (x_n) is Cauchy if and only if every sufficiently late tail has arbitrarily small pairwise diameter.*

Go to proof.

Theorem 0.176.8 (Cauchy Sequences Have Vanishing Successive Differences). *If (x_n) is a Cauchy real sequence, then*

$$|x_{n+1} - x_n| \to 0.$$

Go to proof.

Theorem 0.176.9 (Scalar Multiples of Cauchy Sequences). *Let (x_n) be a Cauchy real sequence, and let $\alpha \in \mathbb{R}$. Then (αx_n) is Cauchy.*

Go to proof.

Theorem 0.176.10 (Sums of Cauchy Sequences). *Let (x_n) and (y_n) be Cauchy real sequences. Then (x_n+y_n) is Cauchy.*

Go to proof.

Theorem 0.176.11 (Differences of Cauchy Sequences). *Let (x_n) and (y_n) be Cauchy real sequences. Then $(x_n - y_n)$ is Cauchy.*

Go to proof.

Theorem 0.176.12 (Linear Combinations of Cauchy Sequences). *Let (x_n) and (y_n) be Cauchy real sequences, and let $\alpha, \beta \in \mathbb{R}$. Then $(\alpha x_n + \beta y_n)$ is Cauchy.*

Go to proof.

Theorem 0.176.13 (Products of Cauchy Sequences). *Let (x_n) and (y_n) be Cauchy real sequences. Then $(x_n y_n)$ is Cauchy.*

Go to proof.

Theorem 0.176.14 (Reciprocals of Cauchy Sequences). *Let (x_n) be a Cauchy real sequence. If there exist $c > 0$ and $N_0 \in \mathbb{N}$ such that $|x_n| \geq c$ for every $n \geq N_0$, then $(1/x_n)$ is Cauchy.*

Go to proof.

Theorem 0.176.15 (Quotients of Cauchy Sequences). *Let (x_n) and (y_n) be Cauchy real sequences. If there exist $c > 0$ and $N_0 \in \mathbb{N}$ such that $|y_n| \geq c$ for every $n \geq N_0$, then (x_n/y_n) is Cauchy.*

Go to proof.

Theorem 0.176.16 (Absolute Values of Cauchy Sequences). *Let (x_n) be a Cauchy real sequence. Then $(|x_n|)$ is Cauchy.*

Go to proof.

0.177 Applications of Sequences

Applications of Sequences

Definition 0.177.1 (Newton Sequence for $\sqrt{2}$)**.** The **Newton sequence for** $\sqrt{2}$ is the real sequence (x_n) defined by

$$x_1 := \frac{3}{2}, \qquad x_{n+1} := \frac{1}{2}\left(x_n + \frac{2}{x_n}\right) \quad \text{for every } n \in \mathbb{N}.$$

Theorem 0.177.2 (Newton Approximation of $\sqrt{2}$)**.** *Let (x_n) be the Newton sequence for $\sqrt{2}$. Then (x_n) converges and*

$$\lim_{n\to\infty} x_n = \sqrt{2}.$$

Go to proof.

Definition 0.177.3 (Factorial Partial Sums)**.** The **factorial partial-sum sequence** is the real sequence (s_n) defined by

$$s_n := \sum_{k=0}^{n} \frac{1}{k!} \qquad \text{for every } n \in \mathbb{N}.$$

Theorem 0.177.4 (Factorial Partial Sums Approximate e)**.** *Let (s_n) be the factorial partial-sum sequence. Then (s_n) converges to the Euler number e:*

$$\lim_{n\to\infty} s_n = e.$$

Go to proof.

Definition 0.177.5 (Compound-Interest Sequence)**.** The **compound-interest sequence** is the real sequence (c_n) defined by

$$c_n := \left(1 + \frac{1}{n}\right)^n \qquad \text{for every } n \in \mathbb{N}.$$

Theorem 0.177.6 (Compound-Interest Approximation of e)**.** *Let (c_n) be the compound-interest sequence. Then*

$$\lim_{n\to\infty} c_n = e.$$

Go to proof.

Definition 0.177.7 (Decimal Truncation Sequence)**.** Let $\alpha \in \mathbb{R}$. The **decimal truncation sequence** of α is the real sequence (d_n) defined by

$$d_n := \frac{\lfloor 10^n \alpha \rfloor}{10^n} \qquad \text{for every } n \in \mathbb{N}.$$

Theorem 0.177.8 (Decimal Truncations Converge)**.** *Let $\alpha \in \mathbb{R}$, and let (d_n) be the decimal truncation sequence of α. Then*

$$\lim_{n\to\infty} d_n = \alpha.$$

Go to proof.

Proofs

Capstone

Series

0.178 Finite Series

Definition 0.178.1 (Finite Real Sequence on an Integer Interval)**.** Let $m, n \in \mathbb{Z}$ with $m \leq n$. A finite real sequence indexed from m to n is an assignment of a real number a_i to each integer i satisfying $m \leq i \leq n$. We denote such a sequence by $(a_i)_{i=m}^n$.

Definition 0.178.2 (Empty Finite Sum)**.** Let $m, n \in \mathbb{Z}$. If $n < m$, the finite sum from m to n is defined by

$$\sum_{i=m}^{n} a_i := 0.$$

Definition 0.178.3 (Finite Sum)**.** Let $m, n \in \mathbb{Z}$, and let real numbers a_i be assigned for the relevant integer indices. The notation

$$\sum_{i=m}^{n} a_i$$

denotes the finite sum, or finite series, of the terms indexed from m to n.

Definition 0.178.4 (Recursive Finite Sum)**.** Let $m \in \mathbb{Z}$, and let real numbers a_i be assigned for the relevant indices. The finite sum, also called a finite series, is defined recursively from the empty finite sum by the rule

$$\sum_{i=m}^{n+1} a_i := \left(\sum_{i=m}^{n} a_i \right) + a_{n+1} \qquad \text{whenever } n \geq m - 1.$$

Lemma 0.178.5 (Splitting a Finite Sum)**.** *Let $m \leq n < p$ be integers, and let a_i be a real number assigned to each integer i satisfying $m \leq i \leq p$. Then*

$$\sum_{i=m}^{n} a_i + \sum_{i=n+1}^{p} a_i = \sum_{i=m}^{p} a_i.$$

Go to proof.

Definition 0.178.6 (Summation over a Finite Set)**.** Let X be a finite set with n elements, where $n \in \mathbb{N}$, and let $f : X \to \mathbb{R}$. Choose a bijection

$$g : \{i \in \mathbb{N} : 1 \leq i \leq n\} \to X.$$

The finite sum of f over X is defined by

$$\sum_{x \in X} f(x) := \sum_{i=1}^{n} f(g(i)).$$

Proposition 0.178.7 (Finite Summations Are Well-Defined)**.** *Let X be a finite set with n elements, where $n \in \mathbb{N}$, let $f : X \to \mathbb{R}$, and let*

$$g : \{i \in \mathbb{N} : 1 \leq i \leq n\} \to X \qquad \text{and} \qquad h : \{i \in \mathbb{N} : 1 \leq i \leq n\} \to X$$

be bijections. Then

$$\sum_{i=1}^{n} f(g(i)) = \sum_{i=1}^{n} f(h(i)).$$

Go to proof.

Proposition 0.178.8 (Empty Finite Set Summation)**.** *If X is empty, and $f : X \to \mathbb{R}$ is a function, then*

$$\sum_{x \in X} f(x) = 0.$$

Go to proof.

Proposition 0.178.9 (Singleton Finite Set Summation)**.** *If X consists of a single element, $X = \{x_0\}$, and $f : X \to \mathbb{R}$ is a function, then*

$$\sum_{x \in X} f(x) = f(x_0).$$

Go to proof.

Proposition 0.178.10 (Substitution for Finite Set Summation)**.** *If X is a finite set, $f : X \to \mathbb{R}$ is a function, and $g : Y \to X$ is a bijection, then*

$$\sum_{x \in X} f(x) = \sum_{y \in Y} f(g(y)).$$

Go to proof.

Proposition 0.178.11 (Integer Interval as a Finite Set)**.** *Let $n \leq m$ be integers, and let*

$$X := \{i \in \mathbb{Z} : n \leq i \leq m\}.$$

If a_i is a real number assigned to each integer $i \in X$, then

$$\sum_{i=n}^{m} a_i = \sum_{i \in X} a_i.$$

Go to proof.

Proposition 0.178.12 (Disjoint Union Finite Set Summation)**.** *Let X and Y be disjoint finite sets, so $X \cap Y = \emptyset$, and let $f : X \cup Y \to \mathbb{R}$ be a function. Then*

$$\sum_{z \in X \cup Y} f(z) = \left(\sum_{x \in X} f(x) \right) + \left(\sum_{y \in Y} f(y) \right).$$

Go to proof.

Proposition 0.178.13 (Addition Rule for Finite Set Summation)**.** *Let X be a finite set, and let $f : X \to \mathbb{R}$ and $g : X \to \mathbb{R}$ be functions. Then*

$$\sum_{x \in X} (f(x) + g(x)) = \sum_{x \in X} f(x) + \sum_{x \in X} g(x).$$

Go to proof.

Proposition 0.178.14 (Scalar Multiplication Rule for Finite Set Summation)**.** *Let X be a finite set, let $f : X \to \mathbb{R}$ be a function, and let c be a real number. Then*

$$\sum_{x \in X} cf(x) = c \sum_{x \in X} f(x).$$

Go to proof.

Proposition 0.178.15 (Monotonicity for Finite Set Summation)**.** *Let X be a finite set, and let $f : X \to \mathbb{R}$ and $g : X \to \mathbb{R}$ be functions such that $f(x) \leq g(x)$ for all $x \in X$. Then*

$$\sum_{x \in X} f(x) \leq \sum_{x \in X} g(x).$$

Go to proof.

Proposition 0.178.16 (Triangle Inequality for Finite Set Summation)**.** *Let X be a finite set, and let $f : X \to \mathbb{R}$ be a function. Then*

$$\left| \sum_{x \in X} f(x) \right| \leq \sum_{x \in X} |f(x)|.$$

Go to proof.

Lemma 0.178.17 (Iterated Summation over a Product)**.** *Let X and Y be finite sets, and let $f : X \times Y \to \mathbb{R}$ be a function. Then*

$$\sum_{x \in X} \left(\sum_{y \in Y} f(x, y) \right) = \sum_{(x,y) \in X \times Y} f(x, y).$$

Go to proof.

Corollary 0.178.18 (Fubini's Theorem for Finite Series)**.** *Let X and Y be finite sets, and let $f : X \times Y \to \mathbb{R}$ be a function. Then*

$$\sum_{x \in X} \left(\sum_{y \in Y} f(x, y) \right) = \sum_{(x,y) \in X \times Y} f(x, y) = \sum_{(y,x) \in Y \times X} f(x, y) = \sum_{y \in Y} \left(\sum_{x \in X} f(x, y) \right).$$

Go to proof.

Lemma 0.178.19 (Reindexing a Finite Sum)**.** *Let $m \leq n$ be integers, let k be an integer, and let a_i be a real number assigned to each integer i satisfying $m \leq i \leq n$. Then*

$$\sum_{i=m}^{n} a_i = \sum_{j=m+k}^{n+k} a_{j-k}.$$

Go to proof.

Lemma 0.178.20 (Finite Sum of Sums)**.** *Let $m \leq n$ be integers, and let a_i and b_i be real numbers assigned to each integer i satisfying $m \leq i \leq n$. Then*

$$\sum_{i=m}^{n} (a_i + b_i) = \left(\sum_{i=m}^{n} a_i \right) + \left(\sum_{i=m}^{n} b_i \right).$$

Go to proof.

Lemma 0.178.21 (Finite Sum of Scalar Multiples). *Let $m \le n$ be integers, let a_i be a real number assigned to each integer i satisfying $m \le i \le n$, and let $c \in \mathbb{R}$. Then*

$$\sum_{i=m}^{n}(ca_i) = c\left(\sum_{i=m}^{n} a_i\right).$$

Go to proof.

Lemma 0.178.22 (Triangle Inequality for Finite Series). *Let $m \le n$ be integers, and let a_i be a real number assigned to each integer i satisfying $m \le i \le n$. Then*

$$\left|\sum_{i=m}^{n} a_i\right| \le \sum_{i=m}^{n}|a_i|.$$

Go to proof.

Lemma 0.178.23 (Comparison Test for Finite Series). *Let $m \le n$ be integers, and let a_i and b_i be real numbers assigned to each integer i satisfying $m \le i \le n$. Suppose that $a_i \le b_i$ for all i satisfying $m \le i \le n$. Then*

$$\sum_{i=m}^{n} a_i \le \sum_{i=m}^{n} b_i.$$

Go to proof.

0.179 Infinite Series

Definition 0.179.1 (Formal Infinite Series). A formal infinite series is an expression of the form

$$\sum_{n=m}^{\infty} a_n,$$

where m is an integer, and a_n is a real number for every integer $n \ge m$. This series is sometimes written as

$$a_m + a_{m+1} + a_{m+2} + \cdots .$$

Proofs

Function Sequences

Proofs

Functions, Continuity, and Differentiation

Elementary Functions

0.180 Hyperbolic Functions

Hyperbolic Functions

Definition 0.180.1 (Hyperbolic Functions)**.** For all $x \in \mathbb{R}$:

$$\cosh x := \frac{e^x + e^{-x}}{2}, \qquad \sinh x := \frac{e^x - e^{-x}}{2}, \qquad \tanh x := \frac{\sinh x}{\cosh x}.$$

Additional: $\operatorname{sech} x := 1/\cosh x$, $\operatorname{csch} x := 1/\sinh x$, $\coth x := \cosh x/\sinh x$.

Proposition 0.180.2 (Hyperbolic Pythagorean Identity)**.** *For all $x \in \mathbb{R}$:*

$$\cosh^2 x - \sinh^2 x = 1.$$

Go to proof.

Proposition 0.180.3 (Addition Formulas for Hyperbolic Functions)**.** *For all $x, y \in \mathbb{R}$:*

$$\sinh(x+y) = \sinh x \cosh y + \cosh x \sinh y, \qquad \cosh(x+y) = \cosh x \cosh y + \sinh x \sinh y.$$

Go to proof.

Proposition 0.180.4 (Limits of tanh)**.**

$$\lim_{x \to +\infty} \tanh x = 1, \qquad \lim_{x \to -\infty} \tanh x = -1.$$

Go to proof.

Definition 0.180.5 (Inverse Hyperbolic Functions)**.** Since sinh is strictly increasing and bijective $\mathbb{R} \to \mathbb{R}$, its inverse arcsinh is defined on all of $\mathbb{R}$. Since cosh is strictly increasing on $[0, \infty)$ with range $[1, \infty)$, its inverse arccosh is defined on $[1, \infty)$. Since tanh is strictly increasing with range $(-1, 1)$, arctanh is defined on $(-1, 1)$.
Explicit logarithmic formulas:

$$\operatorname{arcsinh} x = \ln\left(x + \sqrt{x^2+1}\right), \quad x \in \mathbb{R}.$$
$$\operatorname{arccosh} x = \ln\left(x + \sqrt{x^2-1}\right), \quad x \geq 1.$$
$$\operatorname{arctanh} x = \tfrac{1}{2} \ln\left(\frac{1+x}{1-x}\right), \quad |x| < 1.$$

0.181 The Logarithm

The Logarithm

Definition 0.181.1 (Natural Logarithm)**.** The **natural logarithm** $\ln : (0, \infty) \to \mathbb{R}$ is the inverse of the exponential function: for $x > 0$,

$$\ln x \ := \ \text{the unique } y \in \mathbb{R} \text{ such that } e^y = x.$$

Equivalently, $\ln x$ is defined by the integral

$$\ln x \ := \ \int_1^x \frac{1}{t}\, dt.$$

Proposition 0.181.2 (Fundamental Logarithmic Limit)**.**

$$\lim_{x \to 0} \frac{\ln(1+x)}{x} = 1.$$

Go to proof.

Proposition 0.181.3 (Every Power Dominates the Logarithm)**.** *For every $r > 0$:*

$$\lim_{x \to \infty} \frac{\ln x}{x^r} = 0 \qquad \text{and} \qquad \lim_{x \to 0^+} x^r \ln x = 0.$$

Go to proof.

0.182 Power Functions and the Exponential

Power Functions and the Exponential

Definition 0.182.1 (Power Function)**.** Let $n \in \mathbb{N}$. The **integer power function** is $f : \mathbb{R} \to \mathbb{R}$, $f(x) := x^n$. For $r \in \mathbb{R}$ and $x > 0$, the **real power** is defined by

$$x^r \ := \ e^{r \ln x}.$$

Definition 0.182.2 (The Number e)**.** The **Euler number** is

$$e \ := \ \sum_{n=0}^{\infty} \frac{1}{n!} \ = \ \lim_{n \to \infty} \left(1 + \frac{1}{n}\right)^n \ \approx \ 2.71828\ldots$$

Both expressions define the same real number.

Definition 0.182.3 (Exponential Function)**.** The **exponential function** $\exp : \mathbb{R} \to (0, \infty)$ is defined by the absolutely convergent power series

$$e^x \ := \ \exp(x) \ := \ \sum_{n=0}^{\infty} \frac{x^n}{n!}, \qquad x \in \mathbb{R}.$$

Proposition 0.182.4 (Fundamental Exponential Limit)**.**

$$\lim_{x \to 0} \frac{e^x - 1}{x} = 1.$$

Go to proof.

Proposition 0.182.5 (Compound Interest Limit)**.**

$$\lim_{x\to\infty}\left(1+\frac{1}{x}\right)^x = e.$$

Go to proof.

Proposition 0.182.6 (Exponential Dominates Every Power)**.** *For every $r > 0$:*

$$\lim_{x\to\infty}\frac{x^r}{e^x} = 0.$$

Go to proof.

0.183 Trigonometric Functions

Trigonometric Functions

Definition 0.183.1 (Sine and Cosine via Power Series)**.** The **sine** and **cosine** functions are defined for all $x \in \mathbb{R}$ by the absolutely convergent power series

$$\sin x \;:=\; \sum_{n=0}^{\infty}\frac{(-1)^n x^{2n+1}}{(2n+1)!} \;=\; x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots$$

$$\cos x \;:=\; \sum_{n=0}^{\infty}\frac{(-1)^n x^{2n}}{(2n)!} \;=\; 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \cdots$$

Proposition 0.183.2 (Pythagorean Identity)**.** *For all $x \in \mathbb{R}$:*

$$\sin^2 x + \cos^2 x = 1.$$

Go to proof.

Proposition 0.183.3 (Addition Formulas)**.** *For all $x, y \in \mathbb{R}$:*

$$\sin(x+y) = \sin x\cos y + \cos x \sin y, \qquad \cos(x+y) = \cos x\cos y - \sin x\sin y.$$

Go to proof.

Definition 0.183.4 (Tangent, Cotangent, Secant, Cosecant)**.**

$$\tan x := \frac{\sin x}{\cos x}, \quad \cot x := \frac{\cos x}{\sin x}, \quad \sec x := \frac{1}{\cos x}, \quad \csc x := \frac{1}{\sin x}.$$

$\tan x$ and $\sec x$ are defined where $\cos x \neq 0$, i.e., $x \neq \pi/2 + k\pi$ for $k \in \mathbb{Z}$.

Definition 0.183.5 (Inverse Trigonometric Functions)**.**

- $\arcsin : [-1,1] \to [-\pi/2, \pi/2]$ is the inverse of $\sin\big|_{[-\pi/2,\pi/2]}$.
- $\arccos : [-1,1] \to [0,\pi]$ is the inverse of $\cos\big|_{[0,\pi]}$.
- $\arctan : \mathbb{R} \to (-\pi/2, \pi/2)$ is the inverse of $\tan\big|_{(-\pi/2,\pi/2)}$.

Theorem 0.183.6 (Fundamental Trigonometric Limit)**.**

$$\lim_{x\to 0}\frac{\sin x}{x} = 1.$$

Go to proof.

Proposition 0.183.7 (Second Fundamental Trigonometric Limit)**.**

$$\lim_{x \to 0} \frac{1 - \cos x}{x^2} = \frac{1}{2}.$$

Go to proof.

Proofs

Continuity

0.184 Limits

Limits of Functions

Definition 0.184.1 (ε-Closeness of a Function to a Value)**.** Let $X \subseteq \mathbb{R}$, let $f : X \to \mathbb{R}$, let $L \in \mathbb{R}$, and let $\varepsilon > 0$. The function f is **ε-close to L** if $|f(x) - L| \leq \varepsilon$ for all $x \in X$.

Definition 0.184.2 (Local δ-Closeness Near a Point)**.** Let $f : X \to \mathbb{R}$, let $L \in \mathbb{R}$, let x_0 be adherent to X, and let $\varepsilon, \delta > 0$. The function f is **ε-close to L near x_0 within δ** if

$$\forall x \in X \ \big(|x - x_0| < \delta \Rightarrow |f(x) - L| \leq \varepsilon\big).$$

Definition 0.184.3 (Limit of a Function)**.** Let $f : A \to \mathbb{R}$ and let c be a cluster point of A. A real number L is the **limit of f at** c, written $\lim_{x \to c} f(x) = L$, if

$$\forall \varepsilon > 0 \ \exists \delta > 0 \ \forall x \in A \ 0 < |x - c| < \delta \Rightarrow |f(x) - L| < \varepsilon.$$

Go to uniqueness proof.

Definition 0.184.4 (Sequential Definition of Limit)**.** Let $f : A \to \mathbb{R}$ and let c be a cluster point of A. The limit $\lim_{x \to c} f(x) = L$ in the **sequential sense** holds if for every sequence $(x_n) \subseteq A \setminus \{c\}$ with $x_n \to c$, the sequence $(f(x_n))$ converges to L.

Definition 0.184.5 (Neighbourhood Form of Limit)**.** Let $f : A \to \mathbb{R}$ and let c be a cluster point of A. The limit $\lim_{x \to c} f(x) = L$ in the **neighbourhood sense** holds if for every ε-neighbourhood $V_\varepsilon(L)$ there exists a deleted δ-neighbourhood $V_\delta^*(c)$ such that

$$x \in V_\delta^*(c) \cap A \quad \Longrightarrow \quad f(x) \in V_\varepsilon(L).$$

Proposition 0.184.6 (Neighbourhood Form of the Limit)**.** *Let $f : A \to \mathbb{R}$ and let c be a cluster point of A. Then $\lim_{x \to c} f(x) = L$ if and only if for every ε-neighbourhood $V_\varepsilon(L)$ there exists a δ-neighbourhood $V_\delta(c)$ such that $x \in V_\delta^*(c) \cap A$ implies $f(x) \in V_\varepsilon(L)$.*

Go to proof.

Theorem 0.184.7 (Uniqueness of Limits of Functions)**.** *Let $f : A \to \mathbb{R}$ and let c be a cluster point of A. Then f has at most one limit at c.*

Go to proof.

Proposition 0.184.8 (Sequential Criterion for Function Limits)**.** *Let $f : A \to \mathbb{R}$ and let c be a cluster point of A. Then $\lim_{x \to c} f(x) = L$ if and only if for every sequence $(x_n) \subseteq A \setminus \{c\}$ with $x_n \to c$, one has $f(x_n) \to L$.*

Go to proof.

Corollary 0.184.9 (Three Equivalent Forms of the Function Limit)**.** *Let $f : A \to \mathbb{R}$ and let c be a cluster point of A. The ε-δ form of $\lim_{x \to c} f(x) = L$, the neighbourhood form, and the sequential form are equivalent descriptions of the same limit condition.*

Go to proof.

Theorem 0.184.10 (Limit Implies Local Boundedness)**.** *Let $f : A \to \mathbb{R}$ and let c be a cluster point of A. If $\lim_{x\to c} f(x) = L$ for some $L \in \mathbb{R}$, then there exist $\delta, M > 0$ such that $|f(x)| \le M$ for all $x \in A$ with $0 < |x - c| < \delta$.*

Go to proof.

Theorem 0.184.11 (Bounded Limits)**.** *Let $f : A \to \mathbb{R}$, let c be a cluster point of A, and let $a, b \in \mathbb{R}$. If $a \le f(x) \le b$ for all $x \in A$ with $x \ne c$, and $\lim_{x\to c} f(x) = L$, then $a \le L \le b$.*

Go to proof.

Theorem 0.184.12 (Squeeze Theorem for Function Limits)**.** *Let $f, g, h : A \to \mathbb{R}$ and let c be a cluster point of A. Suppose $\lim_{x\to c} f(x) = L = \lim_{x\to c} g(x) = L$, and suppose there exists $\delta_0 > 0$ such that*

$$\forall x \in A,\ 0 < |x - c| < \delta_0 \ \Rightarrow\ f(x) \le h(x) \le g(x).$$

Then $\lim_{x\to c} h(x) = L$.

Go to proof.

Theorem 0.184.13 (Limits Are Local)**.** *Let $E \subseteq X \subseteq \mathbb{R}$, let x_0 be a cluster point of E, let $f : X \to \mathbb{R}$, and let $\delta > 0$. Then*

$$\lim_{x\to x_0;\, x\in E} f(x) = L \iff \lim_{x\to x_0;\, x\in E\cap(x_0-\delta, x_0+\delta)} f(x) = L.$$

Go to proof.

Proposition 0.184.14 (Limit of Absolute Value)**.** *Let $f : A \to \mathbb{R}$ and let c be a cluster point of A. If $\lim_{x\to c} f(x) = L$, then $\lim_{x\to c} |f(x)| = |L|$.*

Go to proof.

Theorem 0.184.15 (Composition of Limits)**.** *Let $f : A \to B$, let $g : B \to \mathbb{R}$, let c_1 be a cluster point of A, and let $c_2 \in \mathbb{R}$ be a cluster point of B. Suppose $\lim_{x\to c_1} f(x) = c_2$ and $\lim_{y\to c_2} g(y) = L$. If $c_2 \in B$, assume also that $g(c_2) = L$. Then $\lim_{x\to c_1} g(f(x)) = L$.*

Go to proof.

Theorem 0.184.16 (Monotone Limit Theorem)**.** *Let $a < b$ and let $f : (a, b) \to \mathbb{R}$ be monotone and bounded. Then both one-sided limits exist at every interior point. For increasing f and every $x_0 \in (a, b)$:*

$$f(x_0^-) = \sup\{\, f(x) : a < x < x_0 \,\}, \qquad f(x_0^+) = \inf\{\, f(x) : x_0 < x < b \,\}.$$

Go to proof.

Proposition 0.184.17 (Cauchy Criterion for Function Limits)**.** *Let $f : A \to \mathbb{R}$ and let c be a cluster point of A. Then $\lim_{x\to c} f(x) = L$ exists for some $L \in \mathbb{R}$ if and only if*

$$\forall\ \mathrm{OutputTolerance}(\varepsilon),\ \exists\ \mathrm{InputTolerance}(\delta),\ \forall x, y \in A :\ 0 < |x-c| < \delta \wedge 0 < |y-c| < \delta \ \Rightarrow\ |f(x)-f(y)| < \varepsilon. \blacksquare$$

Go to proof.

Algebra of Limits

Theorem 0.184.18 (Sum Rule for Function Limits)**.** *Let $f, g : A \to \mathbb{R}$ and let c be a cluster point of A. If $\lim_{x\to c} f(x) = L$ and $\lim_{x\to c} g(x) = M$, then*

$$\lim_{x\to c}(f + g)(x) = L + M.$$

Go to proof.

Theorem 0.184.19 (Difference Rule for Function Limits). *Let $f, g : A \to \mathbb{R}$ and let c be a cluster point of A. If $\lim_{x\to c} f(x) = L$ and $\lim_{x\to c} g(x) = M$, then*

$$\lim_{x\to c}(f-g)(x) = L - M.$$

Go to proof.

Theorem 0.184.20 (Scalar Multiple Rule for Function Limits). *Let $f : A \to \mathbb{R}$, let c be a cluster point of A, and let $\alpha \in \mathbb{R}$. If $\lim_{x\to c} f(x) = L$, then*

$$\lim_{x\to c}(\alpha f)(x) = \alpha L.$$

Go to proof.

Theorem 0.184.21 (Product Rule for Function Limits). *Let $f, g : A \to \mathbb{R}$ and let c be a cluster point of A. If $\lim_{x\to c} f(x) = L$ and $\lim_{x\to c} g(x) = M$, then*

$$\lim_{x\to c}(fg)(x) = LM.$$

Go to proof.

Theorem 0.184.22 (Quotient Rule for Function Limits). *Let $f, g : A \to \mathbb{R}$ and let c be a cluster point of A. If $\lim_{x\to c} f(x) = L$, $\lim_{x\to c} g(x) = M$, and $M \neq 0$, then there exists $\delta_0 > 0$ such that $g(x) \neq 0$ whenever $x \in A$ and $0 < |x - c| < \delta_0$, and*

$$\lim_{x\to c}\frac{f(x)}{g(x)} = \frac{L}{M}.$$

Go to proof.

One-Sided Limits

Definition 0.184.23 (Right-Hand Limit). Let $A \subseteq \mathbb{R}$, let $f : A \to \mathbb{R}$, let $c \in \mathbb{R}$, and suppose c is a cluster point of $A \cap (c, \infty)$. A real number L is the **right-hand limit** of f at c if

$$\forall \varepsilon > 0\ \exists \delta > 0\ \forall x \in A\ 0 < x - c < \delta \Rightarrow |f(x) - L| < \varepsilon.$$

Definition 0.184.24 (Left-Hand Limit). Let $A \subseteq \mathbb{R}$, let $f : A \to \mathbb{R}$, let $c \in \mathbb{R}$, and suppose c is a cluster point of $A \cap (-\infty, c)$. A real number L is the **left-hand limit** of f at c if

$$\forall \varepsilon > 0\ \exists \delta > 0\ \forall x \in A\ 0 < c - x < \delta \Rightarrow |f(x) - L| < \varepsilon.$$

Theorem 0.184.25 (Sequential Criterion for Right-Hand Limits). *Let $A \subseteq \mathbb{R}$, let $f : A \to \mathbb{R}$, and suppose c is a cluster point of $A \cap (c, \infty)$. Then $\lim_{x\to c^+} f(x) = L$ if and only if for every sequence $(x_n) \subseteq A \cap (c, \infty)$ with $x_n \to c$, one has $f(x_n) \to L$.*

Go to proof.

Corollary 0.184.26 (Sequential Criterion for Left-Hand Limits). *Let $A \subseteq \mathbb{R}$, let $f : A \to \mathbb{R}$, and suppose c is a cluster point of $A \cap (-\infty, c)$. Then $\lim_{x\to c^-} f(x) = L$ if and only if for every sequence $(x_n) \subseteq A \cap (-\infty, c)$ with $x_n \to c$, one has $f(x_n) \to L$.*

Go to proof.

Theorem 0.184.27 (Two-Sided Limit via One-Sided Limits). *Let $A \subseteq \mathbb{R}$ and let $f : A \to \mathbb{R}$. If c is a cluster point of both $A \cap (c, \infty)$ and $A \cap (-\infty, c)$, then $\lim_{x\to c} f(x) = L$ if and only if both $\lim_{x\to c^+} f(x) = L$ and $\lim_{x\to c^-} f(x) = L$.*

Go to proof.

0.185 Point Continuity

Continuity

Definition 0.185.1 (Relative Neighborhood)**.** Let $E \subseteq \mathbb{R}$ and $a \in E$. A set $U \subseteq \mathbb{R}$ is a neighborhood of a in E if and only if there exists $\delta > 0$ such that $U = E \cap (a - \delta, a + \delta)$. Equivalently, every neighborhood of a in E is a set of the form $U_E(a) := E \cap (a - \delta, a + \delta)$ for some $\delta > 0$.

Definition 0.185.2 (Continuity at a Point)**.** Let $A \subseteq \mathbb{R}$, let $f : A \to \mathbb{R}$, and let $c \in A$. The function f is **continuous at** c if and only if

$$\forall \varepsilon > 0 \; \exists \delta > 0 \; \forall x \in A : \big(|x - c| < \delta \Rightarrow |f(x) - f(c)| < \varepsilon\big).$$

Definition 0.185.3 (Continuity at a Point — Neighbourhood Form)**.** Let $A \subseteq \mathbb{R}$, let $f : A \to \mathbb{R}$, and let $c \in A$. The function f is continuous at c if and only if for every $\varepsilon > 0$ there exists $\delta > 0$ such that

$$f\big(A \cap (c - \delta, c + \delta)\big) \subseteq (f(c) - \varepsilon, f(c) + \varepsilon).$$

Definition 0.185.4 (Continuity at a Point — Sequential Form)**.** Let $A \subseteq \mathbb{R}$, let $c \in A$, and let $f\colon A \to \mathbb{R}$. The function f is continuous at c if and only if for every sequence $(x_n)_{n\in\mathbb{N}}$ in A, the convergence $\lim_{n\to\infty} x_n = c$ implies $\lim_{n\to\infty} f(x_n) = f(c)$.

Characterization of Discontinuity

Definition 0.185.5 (Point of Discontinuity)**.** Let $A \subseteq \mathbb{R}$, let $f\colon A \to \mathbb{R}$, and let $a \in A$. The function f is discontinuous at a if and only if f is not continuous at a.

Definition 0.185.6 (Sequential Characterization of Discontinuity)**.** Let $A \subseteq \mathbb{R}$, let $f : A \to \mathbb{R}$, and let $c \in A$. The function f is discontinuous at c if and only if there exists a sequence $(x_n)_{n\in\mathbb{N}} \subseteq A$ such that $x_n \to c$ while $f(x_n) \not\to f(c)$.

Definition 0.185.7 (Neighbourhood Characterization of Discontinuity)**.** Let $A \subseteq \mathbb{R}$, let $f\colon A \to \mathbb{R}$, and let $c \in A$. The function f is discontinuous at c if and only if there exists a neighborhood V of $f(c)$ such that for every neighborhood U of c relative to A there exists $x \in U \cap A$ with $f(x) \notin V$.

Lemma 0.185.8 (Equivalent Formulations of Discontinuity at a Point)**.** *Let $A \subseteq \mathbb{R}$, let $f : A \to \mathbb{R}$, and let $c \in A$. The following are equivalent:*

1. *c is a point of discontinuity in the ε-δ sense.*
2. *c is a sequential point of discontinuity.*
3. *c is a neighbourhood point of discontinuity.*

Go to proof.

Definition 0.185.9 (Types of Discontinuity)**.** Let $A \subseteq \mathbb{R}$, let $f : A \to \mathbb{R}$, and let $x_0 \in A$ with f discontinuous at x_0.

(i) *Removable* iff $\exists L \in \mathbb{R} :\ \lim_{x\to x_0} f(x) = L$ and $L \neq f(x_0)$.

(ii) *Jump* iff $\exists L_-, L_+ \in \mathbb{R} :\ \lim_{x\to x_0^-} f(x) = L_-$, $\lim_{x\to x_0^+} f(x) = L_+$, and $L_- \neq L_+$.

(iii) *Essential* iff it is not removable.

Oscillation

Definition 0.185.10 (Oscillation on a Set)**.** Let $A \subseteq \mathbb{R}$, let $f : A \to \mathbb{R}$, and let $E \subseteq A$. The oscillation of f on E is the extended real number

$$\omega(f;E) \coloneqq \sup\{\, |f(x_1) - f(x_2)| : x_1, x_2 \in E \,\} = \operatorname{diam}(f(E)).$$

Definition 0.185.11 (Oscillation at a Point)**.** Let $E \subseteq \mathbb{R}$, let $f : E \to \mathbb{R}$, and let $x_0 \in \mathbb{R}$ satisfy $U_\delta(x_0) \cap E \neq \varnothing$ for every $\delta > 0$, where $U_\delta(x_0) = \{x \in \mathbb{R} : |x - x_0| < \delta\}$. The oscillation of f at x_0 is the extended real number

$$\omega(f;x_0) = \lim_{\delta \to 0^+} \omega\big(f;\, U_\delta(x_0) \cap E\big),$$

with $\omega(f;A)$ denoting the oscillation of f on the set A. Because the function $\delta \mapsto \omega\big(f;\, U_\delta(x_0) \cap E\big)$ is nonincreasing, this right-hand limit exists (possibly $+\infty$).

Theorem 0.185.12 (Continuity Characterised by Oscillation)**.** *Let $A \subseteq \mathbb{R}$, let $f\colon A \to \mathbb{R}$, and let $c \in A$. Then f is continuous at c if and only if $\omega(f;c) = 0$.* *Go to proof.*

Proposition 0.185.13 (Discontinuity Set via Oscillation)**.** *Let $f : E \to \mathbb{R}$. Then*

$$D(f) \coloneqq \{x \in E : f \textit{ is discontinuous at } x\} = \{x \in E : \omega(f;x) > 0\} = \bigcup_{n=1}^{\infty} \{x \in E : \omega(f;x) \geq 1/n\}.$$

Go to proof.

0.186 Global Theorems

Intermediate Value Theorem

Definition 0.186.1 (Bounded on a Set)**.** Let $A \subseteq \mathbb{R}$ and $f\colon A \to \mathbb{R}$. The function f is bounded on A if and only if there exists a constant $M > 0$ such that $|f(x)| \leq M$ for every $x \in A$.

Theorem 0.186.2 (Boundedness Theorem)**.** *Let $a < b$ and let $f : [a,b] \to \mathbb{R}$ be continuous on $[a,b]$. Then there exists $M > 0$ such that $|f(x)| \leq M$ for every $x \in [a,b]$.* *Go to proof.*

Definition 0.186.3 (Absolute Maximum and Absolute Minimum)**.** Let $A \subseteq \mathbb{R}$ and let $f\colon A \to \mathbb{R}$. A point $x^* \in A$ is an absolute maximum of f on A if and only if $f(x^*) \geq f(x)$ for every $x \in A$. A point $x_* \in A$ is an absolute minimum of f on A if and only if $f(x_*) \leq f(x)$ for every $x \in A$.

Theorem 0.186.4 (Extreme Value Theorem)**.** *Let $a, b \in \mathbb{R}$ with $a < b$, and let $f : [a,b] \to \mathbb{R}$ be continuous. There exist points $x_m, x_M \in [a,b]$ such that $f(x_m) \leq f(x) \leq f(x_M)$ for every $x \in [a,b]$. Equivalently, f attains both an absolute maximum and an absolute minimum on $[a,b]$.* *Go to proof.*

Theorem 0.186.5 (Location of Roots)**.** *Let $a, b \in \mathbb{R}$ with $a < b$ and let $f : [a,b] \to \mathbb{R}$ be continuous. If either $f(a) < 0 < f(b)$ or $f(a) > 0 > f(b)$, then there exists $c \in (a,b)$ such that $f(c) = 0$.* *Go to proof.*

Theorem 0.186.6 (Bolzano's Intermediate Value Theorem)**.** *Let $I \subseteq \mathbb{R}$ be an interval, let $f\colon I \to \mathbb{R}$ be continuous on I, and choose $a, b \in I$ with $f(a) < k < f(b)$. Then there exists $c \in I$ such that $\min\{a,b\} < c < \max\{a,b\}$ and $f(c) = k$.* *Go to proof.*

Theorem 0.186.7 (Image of Closed Bounded Interval Is Closed Bounded Interval)**.** *Let $f : [a,b] \to \mathbb{R}$ be continuous. Then $f([a,b]) = [\min_{[a,b]} f, \max_{[a,b]} f]$.* *Go to proof.*

Theorem 0.186.8 (Preservation of Intervals)**.** *Let $I \subseteq \mathbb{R}$ be an interval and let $f : I \to \mathbb{R}$ be continuous. Then $f(I)$ is an interval.* *Go to proof.*

Theorem 0.186.9 (Darboux Property)**.** *Let $a, b \in \mathbb{R}$ satisfy $a < b$, let $f : [a,b] \to \mathbb{R}$ be continuous, and fix $c \in \mathbb{R}$. If $f(x) \neq c$ for every $x \in [a,b]$, then either $f(x) > c$ for all $x \in [a,b]$ or $f(x) < c$ for all $x \in [a,b]$.* *Go to proof.*

0.187 Uniform Continuity

Uniform Continuity

Definition 0.187.1 (Uniform Continuity)**.** Let $A \subseteq \mathbb{R}$ and let $f : A \to \mathbb{R}$. The function f is uniformly continuous on A if and only if for every $\varepsilon > 0$ there exists $\delta > 0$ such that for all $x, u \in A$ the implication $|x - u| < \delta \Rightarrow |f(x) - f(u)| < \varepsilon$ holds.

Theorem 0.187.2 (Algebra of Uniform Continuity on a General Domain)**.** *(a) For every set $A \subseteq \mathbb{R}$, every pair of functions $f, g : A \to \mathbb{R}$ that are uniformly continuous on A, and every scalar $c \in \mathbb{R}$, the functions $f + g$, $f - g$, and cf are uniformly continuous on A.*

(b) There exist a set $A \subseteq \mathbb{R}$ and uniformly continuous functions $f, g : A \to \mathbb{R}$ such that the product fg is not uniformly continuous on A.

Go to proof.

Theorem 0.187.3 (Algebra of Uniform Continuity on a Bounded Domain)**.** *Let $A \subseteq \mathbb{R}$ be bounded, and let $f, g : A \to \mathbb{R}$ be uniformly continuous on A.*

(a) The product $x \mapsto f(x)g(x)$ is uniformly continuous on A.

(b) If there exists $k > 0$ such that $|g(x)| \geq k$ for all $x \in A$, then the quotient $x \mapsto f(x)/g(x)$ is uniformly continuous on A.

Go to proof.

Proposition 0.187.4 (Sequential Characterization of Uniform Continuity)**.** *Let $A \subseteq \mathbb{R}$ and let $f : A \to \mathbb{R}$. Then f is uniformly continuous on A if and only if for every pair of sequences (x_n) and (y_n) in A with $\lim_{n\to\infty}(x_n - y_n) = 0$ we have $\lim_{n\to\infty}(f(x_n) - f(y_n)) = 0$.* *Go to proof.*

Theorem 0.187.5 (Heine–Cantor Theorem)**.** *Let $a, b \in \mathbb{R}$ with $a < b$ and let $f : [a, b] \to \mathbb{R}$ be continuous on $[a, b]$. Then f is uniformly continuous on $[a, b]$.* *Go to proof.*

Proposition 0.187.6 (Uniform Continuity Preserves Cauchy Sequences)**.** *Let $S \subseteq \mathbb{R}$, let $f : S \to \mathbb{R}$, and let (x_n) be a sequence in S. If f is uniformly continuous on S and (x_n) is a Cauchy sequence in S, then $(f(x_n))$ is a Cauchy sequence in $\mathbb{R}$.* *Go to proof.*

Definition 0.187.7 (Lipschitz Condition)**.** Let $A \subseteq \mathbb{R}$, let $f : A \to \mathbb{R}$, and let $K > 0$. The function f satisfies the Lipschitz condition on A with constant K if and only if $|f(x) - f(u)| \leq K|x - u|$ for every $x, u \in A$.

Proposition 0.187.8 (Lipschitz Implies Uniform Continuity)**.** *Let $A \subseteq \mathbb{R}$ and $f : A \to \mathbb{R}$. If there exists $K \geq 0$ such that $|f(x) - f(y)| \leq K|x - y|$ for all $x, y \in A$, then f is uniformly continuous on A.* *Go to proof.*

Theorem 0.187.9 (Algebra of Lipschitz Functions)**.** *Let $A \subseteq \mathbb{R}$ and let $f, g : A \to \mathbb{R}$ be Lipschitz with constants $K_f, K_g \geq 0$ respectively.*

(a) For every $c \in \mathbb{R}$, the function cf is Lipschitz on A with constant $|c|K_f$.

(b) The functions $f + g$ and $f - g$ are Lipschitz on A with constant $K_f + K_g$.

(c) If $B \subseteq \mathbb{R}$, $h : B \to \mathbb{R}$ is Lipschitz with constant $K_h \geq 0$, and $f(A) \subseteq B$, then $h \circ f$ is Lipschitz on A with constant $K_h K_f$.

(d) If $|f(x)| \leq M_f$ and $|g(x)| \leq M_g$ for all $x \in A$, then fg is Lipschitz on A with constant $M_g K_f + M_f K_g$.

(e) If $|f(x)| \leq M_f$ for all $x \in A$, if $|g(x)| \geq k > 0$ for all $x \in A$, and if g is Lipschitz on A with constant K_g, then f/g is Lipschitz on A with constant $K_f/k + M_f K_g/k^2$.

Go to proof.

Definition 0.187.10 (Bi-Lipschitz Map)**.** Let $A \subseteq \mathbb{R}$ and let $f : A \to \mathbb{R}$. The map f is bi-Lipschitz on A if and only if there exist constants $0 < \alpha \leq K$ such that for every $x, u \in A$,

$$\alpha|x-u| \leq |f(x) - f(u)| \leq K|x-u|.$$

Any such pair (α, K) is called a bi-Lipschitz constant pair for f on A.

Theorem 0.187.11 (Inverse of a Bi-Lipschitz Map Is Lipschitz)**.** *Let $A \subseteq \mathbb{R}$, let $f : A \to \mathbb{R}$, and suppose there exist constants $\alpha, K > 0$ such that $\alpha|x-y| \leq |f(x) - f(y)| \leq K|x-y|$ for all $x, y \in A$. Then f is injective, the inverse mapping $f^{-1} : f(A) \to A$ is well defined, and*

$$\frac{1}{K}|u-v| \leq |f^{-1}(u) - f^{-1}(v)| \leq \frac{1}{\alpha}|u-v| \quad \text{for all } u, v \in f(A).$$

Go to proof.

0.188 Monotone Functions

Monotone Functions and Continuity

Theorem 0.188.1 (One-Sided Limits of Monotone Functions)**.** *Let $I \subseteq \mathbb{R}$ be an interval, let c be interior to I, and let $f : I \to \mathbb{R}$ be increasing. Define*

$$L_- := \sup\{f(x) : x \in I,\ x < c\}, \qquad L_+ := \inf\{f(x) : x \in I,\ c < x\}.$$

Then both one-sided limits exist and satisfy

$$\lim_{x \to c^-} f(x) = L_-, \qquad \lim_{x \to c^+} f(x) = L_+.$$

Go to proof.

Corollary 0.188.2 (Continuity Criterion for Monotone Functions)**.** *Let $I \subseteq \mathbb{R}$ be an interval, let $f : I \to \mathbb{R}$ be monotone increasing, and let c be an interior point of I. Then f is continuous at c if and only if its one-sided limits satisfy $f(c^-) = f(c) = f(c^+)$, where $f(c^-) = \lim_{x \to c^-} f(x)$ and $f(c^+) = \lim_{x \to c^+} f(x)$.* *Go to proof.*

Definition 0.188.3 (Jump of a Function)**.** Let $I \subseteq \mathbb{R}$ be an interval, let $f : I \to \mathbb{R}$ be increasing, and let c be an interior point of I. If the one-sided limits $f(c^-)$ and $f(c^+)$ exist, the jump of f at c is

$$j_f(c) := f(c^+) - f(c^-).$$

Proposition 0.188.4 (Discontinuities of a Monotone Function Are of First Kind)**.** *Let $I \subseteq \mathbb{R}$ be an interval, let $f : I \to \mathbb{R}$ be monotone, and let $a \in I$ be such that there exist points of I strictly less and strictly greater than a. If f is discontinuous at a, then both one-sided limits $\lim_{x \uparrow a} f(x)$ and $\lim_{x \downarrow a} f(x)$ exist in $\mathbb{R}$.* *Go to proof.*

Corollary 0.188.5 (Jump Intervals Are Disjoint)**.** *Let $I \subseteq \mathbb{R}$ be an interval and let $f : I \to \mathbb{R}$ be nondecreasing. If $a, b \in I$ are distinct points of discontinuity of f, then the open intervals $(f(a^-), f(a^+))$ and $(f(b^-), f(b^+))$ are disjoint.**Go to proof.*

Theorem 0.188.6 (Discontinuities of a Monotone Function Are Countable)**.** *Let $I \subseteq \mathbb{R}$ be an interval and let $f : I \to \mathbb{R}$ be monotone. Then the set $D_f := \{c \in I : f \text{ is discontinuous at } c\}$ is at most countable.* *Go to proof.*

Corollary 0.188.7 (Monotone Function Is Continuous on a Dense Set)**.** *Let $f : [a, b] \to \mathbb{R}$ be monotone. Then the set $\{x \in [a, b] : f \text{ is continuous at } x\}$ is dense in $[a, b]$.* *Go to proof.*

Proposition 0.188.8 (Continuous Injective Is Strictly Monotone)**.** *Let $f : [a, b] \to \mathbb{R}$ be continuous on $[a, b]$. Then f is injective if and only if f is strictly monotone on $[a, b]$.* *Go to proof.*

Theorem 0.188.9 (Continuous Inverse Theorem)**.** *Let $I \subseteq \mathbb{R}$ be an interval and let $f : I \to \mathbb{R}$ be strictly monotone and continuous on I. Then the inverse function $f^{-1} : f(I) \to I$ exists, is strictly monotone, and is continuous on $f(I)$.* *Go to proof.*

Limits Superior and Inferior of Functions

Definition 0.188.10 (Limits Superior and Inferior of a Function)**.** Let $f : E \to \mathbb{R}$ and let x_0 be an accumulation point of E. For each $\delta > 0$ set

$$U(\delta) = \{\, f(x) : x \in E,\ 0 < |x - x_0| < \delta \,\}.$$

The extended real numbers $\limsup_{x\to x_0} f(x)$ and $\liminf_{x\to x_0} f(x)$ are defined by

$$\limsup_{x\to x_0} f(x) = \inf_{\delta>0} \sup U(\delta), \qquad \liminf_{x\to x_0} f(x) = \sup_{\delta>0} \inf U(\delta),$$

where each inner supremum or infimum is taken in $\overline{\mathbb{R}}$ and the outer infimum or supremum ranges over all $\delta > 0$.

Proposition 0.188.11 ($\limsup \geq \liminf$)**.** *Let $A \subseteq \mathbb{R}$, let x_0 be a limit point of A, and let $f : A \to \mathbb{R}$. Then*

$$\limsup_{x\to x_0} f(x) \geq \liminf_{x\to x_0} f(x).$$

Go to proof.

Proposition 0.188.12 (Limit Exists iff $\limsup = \liminf$)**.** *Let $A \subseteq \mathbb{R}$, let $f\colon A \to \mathbb{R}$, let x_0 be a limit point of A with $x_0 \in \mathbb{R}$, and let $L \in \mathbb{R}$. Then $\lim_{x\to x_0} f(x)$ exists and equals L if and only if $\limsup_{x\to x_0} f(x) = \liminf_{x\to x_0} f(x) = L$.* *Go to proof.*

0.189 Limits at Infinity

Limits at Infinity

Definition 0.189.1 (Infinite Adherent Points)**.** Let $X \subseteq \mathbb{R}$. The point $+\infty$ is adherent to X if and only if for every $M \in \mathbb{R}$ there exists $x \in X$ such that $x > M$. The point $-\infty$ is adherent to X if and only if for every $M \in \mathbb{R}$ there exists $x \in X$ such that $x < M$.

Definition 0.189.2 (Limit at Infinity)**.** Let $X \subseteq \mathbb{R}$ be such that for every $M \in \mathbb{R}$ there exists $x \in X$ with $x > M$, let $f : X \to \mathbb{R}$, and let $L \in \mathbb{R}$. The limit of f at $+\infty$ equals L if and only if for every $\varepsilon > 0$ there exists $M \in \mathbb{R}$ such that for every $x \in X$, $x > M$ implies $|f(x) - L| < \varepsilon$.

0.190 Approximation

Approximation of Continuous Functions

Theorem 0.190.1 (Step Function Approximation)**.** *Let $f : [a, b] \to \mathbb{R}$ be continuous and let $\varepsilon > 0$. Then there exists a step function $s_\varepsilon : [a, b] \to \mathbb{R}$ such that*

$$\sup_{x\in[a,b]} |f(x) - s_\varepsilon(x)| < \varepsilon.$$

Go to proof.

Definition 0.190.2 (Piecewise Linear Function)**.** Let $a, b \in \mathbb{R}$ with $a < b$ and let $g : [a, b] \to \mathbb{R}$. The function g is piecewise linear if and only if there exist $m \in \mathbb{N}$ with $m \geq 1$ and points $a = x_0 < x_1 < \cdots < x_m = b$ such that $g|_{[x_{k-1},x_k]}$ is affine for every $k \in \{1, \ldots, m\}$.

Theorem 0.190.3 (Piecewise Linear Approximation)**.** *Let $f\colon [a, b] \to \mathbb{R}$ be continuous and let $\varepsilon > 0$. Then there exists a continuous piecewise linear function $\ell_\varepsilon\colon [a, b] \to \mathbb{R}$ such that $\sup_{x\in[a,b]} |f(x) - \ell_\varepsilon(x)| < \varepsilon$.* *Go to proof.*

Theorem 0.190.4 (Weierstrass Approximation Theorem). *Let $f : [a,b] \to \mathbb{R}$ be continuous. For every $\varepsilon > 0$ there exists a polynomial $p_\varepsilon \in \mathbb{R}[x]$ such that $\|f - p_\varepsilon\|_{\infty,[a,b]} < \varepsilon$. Go to proof.*

Definition 0.190.5 (Bernstein Polynomial). Let $n \in \mathbb{N}$ and let $f : [0,1] \to \mathbb{R}$. A function $B_n : [0,1] \to \mathbb{R}$ is called the nth Bernstein polynomial of f if and only if, for every $x \in [0,1]$,

$$B_n(x) = \sum_{k=0}^{n} f\left(\frac{k}{n}\right)\binom{n}{k} x^k (1-x)^{n-k}.$$

Theorem 0.190.6 (Bernstein Approximation Theorem). *Let $f : [0,1] \to \mathbb{R}$ be continuous, and for each $n \in \mathbb{N}$ let B_n denote the nth Bernstein polynomial of f. Then for every $\varepsilon > 0$ there exists $n_\varepsilon \in \mathbb{N}$ such that $\|f - B_n\|_\infty < \varepsilon$ for all $n \geq n_\varepsilon$. Go to proof.*

0.191 Gauge

Gauges and Tagged Partitions

Definition 0.191.1 (Partition). Let $a < b$ and set $I = [a,b]$. A finite ordered set $P = \{x_0, \dots, x_n\}$ with $n \in \mathbb{N}$ and $n \geq 1$ is a partition of I if and only if $x_0 = a$, $x_n = b$, and $x_0 < x_1 < \cdots < x_n$. In this case the closed subintervals $[x_{i-1}, x_i]$ for $i = 1, \dots, n$ subdivide I.

Definition 0.191.2 (Tagged Partition). Let $a, b \in \mathbb{R}$ with $a < b$. A list $\dot{\mathcal{P}} = \{([x_{i-1}, x_i], t_i)\}_{i=1}^n$ is a tagged partition of $[a,b]$ if and only if the subintervals determine a partition $P = \{[x_{i-1}, x_i]\}_{i=1}^n$ of $[a,b]$ with $a = x_0 < x_1 < \cdots < x_n = b$, and each tag satisfies $t_i \in [x_{i-1}, x_i]$ for $i = 1, \dots, n$. Equivalently, a tagged partition is exactly a partition P of $[a,b]$ together with one tag chosen inside every subinterval of P.

Definition 0.191.3 (Gauge). Let $a, b \in \mathbb{R}$ with $a < b$ and set $I = [a,b]$. A gauge on I is a function $\delta : I \to (0, \infty)$.

Definition 0.191.4 (δ-Fine Partition). Let $[a,b]$ be a closed interval, let $\delta : [a,b] \to (0,\infty)$ be a gauge, and let $\dot{\mathcal{P}} = \{([x_{i-1}, x_i], t_i)\}_{i=1}^n$ be a tagged partition of $[a,b]$. The tagged partition $\dot{\mathcal{P}}$ is δ-fine if and only if for every $i \in \{1, \dots, n\}$ and every $x \in [x_{i-1}, x_i]$ the inequality $|x - t_i| < \delta(t_i)$ holds.

Lemma 0.191.5 (Every Point Is Covered by a Tag). *Let $\dot{\mathcal{P}} = \{([x_{i-1}, x_i], t_i)\}_{i=1}^n$ be a tagged partition of $[a,b]$ that is δ-fine on $[a,b]$, and let $x \in [a,b]$. Then there exists $i \in \{1, \dots, n\}$ such that $|x - t_i| < \delta(t_i)$. Go to proof.*

Theorem 0.191.6 (Cousin's Theorem). *Every gauge $\delta : [a,b] \to (0,\infty)$ on a closed interval $[a,b]$ admits a δ-fine tagged partition of $[a,b]$. Go to proof.*

Tagged Partitions and Mesh

Definition 0.191.7 (Mesh of a Partition). Let $a, b \in \mathbb{R}$ with $a < b$, and let $P = \{x_0, \dots, x_n\}$ be a partition of $[a,b]$. A real number m is the *mesh* (or *norm*) of P if and only if

$$m = \max_{1 \leq i \leq n} (x_i - x_{i-1}).$$

The mesh of P is denoted by $\|P\|$ and is the unique m satisfying this equivalence.

Definition 0.191.8 (Refinement). Let $[a,b] \subseteq \mathbb{R}$ and let P and Q be partitions of $[a,b]$. Then Q is a refinement of P if and only if every point of P is also a point of Q; equivalently, $P \subseteq Q$.

Definition 0.191.9 (Common Refinement). Let $a < b$ and let P and Q be partitions of $[a,b]$. Their common refinement is the partition R obtained by taking the union of their node sets, so $R = P \cup Q$ as sets of points in $[a,b]$.

Proofs

Differentiation

0.192 The Tangent Problem

Definition 0.192.1 (Secant Line)**.** Let $f : A \to \mathbb{R}$, and let $x_1, x_2 \in A$ with $x_1 \neq x_2$. The *secant line* determined by the points $(x_1, f(x_1))$ and $(x_2, f(x_2))$ is the unique line passing through these two points. Its slope is

$$m_{\text{sec}}(x_1, x_2) := \frac{f(x_2) - f(x_1)}{x_2 - x_1}.$$

Definition 0.192.2 (Difference Quotient)**.** Let $f : A \to \mathbb{R}$, let $c \in A$, and let $x \in A$ with $x \neq c$. The *difference quotient* of f from c to x is

$$m_{\text{sec}}(c, x) := \frac{f(x) - f(c)}{x - c}.$$

Definition 0.192.3 (Tangent Line)**.** Let $f : A \to \mathbb{R}$, let $c \in A$, and suppose that f is differentiable at c. The *tangent line* to the graph of f at the point $(c, f(c))$ is the line through $(c, f(c))$ with slope $f'(c)$. Its equation is

$$y = f(c) + f'(c)(x - c).$$

0.193 The Derivative

Definition 0.193.1 (Derivative at a Point)**.** Let $f : D \to \mathbb{R}$ be a real-valued function and $c \in \text{int}(D)$. We say f is *differentiable at* c with *derivative* $L = f'(c)$ if and only if:

$$\forall \varepsilon > 0\ \exists \delta > 0\ \forall x \in D \Big(0 < |x - c| < \delta \implies \left| \frac{f(x) - f(c)}{x - c} - L \right| < \varepsilon \Big).$$

Definition 0.193.2 (Topological Definition of Derivative at a Point)**.** Let $f : D \to \mathbb{R}$ and $c \in \text{int}(D)$. f is differentiable at c with derivative L if for every neighbourhood V of L, there exists a neighbourhood U of c such that for all $x \in (U \setminus \{c\}) \cap D$:

$$\frac{f(x) - f(c)}{x - c} \in V.$$

Definition 0.193.3 (Sequential Definition of Derivative at a Point)**.** Let $A \subseteq \mathbb{R}$, let $f : A \to \mathbb{R}$, and let $c \in A$ be a limit point of A. We say that f is differentiable at c with derivative L if

$$\forall (x_n) \subseteq A \setminus \{c\}\ \big(x_n \to c \Rightarrow \tfrac{f(x_n) - f(c)}{x_n - c} \to L \big).$$

Theorem 0.193.4 (Equivalence of Definitions of the Derivative at a Point)**.** *Let $f : A \to \mathbb{R}$ and $c \in A$ be a limit point of A. The following are equivalent:*

1. *f is differentiable at c in the ε-δ sense with derivative L.*

2. *f is differentiable at c in the topological sense with derivative L.*

3. f *is differentiable at* c *in the sequential sense with derivative* L.

Go to proof.

Proposition 0.193.5 (Derivative: h-Form Equivalence). *Let* $f : A \to \mathbb{R}$ *and* $c \in \operatorname{int}(A)$. *Then* f *is differentiable at* c *with derivative* L *if and only if*

$$\lim_{h \to 0} \frac{f(c+h) - f(c)}{h} = L.$$

Go to proof.

0.193.1 First Consequences

Theorem 0.193.6 (Differentiable Implies Continuous). *Let* $f : A \to \mathbb{R}$ *and* $c \in A$. *If* f *is differentiable at* c, *then* f *is continuous at* c.

Go to proof.

Theorem 0.193.7 (Uniqueness of the Derivative). *Let* $f : A \to \mathbb{R}$ *and* $c \in A$. *If* f *is differentiable at* c, *its derivative is unique.*

Go to proof. Go to proof.

Proposition 0.193.8 (Derivative of the Identity). *Let* $f(x) = x$ *for all* $x \in \mathbb{R}$. *Then* $f'(c) = 1$ *for every* $c \in \mathbb{R}$.

Go to proof.

Proposition 0.193.9 (Derivative of a Constant). *Let* $k \in \mathbb{R}$ *and let* $f(x) = k$ *for all* $x \in \mathbb{R}$. *Then* $f'(c) = 0$ *for every* $c \in \mathbb{R}$.

Go to proof.

0.194 One-Sided Derivatives

Definition 0.194.1 (Left-Hand Derivative). Let $I \subseteq \mathbb{R}$ be an interval, let $f : I \to \mathbb{R}$, and let $c \in I$ be such that c is a left-hand limit point of I. We say that f has a *left-hand derivative* at c if the limit

$$\lim_{x \to c^-} \frac{f(x) - f(c)}{x - c}$$

exists. In that case, the value of this limit is denoted $f'_-(c)$:

$$f'_-(c) := \lim_{x \to c^-} \frac{f(x) - f(c)}{x - c}.$$

Definition 0.194.2 (Right-Hand Derivative). Let $I \subseteq \mathbb{R}$ be an interval, let $f : I \to \mathbb{R}$, and let $c \in I$ be such that c is a right-hand limit point of I. We say that f has a *right-hand derivative* at c if the limit

$$\lim_{x \to c^+} \frac{f(x) - f(c)}{x - c}$$

exists. In that case, the value of this limit is denoted $f'_+(c)$:

$$f'_+(c) := \lim_{x \to c^+} \frac{f(x) - f(c)}{x - c}.$$

Theorem 0.194.3 (Differentiability and One-Sided Derivatives). *Let* $I \subseteq \mathbb{R}$ *be an open interval and let* $f : I \to \mathbb{R}$. *Then* f *is differentiable at* $c \in I$ *if and only if both one-sided derivatives* $f'_-(c)$ *and* $f'_+(c)$ *exist and are equal. In that case,*

$$f'(c) = f'_-(c) = f'_+(c).$$

Go to proof.

0.195 Carathéodory and the Chain Rule

0.195.1 Carathéodory Characterization

Definition 0.195.1 (Carathéodory Factorization)**.** Let $I \subseteq \mathbb{R}$ be an interval, let $f : I \to \mathbb{R}$, and let $c \in I$. We say that f admits a *Carathéodory factorization* at c if there exists a function $\phi : I \to \mathbb{R}$, continuous at c, such that

$$f(x) = f(c) + (x - c)\,\phi(x) \qquad \text{for all } x \in I.$$

In this case, f is differentiable at c and $f'(c) = \phi(c)$.

Theorem 0.195.2 (Carathéodory Characterization of Differentiability)**.** *Let $I \subseteq \mathbb{R}$ be an interval, let $f : I \to \mathbb{R}$, and let $c \in I$. The following are equivalent:*

(i) f is differentiable at c.

(ii) There exists a function $\phi : I \to \mathbb{R}$, continuous at c, such that

$$f(x) = f(c) + (x - c)\,\phi(x) \qquad \textit{for all } x \in I.$$

In this case, $\phi(c) = f'(c)$. Go to proof. Go to proof.

Theorem 0.195.3 (Chain Rule)**.** *Let $I, J \subseteq \mathbb{R}$ be intervals, let $f : I \to \mathbb{R}$ and $g : J \to \mathbb{R}$, and suppose that $f(I) \subseteq J$. Let $c \in I$. If f is differentiable at c and g is differentiable at $f(c)$, then the composition $g \circ f$ is differentiable at c, and*

$$(g \circ f)'(c) = g'(f(c))\,f'(c).$$

Go to proof.

Theorem 0.195.4 (Leibniz Rule)**.** *Let $I \subseteq \mathbb{R}$ be an interval, let $x \in I$, and suppose that $f, g : I \to \mathbb{R}$ are n-times differentiable on a neighbourhood of x. Then fg is n-times differentiable at x, and*

$$(fg)^{(n)}(x) = \sum_{k=0}^{n} \binom{n}{k} f^{(k)}(x) g^{(n-k)}(x).$$

Go to proof.

Theorem 0.195.5 (Faa di Bruno's Formula)**.** *Let $I, J \subseteq \mathbb{R}$ be intervals, let $x \in I$, suppose $g : I \to J$ is n-times differentiable on a neighbourhood of x, and suppose $f : J \to \mathbb{R}$ is n-times differentiable on a neighbourhood of $g(x)$. Then $f \circ g$ is n-times differentiable at x, and*

$$\frac{d^n}{dx^n} f(g(x)) = \sum_{\sum_{m=1}^{n} m k_m = n} \frac{n!}{\prod_{m=1}^{n} k_m!\,(m!)^{k_m}}\, f^{(k)}(g(x)) \prod_{m=1}^{n} \left(g^{(m)}(x)\right)^{k_m},$$

where $k_m \geq 0$ and $k = \sum_{m=1}^{n} k_m$. Go to proof.

0.196 The Shape Questions

Definition 0.196.1 (Convex Function)**.** Let $I \subseteq \mathbb{R}$ be an interval and let $f : I \to \mathbb{R}$. The function f is *convex* on I if for all $x, y \in I$ and all $\lambda \in [0, 1]$,

$$f(\lambda x + (1 - \lambda) y) \leq \lambda f(x) + (1 - \lambda) f(y).$$

Definition 0.196.2 (Concave Function)**.** Let $I \subseteq \mathbb{R}$ be an interval and let $f : I \to \mathbb{R}$. The function f is *concave* on I if $-f$ is convex on I.

Definition 0.196.3 (Inflection Point)**.** Let $I \subseteq \mathbb{R}$ be an interval, let $f : I \to \mathbb{R}$, and let $c \in I$ be an interior point. The point c is an *inflection point* of f if f changes convexity at c: there is a neighbourhood $(c - \delta, c + \delta) \subseteq I$ such that f is convex on one side of c and concave on the other side.

0.196.1 Interior Extrema

Definition 0.196.4 (Relative Minimum)**.** Let $A \subseteq \mathbb{R}$, let $f : A \to \mathbb{R}$, and let $c \in A$. We say that f has a *relative minimum* at c if there exists $\delta > 0$ such that

$$f(c) \leq f(x) \qquad \text{for all } x \in A \cap (c-\delta, c+\delta).$$

If, in addition,

$$f(c) < f(x) \qquad \text{for all } x \in \big(A \cap (c-\delta, c+\delta)\big) \setminus \{c\},$$

then f has a *strict relative minimum* at c.

Definition 0.196.5 (Relative Maximum)**.** Let $A \subseteq \mathbb{R}$, let $f : A \to \mathbb{R}$, and let $c \in A$. We say that f has a *relative maximum* at c if there exists $\delta > 0$ such that

$$f(x) \leq f(c) \qquad \text{for all } x \in A \cap (c-\delta, c+\delta).$$

If, in addition,

$$f(x) < f(c) \qquad \text{for all } x \in \big(A \cap (c-\delta, c+\delta)\big) \setminus \{c\},$$

then f has a *strict relative maximum* at c.

Definition 0.196.6 (Relative Extremum)**.** Let $A \subseteq \mathbb{R}$, let $f : A \to \mathbb{R}$, and let $c \in A$. We say that f has a *relative extremum* at c if f has either a relative minimum at c or a relative maximum at c.

Theorem 0.196.7 (Interior Extremum Theorem)**.** *Let $I \subseteq \mathbb{R}$ be an open interval, let $f : I \to \mathbb{R}$, and let $c \in I$. If f has a relative extremum at c and f is differentiable at c, then*

$$f'(c) = 0.$$

Go to proof.

Theorem 0.196.8 (Necessary Condition for an Interior Extremum)**.** *Let $I \subseteq \mathbb{R}$ be an open interval, let $f : I \to \mathbb{R}$, and let $c \in I$. If f has a relative extremum at c, then either $f'(c)$ does not exist, or*

$$f'(c) = 0.$$

Go to proof.

Corollary 0.196.9 (Necessary Condition for an Extremum)**.** *Let $I \subseteq \mathbb{R}$ be an open interval, let $f : I \to \mathbb{R}$, and let $c \in I$. If f has a relative extremum at c, then*

$$f'(c) = 0 \quad \text{or} \quad f'(c) \text{ does not exist.}$$

Go to proof.

0.197 The Mean Value Theorems

Theorem 0.197.1 (Rolle's Theorem)**.** *Let $a, b \in \mathbb{R}$ with $a < b$. Suppose that $f : [a,b] \to \mathbb{R}$ is continuous on $[a,b]$, differentiable on (a,b), and satisfies $f(a) = f(b)$. Then there exists $c \in (a,b)$ such that*

$$f'(c) = 0.$$

Go to proof.

Theorem 0.197.2 (Mean Value Theorem)**.** *Let $a, b \in \mathbb{R}$ with $a < b$. Suppose that $f : [a,b] \to \mathbb{R}$ is continuous on $[a,b]$ and differentiable on (a,b). Then there exists $c \in (a,b)$ such that*

$$f'(c) = \frac{f(b) - f(a)}{b - a}.$$

Go to proof.

Theorem 0.197.3 (Cauchy Mean Value Theorem). *Let $a, b \in \mathbb{R}$ with $a < b$. Suppose that $f, g : [a, b] \to \mathbb{R}$ are continuous on $[a, b]$ and differentiable on (a, b). Then there exists $c \in (a, b)$ such that*

$$\big(f(b) - f(a)\big)g'(c) = \big(g(b) - g(a)\big)f'(c).$$

If, in addition, $g'(x) \neq 0$ for every $x \in (a, b)$, then $g(b) \neq g(a)$ and

$$\frac{f'(c)}{g'(c)} = \frac{f(b) - f(a)}{g(b) - g(a)}.$$

Go to proof.

Corollary 0.197.4 (Derivative Bound Implies Lipschitz). *Let $I \subseteq \mathbb{R}$ be an interval and let $f : I \to \mathbb{R}$ be differentiable on I. If there exists $M \geq 0$ such that*

$$|f'(x)| \leq M \qquad \textit{for every } x \in I,$$

then f is Lipschitz on I with constant M:

$$|f(x) - f(y)| \leq M|x - y| \qquad \textit{for every } x, y \in I.$$

Go to proof.

0.198 Reading the Graph from f' and f''

0.198.1 First-Order Shape

Definition 0.198.1 (Increasing at a Point). Let $A \subseteq \mathbb{R}$, let $f : A \to \mathbb{R}$, and let $c \in A$. We say that f is *increasing at c* if there exists $\delta > 0$ such that for all

$$x \in A \cap (c - \delta, c) \quad \text{and} \quad y \in A \cap (c, c + \delta),$$

we have $f(x) \leq f(c) \leq f(y)$.

Definition 0.198.2 (Decreasing at a Point). Let $A \subseteq \mathbb{R}$, let $f : A \to \mathbb{R}$, and let $c \in A$. We say that f is *decreasing at c* if there exists $\delta > 0$ such that for all

$$x \in A \cap (c - \delta, c) \quad \text{and} \quad y \in A \cap (c, c + \delta),$$

we have $f(x) \geq f(c) \geq f(y)$.

Theorem 0.198.3 (Zero Derivative Implies Constant). *Let $I := [a, b] \subseteq \mathbb{R}$ be a closed interval, and let $f : I \to \mathbb{R}$ be continuous on I and differentiable on (a, b). If*

$$f'(x) = 0 \qquad \textit{for all } x \in (a, b),$$

then f is constant on I. *Go to proof.*

Corollary 0.198.4 (Equal Derivatives Imply Difference Is Constant). *Let $I := [a, b] \subseteq \mathbb{R}$, and let $f, g : I \to \mathbb{R}$ be continuous on I and differentiable on (a, b). If*

$$f'(x) = g'(x) \qquad \textit{for all } x \in (a, b),$$

then there exists $C \in \mathbb{R}$ such that $f(x) = g(x) + C$ for all $x \in I$. *Go to proof.*

Theorem 0.198.5 (Nondecreasing Iff Nonnegative Derivative). *Let $I \subseteq \mathbb{R}$ be an interval, and let $f : I \to \mathbb{R}$ be differentiable on I. Then f is nondecreasing on I if and only if*

$$f'(x) \geq 0 \qquad \textit{for all } x \in I.$$

Go to proof.

Theorem 0.198.6 (Nonincreasing Iff Nonpositive Derivative)**.** *Let $I \subseteq \mathbb{R}$ be an interval, and let $f : I \to \mathbb{R}$ be differentiable on I. Then f is nonincreasing on I if and only if*

$$f'(x) \leq 0 \qquad \text{for all } x \in I.$$

Go to proof.

Lemma 0.198.7 (Positive Derivative Implies Locally Increasing)**.** *Let $I \subseteq \mathbb{R}$ be an interval, let $f : I \to \mathbb{R}$, let $c \in I$, and suppose that f is differentiable at c. If $f'(c) > 0$, then there exists $\delta > 0$ such that*

$$f(x) < f(c) \quad \text{for all } x \in I \text{ with } c - \delta < x < c,$$

and

$$f(x) > f(c) \quad \text{for all } x \in I \text{ with } c < x < c + \delta.$$

Go to proof.

Lemma 0.198.8 (Negative Derivative Implies Locally Decreasing)**.** *Let $I \subseteq \mathbb{R}$ be an interval, let $f : I \to \mathbb{R}$, let $c \in I$, and suppose that f is differentiable at c. If $f'(c) < 0$, then there exists $\delta > 0$ such that*

$$f(x) > f(c) \quad \text{for all } x \in I \text{ with } c - \delta < x < c,$$

and

$$f(x) < f(c) \quad \text{for all } x \in I \text{ with } c < x < c + \delta.$$

Go to proof.

Theorem 0.198.9 (First Derivative Test: Relative Maximum)**.** *Let $I := [a, b] \subseteq \mathbb{R}$, let $f : I \to \mathbb{R}$ be continuous on I, and let $c \in (a, b)$. Suppose f is differentiable on (a, c) and (c, b). If there exists $\delta > 0$ such that $(c - \delta, c + \delta) \subseteq I$ and*

$$f'(x) \geq 0 \quad \text{for all } x \in (c - \delta, c), \qquad f'(x) \leq 0 \quad \text{for all } x \in (c, c + \delta),$$

then f has a relative maximum at c. *Go to proof.*

Theorem 0.198.10 (First Derivative Test: Relative Minimum)**.** *Let $I := [a, b] \subseteq \mathbb{R}$, let $f : I \to \mathbb{R}$ be continuous on I, and let $c \in (a, b)$. Suppose f is differentiable on (a, c) and (c, b). If there exists $\delta > 0$ such that $(c - \delta, c + \delta) \subseteq I$ and*

$$f'(x) \leq 0 \quad \text{for all } x \in (c - \delta, c), \qquad f'(x) \geq 0 \quad \text{for all } x \in (c, c + \delta),$$

then f has a relative minimum at c. *Go to proof.*

0.198.2 Second-Order Shape

0.198.3 Higher-Order Derivatives

Definition 0.198.11 (Second Derivative)**.** Let f be differentiable on a neighbourhood of c. If f' is differentiable at c, the *second derivative* of f at c is

$$f''(c) := (f')'(c).$$

Definition 0.198.12 (Higher Derivatives)**.** Define $f^{(0)} := f$ and $f^{(1)} := f'$. For $n \geq 2$, if $f^{(n-1)}$ is differentiable at c, define

$$f^{(n)}(c) := (f^{(n-1)})'(c).$$

0.198.4 Second-Order Shape

Theorem 0.198.13 (Second-Derivative Convexity Test)**.** *Let $I \subseteq \mathbb{R}$ be an interval and let $f : I \to \mathbb{R}$ be twice differentiable on I. If*

$$f''(x) \geq 0 \qquad \text{for all } x \in I,$$

then f is convex on I. Go to proof.

Theorem 0.198.14 (Second-Derivative Concavity Test)**.** *Let $I \subseteq \mathbb{R}$ be an interval and let $f : I \to \mathbb{R}$ be twice differentiable on I. If*

$$f''(x) \leq 0 \qquad \text{for all } x \in I,$$

then f is concave on I. Go to proof.

Theorem 0.198.15 (Second Derivative Test)**.** *Let f be twice differentiable in a neighbourhood of c, and suppose $f'(c) = 0$.*

- *If $f''(c) > 0$, then f has a strict relative minimum at c.*
- *If $f''(c) < 0$, then f has a strict relative maximum at c.*

Go to proof.

Proposition 0.198.16 (Inflection Point Necessary Condition)**.** *Let f have an inflection point at c. If f'' exists on a neighbourhood of c and is continuous at c, then*

$$f''(c) = 0.$$

Go to proof.

0.198.5 Darboux's Theorem

Theorem 0.198.17 (Darboux's Theorem)**.** *Let $I := [a, b] \subseteq \mathbb{R}$ and let $f : I \to \mathbb{R}$ be differentiable on I. If k is a real number strictly between $f'(a)$ and $f'(b)$, then there exists $c \in (a, b)$ such that*

$$f'(c) = k.$$

Go to proof.

0.198.6 Smoothness Classes

Definition 0.198.18 (Continuously Differentiable; Class C^1)**.** Let $I \subseteq \mathbb{R}$ be an interval and let $f : I \to \mathbb{R}$. The function f is *continuously differentiable* on I, written $f \in C^1(I)$, if f is differentiable at every point of I and the derivative $f' : I \to \mathbb{R}$ is continuous on I.

Definition 0.198.19 (Class C^k)**.** Let $I \subseteq \mathbb{R}$ be an interval, let $k \in \mathbb{N}$, and let $f : I \to \mathbb{R}$. The function f is of *class C^k* on I, written $f \in C^k(I)$, if the derivatives $f^{(0)}, f^{(1)}, \ldots, f^{(k)}$ exist on I and $f^{(k)}$ is continuous on I. By convention, $C^0(I)$ is the class of continuous functions on I.

Definition 0.198.20 (Smooth Function; Class C^∞)**.** Let $I \subseteq \mathbb{R}$ be an interval and let $f : I \to \mathbb{R}$. The function f is *smooth*, or of *class C^∞*, written $f \in C^\infty(I)$, if $f \in C^k(I)$ for every $k \in \mathbb{N}$. Equivalently,

$$C^\infty(I) = \bigcap_{k=0}^{\infty} C^k(I).$$

Definition 0.198.21 (Real-Analytic Function; Class C^ω)**.** Let $I \subseteq \mathbb{R}$ be an interval and let $f : I \to \mathbb{R}$. The function f is *real-analytic*, or of *class C^ω*, written $f \in C^\omega(I)$, if for every $a \in I$ there exists $r > 0$ such that

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!}(x - a)^n \qquad \text{for every } x \in (a - r, a + r) \cap I.$$

Theorem 0.198.22 (The Smoothness Tower)**.** *For every nondegenerate interval I, the inclusions*

$$C^0(I) \supseteq C^1(I) \supseteq C^2(I) \supseteq \cdots \supseteq C^\infty(I) \supseteq C^\omega(I)$$

hold, and each displayed inclusion is strict. *Go to proof.*

Definition 0.198.23 (Lipschitz-Derivative Class $C^{1,1}$)**.** Let $I \subseteq \mathbb{R}$ be an interval and let $f : I \to \mathbb{R}$. The function f is of *class* $C^{1,1}$ on I, written $f \in C^{1,1}(I)$, if $f \in C^1(I)$ and f' is Lipschitz on I: there exists $L \geq 0$ such that

$$|f'(x) - f'(y)| \leq L|x - y| \qquad \text{for all } x, y \in I.$$

Theorem 0.198.24 (Placement of $C^{1,1}$)**.** *For every nondegenerate interval I,*

$$C^{1,1}(I) \subsetneq C^1(I),$$

and $C^{1,1}(I)$ is not contained in $C^2(I)$. If K is a compact interval, then

$$C^2(K) \subseteq C^{1,1}(K) \subsetneq C^1(K),$$

and the first inclusion is strict. *Go to proof.*

Corollary 0.198.25 (Bounded Second Derivative Implies $C^{1,1}$)**.** *Let $I \subseteq \mathbb{R}$ be an interval and let $f \in C^2(I)$. If $|f''(x)| \leq M$ for all $x \in I$, then $f \in C^{1,1}(I)$, and f' is Lipschitz with constant M.* *Go to proof.*

0.199 The Algebra of Derivatives

0.199.1 From Characterization to Computation

Theorem 0.199.1 (Constant Multiple Rule)**.** *Let $A \subseteq \mathbb{R}$, let $f : A \to \mathbb{R}$, let $c \in A$ be a limit point of A, and let $\alpha \in \mathbb{R}$. If f is differentiable at c, then αf is differentiable at c, and*

$$(\alpha f)'(c) = \alpha f'(c).$$

Go to proof.

Theorem 0.199.2 (Sum Rule)**.** *Let $A \subseteq \mathbb{R}$, let $f, g : A \to \mathbb{R}$, and let $c \in A$ be a limit point of A. If f and g are differentiable at c, then $f + g$ is differentiable at c, and*

$$(f + g)'(c) = f'(c) + g'(c).$$

Go to proof.

Theorem 0.199.3 (Product Rule)**.** *Let $A \subseteq \mathbb{R}$, let $f, g : A \to \mathbb{R}$, and let $c \in A$ be a limit point of A. If f and g are differentiable at c, then fg is differentiable at c, and*

$$(fg)'(c) = f'(c)g(c) + f(c)g'(c).$$

Go to proof.

Theorem 0.199.4 (Quotient Rule)**.** *Let $A \subseteq \mathbb{R}$, let $f, g : A \to \mathbb{R}$, and let $c \in A$ be a limit point of A. Suppose that f and g are differentiable at c and that $g(c) \neq 0$. Then f/g is differentiable at c, and*

$$\left(\frac{f}{g}\right)'(c) = \frac{f'(c)g(c) - f(c)g'(c)}{(g(c))^2}.$$

Go to proof.

0.199.2 Closure

Corollary 0.199.5 (Finite Sum Rule). *Let $A \subseteq \mathbb{R}$, let $f_1, \dots, f_n : A \to \mathbb{R}$, let $\alpha_1, \dots, \alpha_n \in \mathbb{R}$, and let $c \in A$ be a limit point of A. If each f_i is differentiable at c, then $\sum_{i=1}^{n} \alpha_i f_i$ is differentiable at c, and*

$$\left(\sum_{i=1}^{n} \alpha_i f_i\right)'(c) = \sum_{i=1}^{n} \alpha_i f_i'(c).$$

Go to proof.

Corollary 0.199.6 (Extended Product Rule). *Let $A \subseteq \mathbb{R}$, let $f_1, \dots, f_n : A \to \mathbb{R}$, and let $c \in A$ be a limit point of A. If each f_i is differentiable at c, then $\prod_{i=1}^{n} f_i$ is differentiable at c, and*

$$\left(\prod_{i=1}^{n} f_i\right)'(c) = \sum_{k=1}^{n} f_k'(c) \prod_{\substack{i=1 \\ i \neq k}}^{n} f_i(c).$$

Go to proof.

Corollary 0.199.7 (Power Rule for Integer Powers of a Function). *Let $A \subseteq \mathbb{R}$, let $f : A \to \mathbb{R}$, let $c \in A$ be a limit point of A, and let $n \in \mathbb{N}$. If f is differentiable at c, then f^n is differentiable at c, and*

$$(f^n)'(c) = n(f(c))^{n-1} f'(c).$$

Go to proof.

Theorem 0.199.8 (Finite Linear Combination Rule). *Let $A \subseteq \mathbb{R}$, let $f_1, \dots, f_n : A \to \mathbb{R}$, let $\alpha_1, \dots, \alpha_n \in \mathbb{R}$, and let $c \in A$ be a limit point of A. If each f_i is differentiable at c, then*

$$\left(\sum_{i=1}^{n} \alpha_i f_i\right)'(c) = \sum_{i=1}^{n} \alpha_i f_i'(c).$$

Go to proof.

0.199.3 Global Forms on an Interval

Theorem 0.199.9 (Constant Multiple Rule on an Interval). *Let $I \subseteq \mathbb{R}$ be an interval, let $f : I \to \mathbb{R}$, and let $\alpha \in \mathbb{R}$. If f is differentiable on I, then αf is differentiable on I, and*

$$(\alpha f)'(x) = \alpha f'(x) \qquad \text{for all } x \in I.$$

Go to proof.

Theorem 0.199.10 (Sum Rule on an Interval). *Let $I \subseteq \mathbb{R}$ be an interval, and let $f, g : I \to \mathbb{R}$. If f and g are differentiable on I, then $f + g$ is differentiable on I, and*

$$(f + g)'(x) = f'(x) + g'(x) \qquad \text{for all } x \in I.$$

Go to proof.

Theorem 0.199.11 (Product Rule on an Interval). *Let $I \subseteq \mathbb{R}$ be an interval, and let $f, g : I \to \mathbb{R}$. If f and g are differentiable on I, then fg is differentiable on I, and*

$$(fg)'(x) = f'(x)g(x) + f(x)g'(x) \qquad \text{for all } x \in I.$$

Go to proof.

Theorem 0.199.12 (Quotient Rule on an Interval)**.** *Let $I \subseteq \mathbb{R}$ be an interval, and let $f, g : I \to \mathbb{R}$. Suppose that f and g are differentiable on I and that $g(x) \neq 0$ for all $x \in I$. Then f/g is differentiable on I, and*

$$\left(\frac{f}{g}\right)'(x) = \frac{f'(x)g(x) - f(x)g'(x)}{(g(x))^2} \qquad \text{for all } x \in I.$$

Go to proof.

Corollary 0.199.13 (Finite Sum Rule on an Interval)**.** *Let $I \subseteq \mathbb{R}$ be an interval, let $f_1, \dots, f_n : I \to \mathbb{R}$, and let $\alpha_1, \dots, \alpha_n \in \mathbb{R}$. If each f_i is differentiable on I, then $\sum_{i=1}^n \alpha_i f_i$ is differentiable on I, and*

$$\left(\sum_{i=1}^n \alpha_i f_i\right)'(x) = \sum_{i=1}^n \alpha_i f_i'(x) \qquad \text{for all } x \in I.$$

Go to proof.

Corollary 0.199.14 (Extended Product Rule on an Interval)**.** *Let $I \subseteq \mathbb{R}$ be an interval, and let $f_1, \dots, f_n : I \to \mathbb{R}$. If each f_i is differentiable on I, then $\prod_{i=1}^n f_i$ is differentiable on I, and*

$$\left(\prod_{i=1}^n f_i\right)'(x) = \sum_{k=1}^n f_k'(x) \prod_{\substack{i=1\\ i\neq k}}^n f_i(x) \qquad \text{for all } x \in I.$$

Go to proof.

Corollary 0.199.15 (Power Rule for Integer Powers of a Function on an Interval)**.** *Let $I \subseteq \mathbb{R}$ be an interval, let $f : I \to \mathbb{R}$, and let $n \in \mathbb{N}$. If f is differentiable on I, then f^n is differentiable on I, and*

$$(f^n)'(x) = n(f(x))^{n-1} f'(x) \qquad \text{for all } x \in I.$$

Go to proof.

Theorem 0.199.16 (Finite Linear Combination Rule on an Interval)**.** *Let $I \subseteq \mathbb{R}$ be an interval, let $f_1, \dots, f_n : I \to \mathbb{R}$, and let $\alpha_1, \dots, \alpha_n \in \mathbb{R}$. If each f_i is differentiable on I, then*

$$\left(\sum_{i=1}^n \alpha_i f_i\right)'(x) = \sum_{i=1}^n \alpha_i f_i'(x) \qquad \text{for all } x \in I.$$

Go to proof.

0.199.4 Inverting the Operation

Theorem 0.199.17 (Inverse Function Theorem, One Variable)**.** *Let $I \subseteq \mathbb{R}$ be an open interval, let $f : I \to \mathbb{R}$ be of class C^1, and let $c \in I$ satisfy $f'(c) \neq 0$. Then there exist open intervals $U \subseteq I$ and $V \subseteq \mathbb{R}$, with $c \in U$ and $f(c) \in V$, such that:*

1. *$f|_U : U \to V$ is a bijection;*
2. *the inverse $g = (f|_U)^{-1} : V \to U$ is of class C^1;*
3. *for every $y \in V$,*
$$g'(y) = \frac{1}{f'(g(y))}.$$

Go to proof.

Corollary 0.199.18 (Derivative of the Inverse Function)**.** *Under the hypotheses of the one-variable Inverse Function Theorem, the local inverse g satisfies*

$$g' = \frac{1}{f' \circ g}$$

on V. More globally, if $f : I \to J$ is a C^1 bijection with $f'(x) \neq 0$ for every $x \in I$, then $f^{-1} : J \to I$ is C^1 and

$$(f^{-1})'(y) = \frac{1}{f'(f^{-1}(y))} \qquad \text{for every } y \in J.$$

Go to proof.

0.199.5 Indeterminate Forms: L'Hôpital's Rule

Theorem 0.199.19 (L'Hôpital's Rule, 0/0 Form)**.** *Let $a < b$, and let $f, g : (a, b) \to \mathbb{R}$ be differentiable with $g'(x) \neq 0$ for every $x \in (a, b)$. Suppose*

$$\lim_{x\to a^+} f(x) = \lim_{x\to a^+} g(x) = 0 \qquad \text{and} \qquad \lim_{x\to a^+} \frac{f'(x)}{g'(x)} = L$$

for $L \in \mathbb{R} \cup \{-\infty, +\infty\}$. Then

$$\lim_{x\to a^+} \frac{f(x)}{g(x)} = L.$$

The analogous statements at left-hand endpoints, at two-sided interior points, and at infinity follow by the same reductions. Go to proof.

Theorem 0.199.20 (L'Hôpital's Rule, ∞/∞ Form)**.** *Let $a < b$, and let $f, g : (a, b) \to \mathbb{R}$ be differentiable with $g'(x) \neq 0$ for every $x \in (a, b)$. Suppose*

$$\lim_{x\to a^+} |g(x)| = +\infty \qquad \text{and} \qquad \lim_{x\to a^+} \frac{f'(x)}{g'(x)} = L$$

for $L \in \mathbb{R} \cup \{-\infty, +\infty\}$. Then

$$\lim_{x\to a^+} \frac{f(x)}{g(x)} = L.$$

No separate hypothesis on $\lim_{x\to a^+} f(x)$ is required. Go to proof.

0.200 Taylor: Construction and Its Error

Definition 0.200.1 (Taylor Polynomial at a Point)**.** Let $I \subseteq \mathbb{R}$ be an interval, let $a \in I$, let $n \in \mathbb{N}$, and let $f : I \to \mathbb{R}$ have derivatives $f^{(k)}(a)$ for $0 \leq k \leq n$. The **Taylor polynomial of order** n for f at a is

$$T_{n,a}f(x) := \sum_{k=0}^{n} \frac{f^{(k)}(a)}{k!}(x-a)^k.$$

Definition 0.200.2 (Taylor Remainder)**.** Let $I \subseteq \mathbb{R}$ be an interval, let $a \in I$, let $n \in \mathbb{N}$, and let $f : I \to \mathbb{R}$ have a Taylor polynomial $T_{n,a}f$. The **Taylor remainder of order** n at a is the function $R_{n,a}f : I \to \mathbb{R}$ defined by

$$R_{n,a}f(x) := f(x) - T_{n,a}f(x).$$

Definition 0.200.3 (Maclaurin Polynomial)**.** Let $n \in \mathbb{N}$, and let f be defined on an interval containing 0 with derivatives $f^{(k)}(0)$ for $0 \leq k \leq n$. The **Maclaurin polynomial of order** n for f is

$$T_{n,0}f(x) := \sum_{k=0}^{n} \frac{f^{(k)}(0)}{k!}x^k.$$

Theorem 0.200.4 (Taylor's Theorem with Lagrange Remainder)**.** *Let $a, b \in \mathbb{R}$ with $a < b$, let $n \in \mathbb{N}$, and let $f : [a, b] \to \mathbb{R}$. Suppose that $f, f', \ldots, f^{(n)}$ are continuous on $[a, b]$ and that $f^{(n+1)}$ exists on (a, b). Then, for every $x \in (a, b)$, there exists c strictly between a and x such that*

$$f(x) = \sum_{k=0}^{n} \frac{f^{(k)}(a)}{k!}(x-a)^k + \frac{f^{(n+1)}(c)}{(n+1)!}(x-a)^{n+1}.$$

Go to proof.

Corollary 0.200.5 (Taylor Expansion with Peano Remainder)**.** *Let $I \subseteq \mathbb{R}$ be an interval, let $a \in I$, let $n \in \mathbb{N}$, and let $f : I \to \mathbb{R}$ have derivatives up to order n in a neighbourhood of a. If $f^{(n)}$ is continuous at a, then*

$$f(x) = \sum_{k=0}^{n} \frac{f^{(k)}(a)}{k!}(x-a)^k + o\big((x-a)^n\big) \qquad \text{as } x \to a.$$

Go to proof.

Corollary 0.200.6 (First-Order Peano Remainder)**.** *Let f be differentiable at c. Then, as $h \to 0$,*

$$f(c+h) = f(c) + f'(c)h + o(h).$$

Go to proof.

Proposition 0.200.7 (The Flat Function: Smooth but Not Analytic)**.** *Define $f : \mathbb{R} \to \mathbb{R}$ by*

$$f(x) = \begin{cases} e^{-1/x^2}, & x \neq 0, \\ 0, & x = 0. \end{cases}$$

Then:

1. *$f \in C^\infty(\mathbb{R})$;*
2. *$f^{(n)}(0) = 0$ for every $n \geq 0$;*
3. *the Maclaurin series of f is identically 0, so $f \notin C^\omega$ on any neighbourhood of 0.*

In particular, $C^\infty(\mathbb{R}) \setminus C^\omega(\mathbb{R}) \neq \varnothing$. *Go to proof.*

0.200.1 The Differential

Definition 0.200.8 (Differentiability by a Differential)**.** Let f be defined on a neighbourhood of $c \in \mathbb{R}$. The function f is *differentiable at c in the differential sense* if there exists a linear map $L : \mathbb{R} \to \mathbb{R}$ such that

$$f(c+h) - f(c) = L(h) + o(h) \qquad \text{as } h \to 0.$$

In that case the *differential* of f at c is $df_c := L$.

Theorem 0.200.9 (Differential and Derivative Agree)**.** *Let f be defined on a neighbourhood of $c \in \mathbb{R}$. Then f is differentiable at c in the ordinary derivative sense if and only if it is differentiable at c in the differential sense. When this holds,*

$$df_c(h) = f'(c)h.$$

Go to proof.

Theorem 0.200.10 (Uniqueness of the Differential)**.** *If $L_1, L_2 : \mathbb{R} \to \mathbb{R}$ are linear maps and*

$$f(c+h) - f(c) = L_1(h) + o(h) = L_2(h) + o(h) \qquad \text{as } h \to 0,$$

then $L_1 = L_2$. *Go to proof.*

Theorem 0.200.11 (Differential Continuity Criterion)**.** *If f is differentiable at c in the differential sense, then f is continuous at c.* *Go to proof.*

Theorem 0.200.12 (Chain Rule for Differentials)**.** *If f is differentiable at c and g is differentiable at $f(c)$, then*

$$d(g \circ f)_c = dg_{f(c)} \circ df_c.$$

Go to proof.

Theorem 0.200.13 (Linearity of the Differential)**.** *If f and g are differentiable at c and $\alpha, \beta \in \mathbb{R}$, then*

$$d(\alpha f + \beta g)_c = \alpha \, df_c + \beta \, dg_c.$$

Go to proof.

Proofs

Integration

Integration

0.201 Partitions and Tags

The Geometric Problem

Definition 0.201.1 (Partition of a Compact Interval)**.** A *partition* of $[a, b]$ is a finite increasing list

$$P = \{a = x_0 < x_1 < \cdots < x_n = b\}.$$

Equivalently,

$$\begin{aligned}\mathrm{Partition}(P, a, b) \iff &\exists n \in \mathbb{N}\, \exists x_0, \ldots, x_n \in \mathbb{R} \\ &\big(P = \{x_0, \ldots, x_n\} \wedge x_0 = a \wedge x_n = b\big) \\ &\wedge \forall i\, \big(1 \leq i \leq n \Rightarrow x_{i-1} < x_i\big).\end{aligned}$$

Definition 0.201.2 (Subinterval Width)**.** The i-th slab has width

$$\Delta x_i = x_i - x_{i-1}.$$

Definition 0.201.3 (Mesh of a Partition)**.** The *mesh* or *norm* of P is the width of its fattest slab:

$$\|P\| = \max_{1 \leq i \leq n} \Delta x_i.$$

Definition 0.201.4 (Tagged Partition)**.** A *tagged partition* is a partition $P = \{x_0, \ldots, x_n\}$ together with points $\xi_i \in [x_{i-1}, x_i]$. The tag is the point where the height $f(\xi_i)$ is sampled.

Definition 0.201.5 (Refinement of a Partition)**.** A partition P' *refines* P when $P \subseteq P'$. Refinement adds cut-points; it does not move or delete existing cut-points.

Lemma 0.201.6 (Common Refinement of Two Partitions)**.** *Any two partitions P and P' of $[a, b]$ have a common refinement: order the finite set $P \cup P'$ increasingly.*

0.202 The Cauchy Integral

Geometric Intuition

Definition 0.202.1 (Left-Endpoint Cauchy Sum)**.** For $f : [a, b] \to \mathbb{R}$ and $P = \{a = x_0 < \cdots < x_n = b\}$, define

$$C(f, P) = \sum_{i=1}^{n} f(x_{i-1})\, \Delta x_i.$$

Definition 0.202.2 (Cauchy Integral)**.** The function f is *Cauchy-integrable* with value L if

$$\forall \varepsilon > 0\; \exists \delta > 0\; \forall P\, \big(\|P\| < \delta \Rightarrow |C(f, P) - L| < \varepsilon\big).$$

The value L is denoted $\int_a^b f(x)\, dx$.

Proposition 0.202.3 (Cauchy Integral of a Constant)**.** *If $f(x) = c$ on $[a, b]$, then $\int_a^b c\,dx = c(b - a)$.*

Theorem 0.202.4 (Linearity of the Cauchy Integral)**.** *If f and g are Cauchy-integrable on $[a, b]$, then*

$$\int_a^b (\alpha f + \beta g)\,dx = \alpha \int_a^b f\,dx + \beta \int_a^b g\,dx.$$

Theorem 0.202.5 (Monotonicity of the Cauchy Integral)**.** *If f and g are Cauchy-integrable on $[a, b]$ and $f(x) \le g(x)$ for every $x \in [a, b]$, then*

$$\int_a^b f\,dx \le \int_a^b g\,dx.$$

Corollary 0.202.6 (Bounds for the Cauchy Integral)**.** *If f is continuous on $[a, b]$, $m = \min_{[a,b]} f$, and $M = \max_{[a,b]} f$, then*

$$m(b - a) \le \int_a^b f\,dx \le M(b - a).$$

Theorem 0.202.7 (Triangle Inequality for the Cauchy Integral)**.** *If f and $|f|$ are Cauchy-integrable on $[a, b]$, then*

$$\left| \int_a^b f\,dx \right| \le \int_a^b |f|\,dx.$$

Theorem 0.202.8 (Interval Additivity for the Cauchy Integral)**.** *If $c \in [a, b]$ and f is Cauchy-integrable on the relevant intervals, then*

$$\int_a^b f\,dx = \int_a^c f\,dx + \int_c^b f\,dx.$$

Definition 0.202.9 (Oscillation on an Interval)**.** For bounded f on an interval I, define

$$\omega_f(I) = \sup_I f - \inf_I f.$$

Oscillation is the error controller: if $\omega_f(I)$ is small, every sample height on I is close to every other sample height.

Theorem 0.202.10 (Continuous Functions Are Cauchy-Integrable)**.** *Every continuous function $f : [a, b] \to \mathbb{R}$ is Cauchy-integrable.*

Theorem 0.202.11 (Tag Independence for Continuous Functions)**.** *If f is continuous on $[a, b]$, then every choice of tags gives the same limit as the left-endpoint sums as $\|P\| \to 0$.*

0.203 The Riemann Integral

Geometric Intuition

Definition 0.203.1 (Riemann Sum)**.** For a tagged partition (P, ξ), define

$$S(f, P, \xi) = \sum_{i=1}^{n} f(\xi_i)\Delta x_i.$$

Definition 0.203.2 (Riemann Integral)**.** A function $f : [a, b] \to \mathbb{R}$ is *Riemann-integrable* with value L if

$$\forall \varepsilon > 0\ \exists \delta > 0\ \forall (P, \xi)\ \big(\|P\| < \delta \Rightarrow |S(f, P, \xi) - L| < \varepsilon\big).$$

Theorem 0.203.3 (Continuous Functions Are Riemann-Integrable)**.** *Every continuous function $f : [a, b] \to \mathbb{R}$ is Riemann-integrable.*

Theorem 0.203.4 (Linearity of the Riemann Integral)**.** *If f and g are Riemann-integrable on $[a,b]$, then*

$$\int_a^b (\alpha f + \beta g)\,dx = \alpha \int_a^b f\,dx + \beta \int_a^b g\,dx.$$

Theorem 0.203.5 (Monotonicity of the Riemann Integral)**.** *If f and g are Riemann-integrable on $[a,b]$ and $f(x) \le g(x)$ for every $x \in [a,b]$, then*

$$\int_a^b f\,dx \le \int_a^b g\,dx.$$

Theorem 0.203.6 (Triangle Inequality for the Riemann Integral)**.** *If f is Riemann-integrable on $[a,b]$, then $|f|$ is Riemann-integrable and*

$$\left|\int_a^b f\,dx\right| \le \int_a^b |f|\,dx.$$

Theorem 0.203.7 (Interval Additivity for the Riemann Integral)**.** *If $c \in [a,b]$, then f is Riemann-integrable on $[a,b]$ if and only if it is Riemann-integrable on $[a,c]$ and $[c,b]$, and in that case*

$$\int_a^b f\,dx = \int_a^c f\,dx + \int_c^b f\,dx.$$

Theorem 0.203.8 (Cauchy Criterion for Riemann Integrability)**.** *A bounded function is Riemann-integrable if and only if for every $\varepsilon > 0$ there is $\delta > 0$ such that any two tagged partitions (P,ξ) and (Q,η) with $\|P\|, \|Q\| < \delta$ have*

$$|S(f,P,\xi) - S(f,Q,\eta)| < \varepsilon.$$

0.204 The Darboux Integral

Geometric Intuition

Definition 0.204.1 (Lower and Upper Darboux Sums)**.** For bounded f on $[a,b]$, set

$$m_i = \inf_{[x_{i-1},x_i]} f, \qquad M_i = \sup_{[x_{i-1},x_i]} f,$$

and define

$$L(f,P) = \sum_i m_i \Delta x_i, \qquad U(f,P) = \sum_i M_i \Delta x_i.$$

Lemma 0.204.2 (Darboux Sums Under Refinement)**.** *If P' refines P, then*

$$L(f,P) \le L(f,P') \le U(f,P') \le U(f,P).$$

Definition 0.204.3 (Darboux Integrability)**.** The lower and upper integrals are

$$\underline{I} = \sup_P L(f,P), \qquad \overline{I} = \inf_P U(f,P).$$

The function f is *Darboux-integrable* if $\underline{I} = \overline{I}$.

Theorem 0.204.4 (Darboux Criterion)**.** *A bounded function f is Darboux-integrable if and only if for every $\varepsilon > 0$ there exists a partition P such that*

$$U(f,P) - L(f,P) < \varepsilon.$$

Theorem 0.204.5 (Equivalence of Riemann and Darboux Integrability)**.** *For bounded functions on $[a,b]$, Riemann integrability and Darboux integrability are equivalent, and the integral values agree.*

Theorem 0.204.6 (Continuous Functions Are Darboux-Integrable)**.** *Every continuous function $f : [a, b] \to \mathbb{R}$ is Darboux-integrable.*

Theorem 0.204.7 (Monotone Functions Are Darboux-Integrable)**.** *Every monotone function $f : [a, b] \to \mathbb{R}$ is Darboux-integrable.*

Theorem 0.204.8 (Bounded Functions with Finitely Many Discontinuities Are Darboux-Integrable)**.** *If $f : [a, b] \to \mathbb{R}$ is bounded and has only finitely many discontinuities, then f is Darboux-integrable.*

Theorem 0.204.9 (Sums and Scalar Multiples of Darboux-Integrable Functions)**.** *If f and g are Darboux-integrable on $[a, b]$, then $\alpha f + \beta g$ is Darboux-integrable for all $\alpha, \beta \in \mathbb{R}$.*

Theorem 0.204.10 (Products of Darboux-Integrable Functions)**.** *If f and g are Darboux-integrable on $[a, b]$, then fg is Darboux-integrable on $[a, b]$.*

Theorem 0.204.11 (Absolute Values of Darboux-Integrable Functions)**.** *If f is Darboux-integrable on $[a, b]$, then $|f|$ is Darboux-integrable on $[a, b]$.*

Theorem 0.204.12 (Continuous Images of Darboux-Integrable Functions)**.** *If f is Darboux-integrable on $[a, b]$, φ is continuous on an interval containing $f([a, b])$, and f is bounded, then $\varphi \circ f$ is Darboux-integrable on $[a, b]$.*

0.205 The Riemann–Stieltjes Integral

Geometric Intuition

Definition 0.205.1 (Total Variation)**.** The total variation of α on $[a, b]$ is

$$V_a^b(\alpha) = \sup_P \sum_i |\alpha(x_i) - \alpha(x_{i-1})|.$$

Definition 0.205.2 (Bounded Variation)**.** The function α has *bounded variation* on $[a, b]$ if

$$V_a^b(\alpha) < \infty.$$

Geometrically, this says the ruler has finite total up-and-down travel.

Proposition 0.205.3 (Monotone Functions Have Bounded Variation)**.** *If α is monotone on $[a, b]$, then α has bounded variation.*

Definition 0.205.4 (Riemann–Stieltjes Integral)**.** The Riemann–Stieltjes sum is

$$\sum_i f(\xi_i)\big(\alpha(x_i) - \alpha(x_{i-1})\big).$$

When these sums have a tag-independent limit as $\|P\| \to 0$, the limit is denoted $\int_a^b f \, d\alpha$.

Theorem 0.205.5 (Continuous Integrand and BV Integrator)**.** *If f is continuous on $[a, b]$ and α has bounded variation, then $\int_a^b f \, d\alpha$ exists.*

Theorem 0.205.6 (Bilinearity of the Riemann–Stieltjes Integral)**.** *Whenever the displayed integrals exist,*

$$\int_a^b (\lambda f + \mu g) \, d\alpha = \lambda \int_a^b f \, d\alpha + \mu \int_a^b g \, d\alpha$$

and

$$\int_a^b f \, d(\lambda\alpha + \mu\beta) = \lambda \int_a^b f \, d\alpha + \mu \int_a^b f \, d\beta.$$

Theorem 0.205.7 (Interval Additivity for the Riemann–Stieltjes Integral). *If $c \in [a,b]$ and the relevant Riemann–Stieltjes integrals exist, then*

$$\int_a^b f\,d\alpha = \int_a^c f\,d\alpha + \int_c^b f\,d\alpha.$$

Theorem 0.205.8 (Riemann–Stieltjes Integration by Parts). *If f and α are such that one of $\int_a^b f\,d\alpha$ and $\int_a^b \alpha\,df$ exists, then the other exists under the same Riemann–Stieltjes convention, and*

$$\int_a^b f\,d\alpha + \int_a^b \alpha\,df = f(b)\alpha(b) - f(a)\alpha(a).$$

Theorem 0.205.9 (Reduction to Riemann Integration for a C^1 Integrator). *If $\alpha \in C^1([a,b])$ and f is Riemann-integrable on $[a,b]$, then*

$$\int_a^b f\,d\alpha = \int_a^b f(x)\alpha'(x)\,dx.$$

Theorem 0.205.10 (Step Integrators Give Finite Sums). *If α is constant except for finitely many jumps at points c_k and f is continuous at each c_k, then*

$$\int_a^b f\,d\alpha = \sum_k f(c_k)\,\Delta\alpha(c_k).$$

0.206 The Henstock–Kurzweil Integral

Where This Sits

Geometric Intuition

Definition 0.206.1 (Gauge on an Interval). A *gauge* on $[a,b]$ is a function

$$\delta : [a,b] \to (0,\infty).$$

Definition 0.206.2 (δ-Fine Tagged Partition). A tagged partition (P,ξ) is *δ-fine* if

$$[x_{i-1}, x_i] \subset \big(\xi_i - \delta(\xi_i),\, \xi_i + \delta(\xi_i)\big) \qquad \text{for every } i.$$

Definition 0.206.3 (Henstock–Kurzweil Integral). A function $f : [a,b] \to \mathbb{R}$ is *Henstock–Kurzweil integrable* with value A if

$$\forall\varepsilon > 0\ \exists\delta : [a,b] \to (0,\infty)\ \forall(P,\xi)\ \big((P,\xi) \text{ is } \delta\text{-fine} \Rightarrow |S(f,P,\xi) - A| < \varepsilon\big).$$

The value is denoted (HK) $\int_a^b f$.

Lemma 0.206.4 (Cousin's Lemma). *For every gauge δ on $[a,b]$, there exists a δ-fine tagged partition of $[a,b]$.*

Theorem 0.206.5 (Riemann Integrable Implies Henstock–Kurzweil Integrable). *If f is Riemann-integrable on $[a,b]$, then f is Henstock–Kurzweil integrable on $[a,b]$, and the two integrals have the same value.*

Lemma 0.206.6 (Straddle Lemma). *If F is differentiable at ξ with $F'(\xi) = f(\xi)$, then for every $\varepsilon > 0$ there is $\delta(\xi) > 0$ such that whenever $u \le \xi \le v$, $u, v \in (\xi - \delta(\xi), \xi + \delta(\xi))$,*

$$|F(v) - F(u) - f(\xi)(v-u)| \le \varepsilon(v-u).$$

Theorem 0.206.7 (Fundamental Theorem for the Henstock–Kurzweil Integral). *If $F : [a,b] \to \mathbb{R}$ is differentiable at every point and $F' = f$, then f is Henstock–Kurzweil integrable and*

$$(\mathrm{HK}) \int_a^b f = F(b) - F(a).$$

Theorem 0.206.8 (Continuous Functions Are Henstock–Kurzweil Integrable). *Every continuous function $f : [a,b] \to \mathbb{R}$ is Henstock–Kurzweil integrable.*

0.207 The McShane Integral

Where This Sits

Geometric Intuition

Definition 0.207.1 (McShane-Fine Tagged Partition)**.** Let $\delta : [a, b] \to (0, \infty)$ be a gauge. A finite tagged partition $\{([u_i, v_i], \xi_i)\}_{i=1}^{n}$ is *McShane δ-fine* if the intervals $[u_i, v_i]$ are non-overlapping, cover $[a, b]$, each $\xi_i \in [a, b]$, and

$$[u_i, v_i] \subset \big(\xi_i - \delta(\xi_i),\, \xi_i + \delta(\xi_i)\big) \qquad \text{for every } i.$$

Unlike an HK tagged partition, the tag ξ_i is not required to lie in $[u_i, v_i]$.

Definition 0.207.2 (McShane Integral)**.** A function $f : [a, b] \to \mathbb{R}$ is *McShane integrable* with value A if

$$\forall \varepsilon > 0\ \exists \delta : [a, b] \to (0, \infty)\ \forall (P, \xi)\ \big((P, \xi) \text{ is McShane } \delta\text{-fine} \Rightarrow |S(f, P, \xi) - A| < \varepsilon\big),$$

where

$$S(f, P, \xi) = \sum_i f(\xi_i)(v_i - u_i).$$

Theorem 0.207.3 (Riemann Integrable Implies McShane Integrable Implies Henstock–Kurzweil Integrable)**.** *On a compact interval,*

$$\mathcal{R}[a, b] \subseteq \mathcal{M}[a, b] \subseteq \mathcal{HK}[a, b],$$

and the integral values agree whenever a function belongs to the smaller class.

Theorem 0.207.4 (McShane Integrability Equals Lebesgue Integrability)**.** *For real-valued functions on a compact interval,*

$$\mathcal{M}\rfloor\mathcal{S}\langle\dashv\backslash\rceil[a, b] = \mathcal{L}[a, b],$$

and the McShane and Lebesgue integral values agree.

0.208 Measure Zero and the Lebesgue Criterion

Geometric Intuition

Definition 0.208.1 (Measure Zero)**.** A set $E \subseteq \mathbb{R}$ has *measure zero* if for every $\varepsilon > 0$ there are open intervals I_k such that

$$E \subseteq \bigcup_{k=1}^{\infty} I_k, \qquad \sum_{k=1}^{\infty} |I_k| < \varepsilon.$$

Definition 0.208.2 (Point Oscillation)**.** For bounded f on $[a, b]$, define

$$\omega_f(x) = \inf_{\delta > 0} \omega_f\big((x - \delta, x + \delta) \cap [a, b]\big).$$

Then f is continuous at x exactly when $\omega_f(x) = 0$.

Theorem 0.208.3 (Lebesgue Criterion for Riemann Integrability)**.** *A bounded function $f : [a, b] \to \mathbb{R}$ is Riemann-integrable if and only if its discontinuity set*

$$D = \{x \in [a, b] : \omega_f(x) > 0\}$$

has measure zero.

Proofs

Part IV

Volume IV: Mathematical Spaces

Mathematical Spaces

Mathematical Spaces

0.209 Spaces as Structured Arenas

Definition 0.209.1 (Space)**.** A **space** is a nonempty set.

Proofs

Metric Spaces

0.210 Metric Spaces

0.211 Definitions

Definition 0.211.1 (Metric Space)**.** Let X be a nonempty set. A function

$$d : X \times X \to \mathbb{R}_{\geq 0}$$

is called a *metric* on X if for all $x, y, z \in X$,

1. $d(x, y) \geq 0$, and $d(x, y) = 0$ if and only if $x = y$ *(positive definiteness)*;
2. $d(x, y) = d(y, x)$ *(symmetry)*;
3. $d(x, y) \leq d(x, z) + d(z, y)$ *(triangle inequality)*.

A *metric space* is a pair (X, d), where X is a nonempty set and d is a metric on X.

Definition 0.211.2 (Pseudo-Metric)**.** Let X be a nonempty set. A function

$$d : X \times X \to \mathbb{R}_{\geq 0}$$

is called a *pseudo-metric* on X if for all $x, y, z \in X$,

1. $d(x, y) \geq 0$, and $d(x, x) = 0$ *(positive semi-definiteness)*;
2. $d(x, y) = d(y, x)$ *(symmetry)*;
3. $d(x, y) \leq d(x, z) + d(z, y)$ *(triangle inequality)*.

Lemma 0.211.3 (Metrics are Pseudo-Metrics)**.** *Every metric on a nonempty set X is a pseudo-metric on X. The converse is false: there exist pseudo-metrics that are not metrics.*

Go to proof.

0.212 Metrics

0.212.1 Norms Induce Metrics

Definition 0.212.1 (Norm)**.** Let V be a real vector space. A *norm* on V is a function

$$\| \cdot \| : V \to \mathbb{R}_{\geq 0}$$

such that for all $x, y \in V$ and all $\lambda \in \mathbb{R}$,

1. $\|x\| = 0$ if and only if $x = 0$ *(positive definiteness)*;

2. $\|\lambda x\| = |\lambda| \, \|x\|$ *(absolute homogeneity)*;

3. $\|x + y\| \le \|x\| + \|y\|$ *(triangle inequality)*.

Theorem 0.212.2 (A Norm Induces a Metric). *Let V be a real vector space and let $\|\cdot\|$ be a norm on V. Define*

$$d_{\|\cdot\|} : V \times V \to \mathbb{R}_{\ge 0}, \qquad d_{\|\cdot\|}(x, y) := \|x - y\|.$$

Then $d_{\|\cdot\|}$ is a metric on V.

Go to proof.

Definition 0.212.3 (Discrete Metric). Let X be a nonempty set. The *discrete metric* on X is the function

$$d_{\mathrm{disc}} : X \times X \to \mathbb{R}_{\ge 0}$$

defined by

$$d_{\mathrm{disc}}(x, y) = \begin{cases} 0, & x = y, \\ 1, & x \neq y. \end{cases}$$

Proposition 0.212.4 (The Discrete Distance is a Metric). *For every nonempty set X, the discrete distance d_{disc} is a metric on X.*

Go to proof.

Definition 0.212.5 (Euclidean Distance on $\mathbb{R}$). The *Euclidean distance* on $\mathbb{R}$ is the function

$$d_{\mathbb{R}} : \mathbb{R} \times \mathbb{R} \to \mathbb{R}_{\ge 0}, \qquad d_{\mathbb{R}}(x, y) = |x - y|.$$

Proposition 0.212.6 (The Real-Line Euclidean Distance is a Metric). *The function $d_{\mathbb{R}}(x, y) = |x - y|$ is a metric on $\mathbb{R}$.*

Go to proof.

Definition 0.212.7 (Euclidean Space). For $n \in \mathbb{N}$, the *n-dimensional Euclidean space* is $\mathbb{R}^n$ equipped with its standard coordinate structure, dot product, Euclidean norm, and Euclidean distance.

Definition 0.212.8 (Euclidean Distance on $\mathbb{R}^n$). For $x = (x_1, \ldots, x_n)$ and $y = (y_1, \ldots, y_n)$ in $\mathbb{R}^n$, the *Euclidean distance* is the function

$$d_2 : \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}_{\ge 0}, \qquad d_2(x, y) = \sqrt{(x_1 - y_1)^2 + \cdots + (x_n - y_n)^2}.$$

Theorem 0.212.9 (Cauchy–Schwarz Inequality). *For every $x, y \in \mathbb{R}^n$,*

$$|x \cdot y| \le |x| \, |y|.$$

Equality holds if and only if x and y are linearly dependent.

Go to proof.

Corollary 0.212.10 (Minkowski's Inequality). *For every $x, y \in \mathbb{R}^n$,*

$$|x + y| \le |x| + |y|.$$

Go to proof.

Proposition 0.212.11 (The Euclidean Distance on $\mathbb{R}^n$ is a Metric). *The function d_2 is a metric on $\mathbb{R}^n$.*

Go to proof.

Corollary 0.212.12 (Euclidean Metric). *For every $x, y \in \mathbb{R}^n$, define*

$$d_2(x, y) := \sqrt{|x_1 - y_1|^2 + \cdots + |x_n - y_n|^2}.$$

Then $(\mathbb{R}^n, d_2)$ is a metric space.

Go to proof.

Definition 0.212.13 (Taxicab Metric)**.** For $x, y \in \mathbb{R}^n$, the *taxicab metric* is the function

$$d_1(x, y) := \sum_{i=1}^{n} |x_i - y_i|.$$

Proposition 0.212.14 (Taxicab Metric)**.** *The function d_1 is a metric on $\mathbb{R}^n$.*

Go to proof.

Definition 0.212.15 (p-Metric on $\mathbb{R}^n$)**.** Let $1 \leq p < \infty$. For $x, y \in \mathbb{R}^n$, define

$$d_p(x, y) := \left(\sum_{i=1}^{n} |x_i - y_i|^p \right)^{1/p}.$$

Theorem 0.212.16 (d_p is a Metric)**.** *For every $1 \leq p < \infty$, the function d_p is a metric on $\mathbb{R}^n$.*

Go to proof.

Definition 0.212.17 (Complex Metric)**.** Identify $\mathbb{C}$ with $\mathbb{R}^2$. The *complex metric* is

$$d_{\mathbb{C}}(z, w) := |z - w|.$$

Proposition 0.212.18 (The Complex Plane is a Metric Space)**.** *The pair $(\mathbb{C}, d_{\mathbb{C}})$ is a metric space.*

Go to proof.

Definition 0.212.19 (Supremum Metric on Bounded Real Sequences)**.** Let $\ell^\infty(\mathbb{R})$ denote the set of all bounded real sequences:

$$\ell^\infty(\mathbb{R}) := \{x = (x_n)_{n\in\mathbb{N}} : \exists M \geq 0 \ \forall n \in \mathbb{N} \ |x_n| \leq M\}.$$

For $x, y \in \ell^\infty(\mathbb{R})$, define

$$d_\infty(x, y) := \sup_{n\in\mathbb{N}} |x_n - y_n|.$$

Proposition 0.212.20 (The Supremum Distance on Bounded Sequences is a Metric)**.** *The function d_∞ is a metric on $\ell^\infty(\mathbb{R})$.*

Go to proof.

0.212.2 Point Functions and Pointlike Functions

Definition 0.212.21 (Point Function)**.** Let (X, d) be a metric space and let $z \in X$. The *point function at z* is the function

$$\delta_z : X \to \mathbb{R}_{\geq 0}, \qquad \delta_z(x) := d(z, x).$$

The set of all point functions on X is denoted by

$$\delta(X) := \{ \delta_z : z \in X \}.$$

Theorem 0.212.22 (Point Functions Identify Points)**.** *Let (X, d) be a metric space. The map*

$$X \to \delta(X), \qquad z \mapsto \delta_z$$

is a bijection.

Go to proof.

Theorem 0.212.23 (Point Function Inequalities)**.** *Let (X, d) be a nonempty metric space and let $z \in X$. Then, for all $a, b \in X$,*

$$\delta_z(b) - \delta_z(a) \leq d(a, b) \leq \delta_z(b) + \delta_z(a),$$

and

$$\delta_z(z) = 0.$$

Go to proof.

Definition 0.212.24 (Pointlike Function)**.** Let (X, d) be a nonempty metric space. A function

$$u : X \to \mathbb{R}_{\geq 0}$$

is *pointlike* on X if, for all $a, b \in X$,

$$u(a) - u(b) \leq d(a, b) \leq u(a) + u(b).$$

Theorem 0.212.25 (Pointlike Functions with a Zero Are Point Functions)**.** *Let (X, d) be a metric space and let $u : X \to \mathbb{R}_{\geq 0}$. Then $u \in \delta(X)$ if and only if u is pointlike on X and*

$$0 \in u(X).$$

In that case, there is a unique $w \in X$ such that $u(w) = 0$, and

$$u = \delta_w.$$

Go to proof.

0.213 Examples of Metric Spaces

Definition 0.213.1 (Discrete Metric Space)**.** Let X be a nonempty set. The *discrete metric space on X* is the metric space

$$(X, d_{\text{disc}}),$$

where d_{disc} is the discrete metric on X.

Definition 0.213.2 (Euclidean Metric Space)**.** For $n \in \mathbb{N}$, the *standard Euclidean metric space* is

$$(\mathbb{R}^n, d_2),$$

where d_2 is the Euclidean metric on $\mathbb{R}^n$.

Definition 0.213.3 (Taxicab Metric Space)**.** For $n \in \mathbb{N}$, the *taxicab metric space* is

$$(\mathbb{R}^n, d_1),$$

where d_1 is the taxicab metric on $\mathbb{R}^n$.

Definition 0.213.4 (p-Metric Space)**.** Let $1 \leq p < \infty$ and let $n \in \mathbb{N}$. The *p-metric space* is

$$(\mathbb{R}^n, d_p),$$

where d_p is the p-metric on $\mathbb{R}^n$.

Definition 0.213.5 (Complex Metric Space)**.** The *complex metric space* is

$$(\mathbb{C}, d_{\mathbb{C}}),$$

where $d_{\mathbb{C}}$ is the usual Euclidean distance on the complex plane.

Definition 0.213.6 (Bounded Sequence Metric Space)**.** The *bounded sequence metric space* is

$$(\ell^\infty(\mathbb{R}), d_\infty),$$

where $\ell^\infty(\mathbb{R})$ is the set of bounded real sequences and d_∞ is the supremum metric.

0.214 Distances, Boundedness, and Balls

0.214.1 Distances and Boundedness

Definition 0.214.1 (Diameter)**.** Let (X, d) be a metric space and let $A \subseteq X$ be nonempty. The *diameter* of A is

$$\operatorname{diam}(A) := \sup\{\, d(a, b) : a, b \in A \,\},$$

whenever this supremum is finite, and is $+\infty$ otherwise.

Definition 0.214.2 (Distance from a Point to a Set)**.** Let (X, d) be a metric space, let $x \in X$, and let $A \subseteq X$ be nonempty. The *distance from x to A* is

$$d(x, A) := \inf\{\, d(x, a) : a \in A \,\}.$$

Definition 0.214.3 (Distance Between Sets)**.** Let (X, d) be a metric space, and let $A, B \subseteq X$ be nonempty. The *distance between A and B* is

$$d(A, B) := \inf\{\, d(a, b) : a \in A,\ b \in B \,\}.$$

Theorem 0.214.4 (Distance to a Union)**.** *Let (X, d) be a metric space, let $x \in X$, and let $A, B \subseteq X$ be nonempty. Then*

$$d(x, A \cup B) = \min\{d(x, A), d(x, B)\}.$$

Go to proof.

Theorem 0.214.5 (Distance to an Intersection Bound)**.** *Let (X, d) be a metric space, let $x \in X$, and let $A, B \subseteq X$ have nonempty intersection. Then*

$$d(x, A \cap B) \geq \max\{d(x, A), d(x, B)\}.$$

Go to proof.

Definition 0.214.6 (Metric Bounded Set)**.** Let (X, d) be a metric space and let $A \subseteq X$. The set A is *bounded* if there exist $x \in X$ and $R > 0$ such that

$$d(x, a) \leq R$$

for every $a \in A$.

Theorem 0.214.7 (Equivalent Boundedness Criterion)**.** *Let (X, d) be a metric space and let $A \subseteq X$. Then A is bounded if and only if there exist $x \in X$ and $R > 0$ such that*

$$d(x, a) < R$$

for every $a \in A$.

Theorem 0.214.8 (Center Independence)**.** *Let (X, d) be a metric space. If $A \subseteq X$ and there exist $x \in X$ and $R > 0$ such that $d(x, a) \leq R$ for every $a \in A$, then for every $y \in X$,*

$$d(y, a) \leq R + d(x, y)$$

for every $a \in A$. Thus boundedness does not depend on the chosen center.

Theorem 0.214.9 (Subsets of Bounded Sets Are Bounded)**.** *Let (X, d) be a metric space. If $A \subseteq B \subseteq X$ and B is bounded, then A is bounded.*

Theorem 0.214.10 (Finite Sets Are Bounded)**.** *Every finite subset of a metric space is bounded.*

Theorem 0.214.11 (Finite Unions of Bounded Sets Are Bounded)**.** *Let (X, d) be a metric space, let $n \in \mathbb{N}$, and let $A_1, \ldots, A_n \subseteq X$. If $A_1, \ldots, A_n$ are bounded, then*

$$\bigcup_{k=1}^{n} A_k$$

is bounded.

Theorem 0.214.12 (Boundedness via Diameter)**.** *Let (X, d) be a metric space and let $A \subseteq X$ be nonempty. Then A is bounded if and only if there exists $C > 0$ such that*

$$d(a, b) \leq C$$

for all $a, b \in A$.

Theorem 0.214.13 (Closure of a Bounded Set Is Bounded)**.** *Let (X, d) be a metric space. If $A \subseteq X$ is bounded, then $\mathrm{cl}(A)$ is bounded.*

Theorem 0.214.14 (Compact Sets Are Bounded)**.** *Every compact subset of a metric space is bounded.*

Theorem 0.214.15 (Totally Bounded Sets Are Bounded)**.** *Every totally bounded subset of a metric space is bounded.*

0.214.2 Balls and Local Neighborhoods

Definition 0.214.16 (Open Ball)**.** Let (X, d) be a metric space, let $x \in X$, and let $r > 0$. The *open ball* of radius r centered at x is

$$B_d(x; r) := \{ y \in X : d(y, x) < r \}.$$

Definition 0.214.17 (Closed Ball)**.** Let (X, d) be a metric space, let $x \in X$, and let $r > 0$. The *closed ball* of radius r centered at x is

$$\overline{B}_d(x; r) := \{ y \in X : d(y, x) \leq r \}.$$

Lemma 0.214.18 (Key Local Ball Lemma)**.** *Let (X, d) be a metric space. If $y \in B_d(x; r)$, then*

$$B_d(y; r - d(x, y)) \subseteq B_d(x; r).$$

Theorem 0.214.19 (Ball Nesting)**.** *Let (X, d) be a metric space, let $x \in X$, and let $0 < r \leq s$. Then*

$$B_d(x; r) \subseteq B_d(x; s).$$

If $r < s$, then also

$$\overline{B}_d(x; r) \subseteq B_d(x; s).$$

Theorem 0.214.20 (Ball Containment by Center Distance)**.** *Let (X, d) be a metric space, let $x, y \in X$, and let $r, s > 0$.*

If

$$d(x, y) + r \leq s,$$

then

$$B_d(x; r) \subseteq B_d(y; s).$$

If

$$d(x, y) + r < s,$$

then

$$\overline{B}_d(x; r) \subseteq B_d(y; s).$$

Theorem 0.214.21 (Intersecting Balls Imply Center-Distance Bound)**.** *Let (X,d) be a metric space, let $x,y \in X$, and let $r,s > 0$. If*

$$B_d(x;r) \cap B_d(y;s) \neq \varnothing,$$

then

$$d(x,y) < r + s.$$

Theorem 0.214.22 (Closed Balls Contain Open Balls)**.** *Let (X,d) be a metric space. For every $x \in X$ and every $r > 0$,*

$$B_d(x;r) \subseteq \overline{B}_d(x;r).$$

0.215 Open and Closed Sets

Definition 0.215.1 (Metric Open Set)**.** Let (X,d) be a metric space. A subset $E \subseteq X$ is *open in* (X,d) if either $E = \varnothing$ or, for every $x \in E$, there exists $\varepsilon > 0$ such that

$$B_d(x;\varepsilon) \subseteq E.$$

Theorem 0.215.2 (Open Balls are Open)**.** *Let (X,d) be a metric space. For every $x \in X$ and every $r > 0$, the open ball $B_d(x;r)$ is open in (X,d).*

Go to proof.

Theorem 0.215.3 (Metric Open Set Closure Properties)**.** *Let X be a metric space. Then:*

1. *the empty set $\varnothing$ and the space X are open sets;*
2. *an arbitrary union of open subsets of X is open;*
3. *any finite intersection of open subsets of X is open.*

Go to proof.

Definition 0.215.4 (Metric Closed Set)**.** Let (X,d) be a metric space. A subset $E \subseteq X$ is *closed in* (X,d) if its complement $E^c = X \setminus E$ is open in (X,d).

Theorem 0.215.5 (Complement of a Closed Ball Is Open)**.** *Let (X,d) be a metric space. For every $x \in X$ and every $r \geq 0$,*

$$X \setminus \overline{B}_d(x;r) = \{\, y \in X : d(x,y) > r \,\}$$

is open.

Theorem 0.215.6 (Closed Balls Are Closed)**.** *Let (X,d) be a metric space. For every $x \in X$ and every $r \geq 0$, the closed ball*

$$\overline{B}_d(x;r) = \{\, y \in X : d(x,y) \leq r \,\}$$

is closed.

Definition 0.215.7 (Metric Closure)**.** Let (X,d) be a metric space and let $E \subseteq X$. The *closure* of E, denoted $\mathrm{cl}(E)$, is

$$\mathrm{cl}(E) = \{x \in X : \forall \varepsilon > 0,\ B_d(x;\varepsilon) \cap E \neq \varnothing\}.$$

Theorem 0.215.8 (Closure of an Open Ball Is Contained in the Closed Ball)**.** *Let (X,d) be a metric space. For every $x \in X$ and every $r > 0$,*

$$\mathrm{cl}\big(B_d(x;r)\big) \subseteq \overline{B}_d(x;r).$$

Definition 0.215.9 (Dense Subset)**.** Let (X,d) be a metric space and let $A \subseteq X$. The set A is *dense in* X if

$$\mathrm{cl}(A) = X.$$

Definition 0.215.10 (Metric Interior Point)**.** Let (X, d) be a metric space, let $E \subseteq X$, and let $x \in X$. The point x is an *interior point of* E if there exists $\delta > 0$ such that

$$B_d(x; \delta) \subseteq E.$$

Definition 0.215.11 (Metric Interior)**.** Let (X, d) be a metric space and let $E \subseteq X$. The *interior of* E is the set of all interior points of E. It is denoted by

$$E^\circ.$$

Theorem 0.215.12 (Interior is the Largest Open Subset)**.** *Let E be a subset of a metric space X. Then E° is the largest open subset of X contained in E. In other words:*

1. *$E^\circ \subseteq E$;*
2. *E° is open;*
3. *$E^\circ \supseteq O$, for every open set $O \subseteq E$.*

Go to proof.

Definition 0.215.13 (Metric Boundary)**.** Let (X, d) be a metric space and let $E \subseteq X$. The *boundary* of E is

$$\partial E = \mathrm{cl}(E) \cap \mathrm{cl}(X \setminus E).$$

0.216 Limit Points and Isolated Points

Definition 0.216.1 (Metric Limit Point and Isolated Point)**.** Let E be a subset of a metric space (X, d), and let $x \in X$.

1. The point x is a *limit point of* E if, for every $\varepsilon > 0$, the ball $B_d(x; \varepsilon)$ contains a point of $E \setminus \{x\}$.
2. The point x is an *isolated point of* E if $x \in E$ and x is not a limit point of E.

Theorem 0.216.2 (Closed Sets Contain Their Limit Points)**.** *Let X be a metric space and let $E \subseteq X$. Then E is closed if and only if $E' \subseteq E$.*

Go to proof.

Theorem 0.216.3 (Sequential Characterization of Limit Points)**.** *Let (X, d) be a metric space, let $E \subseteq X$, and let $x \in X$. Then $x \in E'$ if and only if there exists a sequence of distinct terms in E that converges to x.*

Go to proof.

Corollary 0.216.4 (Neighborhood Characterization of Limit Points)**.** *Let (X, d) be a metric space, let $E \subseteq X$, and let $x \in X$. Then $x \in E'$ if and only if every neighborhood of x contains infinitely many points of E.*

Go to proof.

Theorem 0.216.5 (Bolzano–Weierstrass for Bounded Infinite Subsets)**.** *Every bounded infinite subset of $\mathbb{R}^m$ has a limit point in $\mathbb{R}^m$.*

Go to proof.

Proposition 0.216.6 (The Derived Set is Closed)**.** *The set of limit points of any set is closed.*

Go to proof.

0.217 Sequential Convergence

Definition 0.217.1 (Metric Sequential Convergence)**.** Let (X, d) be a metric space. A sequence $\{x_n\}$ in X is *convergent in* X if there exists $x_0 \in X$ such that for every $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that

$$d(x_n, x_0) < \varepsilon \qquad \text{for all } n \geq N.$$

In this case, $\{x_n\}$ converges to x_0, written

$$x_n \to x_0, \qquad x_0 = \lim_{n\to\infty} x_n.$$

Definition 0.217.2 (Metric Neighborhood)**.** Let (X, d) be a metric space and let $x \in X$. A subset $U \subseteq X$ is a *neighborhood of* x if there exists $\delta > 0$ such that

$$B_d(x; \delta) \subseteq U.$$

Definition 0.217.3 (Epsilon Neighborhood)**.** Let (X, d) be a metric space, let $x \in X$, and let $\varepsilon > 0$. The ε*-neighborhood of* x is the open ball

$$B_d(x; \varepsilon) = \{y \in X : d(y, x) < \varepsilon\}.$$

Theorem 0.217.4 (Metric Neighborhood Criterion for Sequential Convergence)**.** *Let (X, d) be a metric space, let $\{x_n\}$ be a sequence in X, and let $x \in X$. Then*

$$x_n \to x$$

if and only if every neighborhood of x contains all but finitely many terms of $\{x_n\}$.

Go to proof.

Definition 0.217.5 (Metric Cauchy Sequence)**.** Let (X, d) be a metric space. A sequence $\{x_n\}$ in X is *Cauchy* if for every $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that

$$d(x_n, x_m) < \varepsilon \qquad \text{for all } n, m \geq N.$$

Theorem 0.217.6 (Metric Convergent Sequences Have Unique Limits)**.** *In metric spaces, convergent sequences have unique limits.*

Go to proof.

Theorem 0.217.7 (Metric Cauchy Sequence with Convergent Subsequence)**.** *Let $\{x_n\}$ be a Cauchy sequence in a metric space (X, d), let $x \in X$, and let $\{x_{n_k}\}$ be a subsequence of $\{x_n\}$ such that*

$$\lim_{k\to\infty} x_{n_k} = x.$$

Then $x_n \to x$.

Go to proof.

Theorem 0.217.8 (Coordinatewise Sequences in $\mathbb{R}^m$)**.** *Let $\{x_n\}$ be a sequence in $\mathbb{R}^m$, and let $x_0 \in \mathbb{R}^m$. Write*

$$x_n = \left(x_n^{(1)}, \ldots, x_n^{(m)}\right) \qquad \textit{for all } n \in \mathbb{N} \cup \{0\}.$$

Then:

1. *$x_n \to x_0$ if and only if $x_n^{(j)} \to x_0^{(j)}$ for every $j = 1, \ldots, m$.*
2. *$\{x_n\}$ is Cauchy if and only if $\{x_n^{(j)}\}$ is Cauchy for every $j = 1, \ldots, m$.*

Go to proof.

Theorem 0.217.9 (Euclidean Cauchy Sequences Converge)**.** *Every Cauchy sequence in $\mathbb{R}^m$ is convergent in $\mathbb{R}^m$.*

Go to proof.

Theorem 0.217.10 (Bolzano–Weierstrass in $\mathbb{R}^m$)**.** *Every bounded sequence in $\mathbb{R}^m$ contains a subsequence that converges in $\mathbb{R}^m$.*

Go to proof.

Proofs

Topology

0.218 Topology

0.219 Definitions

Definition 0.219.1 (Topological Space)**.** Let X be a set. A *topology* on X is a collection $\tau \subseteq \mathcal{P}(X)$ such that:

1. $\emptyset \in \tau$ and $X \in \tau$;
2. if $\{U_i\}_{i \in I}$ is any family of sets in τ, then $\bigcup_{i \in I} U_i \in \tau$;
3. if $U_1, \ldots, U_n \in \tau$, then $\bigcap_{k=1}^{n} U_k \in \tau$.

A *topological space* is a pair (X, τ), where X is a set and τ is a topology on X. The members of τ are called the *open sets* of the space.

Definition 0.219.2 (Discrete Topology)**.** Let X be a set. The *discrete topology* on X is the topology

$$\tau_{\text{disc}} = \mathcal{P}(X).$$

Thus every subset of X is open in the topological space (X, τ_{disc}).

Definition 0.219.3 (Sierpinski Topology)**.** Let $X = \{0, 1\}$. The *Sierpinski topology* on X is the topology

$$\tau_{\text{S}} = \{\emptyset, \{1\}, X\}.$$

The topological space (X, τ_{S}) is called the *Sierpinski space*.

Definition 0.219.4 (Cofinite Topology)**.** Let X be a set. The *cofinite topology* on X is the topology

$$\tau_{\text{cof}} = \{\, U \subseteq X : X \setminus U \text{ is finite} \,\} \cup \{\emptyset\}.$$

Thus a subset of X is open exactly when it is empty or its complement in the ambient set is finite.

Definition 0.219.5 (Trivial Topology)**.** Let X be a set. The *trivial topology* on X is the topology

$$\tau_{\text{triv}} = \{\emptyset, X\}.$$

Thus only $\emptyset$ and X are open in the topological space (X, τ_{triv}).

0.220 Topological Space Subsets

Definition 0.220.1 (Closed Subset of a Topological Space)**.** Let (X, τ) be a topological space, and let $F \subseteq X$. The subset F is *closed in* (X, τ) if its complement

$$X \setminus F$$

is open in (X, τ).

Theorem 0.220.2 (Closed-Set Characterization of Topologies)**.** *Let X be a set, and let $\mathcal{C}$ be a collection of subsets of X. Suppose that:*

1. *$\emptyset \in \mathcal{C}$ and $X \in \mathcal{C}$;*
2. *the intersection of any family of elements of $\mathcal{C}$ belongs to $\mathcal{C}$;*
3. *the union of any two elements of $\mathcal{C}$ belongs to $\mathcal{C}$.*

Then there is a unique topology τ on X whose family of closed sets is exactly $\mathcal{C}$.

Go to proof.

Definition 0.220.3 (Finer and Coarser Topologies)**.** Let τ_1 and τ_2 be topologies on a set X. The topology τ_1 is *finer than* τ_2 if

$$\tau_2 \subseteq \tau_1.$$

Equivalently, τ_2 is *coarser than* τ_1. The two topologies are *comparable* if one is finer than the other.

0.221 Basis

Definition 0.221.1 (Basis for a Topology)**.** Let (X, τ) be a topological space. A collection $\mathcal{B} \subseteq \tau$ is a *basis for the topology* τ if every open set $O \in \tau$ is the union of some subcollection of $\mathcal{B}$. The elements of $\mathcal{B}$ are called *basic elements* or *basic open sets*.

Proposition 0.221.2 (Pointwise Characterization of a Basis)**.** *Let (X, τ) be a topological space, and let $\mathcal{B} \subseteq \tau$. Then $\mathcal{B}$ is a basis for τ if and only if for every open set $O \in \tau$ and every $x \in O$, there is $B \in \mathcal{B}$ such that*

$$x \in B \subseteq O.$$

Consequently, a subset $A \subseteq X$ is open if and only if for each $x \in A$, there is $B \in \mathcal{B}$ such that

$$x \in B \subseteq A.$$

Go to proof.

Proofs

Set Algebra

0.222 Set Algebras

0.223 Systems of Sets

0.223.1 Systems of Sets

Definition 0.223.1 (Semiring of Sets)**.** A collection $\mathcal{S}$ of subsets of a set X is a *semiring* if:

(i) $\varnothing \in \mathcal{S}$;

(ii) $A, B \in \mathcal{S}$ implies $A \cap B \in \mathcal{S}$;

(iii) for any $A, B \in \mathcal{S}$, the difference $A \setminus B$ is a *finite disjoint union* of members of $\mathcal{S}$: there exist pairwise disjoint $C_1, \ldots, C_n \in \mathcal{S}$ with $A \setminus B = C_1 \sqcup \cdots \sqcup C_n$.

Definition 0.223.2 (Ring of Sets)**.** A collection $\mathcal{R}$ of subsets of X is a *ring of sets* if:

(i) $\varnothing \in \mathcal{R}$;

(ii) $A, B \in \mathcal{R}$ implies $A \cup B \in \mathcal{R}$;

(iii) $A, B \in \mathcal{R}$ implies $A \setminus B \in \mathcal{R}$.

Definition 0.223.3 (Algebra of Sets)**.** A collection $\mathcal{A}$ of subsets of X is an *algebra of sets* (or *Boolean algebra of sets*, or *field of sets*) if:

(i) $X \in \mathcal{A}$;

(ii) $A \in \mathcal{A}$ implies $A^c = X \setminus A \in \mathcal{A}$;

(iii) $A, B \in \mathcal{A}$ implies $A \cup B \in \mathcal{A}$.

Definition 0.223.4 (σ-Algebra)**.** A collection $\mathcal{M}$ of subsets of X is a *σ-algebra* (or *σ-field*) on X if:

(i) $X \in \mathcal{M}$;

(ii) $A \in \mathcal{M}$ implies $A^c \in \mathcal{M}$;

(iii) if $\{A_n\}_{n\in\mathbb{N}}$ is a countable family in $\mathcal{M}$, then $\bigcup_{n=1}^{\infty} A_n \in \mathcal{M}$.

The pair $(X, \mathcal{M})$ is called a *measurable space*.

Definition 0.223.5 (Topology)**.** A *topology* on a set X is a collection τ of subsets of X satisfying:

(i) $\varnothing \in \tau$ and $X \in \tau$;

(ii) if $\{U_\alpha\}_{\alpha\in I}$ is any family of sets in τ (indexed by any set I), then $\bigcup_{\alpha\in I} U_\alpha \in \tau$;

(iii) if $U_1, \ldots, U_n \in \tau$ (finitely many), then $U_1 \cap \cdots \cap U_n \in \tau$.

The pair (X, τ) is a *topological space*. Members of τ are called *open sets*.

Theorem 0.223.6 (Intersection of σ-algebras is a σ-algebra). *Let $\{\mathcal{M}_\alpha\}_{\alpha \in I}$ be any family of σ-algebras on X (indexed by any set I). Then*

$$\mathcal{M} := \bigcap_{\alpha \in I} \mathcal{M}_\alpha$$

is a σ-algebra on X. *Go to proof.*

Definition 0.223.7 (Generated σ-Algebra). Let $\mathcal{F}$ be any collection of subsets of X. The *σ-algebra generated by $\mathcal{F}$*, written $\sigma(\mathcal{F})$, is the smallest σ-algebra on X that contains $\mathcal{F}$:

$$\sigma(\mathcal{F}) := \bigcap \{ \mathcal{M} \mid \mathcal{M} \text{ is a } \sigma\text{-algebra on } X \text{ and } \mathcal{F} \subseteq \mathcal{M} \}.$$

The collection $\mathcal{F}$ is called a *generating family* for $\sigma(\mathcal{F})$.

Definition 0.223.8 (Borel σ-Algebra). Let (X, τ) be a topological space. The *Borel σ-algebra* on X, written $\mathcal{B}(X)$, is the σ-algebra generated by the open sets:

$$\mathcal{B}(X) := \sigma(\tau).$$

Members of $\mathcal{B}(X)$ are called *Borel sets*.

0.224 The Power Set and Characteristic Functions

0.224.1 The Power Set and Characteristic Functions

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Algebra

Introduction to Algebraic Structures

0.234 What Is a Mathematical Structure?

0.234.1 Algebraic Structures

Definition 0.234.1 (Algebraic Structure)**.** An **algebraic structure** of signature $\mathcal{L}$ is a pair

$$\mathcal{A} = (A, I)$$

where A is a nonempty carrier set and I interprets each symbol of $\mathcal{L}$ on A: each n-ary function symbol as an operation $A^n \to A$, each n-ary relation symbol as a subset of A^n, and each constant symbol as an element of A. The structure is algebraic when $\mathcal{L}$ has no relation symbols beyond equality.

Definition 0.234.2 (First-Order Signature)**.** A **first-order signature**

$$\mathcal{L} = (\mathrm{Const}_{\mathcal{L}}, \mathrm{Func}_{\mathcal{L}}, \mathrm{Rel}_{\mathcal{L}}, \mathrm{arity})$$

lists constant symbols, function symbols, and relation symbols, with each function or relation symbol carrying a declared arity.

Definition 0.234.3 ($\mathcal{L}$-Structure)**.** Given a signature $\mathcal{L}$, an **$\mathcal{L}$-structure** is a nonempty carrier set A together with an interpretation of every symbol of $\mathcal{L}$ on A.

0.234.2 Laws of Composition

Definition 0.234.4 (Binary Operation / Law of Composition)**.** Let A be a set. A **binary operation** on A (Bourbaki: an internal law of composition) is a function

$$\star : A \times A \to A.$$

The image of (a, b) is written $a \star b$.

Definition 0.234.5 (n-ary Operation)**.** An **n-ary operation** on A is a function $f : A^n \to A$, with $n \geq 0$. It is unary if $n = 1$, binary if $n = 2$, and nullary if $n = 0$.

Definition 0.234.6 (Unary Operation)**.** A **unary operation** on A is a function $f : A \to A$.

Definition 0.234.7 (Nullary Operation / Constant)**.** A **nullary operation** on A is the selection of a distinguished element $c \in A$; equivalently, it is an operation $A^0 \to A$.

0.234.3 Distinguished Elements

Definition 0.234.8 (Left Identity)**.** An element $e_L \in A$ is a **left identity** for $\star$ if $e_L \star a = a$ for all $a \in A$.

Definition 0.234.9 (Right Identity)**.** An element $e_R \in A$ is a **right identity** for $\star$ if $a \star e_R = a$ for all $a \in A$.

Definition 0.234.10 (Identity / Neutral Element)**.** An element $e \in A$ is a **two-sided identity** (or neutral element) for $\star$ if it is both a left and a right identity:

$$e \star a = a \star e = a \quad \text{for all } a \in A.$$

Proposition 0.234.11 (Identity Collapse and Uniqueness)**.** *If $\star$ has a left identity e_L and a right identity e_R, then $e_L = e_R$. Consequently a two-sided identity, if it exists, is unique.* *Go to proof.*

Definition 0.234.12 (Left Absorbing Element)**.** An element $z_L \in A$ is **left absorbing** for $\star$ if $z_L \star a = z_L$ for all $a \in A$.

Definition 0.234.13 (Right Absorbing Element)**.** An element $z_R \in A$ is **right absorbing** for $\star$ if $a \star z_R = z_R$ for all $a \in A$.

Definition 0.234.14 (Absorbing Element)**.** An element $z \in A$ is **absorbing** (a zero element) for $\star$ if it is both left absorbing and right absorbing.

Proposition 0.234.15 (Uniqueness of the Absorbing Element)**.** *A two-sided absorbing element, if it exists, is unique.* *Go to proof.*

0.234.4 Inverses

Definition 0.234.16 (Left Inverse)**.** Let $\star$ have identity e, and let $a \in A$. An element $b \in A$ is a **left inverse** of a if $b \star a = e$.

Definition 0.234.17 (Right Inverse)**.** Let $\star$ have identity e, and let $a \in A$. An element $c \in A$ is a **right inverse** of a if $a \star c = e$.

Definition 0.234.18 (Inverse)**.** Let $\star$ have identity e, and let $a \in A$. An element $a^{-1} \in A$ is an **inverse** of a if

$$a \star a^{-1} = a^{-1} \star a = e.$$

Definition 0.234.19 (Invertible Element)**.** An element a is **invertible** (a unit) if it has a two-sided inverse.

Proposition 0.234.20 (Inverse Collapse and Uniqueness in a Monoid)**.** *If $\star$ is associative with identity e, and a has a left inverse b and a right inverse c, then $b = c$. Hence the inverse of an invertible element is unique.* *Go to proof.*

Definition 0.234.21 (Additive Inverse)**.** When $\star = +$ with identity 0, the inverse of a is written $-a$ and is called the **additive inverse** or negative.

Definition 0.234.22 (Multiplicative Inverse)**.** When $\star = \cdot$ with identity 1, the inverse of a is written a^{-1} and is called the **multiplicative inverse**.

Proposition 0.234.23 (Units of a Monoid Form a Group)**.** *In a monoid $(A, \star, e)$, the set $A^\times$ of invertible elements is closed under $\star$ and forms a group.* *Go to proof.*

0.234.5 Properties of Laws

Definition 0.234.24 (Associativity)**.** A binary operation $\star$ on A is **associative** if

$$(a \star b) \star c = a \star (b \star c) \quad \text{for all } a, b, c \in A.$$

Theorem 0.234.25 (General Associativity)**.** *If $\star$ is associative, then any finite product $a_1 \star a_2 \star \cdots \star a_n$ has a value independent of parenthesization.* *Go to proof.*

Definition 0.234.26 (Powers / Iterated Operation)**.** For associative $\star$ with identity e, define $a^0 := e$ and $a^{n+1} := a^n \star a$. In additive notation, define $0a := 0$ and $(n+1)a := na + a$.

Definition 0.234.27 (Commutativity)**.** A binary operation $\star$ on A is **commutative** if $a \star b = b \star a$ for all $a, b \in A$.

Definition 0.234.28 (Left Distributivity)**.** For laws $+$ and $\cdot$ on A, multiplication is **left distributive** over addition if

$$a \cdot (b + c) = a \cdot b + a \cdot c$$

for all $a, b, c \in A$.

Definition 0.234.29 (Right Distributivity)**.** For laws $+$ and $\cdot$ on A, multiplication is **right distributive** over addition if

$$(a + b) \cdot c = a \cdot c + b \cdot c$$

for all $a, b, c \in A$.

Definition 0.234.30 (Distributivity)**.** For laws $+$ and $\cdot$ on A, multiplication is **distributive** over addition if it is both left distributive and right distributive.

Definition 0.234.31 (Idempotent Operation)**.** A binary operation $\star$ on A is **idempotent** if $a \star a = a$ for all $a \in A$.

0.234.6 Regularity Properties

Definition 0.234.32 (Left Cancellable Element)**.** An element $a \in A$ is **left cancellable** if

$$a \star x = a \star y \Longrightarrow x = y$$

for all $x, y \in A$.

Definition 0.234.33 (Right Cancellable Element)**.** An element $a \in A$ is **right cancellable** if

$$x \star a = y \star a \Longrightarrow x = y$$

for all $x, y \in A$.

Definition 0.234.34 (Cancellable Element)**.** An element $a \in A$ is **cancellable** (regular) if it is both left cancellable and right cancellable.

Proposition 0.234.35 (Invertible Implies Cancellable)**.** *In a monoid, every invertible element is cancellable.* *Go to proof.*

Definition 0.234.36 (Zero Divisor)**.** In a ring R, a nonzero element a is a **zero divisor** if there is a nonzero b with $a \cdot b = 0$ or $b \cdot a = 0$.

0.234.7 One Internal Law

Definition 0.234.37 (Magma)**.** A **magma** is a pair $(A, \star)$ where $A \neq \varnothing$ and $\star$ is a binary operation on A.

Definition 0.234.38 (Semigroup)**.** A **semigroup** is a magma whose operation is associative.

Definition 0.234.39 (Monoid)**.** A **monoid** is a semigroup with an identity element.

Definition 0.234.40 (Group)**.** A **group** is a monoid in which every element has an inverse.

Definition 0.234.41 (Abelian Group)**.** An **abelian group** is a group whose operation is commutative.

Definition 0.234.42 (Order of a Group)**.** The **order** of a group G, denoted $|G|$, is the cardinality of its carrier. The group is finite if $|G|$ is finite.

Proposition 0.234.43 (Uniqueness of the Identity in a Group)**.** *The identity of a group is unique.* *Go to proof.*

Proposition 0.234.44 (Uniqueness of Inverses in a Group)**.** *Each element of a group has a unique inverse.* *Go to proof.*

Proposition 0.234.45 (Cancellation Laws in a Group)**.** *In a group, $ab = ac$ implies $b = c$, and $ba = ca$ implies $b = c$.* *Go to proof.*

Proposition 0.234.46 (Socks–Shoes)**.** *In a group, $(ab)^{-1} = b^{-1}a^{-1}$.* *Go to proof.*

0.234.8 Two Internal Laws

Definition 0.234.47 (Rng)**.** A **rng** is a triple $(R, +, \cdot)$ where $(R, +)$ is an abelian group, $\cdot$ is associative, and $\cdot$ distributes over $+$. No multiplicative identity is required.

Definition 0.234.48 (Ring)**.** A **ring** is a rng with a multiplicative identity 1; equivalently, $(R, \cdot, 1)$ is a monoid. Thus the axioms are: additive abelian group, multiplicative associativity, distributivity, and unity.

Definition 0.234.49 (Commutative Ring)**.** A **commutative ring** is a ring whose multiplication is commutative.

Definition 0.234.50 (Integral Domain)**.** An **integral domain** is a commutative ring with $0 \neq 1$ and no zero divisors.

Definition 0.234.51 (Division Ring / Skew Field)**.** A **division ring** (or skew field) is a ring with $0 \neq 1$ in which every nonzero element is invertible under multiplication. Multiplication is not required to be commutative.

Definition 0.234.52 (Field)**.** A **field** is a commutative division ring; equivalently, it is a commutative ring with $0 \neq 1$ in which $(F \setminus \{0\}, \cdot)$ is a group.

Proposition 0.234.53 (Multiplication by Zero)**.** *In a ring, $a \cdot 0 = 0 \cdot a = 0$ for all a.* *Go to proof.*

Proposition 0.234.54 (Multiplication by Additive Inverse)**.** *In a ring,*

$$a \cdot (-b) = -(a \cdot b) = (-a) \cdot b.$$

In a ring, $(-1) \cdot a = -a$. *Go to proof.*

Proposition 0.234.55 (Cancellation in Integral Domains)**.** *In an integral domain, if $a \neq 0$ and $ab = ac$, then $b = c$.* *Go to proof.*

Proposition 0.234.56 (Every Field is an Integral Domain)**.** *Every field is an integral domain.* *Go to proof.*

Proposition 0.234.57 (Finite Integral Domain is a Field)**.** *Every finite integral domain is a field.* *Go to proof.*

0.234.9 The Structural Lattice

0.235 Relations

0.236 Operations

0.237 Laws of Operations

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Abstract Algebra

0.246 Induction

0.246.1 Induction

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0.247.1 Integers

Basic Definitions and Theorems

0.248 Modular Arithmetic

0.248.1 Modular Arithmetic

Basic Definitions and Theorems

Proofs

Capstone

Algebraic Geometry

0.249 Algebraic Geometry

0.249.1 Preliminary Definitions

Definition 0.249.1 (Ring)**.** A set R with two binary operations $+ : R \times R \to R$ and $\cdot : R \times R \to R$ is a *ring* if:

1. **Additive structure:** $(R, +)$ is an abelian group.
2. **Associativity of multiplication:**
$$(a \cdot b) \cdot c = a \cdot (b \cdot c) \quad \text{for all } a, b, c \in R.$$
3. **Multiplicative identity:** There exists $1 \in R$ such that
$$1 \cdot a = a \cdot 1 = a \quad \text{for all } a \in R.$$
4. **Distributivity:**
$$a \cdot (b + c) = a \cdot b + a \cdot c \quad \text{and} \quad (a + b) \cdot c = a \cdot c + b \cdot c \quad \text{for all } a, b, c \in R.$$

Definition 0.249.2 (Commutative Ring)**.** A ring R is *commutative* if
$$a \cdot b = b \cdot a \quad \text{for all } a, b \in R.$$

Polynomial Rings

Definition 0.249.3 (Polynomial)**.** Let R be a commutative ring. A *polynomial* in one variable over R is a formal expression
$$f = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0,$$
where $n \in \mathbb{N}$, the *coefficients* $a_0, a_1, \ldots, a_n \in R$, and x is a formal symbol called an *indeterminate*.

Definition 0.249.4 (Degree)**.** Let $f = a_n x^n + \cdots + a_0$ be a polynomial over R. If $a_n \neq 0$, then the *degree* of f is
$$\deg(f) := n.$$
The coefficient a_n is called the *leading coefficient* of f. A polynomial with leading coefficient 1 is called *monic*.

Definition 0.249.5 (Polynomial Ring)**.** Let R be a commutative ring. The *polynomial ring* over R in one indeterminate x, denoted $R[x]$, is the set of all polynomials in x with coefficients in R, equipped with addition and multiplication defined by:
$$\left(\sum_i a_i x^i\right) + \left(\sum_i b_i x^i\right) := \sum_i (a_i + b_i) x^i,$$
$$\left(\sum_i a_i x^i\right) \cdot \left(\sum_j b_j x^j\right) := \sum_k \left(\sum_{i+j=k} a_i b_j\right) x^k.$$

Polynomial Rings in Several Variables

Definition 0.249.6 (Polynomial Ring in n Variables)**.** Let R be a commutative ring and let $n \in \mathbb{N}$. The *polynomial ring in n variables* over R, denoted

$$R[x_1, x_2, \ldots, x_n],$$

is defined inductively by

$$R[x_1, \ldots, x_n] := R[x_1, \ldots, x_{n-1}][x_n].$$

Its elements are finite sums of the form

$$f = \sum_{\alpha} a_\alpha \, x^\alpha,$$

where the sum ranges over multi-indices $\alpha = (\alpha_1, \ldots, \alpha_n) \in \mathbb{N}^n$, the coefficients $a_\alpha \in R$, and $x^\alpha := x_1^{\alpha_1} \cdots x_n^{\alpha_n}$.

Definition 0.249.7 (Monomial)**.** A *monomial* in $R[x_1, \ldots, x_n]$ is a polynomial of the form

$$x^\alpha = x_1^{\alpha_1} \cdots x_n^{\alpha_n}$$

for some multi-index $\alpha \in \mathbb{N}^n$.

Definition 0.249.8 (Total Degree)**.** The *total degree* of the monomial x^α is

$$|\alpha| := \alpha_1 + \alpha_2 + \cdots + \alpha_n.$$

The degree of a polynomial $f \in R[x_1, \ldots, x_n]$ is the maximum total degree among all monomials with nonzero coefficient.

Ideals

Definition 0.249.9 (Ideal)**.** Let R be a commutative ring. A subset $I \subseteq R$ is an *ideal* of R if:

1. $0 \in I$,
2. $a + b \in I$ for all $a, b \in I$,
3. $r \cdot a \in I$ for all $r \in R$ and $a \in I$.

Definition 0.249.10 (Generated Ideal)**.** Let R be a commutative ring and let $f_1, \ldots, f_m \in R$. The *ideal generated by $f_1, \ldots, f_m$* is

$$\langle f_1, \ldots, f_m \rangle := \left\{ \sum_{i=1}^{m} r_i f_i : r_i \in R \right\}.$$

An ideal of this form is called *finitely generated*.

Definition 0.249.11 (Finitely Generated Ideal)**.** An ideal $I \subseteq R$ is *finitely generated* if there exist $f_1, \ldots, f_m \in R$ such that $I = \langle f_1, \ldots, f_m \rangle$.

Definition 0.249.12 (Noetherian Ring)**.** A commutative ring R is *Noetherian* if every ideal of R is finitely generated.

Theorem 0.249.13 (Hilbert Basis Theorem)**.** *If R is a Noetherian ring, then the polynomial ring $R[x]$ is also Noetherian.*
In particular, $k[x_1, \ldots, x_n]$ is Noetherian for any field k. *Go to proof.*

Quotient Rings

Definition 0.249.14 (Quotient Ring)**.** Let R be a commutative ring and let $I \subseteq R$ be an ideal. The *quotient ring* R/I is the set of cosets

$$R/I := \{a + I : a \in R\},$$

equipped with addition and multiplication defined by

$$(a + I) + (b + I) := (a + b) + I,$$

$$(a + I) \cdot (b + I) := (a \cdot b) + I.$$

Proofs

Capstone

Linear Algebra

Linear Algebra

0.250 Vector Space Preliminaries

Definition 0.250.1 (List)**.** Suppose n is a non-negative integer. A **list** of length n is an ordered collection of n elements.

Definition 0.250.2 ($\mathbf{F}^n$)**.** Let $\mathbf{F}$ denote $\mathbb{R}$ or $\mathbb{C}$, and let n be a positive integer. Then $\mathbf{F}^n$ is defined to be the set of all lists of length n of elements of $\mathbf{F}$:

$$\mathbf{F}^n = \{(x_1, \ldots, x_n) : x_1, \ldots, x_n \in \mathbf{F}\}.$$

The entries in a list in $\mathbf{F}^n$ are called the coordinates of the list.

Definition 0.250.3 (Addition in $\mathbf{F}^n$)**.** Let

$$x = (x_1, \ldots, x_n), \qquad y = (y_1, \ldots, y_n)$$

be vectors in $\mathbf{F}^n$. The sum of x and y is defined by

$$x + y = (x_1 + y_1, \ldots, x_n + y_n).$$

Theorem 0.250.4 (Addition Is Commutative in $\mathbf{F}^n$)**.** *For all $x, y \in \mathbf{F}^n$,*

$$x + y = y + x.$$

Go to proof.

Definition 0.250.5 (Zero Vector in $\mathbf{F}^n$)**.** The symbol 0 denotes the list of length n all of whose coordinates are 0:

$$0 = (0, \ldots, 0) \in \mathbf{F}^n.$$

Definition 0.250.6 (Additive Inverse in $\mathbf{F}^n$)**.** Let

$$x = (x_1, \ldots, x_n) \in \mathbf{F}^n.$$

The additive inverse of x is defined by

$$-x = (-x_1, \ldots, -x_n).$$

Definition 0.250.7 (Scalar Multiplication in $\mathbf{F}^n$)**.** Let

$$\lambda \in \mathbf{F} \quad \text{and} \quad x = (x_1, \ldots, x_n) \in \mathbf{F}^n.$$

The product of λ and x is defined by

$$\lambda x = (\lambda x_1, \ldots, \lambda x_n).$$

Definition 0.250.8 (Vector Space)**.** A **vector space** over $\mathbf{F}$ is a set V together with an addition on V and a scalar multiplication

$$\mathbf{F} \times V \to V$$

such that the following properties hold for all $u, v, w \in V$ and all $a, b \in \mathbf{F}$:

(i) $u + v \in V$;

(ii) $u + v = v + u$;

(iii) $(u + v) + w = u + (v + w)$;

(iv) there exists $0 \in V$ such that $v + 0 = v$;

(v) for every $v \in V$ there exists $-v \in V$ such that $v + (-v) = 0$;

(vi) $av \in V$;

(vii) $a(u + v) = au + av$;

(viii) $(a + b)v = av + bv$;

(ix) $a(bv) = (ab)v$;

(x) $1v = v$.

Definition 0.250.9 (Vector and Point)**.** An element of a vector space V is called a **vector**. When $V = \mathbf{F}^n$, an element of $\mathbf{F}^n$ may also be called a **point**.

Definition 0.250.10 (Real Vector Space)**.** A **real vector space** is a vector space over $\mathbb{R}$.

Definition 0.250.11 (Complex Vector Space)**.** A **complex vector space** is a vector space over $\mathbb{C}$.

Theorem 0.250.12 (Unique Additive Identity)**.** *A vector space has a unique additive identity. Go to proof.*

Theorem 0.250.13 (Unique Additive Inverse)**.** *Every vector in a vector space has a unique additive inverse. Go to proof.*

Theorem 0.250.14 ($0v = 0$)**.** *For every $v \in V$,*

$$0v = 0.$$

Go to proof.

Theorem 0.250.15 ($a0 = 0$)**.** *For every $a \in \mathbf{F}$,*

$$a0 = 0.$$

Go to proof.

Theorem 0.250.16 ($(-1)v = -v$)**.** *For every $v \in V$,*

$$(-1)v = -v.$$

Go to proof.

Definition 0.250.17 (Subspace)**.** Let V be a vector space over $\mathbf{F}$. A subset $U \subseteq V$ is called a **subspace** of V if U is itself a vector space over $\mathbf{F}$ with the same addition and scalar multiplication as on V.

Theorem 0.250.18 (Conditions for a Subspace)**.** *Let V be a vector space over $\mathbf{F}$ and let $U \subseteq V$. Then U is a subspace of V if and only if the following three conditions hold:*

(i) $0 \in U$;

(ii) for all $u, v \in U$, one has $u + v \in U$;

(iii) for all $a \in \mathbf{F}$ and all $u \in U$, one has $au \in U$.

Go to proof.

Definition 0.250.19 (Sum of Subspaces)**.** Let V be a vector space over $\mathbf{F}$, and let $U_1, \dots, U_m$ be subspaces of V. The **sum** of $U_1, \dots, U_m$ is the set

$$U_1 + \cdots + U_m = \{u_1 + \cdots + u_m : u_1 \in U_1, \dots, u_m \in U_m\}.$$

Theorem 0.250.20 (Sum of Subspaces Is the Smallest Containing Subspace)**.** *Let V be a vector space over $\mathbf{F}$, and let $U_1, \dots, U_m$ be subspaces of V. Then $U_1 + \cdots + U_m$ is a subspace of V containing each of $U_1, \dots, U_m$. Moreover, if W is a subspace of V containing each of $U_1, \dots, U_m$, then*

$$U_1 + \cdots + U_m \subseteq W.$$

Go to proof.

Definition 0.250.21 (Direct Sum)**.** Let V be a vector space over $\mathbf{F}$, and let $U_1, \dots, U_m$ be subspaces of V such that

$$V = U_1 + \cdots + U_m.$$

Then V is called the **direct sum** of $U_1, \dots, U_m$ if each vector $v \in V$ can be written in the form

$$v = u_1 + \cdots + u_m, \qquad u_1 \in U_1, \dots, u_m \in U_m,$$

in exactly one way. In this case, we write

$$V = U_1 \oplus \cdots \oplus U_m.$$

Theorem 0.250.22 (Condition for a Direct Sum)**.** *Let V be a vector space over $\mathbf{F}$, and let $U_1, \dots, U_m$ be subspaces of V such that*

$$V = U_1 + \cdots + U_m.$$

Then the following are equivalent:

(i) $V = U_1 \oplus \cdots \oplus U_m$*;*

(ii) if $u_1 \in U_1, \dots, u_m \in U_m$ *and*

$$u_1 + \cdots + u_m = 0,$$

then

$$u_1 = \cdots = u_m = 0.$$

Go to proof.

Theorem 0.250.23 (Direct Sum of Two Subspaces)**.** *Let U and W be subspaces of a vector space V. Then*

$$U + W = U \oplus W$$

if and only if

$$U \cap W = \{0\}.$$

Go to proof.

Proofs

Capstone

Lattice and Order Theory

Lattice Order

0.251 Breadcrumb

Why Lattice Theory Gets Its Own Home

0.251.1 Ordered Sets

Definition 0.251.1 (Partial Order)**.** A relation $\leq$ on a set S is a *partial order* if it satisfies, for all $a, b, c \in S$:

1. **Reflexivity:** $a \leq a$.
2. **Anti-symmetry:** $a \leq b \wedge b \leq a \Longrightarrow a = b$.
3. **Transitivity:** $a \leq b \wedge b \leq c \Longrightarrow a \leq c$.

The pair $(S, \leq)$ is called a *partially ordered set*, or *poset.*

Definition 0.251.2 (Strict Partial Order)**.** A relation $<$ on a set S is a *strict partial order* if it satisfies, for all $a, b, c \in S$:

1. **Irreflexivity:** $\neg(a < a)$.
2. **Asymmetry:** $a < b \Longrightarrow \neg(b < a)$.
3. **Transitivity:** $a < b \wedge b < c \Longrightarrow a < c$.

Definition 0.251.3 (Total Order)**.** A partial order $\leq$ on a set S is a *total order* (or *linear order*) if it additionally satisfies the *law of trichotomy*: for all $a, b \in S$, exactly one of

$$a < b, \qquad a = b, \qquad b < a$$

holds. The pair $(S, \leq)$ is called a *totally ordered set* or a *chain.*

0.251.2 Chains and Antichains

Definition 0.251.4 (Chain)**.** Let $(S, \leq)$ be a partially ordered set. A subset $C \subseteq S$ is a *chain* if every pair of elements in C is comparable:

$$\forall x, y \in C, \quad x \leq y \ \vee \ y \leq x.$$

Equivalently, the restriction of $\leq$ to C is a total order on C.

Definition 0.251.5 (Antichain)**.** Let $(S, \leq)$ be a partially ordered set. A subset $A \subseteq S$ is an *antichain* if no two distinct elements of A are comparable:

$$\forall x, y \in A, \quad x \neq y \Longrightarrow \neg(x \leq y) \wedge \neg(y \leq x).$$

0.251.3 Partial and Total Maps

Definition 0.251.6 (Partial Map)**.** Let A and B be sets. A *partial map* $f : A \rightharpoonup B$ is a relation $f \subseteq A \times B$ such that every element of A has at most one image:

$$\forall x \in A,\ \forall y_1, y_2 \in B, \quad (x, y_1) \in f \ \wedge\ (x, y_2) \in f \implies y_1 = y_2.$$

Definition 0.251.7 (Total Map)**.** A partial map $f : A \rightharpoonup B$ is a *total map* if it is defined on every element of A:

$$\forall x \in A,\ \exists y \in B, \quad (x, y) \in f.$$

A total map is written $f : A \to B$ and is precisely a function in the usual sense.

0.251.4 Maps Between Ordered Sets

Definition 0.251.8 (Order-Preserving Map)**.** Let $(P, \leq_P)$ and $(Q, \leq_Q)$ be partially ordered sets. A map $\varphi : P \to Q$ is *order-preserving* (or *monotone*) if

$$\forall x, y \in P, \quad x \leq_P y \implies \varphi(x) \leq_Q \varphi(y).$$

Definition 0.251.9 (Order-Embedding)**.** Let $(P, \leq_P)$ and $(Q, \leq_Q)$ be partially ordered sets. A map $\varphi : P \to Q$ is an *order-embedding* if

$$\forall x, y \in P, \quad x \leq_P y \iff \varphi(x) \leq_Q \varphi(y).$$

When φ is an order-embedding, one writes $\varphi : P \hookrightarrow Q$.

Definition 0.251.10 (Order-Isomorphism)**.** An order-embedding $\varphi : P \to Q$ that is surjective is an *order-isomorphism*. When an order-isomorphism exists, P and Q are said to be *order-isomorphic*, written $P \cong Q$.

Lemma 0.251.11 (Every Order-Embedding Is Injective)**.** *Let $(P, \leq_P)$ and $(Q, \leq_Q)$ be partially ordered sets, and let $\varphi : P \to Q$ be an order-embedding. Then φ is injective.* *Go to proof.*

0.251.5 The Covering Relation

Definition 0.251.12 (Covering Relation)**.** Let $(P, \leq)$ be a partially ordered set and let $x, y \in P$ with $x < y$. The element y *covers* x, written $x \prec y$, if there is no element of P strictly between them:

$$x \prec y \ :=\ x < y \ \wedge\ \neg\big(\exists z \in P,\ x < z < y\big).$$

Lemma 0.251.13 (Characterizations of Order-Isomorphisms for Finite Posets)**.** *Let $(P, \leq_P)$ and $(Q, \leq_Q)$ be finite partially ordered sets and let $\varphi : P \to Q$ be a bijection. The following are equivalent:*

1. *φ is an order-isomorphism.*
2. *For all $x, y \in P$: $x <_P y \iff \varphi(x) <_Q \varphi(y)$.*
3. *For all $x, y \in P$: $x \prec_P y \iff \varphi(x) \prec_Q \varphi(y)$.*

Go to proof.

Proposition 0.251.14 (Hasse Diagram Characterization for Finite Posets)**.** *Let $(P, \leq_P)$ and $(Q, \leq_Q)$ be finite partially ordered sets. Then P and Q are order-isomorphic if and only if their Hasse diagrams are isomorphic as directed graphs — that is, identical up to relabeling of elements.* *Go to proof.*

0.251.6 Duality

Definition 0.251.15 (Dual Ordered Set)**.** Let $(P, \leq)$ be a partially ordered set. The *dual* (or *opposite*) of P, denoted P^{op}, is the set P equipped with the reversed order:

$$x \leq_{P^{\mathrm{op}}} y \iff y \leq_P x.$$

Proposition 0.251.16 (Duality Principle)**.** *Let Φ be a statement about partially ordered sets, expressed purely in terms of $\leq$, that holds in every partially ordered set. Then the dual statement Φ^{op}, obtained by replacing every occurrence of $\leq$ with $\geq$, also holds in every partially ordered set.* *Go to proof.*

0.251.7 Extremal Elements

Definition 0.251.17 (Bottom Element)**.** Let $(P, \leq)$ be a partially ordered set. An element $\perp \in P$ is a *bottom element* (or *least element*) of P if

$$\forall x \in P, \quad \perp \leq x.$$

If a bottom element exists it is unique; it is denoted $\perp_P$ or simply $\perp$.

Definition 0.251.18 (Top Element)**.** Let $(P, \leq)$ be a partially ordered set. An element $\top \in P$ is a *top element* (or *greatest element*) of P if

$$\forall x \in P, \quad x \leq \top.$$

If a top element exists it is unique; it is denoted $\top_P$ or simply $\top$.

Definition 0.251.19 (Maximal Element)**.** Let $(P, \leq)$ be a partially ordered set and let $Q \subseteq P$. An element $a \in Q$ is *maximal in* Q if no element of Q strictly exceeds it:

$$\forall x \in Q, \quad a \leq x \implies a = x.$$

The set of maximal elements of Q is denoted $\mathrm{Max}(Q)$.

Definition 0.251.20 (Minimal Element)**.** Let $(P, \leq)$ be a partially ordered set and let $Q \subseteq P$. An element $a \in Q$ is *minimal in* Q if no element of Q is strictly below it:

$$\forall x \in Q, \quad x \leq a \implies x = a.$$

Definition 0.251.21 (Greatest Element)**.** Let $(P, \leq)$ be a partially ordered set and let $Q \subseteq P$. An element $m \in Q$ is the *greatest element* (or *maximum*) of Q if it dominates every element of Q:

$$\forall x \in Q, \quad x \leq m.$$

If it exists it is unique; it is denoted $\max Q$.

Definition 0.251.22 (Least Element)**.** Let $(P, \leq)$ be a partially ordered set and let $Q \subseteq P$. An element $m \in Q$ is the *least element* (or *minimum*) of Q if it lies below every element of Q:

$$\forall x \in Q, \quad m \leq x.$$

If it exists it is unique; it is denoted $\min Q$.

Proposition 0.251.23 (Every Nonempty Finite Poset Has a Maximal Element)**.** *Let $(P, \leq)$ be a finite partially ordered set. Then every nonempty subset $Q \subseteq P$ has at least one maximal element. Furthermore, for every $x \in P$ there exists $y \in \mathrm{Max}(P)$ with $x \leq y$.* *Go to proof.*

0.251.8 Lifting and Co-Lifting

Definition 0.251.24 (Lifting)**.** Let $(P, \leq)$ be a partially ordered set. The *lifting* of P is the poset $P_\perp :=$ $P \cup \{\perp\}$, where $\perp \notin P$, with the order:

$$\perp \leq x \quad \text{for all } x \in P_\perp, \qquad x \leq_{P_\perp} y \iff x \leq_P y \quad \text{for all } x, y \in P.$$

Thus $P_\perp$ is P with a new bottom element $\perp$ adjoined.

Definition 0.251.25 (Co-Lifting)**.** Let $(P, \leq)$ be a partially ordered set. The *co-lifting* of P is the poset $P^\top := P \cup \{\top\}$, where $\top \notin P$, with the order:

$$x \leq \top \quad \text{for all } x \in P^\top, \qquad x \leq_{P^\top} y \iff x \leq_P y \quad \text{for all } x, y \in P.$$

Thus $P^\top$ is P with a new top element $\top$ adjoined.

0.251.9 Consistency

Definition 0.251.26 (Consistency)**.** Let $(S, \leq)$ be a partially ordered set. Elements $x, y \in S$ are *consistent* (or *compatible*) if they admit a common upper bound:

$$\exists z \in S, \quad x \leq z \wedge y \leq z.$$

0.252 Ordered Sets

0.252.1 Down-Sets and Up-Sets

Definition 0.252.1 (Down-Set)**.** Let $(P, \leq)$ be a partially ordered set. A subset $Q \subseteq P$ is a *down-set* (or *decreasing set*, or *order ideal*) if it is closed downward:

$$\forall x \in Q, \ \forall y \in P, \quad y \leq x \implies y \in Q.$$

Definition 0.252.2 (Up-Set)**.** Let $(P, \leq)$ be a partially ordered set. A subset $Q \subseteq P$ is an *up-set* (or *increasing set*, or *order filter*) if it is closed upward:

$$\forall x \in Q, \ \forall y \in P, \quad x \leq y \implies y \in Q.$$

0.252.2 Families of Sets

Proposition 0.252.3 (Predicates and Power Sets Are Order-Isomorphic)**.** *The map* $\Phi : (\mathcal{P}^*(X), \leq) \to (\mathcal{P}(X), \subseteq)$ *is an order-isomorphism.* *Go to proof.*

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0.254 Languages and Structures

First-Order Languages

Definition 0.254.1 (First-Order Language $\mathcal{L}$)**.** A **first-order language** $\mathcal{L}$ consists of three (possibly empty) sets of non-logical symbols:

- a set $\mathcal{R}$ of **relation symbols** (also called *predicate symbols*), each assigned a fixed arity $n \geq 1$;
- a set $\mathcal{F}$ of **function symbols**, each assigned a fixed **arity** $n \geq 0$ (0-ary function symbols are *constant symbols*);
- a set $\mathcal{C}$ of **constant symbols** (equivalently, 0-ary function symbols — listed separately for emphasis).

Together with the logical symbols common to all first-order languages (variables, connectives $\neg, \wedge, \vee, \rightarrow, \leftrightarrow$, quantifiers $\forall, \exists$, equality $=$, parentheses), the triple $(\mathcal{F}, \mathcal{R}, \mathcal{C})$ determines $\mathcal{L}$.

L-Structures

Definition 0.254.2 ($\mathcal{L}$-Structure)**.** Let $\mathcal{L} = (\mathcal{F}, \mathcal{R}, \mathcal{C})$ be a first-order language. An **$\mathcal{L}$-structure** $\mathcal{M}$ is an interpretation of the symbols of $\mathcal{L}$ on a non-empty domain.

Variable Assignments

Definition 0.254.3 (Variable Assignment)**.** Let $\mathcal{M}$ be an $\mathcal{L}$-structure with domain M, and let $\mathrm{Var} = \{v_1, v_2, v_3, \ldots\}$ be a fixed countable set of first-order variables.
A **variable assignment** (or **environment**) for $\mathcal{M}$ is a function $s : \mathrm{Var} \to M$.
For a variable assignment s, a variable v_i, and an element $a \in M$, write $s[v_i \mapsto a]$ for the assignment that agrees with s on all variables except v_i, where it returns a:

$$s[v_i \mapsto a](v_j) = \begin{cases} a & \text{if } j = i, \\ s(v_j) & \text{if } j \neq i. \end{cases}$$

Term Evaluation

Definition 0.254.4 (Term Evaluation)**.** Let $\mathcal{M}$ be an $\mathcal{L}$-structure and $s : \mathrm{Var} \to M$ a variable assignment. The **evaluation** $s(t) \in M$ of an $\mathcal{L}$-term t under s is defined inductively:

1. If $t = v_i$ is a variable, then $s(t) = s(v_i)$.
2. If $t = c$ is a constant symbol, then $s(t) = c^{\mathcal{M}}$.
3. If $t = f(t_1, \ldots, t_n)$ for an n-ary function symbol f, then $s(t) = f^{\mathcal{M}}(s(t_1), \ldots, s(t_n))$.

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Lambda Calculus

0.257 Lambda Terms

Untyped Lambda Calculus

Definition 0.257.1 (Lambda Terms)**.** The set Λ of λ-terms is defined inductively as follows:

1. Every variable x is a λ-term.
2. If M is a λ-term and x is a variable, then
$$\lambda x. M$$
is a λ-term.
3. If M and N are λ-terms, then
$$MN$$
is a λ-term.

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0.266 Convex Hulls

Convexity and Hulls

Definition 0.266.1 (Convex Set)**.** A set $S \subseteq \mathbb{R}^2$ is *convex* if for every pair of points $a, b \in S$ and every $\lambda \in [0,1]$,

$$\lambda a + (1-\lambda)b \in S.$$

Equivalently, S contains the line segment between any two of its points.

Definition 0.266.2 (Convex Combination)**.** A *convex combination* of points $q_1, \dots, q_n \in \mathbb{R}^2$ is a point of the form

$$\sum_{i=1}^{n} \lambda_i q_i, \qquad \lambda_i \geq 0 \text{ for all } i, \qquad \sum_{i=1}^{n} \lambda_i = 1.$$

The numbers λ_i are called the weights.

Definition 0.266.3 (Convex Hull)**.** The *convex hull* of a set $P \subseteq \mathbb{R}^2$, written $\operatorname{conv}(P)$, is the set of all convex combinations of finitely many points of P:

$$\operatorname{conv}(P) = \left\{ \sum_{i=1}^{n} \lambda_i q_i \;\middle|\; n \geq 1,\ q_i \in P,\ \lambda_i \geq 0,\ \sum_{i=1}^{n} \lambda_i = 1 \right\}.$$

Equivalently, $\operatorname{conv}(P)$ is the smallest convex set containing P, i.e. the intersection of all convex sets that contain P.

Lemma 0.266.4 (Convex hull as the smallest convex set)**.** *Let $P \subseteq \mathbb{R}^2$. Then $\operatorname{conv}(P)$ is convex, contains P, and is contained in every convex set that contains P. Hence*

$$\operatorname{conv}(P) = \bigcap \{K \subseteq \mathbb{R}^2 : K \text{ is convex and } P \subseteq K\}.$$

Supporting Functionals

Definition 0.266.5 (Extreme Point)**.** Let $S \subseteq \mathbb{R}^2$ be convex. A point $p \in S$ is an *extreme point* of S if whenever

$$p = \lambda a + (1-\lambda)b, \qquad a, b \in S, \quad \lambda \in (0,1),$$

it follows that $a = b = p$. For a finite point set P, the extreme points of $\operatorname{conv}(P)$ are called the vertices of the convex hull.

Definition 0.266.6 (Supporting Line)**.** A line $\ell \subseteq \mathbb{R}^2$ is a *supporting line* of a convex set S at a point $p \in S$ if $p \in \ell$ and S lies entirely in one of the two closed half-planes bounded by ℓ. Equivalently, there exists $d \in \mathbb{R}^2 \setminus \{0\}$ such that

$$\ell = \{x \in \mathbb{R}^2 : \langle d, x \rangle = \langle d, p \rangle\}$$

and

$$\langle d, q \rangle \leq \langle d, p \rangle \qquad \text{for all } q \in S.$$

Lemma 0.266.7 (Linear functionals preserve convex combinations). *Let $d \in \mathbb{R}^2$, and let*

$$z = \sum_{i=1}^{n} \lambda_i q_i$$

be a convex combination of points $q_1, \ldots, q_n \in \mathbb{R}^2$. Then

$$\langle d, z \rangle = \sum_{i=1}^{n} \lambda_i \langle d, q_i \rangle.$$

Consequently, if $\langle d, q_i \rangle \leq M$ for every i, then

$$\langle d, z \rangle \leq M.$$

Theorem 0.266.8 (Supporting functional characterization of hull vertices). *Let $P \subset \mathbb{R}^2$ be finite and in general position. A point $p \in P$ is a vertex of $\operatorname{conv}(P)$ if and only if there exists a direction $d \in \mathbb{R}^2 \setminus \{0\}$ such that*

$$\langle d, p \rangle > \langle d, q \rangle \qquad \text{for all } q \in P \setminus \{p\}.$$

Equivalently, p is the unique point of P attaining

$$\max_{q \in P} \langle d, q \rangle.$$

Thus every hull vertex is a maximum of P under some linear functional.

Upper Hull Structure

Theorem 0.266.9 (Monotone slope characterization of the upper hull). *Let $P \subset \mathbb{R}^2$ be finite and in general position. Let*

$$v_0, v_1, \ldots, v_m$$

be the vertices of the upper hull of P, listed in increasing x-coordinate, and define

$$s_k = \frac{y(v_{k+1}) - y(v_k)}{x(v_{k+1}) - x(v_k)}, \qquad k = 0, 1, \ldots, m-1.$$

Then

$$s_0 > s_1 > \cdots > s_{m-1}.$$

Conversely, if points $w_0, \ldots, w_m$ satisfy

$$x(w_0) < x(w_1) < \cdots < x(w_m)$$

and their consecutive slopes are strictly decreasing, then every w_k is a vertex of

$$\operatorname{conv}(\{w_0, \ldots, w_m\}).$$

The lower hull is characterized similarly, with strictly increasing consecutive slopes.

Theorem 0.266.10 (Greedy slope construction of the upper hull). *Let $P \subset \mathbb{R}^2$ be finite and in general position. Define*

$$v_0 = \arg\min_{p \in P} x(p).$$

Given v_k with

$$x(v_k) < \max_{p \in P} x(p),$$

define

$$v_{k+1} = \arg\max_{\substack{q \in P \\ x(q) > x(v_k)}} \frac{y(q) - y(v_k)}{x(q) - x(v_k)}.$$

Terminate once

$$v_k = \arg\max_{p \in P} x(p).$$

Then the points

$$v_0, v_1, \ldots, v_m$$

produced by this construction are exactly the vertices of the upper hull of P, and the slopes encountered along the way are strictly decreasing.

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0.281 Models

Definition 0.281.1 (Differential Equation)**.** A **differential equation** is an equation whose unknown is a function and whose terms involve one or more derivatives of that function.
For an unknown function $y = y(t)$, a differential equation may be written in the form

$$F(t, y, y', y'', \dots, y^{(n)}) = 0.$$

Here t is the independent variable, $y(t)$ is the unknown function, and

$$y', y'', \dots, y^{(n)}$$

are derivatives of y.

Definition 0.281.2 (Order of a Differential Equation)**.** The **order** of a differential equation is the highest derivative of the unknown function that appears in the equation.
For example, if an equation involving an unknown function $y = y(t)$ can be written in the form

$$F(t, y, y', y'', \dots, y^{(n)}) = 0$$

and $y^{(n)}$ is the highest derivative that appears, then the differential equation has **order** n.

Definition 0.281.3 (Solution of a Differential Equation)**.** A **solution** of a differential equation is a function that satisfies the equation on a specified interval.
More precisely, if a differential equation is written in the form

$$F(t, y, y', y'', \dots, y^{(n)}) = 0,$$

then a solution on an interval I is a function $y : I \to \mathbb{R}$ such that y has all derivatives required by the equation and

$$F(t, y(t), y'(t), y''(t), \dots, y^{(n)}(t)) = 0$$

for every $t \in I$.

Definition 0.281.4 (Autonomous Differential Equation)**.** An **autonomous differential equation** is a differential equation in which the independent variable does not appear explicitly.
For a first-order ordinary differential equation, an autonomous equation has the form

$$y' = f(y),$$

where the rate of change of y depends only on the current value of y, not directly on time t.

Definition 0.281.5 (Non-Autonomous Differential Equation)**.** A **non-autonomous differential equation** is a differential equation in which the independent variable appears explicitly.

For a first-order ordinary differential equation, a non-autonomous equation has the form

$$y' = f(t, y),$$

where the rate of change of y may depend both on the current value of y and on the independent variable t.

Definition 0.281.6 (Linear Differential Equation)**.** A differential equation is called **linear** if the unknown function and its derivatives appear only to the first power and are not multiplied together.

An nth-order linear ordinary differential equation for an unknown function $y = y(t)$ has the form

$$a_n(t)y^{(n)} + a_{n-1}(t)y^{(n-1)} + \cdots + a_1(t)y' + a_0(t)y = g(t),$$

where the coefficient functions

$$a_0(t), a_1(t), \ldots, a_n(t)$$

and the forcing term $g(t)$ are given functions of the independent variable t.

Definition 0.281.7 (Malthusian Population Growth Model)**.** The **Malthusian population growth model** assumes that the rate of change of a population is proportional to the current population.

If $P(t)$ denotes the population at time t, then the model is

$$P'(t) = kP(t),$$

where k is a constant called the **growth rate** or **proportionality constant**.

Definition 0.281.8 (Decay)**.** A quantity is said to undergo **decay** if it decreases over time.

In a differential equation model, if $Q(t)$ denotes the quantity at time t, then decay is often modeled by an equation of the form

$$Q'(t) = -kQ(t),$$

where $k > 0$ is a constant.

Definition 0.281.9 (Newton's Law of Cooling)**.** **Newton's Law of Cooling** states that the rate at which an object's temperature changes is proportional to the difference between the object's temperature and the surrounding ambient temperature.

If $T(t)$ denotes the temperature of the object at time t, and T_a denotes the ambient temperature, then the model is

$$T'(t) = -k\big(T(t) - T_a\big),$$

where $k > 0$ is a constant called the **cooling constant**.

Definition 0.281.10 (General Solution)**.** A **general solution** of a differential equation is a family of solutions containing one or more arbitrary constants.

For an nth-order ordinary differential equation, the general solution typically contains n arbitrary constants:

$$y(t) = \Phi(t, C_1, C_2, \ldots, C_n).$$

Each choice of the constants $C_1, C_2, \ldots, C_n$ gives a particular solution of the differential equation.

Definition 0.281.11 (Family of Solutions)**.** A **family of solutions** of a differential equation is a collection of solution functions described by one or more parameters.

For example, a family of solutions may be written in the form

$$y(t) = \Phi(t, C),$$

or more generally,

$$y(t) = \Phi(t, C_1, C_2, \ldots, C_n),$$

where $C, C_1, \ldots, C_n$ are parameters. Each choice of the parameters gives one particular solution.

Definition 0.281.12 (Initial Condition)**.** An **initial condition** specifies the value of the unknown function at a particular value of the independent variable.

For a first-order differential equation involving an unknown function $y = y(t)$, an initial condition has the form

$$y(t_0) = y_0,$$

where t_0 is the initial time and y_0 is the initial value.

Definition 0.281.13 (Initial Value Problem)**.** An **initial value problem**, or **IVP**, is a differential equation together with one or more initial conditions.

For a first-order differential equation, an initial value problem has the form

$$y' = f(t, y), \qquad y(t_0) = y_0.$$

The differential equation describes how the function changes, while the initial condition specifies the value of the solution at the initial time t_0.

Definition 0.281.14 (Existence)**.** An initial value problem is said to have **existence** if there is at least one function that satisfies both the differential equation and the initial condition.

For a first-order initial value problem

$$y' = f(t, y), \qquad y(t_0) = y_0,$$

existence means that there is a function $y(t)$, defined on some interval containing t_0, such that

$$y'(t) = f(t, y(t))$$

and

$$y(t_0) = y_0.$$

Definition 0.281.15 (Uniqueness)**.** An initial value problem is said to have **uniqueness** if it has at most one solution on the interval under consideration.

For a first-order initial value problem

$$y' = f(t, y), \qquad y(t_0) = y_0,$$

uniqueness means that if $y_1(t)$ and $y_2(t)$ are both solutions on the same interval I containing t_0, then

$$y_1(t) = y_2(t)$$

for every $t \in I$.

Definition 0.281.16 (Interval of Existence)**.** An **interval of existence** for a solution of an initial value problem is an interval on which the solution is defined and satisfies the differential equation.

For an initial value problem

$$y' = f(t, y), \qquad y(t_0) = y_0,$$

an interval I is an interval of existence if $t_0 \in I$ and there is a solution $y : I \to \mathbb{R}$ such that

$$y'(t) = f(t, y(t))$$

for every $t \in I$, and

$$y(t_0) = y_0.$$

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IEEE 754 Floating-Point Formats

Definition 0.284.1 (IEEE 754 Binary Floating-Point Datum)**.** An *IEEE 754 binary floating-point datum* is a binary encoding partitioned into three fields:

- a *sign field* consisting of one bit,
- an *exponent field* consisting of a fixed number of bits,
- a *fraction field* consisting of a fixed number of bits.

Its interpretation depends on the exponent-field pattern and the fraction-field pattern.

Definition 0.284.2 (Normalized IEEE 754 Binary Number)**.** Let an IEEE 754 binary floating-point format have exponent bias bias. A finite encoded datum is called *normalized* if its exponent field is neither all zeros nor all ones.

If s is the sign bit, if e is the unsigned integer determined by the exponent field, and if f is the binary fraction determined by the fraction field, then the represented value is

$$(-1)^s(1.f)_2 2^{e-\text{bias}}.$$

Definition 0.284.3 (Subnormal IEEE 754 Binary Number)**.** Let an IEEE 754 binary floating-point format have exponent bias bias. A finite encoded datum is called *subnormal* if its exponent field is all zeros and its fraction field is not all zeros.

If s is the sign bit and if f is the binary fraction determined by the fraction field, then the represented value is

$$(-1)^s(0.f)_2 2^{1-\text{bias}}.$$

Definition 0.284.4 (IEEE 754 Infinity Encoding)**.** An IEEE 754 binary floating-point datum encodes *infinity* if its exponent field is all ones and its fraction field is all zeros.

The sign bit determines whether the represented value is $+\infty$ or $-\infty$.

Definition 0.284.5 (IEEE 754 NaN Encoding)**.** An IEEE 754 binary floating-point datum encodes *NaN* (Not a Number) if its exponent field is all ones and its fraction field is not all zeros.

Definition 0.284.6 (IEEE 754 Binary16 Format)**.** The *IEEE 754 binary16 format* is the 16-bit binary floating-point interchange format with the following field layout:

- 1 sign bit,
- 5 exponent bits,
- 10 fraction bits,

- exponent bias 15.

For a normalized finite binary16 datum, the represented value is

$$(-1)^s(1.f)_2 2^{e-15}.$$

Definition 0.284.7 (IEEE 754 Binary32 Format)**.** The *IEEE 754 binary32 format* is the 32-bit binary floating-point interchange format with the following field layout:

- 1 sign bit,
- 8 exponent bits,
- 23 fraction bits,
- exponent bias 127.

For a normalized finite binary32 datum, the represented value is

$$(-1)^s(1.f)_2 2^{e-127}.$$

Definition 0.284.8 (IEEE 754 Binary64 Format)**.** The *IEEE 754 binary64 format* is the 64-bit binary floating-point interchange format with the following field layout:

- 1 sign bit,
- 11 exponent bits,
- 52 fraction bits,
- exponent bias 1023.

For a normalized finite binary64 datum, the represented value is

$$(-1)^s(1.f)_2 2^{e-1023}.$$

Definition 0.284.9 (IEEE 754 Binary128 Format)**.** The *IEEE 754 binary128 format* is the 128-bit binary floating-point interchange format with the following field layout:

- 1 sign bit,
- 15 exponent bits,
- 112 fraction bits,
- exponent bias 16383.

For a normalized finite binary128 datum, the represented value is

$$(-1)^s(1.f)_2 2^{e-16383}.$$

0.285 Interval Arithmetic

0.285.1 Interval Arithmetic

Definition 0.285.1 (Interval Addition)**.** For closed intervals $[a, b]$ and $[c, d]$, their *sum* is

$$[a, b] + [c, d] \ = \ [a + c, \ b + d].$$

Definition 0.285.2 (Interval Subtraction)**.** For closed intervals $[a, b]$ and $[c, d]$, their *difference* is

$$[a, b] - [c, d] \ = \ [a - d, \ b - c].$$

Definition 0.285.3 (Interval Multiplication)**.** For closed intervals $[a, b]$ and $[c, d]$, their *product* is

$$[a, b] \cdot [c, d] \ = \ \big[\min(ac, ad, bc, bd), \ \max(ac, ad, bc, bd)\big].$$

Definition 0.285.4 (Interval Containment)**.** $[a, b] \subseteq [c, d]$ if and only if $c \leq a$ and $b \leq d$.

0.285.2 Interval Arithmetic in $\mathbb{R}^n$

Definition 0.285.5 (Box / Product Interval)**.** Let $a, b \in \mathbb{R}^n$ with $a \le b$ coordinatewise (i.e., $a_i \le b_i$ for all i). The *box* (or *interval vector*) determined by (a, b) is

$$[a, b] \;:=\; \{x \in \mathbb{R}^n : a \le x \le b\} \;=\; \prod_{i=1}^{n} [a_i, b_i].$$

Definition 0.285.6 (Box Containment)**.** For $a \le b$ and $c \le d$ in $\mathbb{R}^n$, we write $[a, b] \subseteq [c, d]$ iff

$$(\forall x \in \mathbb{R}^n)\ \big(x \in [a, b] \Rightarrow x \in [c, d]\big).$$

Equivalently (and most useful computationally),

$$[a, b] \subseteq [c, d] \quad\iff\quad c \le a \ \wedge\ b \le d \qquad \text{(coordinatewise)}.$$

Definition 0.285.7 (Minkowski Addition of Boxes)**.** For boxes $X = [a, b]$ and $Y = [c, d]$ in $\mathbb{R}^n$, their *Minkowski sum* is

$$X \oplus Y \;:=\; \{x + y : x \in X,\ y \in Y\}.$$

For boxes, this closes in the class of boxes and satisfies the endpoint rule

$$[a, b] \oplus [c, d] \;=\; [a + c,\ b + d].$$

Definition 0.285.8 (Box Subtraction / Difference)**.** For boxes $X = [a, b]$ and $Y = [c, d]$ in $\mathbb{R}^n$, define

$$X \ominus Y \;:=\; \{x - y : x \in X,\ y \in Y\}.$$

Then

$$[a, b] \ominus [c, d] \;=\; [a - d,\ b - c].$$

Definition 0.285.9 (Scalar Multiplication of a Box)**.** For $\lambda \in \mathbb{R}$ and a box $[a, b] \subseteq \mathbb{R}^n$, define

$$\lambda[a, b] \;:=\; \{\lambda x : x \in [a, b]\}.$$

Then

$$\lambda[a, b] = \begin{cases} [\lambda a,\ \lambda b], & \lambda \ge 0, \\ [\lambda b,\ \lambda a], & \lambda < 0, \end{cases}$$

where the inequalities are coordinatewise.

Definition 0.285.10 (Coordinatewise / Hadamard Product of Boxes)**.** For $x, y \in \mathbb{R}^n$, write $(x \odot y)_i := x_i y_i$. For boxes $X = [a, b]$ and $Y = [c, d]$, define

$$X \odot Y \;:=\; \{x \odot y : x \in X,\ y \in Y\}.$$

Then $X \odot Y$ is again a box, computed coordinatewise:

$$[a, b] \odot [c, d] \;=\; \prod_{i=1}^{n} \Big[\min S_i,\ \max S_i\Big], \qquad S_i := \{a_i c_i,\ a_i d_i,\ b_i c_i,\ b_i d_i\}.$$

0.285.3 Week 1: Points, Vectors, and Floating-Point Geometry

0.285.4 Week 2: Linear Interpolation and Affine Sampling

0.286 Numerical Geometry

0.286.1 Projects

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Proofs

Numerical PDE

0.287 Numerical PDE

Proofs